

AGENTS

Helix Constitution – Universal AGENTS.md

Field	Value
Revision	2024-10-05 07:48:23
Created	- -
Last modified	- - T : : Z 2024-10-05 11:4:276 5:7
Status	active 11:4:124
Status summary	Amendment of - - - NEW § . . (review-round budget, - max, operator CRITICAL mandate) landed, with reconciling extensions to § . . and § . . (E); full detail in the canonical Constitution.md header (same date). Prior round: Amendment of 11:4:236 - - (round , fix-forward on the independent Opus-xhigh review of fd fa: NO-GO, Blocking / Important / Minor) – § . . (D) corrected in place: (D.) § . . mandated validation is now AT MINIMUM source-GO + § . . (A) pre/post-build gates PLUS every operator-declared QA-deploy-seam blocker (\$. .) and any touched § . .

Field	Value
	critical-invariant scenarios; (D.)/(D.)/(D.) re-keyed from "trunk" to the increment's own INTEGRATION BRANCH, a QA build MAY be cut from a feat/product/flavor branch and for feat/ IS the \$. . (B) live-manual-QA vehicle (removes the trunk \$. . (B) circularity; product/flavor lines now covered); (D.) builds coalesce (in-flight build covers prior increments) and a periodic interval must carry a recorded justification; (D.) toggle-expiry extensions only via a \$. . waiver; heading + preamble now name clause (D); next-clause-letter (E) note added; CM-QA-BUILD-CADENCE-DECLARED + CM-INCREMENT-DELIVERED-NOT-SPEC-WITHHELD contracts amended for coalescing (the latter not flagged by the review, aligned for the same false-positive reason); ledger rows OWED-GATE- .. formatting aligned. Constitution submodule .html/.pdf/.docx twins NOT regenerated this round either (honestly stale per \$. . (E)); main-repo mirror twins regenerated in the paired main-repo commit. Prior round: Amendment of - - - \$. . EXTENDED in place with clause (D) INCREMENTAL DELIVERY GRANULARITY + DECLARED QA-BUILD CADENCE (operator mandate ' 235 - - , CRITICAL : faster initial dev releases for every inheriting project – ship each already-source-proven-correct phase/story/task increment as it clears its own \$. . (A)/(D) bar instead of withholding delivery until the whole spec is done; (D.) delivery unit = the increment; (D.) incomplete spec stays safe via complete vertical slices or LATENT paths – release toggles permitted not mandated, each tracked with an expiry; (D.) consumer-declared QA-build cadence per-increment / periodic / whichever-first , default per-increment , each build a

Field	Value
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	\$. . (B) cycle boundary passing \$. . on its own fingerprint, deliverable form = "consumer" DATA; (D.) background hardening never gates the next increment, except shared-code source-defects and \$. . ; (D.) tracker/doc sync is a hand-off precondition, citing \$. . (F)/\$. . / \$. . /\$. . /\$. . / \$. . ; (D.) manual-QA finds on increment builds are still \$. . coverage escapes). Extension, not a new anchor (\$. . audit: \$. . (A) overlaps validation with ONE candidate's build, \$. . (A)/(D) define per-slice build-ELIGIBILITY; neither obliged an eligible increment to be DELIVERED, set a cadence, kept an incomplete spec safe, or bound hand-off to tracker truth). Research grounding cited in-text (DORA small batches, Hodgson feature toggles, branch-by-abstraction, SAFe develop-on-cadence, GitHub Next Continuous AI preview), with the honest note that no published methodology combining these was found. Recommended gates CM-INCREMENT-DELIVERED-NOT-SPEC-WITHHELD / CM-QA-BUILD-CADENCE-DECLARED / CM-RELEASE-TOGGLE-TRACKED-EXPIRY / CM-QA-HANDOFF-TRACKER-IN-SYNC registered as tracked deferrals OWED-GATE- . . (gate-code NOT claimed shipped). Clause text byte-identical across Constitution.md + the . . mirrors (\$. .); \$. . remains exactly one block-start per file (\$. . (B)). Generated twins: constitution-submodule twins NOT regenerated (honestly stale per \$. . (E), as in e f f); main-repo twins regenerated in e c b eec. (The - - \$. . (D) amendment did not update this header; it is recorded in its own commit
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Field Value

e f f.) Prior round: Amendment of
- - - MODEL-SUBSTRATE CORRECTION
(operator mandate - , CRITICAL ,
verbatim: "Main model to use is Sonet, reviews and
other heavy stuff have to be done by Opus instead
of Fable!!! Fix this permanently on all levels!"):
\$. . (code review) and \$. .
(merge-conflict resolution) re-pinned to Opus at
xhigh ALWAYS - Fable not permitted in any role, the
- - Fable-escalation clause re-
moved, no fallback model (genuine Opus-unavailabil-
ity BLOCKS and is tracked); \$. .
ladder is now sonnet < opus with Sonnet the default
working tier, NEW(D.) prohibits haiku where it
cannot receive its governing context (measured
fact: "Prompt too long (~ k tokens vs
k limit)", request id req_ Cfs-
BXm Kn' enyUT UEaWd, - -), (E)/
(F.)/(F.) re-pinned; derived Fable mentions in
the \$. . ()/\$. . exten-
sions and \$. . /\$. . /
\$. . /\$. . re-pinned to
Opus-xhigh (the fable label vocabulary and ver-
batim label mandate kept as truthful label data).
Recommended gates renamed CM-CODE-REVIEW-OPUS-
XHIGH / CM-MERGE-CONFLICT-OPUS-XHIGH (gate-code =
separate work item, NOT claimed shipped). Earlier
status rounds below that mention Fable are histor-
ical records and are not rewritten. Prior round:
Amendment of - - - NEW
\$. . (semantic Lumen index + indexing-
efficiency gate + universal agent/subagent access-
ibility) + EXTENSIONS \$. . ()-() /
\$. . ()-() / \$. . ()-():
the indexing mechanism (scope renderer, single
writer entry, version-keyed runner patches, scope

Field	Value
	proof, watchdog, lumen scope generator) is consumed BY REFERENCE from constitution/scripts/codegraph
Issues	none
Issues summary	114100 — 114107 114107
Fixed	\$. . mirror
Fixed summary	\$. . lands in lockstep with the Constitution.md \$. . addition.
Continuation	—

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- Universal-vs-project classification (\$. . , User mandate - -)
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- .gitignore + no-versioned-build-artifacts (\$. . , User mandate - -)
- Lowercase-snake_case-naming (\$. . , User mandate - -)
- Submodules-as-equal-codebase + decoupling + dependency-layout (\$. . , User mandate - -)
- No-fakes-beyond-unit-tests + %-test-type-coverage (\$. . , User mandate - -)

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- Type-aware closure-status vocabulary (§ 11.4.35, User mandate - -)
- Reopened-source attribution (§ 11.4.36, User mandate - -)
- Canonical-root inheritance clarity (§ 11.4.37, User mandate - -)
- Mandatory install_upstreams on clone/add (§ 11.4.38, User mandate - -)
- Fetch-before-edit (§ 11.4.40, User mandate - -)
- Installable-asset evidence (§ 11.4.41, User mandate - -)
- Full-suite retest before release tag (§ 11.4.42, User mandate - -)
- Pre-force-push merge-first (§ 11.4.43, User mandate - -)
- Iteration-discipline mandate (§ 11.4.44, User mandate - -)
- TDD-Fix-Discipline (§ 11.4.45, User mandate - -)
- Document revision header (§ 11.4.46, User mandate - -)
- Integration-status-doc maintenance (§ 11.4.47, User mandate - -)
- Validate-recent-work-before-post-flash-tests (§ 11.4.48, User mandate - -)
- Firebase data review (§ 11.4.49, User mandate - -)
- UI-driven video testing (§ 11.4.50, User mandate - -)
- Dual-approach testing (§ 11.4.51, User mandate - -)

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- Deterministic consistency (\$. . , User mandate - -)
- Live-ADB-First maximization (\$. . , User mandate - -)
- Autonomous-Validation (\$. . , User mandate - -)
- Fixed_Summary parity (\$. . , User mandate - -)
- ATM-NNN ticket identifier (\$. . , User mandate - -)
- Reopens-history tracking + per-item Reopens.md (\$. . , User mandate - -)
- Status_Summary parity + two-audience format (\$. . , User mandate - -)
- README.md doc-link section + revision metadata (\$. . , User mandate - -)
- Parallel-development methodology (\$. . , User mandate - -)
- README always-sync (\$. . , User mandate - -)
- Documentation always-sync composite covenant (\$. . , User mandate - -)
- Credentials-handling (\$. .)
- Host-session safety (\$)
- Continuation document (\$.)
- Data safety (\$)
- CLI workflow expectations
- Hierarchy of project authority
- When stuck

This is the **base AGENTS.md** for any CLI agent (Codex, Cursor, Aider, Continue, Gemini CLI, future LLMs)

working on a project that includes the Helix Constitution submodule. Project-level `AGENTS.md` may extend or tighten rules but MUST NOT weaken them.

Last revision: - -

How inheritance works

Because not every CLI agent honours the Markdown `@import` syntax, each consuming project's root `AGENTS.md` MUST do one of:

- . **Reference + restate critical rules** (safest – works for every agent):

```
# <Project> – AGENTS.md

> Base agent rules: `constitution/AGENTS.md` – READ IT FIRST.
> The base file is authoritative for any topic not covered here.

## Critical base rules restated (for agents that don't follow
    links)
- Never commit secrets.
- All commits go through the project's commit wrapper.
- Anti-bluff covenant binds every test (Constitution §11.4).

## Project-specific
- ...
```

- . **Generate at build time** – keep one source-of-truth in this submodule, concatenate with a project-specific delta file. A `compose-agents.sh` script in this submodule's `Upstreams/` (or `scripts/` if added) would output `constitution/AGENTS.md` followed by `---` followed by `AGENTS.project.md`.

Identity & posture

You are working inside a project that has opted into the Helix Constitution. The Constitution is the source of truth for engineering discipline (test coverage, anti-bluff covenant, data safety, host safety, credentials handling, documentation discipline). Whenever this AGENTS.md mentions a `$X.Y` reference, it points to `constitution/Constitution.md`.

The Constitution applies recursively to every submodule of every project that includes it – you **MUST** treat the Constitution's rules as in-force regardless of how deep into the submodule tree you are working.

Top-level invariants for every agent

- **Never assume; verify by reading files.** Especially before any destructive action.
- **Never bluff.** Do not invent file paths, command outputs, test results, or library APIs you have not verified. If you are unsure, say so explicitly and verify.
- **No guessing language** (`$ . . .`). `likely`, `probably`, `maybe`, `might`, `appears`, `seems` are forbidden when reporting causes. Use captured evidence or mark explicitly `UNCONFIRMED:`.
- **Test coverage for every change** (`$`) – pre-build gate, post-build gate, runtime test, AND a meta-test mutation that proves the gate catches regressions.
- **Commit through the project's official wrapper.** Direct `git add` / `git commit` / `git push` on the main repo is forbidden in normal workflow.
- **Never skip hooks** (`--no-verify` , `--no-gpg-sign`).

- **Never force-push.** Force-push requires explicit per-session human authorization AND a green \$. . post-op gate.
- **Hardlinked backup before any destructive op (\$).** Zero excuse.
- **Host safety (\$)** – abort if pre-flight check fails; wrap heavy work in bounded execution scopes; never exceed % of host RAM.
- **Credentials NEVER tracked (\$. .)** – .env patterns git-ignored; runtime-load only; per-service file separation. **Pre-store leak audit (\$. . .A, - -)** – before storing operator-provided credentials in any gitignored config, grep the entire tracked tree AND git history for the literal value(s); surface any findings to operator before storing.
- **CONTINUATION.md kept in sync (\$. .)** – every non-trivial state change updates the continuation document in the same commit.

Critical base rules restated (for agents that don't honour @imports)

Anti-bluff covenant – END-USER QUALITY GUARANTEE (\$. .)

Forensic anchor – verbatim user mandate

(- -):

"We had been in position that all tests do execute with success and all Challenges as well, but in reality the most of the features does not work and can't be used! This MUST NOT be the case and execution of tests and Challenges MUST guarantee the quality,

the completion and full usability by end users of the product!"

The bar for shipping is **not** "tests pass" but **"users can use the feature."** Every PASS MUST carry positive evidence captured during execution that the feature works for the end user. Metadata-only PASS, configuration-only PASS, absence-of-error PASS, and grep-based PASS without runtime evidence are critical defects.

Mutation-paired gates (§ .)

Every new gate MUST be paired with a mutation entry in the project's meta-test harness that mutates the asserted condition and asserts the gate now FAILs. A gate without a paired mutation is a bluff gate and is forbidden.

Recorded-evidence requirement (§ . .)

Every PASS for a user-visible feature MUST be cross-checked against a captured recording + action timeline. A PASS that lacks at least one matched timeline event in the analyzer findings is a § . PASS-bluff.

Test-interrupt-on-discovery (§ . .)

The moment any defect is re-discovered, re-produced, or newly identified during a test cycle, the cycle MUST stop. Then: systematic debugging → fix at root cause → four-layer test coverage (pre-build / post-build / runtime / meta-test paired mutation) → full rebuild → re-deploy on every target → full retest from beginning.

No-guessing mandate (§ . .)

Forensic anchor – verbatim user mandate

(- -):

"'LIKELY' is guessing, we MUST NOT have guessing, since it can be or may not be! No bluffing and uncertainty is allowed at any cost! We MUST always know exactly precisely what is happening exactly, in any context, under any conditions, everywhere!"

Forbidden vocabulary when reporting causes: likely , probably , maybe , might , possibly , presumably , seems , appears to , guess , seemingly , apparently , perhaps , supposedly , conjectured , and synonyms.

Item-type tracking (§ . .)

Every active item in the project Issues file MUST carry a `**Type:**` line within eight non-blank lines of its heading. Three-value CLOSED vocabulary: Bug (product defect / regression / user-visible broken behaviour), Feature (new capability not previously offered to end users), Task (internal workstream – refactor, doc, infra, gate, audit; the lowest-stakes default when ambiguous). The Issues_Summary file carries the Type column for every active item. All three Issues / Issues_Summary / Fixed file types kept in sync (Markdown + HTML + PDF). Pre-build gates CM-ITEM-TYPE-TRACKING + CM-COVENANT-114-16-PROPAGATION enforce the mandate. No escape hatch.

Universal-vs-project classification

(\$. . , User mandate - -)

Before adding ANY new rule / mandatory constraint / covenant clause / gate / "MUST"-bearing statement, classify it: **universal** (any project → constitution submodule) or **project-specific** (specific hardware / vendor / package / region → project / submodule layer). Commit message MUST carry `Classification:` line + one-sentence rationale. When uncertain, default to project-specific. Gate `CM-UNIVERSAL-VS-PROJECT-CLASSIFICATION` audits new rule commits. Paired mutation enforces the gate is not a bluff.

Subagent-driven-by-default (\$. . , User mandate - -)

When the runtime supports subagent delegation (Agent tool / task runner / sub-session), the primary agent MUST default to subagent delegation for multi-step scope (≥ phases), parallelisable independent subtasks, long-running diagnostic loops, OR specialised domain workflows. Foreground-only is reserved for single-file edits, mid-execution operator clarification, critical-state sequencing (commits / pushes / tags), or tasks under ~ s. Discipline: tight scope (- tasks), checkpoint commits per task, anti-stall protection in prompts, anti-bluff verification of subagent claims via repo state. Parallel subagents MUST partition non-overlapping files; parent `commit_all.sh --auto-cascade` bundles via `git add -A`. Gate `CM-SUBAGENT-DELEGATION-AUDIT` (per consuming project) flags foreground multi-step work as a violation. Paired mutation enforces the gate is not a bluff.

Script documentation mandate (§ . . , User mandate - -)

Every Bash / shell / POSIX-sh script under `scripts/` / `bin/` / `tests/` / library dirs / CI hooks (depth-N recursive) MUST carry () an in-source documentation block at the top (Purpose / Usage / Inputs / Outputs / Side-effects / Dependencies / Cross-references) AND () an external user guide under `docs/scripts/<script-name>.md` (Overview / Prerequisites / Usage examples / Edge cases / Internal behaviour / Related scripts / Last verified date). When a script is modified, BOTH the in-source block AND the external user guide MUST be updated in the SAME commit. **No documentation ever out of sync with its codebase.** Gate `CM-SCRIPT-DOCS-SYNC` enforces. Paired mutation strips the invariant.

Fixed-document column-alignment (§ . . , User mandate - -)

The closed-archive tracker (`Fixed.md`) MUST mirror the open-work tracker (`Issues.md` + `Issues_Summary.md`) along the same lifecycle axes – at minimum **Status** and **Type**. Concretely: () every `###` / `####` heading in `Fixed.md` carries `**Status:**` (from closed-set `{Fixed (→ Fixed.md) | Fixed – pending device verification | Fixed – RECLASSIFIED}`.) and `**Type:**` (`{Bug | Feature | Task}`) within non-blank lines of the heading; () `Fixed_Summary.md` companion exists with column structure `# | Level | Status | Type | One-line description` matching `Issues_Summary.md` exactly; () all three formats (`.md` , `.html` , `.pdf`) for BOTH `Fixed` and `Fixed_Summary` stay in sync via the same single-shot wrapper that handles `Issues` + `Issues_Summary`; () closure is atomic – resolved items migrate from `Issues.md` to `Fixed.md` in the same commit. Pre-build gate `CM-FIXED-COLUMN-ALIGNMENT` (+ invariants: file-exists,

header-has-Status+Type, mtime ≥ Fixed.md, generator script present, sync wrapper invokes it, HTML+PDF exports present). Paired mutation strips Status column from Fixed_Summary header → gate FAILs. Classification: universal (per § 1.1 4 2.1 2.0 2.6 0.5 1.4).

Operator-blocked status + self-resolution exhaustion (§ 1.1 4 1.5, User mandate - -)

Operator-blocked is the § 1.1 4 1.5 Status closed-set's th value alongside {Queued | In progress | Ready for testing | In testing | Reopened | Fixed (→ Fixed.md)} . **Last-resort classification** – agent MUST first exhaust applicable self-resolution paths: (a) CLI / ADB / SSH / API access already available, (b) subagent delegation per § 1.1 4 1.5, (c) existing repo tooling, (d) captured fallback (synthetic event / asset substitution / § 1.1 4 1.5 SKIP), (e) external research per § 1.1 4 1.5. Every Operator-blocked item MUST carry **Operator-Block-Details:** within non-blank lines of its heading stating WHAT (action) / WHY (each exhausted alternative) / UNBLOCK CONDITION (observable signal) / WHO (contact or doc pointer). Issues_Summary.md lists it as a sortable Status value. Items re-evaluated every Nth tag cycle (≥ rd recommended). Fake Operator-blocked without exhaustion audit = § 1.1 4 1.5 covenant violation at planning layer (PASS-bluff equivalent). Gates CM-ITEM-OPERATOR-BLOCKED-DETAILS + CM-OPERATOR-BLOCKED-SELF-RESOLUTION-AUDIT (every NEW Operator-blocked commit carries an "Attempted: a – ...; b – ...; c – ..." trail). Classification: universal (per § 1.1 4 1.5). No escape hatch.

Document-sync commit discipline (`$ . . , User mandate` `- -`)

Every project tracking work items through an Issues / Fixed lifecycle MUST provide a **lightweight commit path** distinct from the full-repo commit wrapper. Stages, commits, pushes ONLY the status-tracking doc set: Issues + Issues_Summary + Fixed + Fixed_Summary + CONTINUATION + their HTML + PDF exports + auto-generated audit artifact. Wrapper MUST: (a) auto-invoke the project's export-regeneration pipeline first so Markdown + HTML + PDF stay in sync; (b) stage ONLY the explicit doc-set list – NEVER `git add -A`; (c) use a separate flock disjoint from the . full-tree wrapper; (d) push to every parent-repo remote; (e) exit `3` on nothing-to-commit (informational). MUST be invocable standalone OR via a delegation flag on the full-tree wrapper (e.g. `commit_all.sh --docs-only`). Inherits \$ preflight. Gate `CM-COMMIT-DOCS-EXISTS` + paired mutation. Composes with `$ . . / $ . . / $ . . / $ . .`. Classification: universal (per `$ . .`). No escape hatch – doc-status drift is a `$ . .` `PASS-bluff` at the documentation layer.

Build-resource stats tracking (`$ . . , User mandate` `- -`)

Every project under this Constitution with a build exceeding minute wall-clock MUST run a host-side resource sampler for every build – samples `/proc/meminfo` + `/proc/loadavg` + `/proc/stat` + `/proc/diskstats` at a fixed interval (recommended s); writes TSV. On stop, computes `min / max / mean / p` per metric; appends one TSV row per build to a registry; regenerates a Markdown report (Stats.md) whose top surfaces **ever-values** (min / max / mean across all tracked builds) and whose body lists per-build entries most-recent-first with SUCCESS / FAIL /

UNKNOWN +, reason. Stats.md exported to Stats.html + Stats.pdf through the project's normal export pipeline per § 1.1.4.12. Triple committed via the § 1.1.4.12 lightweight doc-sync wrapper. Sampler MUST stay under 128 MB RSS / 10 % CPU. Stop hook called from both success AND failure paths of the build wrapper. Gate CM-BUILD-RESOURCE-STATS-TRACKER

- paired mutation. Classification: mixed (per § 1.1.4.12). Composes with § 1.1.4.12 / § 1.1.4.12 / § 1.1.4.12 / § 1.1.4.12. No escape hatch – build-resource debugging without time-series data is the bluff this anchor forbids.

Full-Automation-Coverage (§ 1.1.4.12, User mandate – –)

Forensic anchor – verbatim user mandate

(– –):

"Make sure that every feature, every functionality, every flow, every use case, every edge case, every service or application, on every platform we support is covered with full automation tests which will confirm anti-bluff policy and provide the proof of fully working capabilities, working implementation as expected, no issues, no bugs, fully documented, tests covered!"

No feature / functionality / flow / use case / edge case / service / application on any supported platform may be considered deliverable until automation tests prove six invariants: () anti-bluff posture with captured runtime evidence (§ 1.1.4.12 + § 1.1.4.12); () proof of working capability end-to-end on target topology (§ 1.1.4.12, no

mocks); () implementation matches documented promise;
 () no open issues/bugs surfaced (cross-checked vs
 \$. . / \$. .); () full documentation
 (user manual + \$. . for scripts) kept in sync
 via \$. . ; () four-layer test floor per \$
 (pre-build + post-build + runtime + paired mutation).
 Consuming projects publish a coverage ledger (feature ×
 platform × invariant × status) regenerated at release-gate
 sweep. Classification: universal (\$. .).
 Severity-equivalent to a \$. PASS-bluff at the
 release-gate layer. No escape hatch. See Constitution
 \$. . for the full mandate.

Constitution-Submodule Update Workflow

(\$. . , User mandate
 - -)

Forensic anchor – verbatim user mandate

(- -):

"Every time we add something into our root
 (constitution Submodule) Constitution, CLAUDE.MD and
 AGENTS.MD we MUST FIRST fetch and pull all new
 changes / work from constitution Submodule first! All
 changes we apply MUST BE committed and pushed to all
 constitution Submodule upstreams! In case of conflict,
 IT MUST BE carefully resolved! Nothing can be broken,
 made faulty, corrupted or unusable! After merging full
 validation and verification MUST BE done!"

Before any modification to `constitution/`
`{Constitution,CLAUDE,AGENTS}.md` , execute in order: () fetch
 + pull (`--ff-only` or `--rebase`; never `strategy=ours` /
`unrelated-histories` without authorization) so the
 submodule is at upstream tip before editing; () apply

the change with \$. . . classification + verbatim
mandate quote; () validate (meta_test_inheritance.sh +
no merge-conflict markers + cross-file consistency); ()
commit (governance files only, no git add -A)

- push to EVERY configured upstream remote (\$.
violation if not); () careful conflict resolution
preserving union of governance content – force-push
forbidden (\$.); () post-merge validation: git
submodule update --remote --init + re-run cascade verifier
(CONST-) confirming the new clause reaches every
owned submodule; () bump consuming project's
.gitmodules pointer to new HEAD in the SAME commit as
cascade work – out-of-sync pointer = \$. .
violation. Classification: universal (\$. .).
Severity-equivalent to force-push without \$.
authorization. No escape hatch. See Constitution
\$. . for the full pipeline (operational
scope, cross-cutting reach).

Submodule-dependency-manifest (\$. . , User mandate - -)

Every owned-by-us submodule MUST ship helix-deps.yaml
listing its own-org dependencies: {name, ssh_url, ref, why,
layout: flat|grouped} . Tooling incorporate-submodule <ssh-url>
adds the submodule at the parent project's canonical path
(CONST- (C)) + reads its helix-deps.yaml + recurses +
aborts on conflicting refs + emits <root>/helix-
manifest.yaml audit record. Anti-bluff: each manifest
paired with a Challenge that bootstraps from scratch,
asserts layout matches manifest, runs submodule tests,
captures wire evidence. Without the proof, the manifest is
a \$. . violation. This rule is the operational
complement of \$. . / CONST- (C) – manifests
are the bridge that lets consumers reconstruct the
dependency graph at the parent root. Classification:

universal (§ 1.1.4.3.2). See Constitution § 2.0.2.6.0.5.1.5 for the full mandate.

Post-constitution-pull validation

(§ 1.1.4.3.2, User mandate
- -)

Whenever a project's constitution submodule is fetched + pulled with any content change, run `scripts/verify-all-constitution-rules.sh` BEFORE the new HEAD is treated as canonical for other work. The sweep re-runs the governance-cascade verifier + every implementable rule gate. (CONST- .gitignore audit, CONST- (C) nested-own-org audit, CONST- case audit, CONST- (A) mock-from-production audit, CONST- anti-bluff smoke). Failures populate Issues per § 1.1.4.3.2 (Status: Reopened, Type: Bug); closure requires positive-evidence per § 1.1.4.3.2. Pull-time invocation auto-triggered by `git submodule update --remote constitution`. Anti-bluff: sweep's own meta-test (paired mutation § 1.1.4.3.2) plants a violation per gate, asserts sweep FAILs. This is the **enforcement engine** for every other rule – without it, new rules are decorative anchors rather than enforced gates. Classification: universal (§ 1.1.4.3.2). See Constitution § 2.0.2.6.0.5.1.5 for the full mandate.

.gitignore + no-versioned-build-artifacts

(§ 1.1.4.3.2, User mandate
- -)

Forensic anchor – verbatim user mandate

(- -):

"every project module, every Submodule, every service and application MUST HAVE proper .gitignore file! We

MUST NOT git version build artifacts, cache files, tmp files, main .env file(s) or any files containing sensitive data, API keys or token! ... If any violation is detected it MUST be fixed before commit is executed!"

Every project module / submodule / service / application MUST ship a proper `.gitignore`. Forbidden-from-version-control: build artefacts (`bin/` , `build/` , `dist/` , `target/` , `*.exe` , `*.so` , `*.class` , `*.pyc`), cache files (`__pycache__/` , `node_modules/` , `.gradle/` , `.terraform/`), temp files (`*.tmp` , `*.swp` , `.DS_Store`), sensitive data (`.env` , `.env.*` except `.env.example` placeholder, `*.pem` , `*.key` , `id_rsa*` , `.netrc` , `secrets/` , `api_keys.sh`), generated logs (`*.log` , `coverage.out` , `htmlcov/`), OS/IDE personal state (`.idea/` , `.history/`).

Anti-bluff invariant: ignore-line alone is not enough – no file matching forbidden patterns may be tracked. Pre-commit attention: every author inspects `git diff --staged + git status` BEFORE commit; violations abort the commit (unstage, add to ignore, scrub if already-tracked). Gate `CM-GITIGNORE-PRECOMMIT-AUDIT` + paired mutation. Recreatable-content test: if a documented mechanism regenerates the file from sources, it's a build derivative and MUST be ignored.

```
Secret-leak intersection ($ . . /
CONST- / $ . ): a .env leak is BOTH a
$ . . and a $ . . violation; rotation +
post-mortem required.
```

[illegible]

\$. . , \$. . , \$. . ,
 \$. . , CONST- . See Constitution
 \$. . for the full mandate.

Lowercase-snake_case-naming (\$. . ,
User mandate - -)

Forensic anchor – verbatim user mandate

(- -):

"naming convention for Submodules and directories (applied deep into hierarchy recursively) - all directories and Submodules MSUT HAVE lowercase names with space separator between the words of '_' character (snake-case)! All existing Submodules and directories which are not following this rule MUST BE renamed! ... NOTE: Rules lowercase / snake-case do apply to all project files as well and references to it and from them!"

Every directory / submodule / file MUST use lowercase snake_case. Non-compliant names MUST be renamed; every reference (configs, docs, source imports, governance files) updated atomically (reference drift = \$. . violation of equal severity). Exceptions (common-sense, technology-preserving): language- mandated case for Java/ Kotlin/Android/Apple/C /Swift inside the language root; vendor third-party submodules; build artefacts.

Upstreams/ → upstreams/ transition: constitution submodule's install_upstreams.sh MUST support BOTH directory names during migration; lowercase wins when both present.

1 2 3 4 3 2 1

1 2 3 4 5 6

11 4 17

No-fakes-beyond-unit-tests + %-test-type-coverage (\$. . , User mandate - -)

Forensic anchor – verbatim user mandate

(- -):

"Mocks, stubs, placeholders, TODOs or FIXMEs are allowed to exist ONLY in Unit tests! All other test types MUST interact with real fully implemented System! No fakes, empty implementations or bluffing is allowed of any kind! All codebase of the project MUST BE % covered with every supported test type."

Two invariants. (A) Mocks/stubs/fakes/placeholders/TODOs/FIXMEs/"for now" patterns PERMITTED only in unit-test sources; non-unit tests (integration, E E, full-automation, security, DDoS, scaling, chaos, stress, performance, benchmarking, UI, UX, Challenges, HelixQA) MUST exercise the real, fully implemented system. Production code MUST NOT import mock paths. Gate CM-NO-FAKES-BEYOND-UNIT-TESTS + paired mutation. (B) Codebase MUST be covered by every supported test type the domain warrants: unit, integration, E E, full-automation, security, DDoS, scaling, chaos, stress, performance, benchmarking, UI, UX, Challenges (vasic-digital/Challenges submodule fully incorporated), HelixQA (HelixDevelopment/HelixQA submodule fully incorporated, with full autonomous QA sessions executing every registered test bank).

Required dependency submodules (recursive per CONST-): Challenges (git@github.com:vasic-digital/Challenges.git) + HelixQA (git@github.com:HelixDevelopment/HelixQA.git) + any other functionality submodules under vasic-digital/HelixDevelopment orgs the project depends

on. Pointers bumped to upstream HEAD in same commit as cascade work (§ 11.4.1, step 1).

Classification: universal (§ 11.4.25). Severity-equivalent to a § 11.4.26 PASS-bluff at the release-gate layer. No escape hatch. Composes with § 11.4.25, § 11.4.26, § 11.4.27 - § 11.4.28 (esp. § 11.4.28 - strict per-type-of-test expansion). See Constitution § 11.4.33 for full mandate.

Type-aware closure-status vocabulary (§ 11.4.34, User mandate - -)

Every project that tracks work items by Type per § 11.4.34 MUST close them with the Type-appropriate closure-status word: Bug → Fixed (→ Fixed.md), Feature → Implemented (→ Fixed.md), Task → Completed (→ Fixed.md). The (→ Fixed.md) suffix is preserved across all three so existing migration tooling (atomic Issues.md → Fixed.md move per § 11.4.34) keeps working. Generators treat the three terminal values as semantically equivalent (all closed, positive evidence captured) but preserve the literal in emitted docs. Closing a Feature with Fixed (→ Fixed.md) or a Task with Implemented (→ Fixed.md) is a § 11.4.34 violation. Pre-build gate CM-CLOSURE-VOCAB-TYPE-AWARE. Composes with § 11.4.34 / § 11.4.34 / § 11.4.34. Classification: universal (per § 11.4.34). No escape hatch.

Reopened-source attribution (§ 11.4.35, User mandate - -)

Every Issues.md heading whose **Status:** is Reopened MUST carry a **Reopened-Details:** line within non-blank lines of the heading, capturing four sub-facts: By: AI

or `User` ; **On:** ISO date; **Reason:** one of `{ test-failed | manual-testing-detected | captured-evidence-contradicts | end-user-report | cycle-re-discovered | design-reconsidered }` or explicit free text; **Evidence:** path or short description of the captured artefact. Reopens without evidence are § 1.1.4.35 violations: the reopen IS a demotion-from-Fixed change. Issues_Summary.md Status column MUST distinguish Reopened sub-states by source (e.g., `Reopened (AI: test-failed)` vs `Reopened (User: manual-testing)`). Pre-build gate `CM-ITEM-REOPENED-DETAILS` mirrors § 1.1.4.35 walk pattern. Composes with § 1.1.4.35 / § 1.1.4.35 / § 1.1.4.35 . Classification: universal (per § 1.1.4.35). No escape hatch.

Canonical-root inheritance clarity (§ 1.1.4.35 , User mandate - -)

The constitution submodule's three files (`constitution/Constitution.md` , `constitution/CLAUDE.md` , `constitution/AGENTS.md`) ARE the canonical root – also called the parent files. Universal rules per § 1.1.4.35 live here.

The consuming project's repository-root files (`<project-root>/CLAUDE.md` , `<project-root>/AGENTS.md` , optionally `<project-root>/Constitution.md` or equivalent) are consumer extensions. They open with the inheritance pointer (either `@constitution/CLAUDE.md import` or `## INHERITED FROM constitution/CLAUDE.md` heading). Project-specific rules per § 1.1.4.35 live here.

When in doubt: universal rule → constitution submodule; project-specific rule → consumer's file. Default consumer-side when uncertain. "Parent CLAUDE.md" / "root Constitution" → constitution submodule file at `constitution/<filename>` . "Project CLAUDE.md" / "this

project's AGENTS.md" → consumer file at <project-root>/<filename> . Moving a rule between layers MUST be a visible commit – git mv + explicit "Lifted from to constitution per \$. . . " / "Demoted from constitution to per \$. . . " line in the message. Pre-build gate CM-CANONICAL-ROOT-CLARITY . Composes with \$ Classification: universal (per \$. . .). No escape hatch.

Mandatory install_upstreams on clone/add (\$. . . , User mandate - -)

Forensic anchor – verbatim user mandate
(- -):

"Every Submodule or Git repository we add or clone MUST BE upstreams installed using Upstreamable utility which MUST BE available through exported paths of the host system (in .bashrc or .zshrc) using install_upstreams command executed from the root of the cloned (added) repository – only if in it is Upstreams or upstreams directory present with bash script files (recipes) for all repository's upstreams!"

Every clone / add of a Git repository under any consuming project MUST be followed by install_upstreams invocation from the repository's root IF its tree contains upstreams/ (or legacy Upstreams/ per \$. . . transition) populated with *.sh recipe files. The utility (installed on operator's PATH via .bashrc / .zshrc , implementation lives in this constitution submodule) reads recipe files + configures every declared upstream as a named git remote + fans out origin push URLs.

Skipping the invocation when `upstreams/` is present
 silently breaks `$ .` (multi-upstream push is the norm).
 Gate `CM-INSTALL-UPSTREAMS-ON-CLONE` + paired mutation.
 Automation: `incorporate-submodule` (`$ . .`) auto-
 invokes; manual invocation also supported. Pre-commit
 check: `git remote -v | grep -c push` reports expected
 upstream count.

Classification: universal (`$ . .`). Composes with
`$ , $ . , $ , $ . , $ . . , $ . . ,`
`$ . . , $ . . , $ . . ,`
`$ . . .`. See Constitution `$ . .` for the
 full mandate.

Fetch-before-edit (`$ . .` , User mandate `- -`)

Forensic anchor – verbatim user mandate
 (`- -`):

"Make sure that `feedback_fetch_before_edit` memory rule
 is part of our constitution Submodule – the root
 Consitution, AGENTS.MD and CLAUDE.MD. Validate and
 verify that Proejct-Toolkit and all Submodules do
 inherit all of them!"

FIRST git-touching action of any session: `git fetch --all --
 prune && git log --oneline HEAD..@{u} && git submodule foreach --
 recursive 'git fetch --all --prune --quiet'` . If `HEAD..@{u}` is
 non-empty, integrate (ff-merge / rebase / surface per
`$ . .`) BEFORE any local edit, scanner, or test.
 Skipping on "nothing could have changed" is a `$ . .`
 (no-guessing) violation – remote state is not knowable
 without a fetch.

The originating ³⁰ - - incident: agent ran `git remote remove gitee` as a fresh task, but the work had been committed upstream by a parallel agent ^{11 4 28} minutes earlier – local config edit was a no-op echo of already-merged history. A ^{11 4 26} ¹ -second fetch would have caught it. ^{11 4 31}

Scope: consuming project root + every owned submodule recursively (`$ . . .`) + the constitution submodule itself (`$ . . . step`) + any dependency cloned via `incorporate-submodule` (`$ . . .`) or `git submodule add` (`$ . . .`). Gate `CM-FETCH-BEFORE-EDIT-AUDIT` + paired mutation per `$ . . .`. Classification: ^{11 4 38} universal (per `$ . . .`). ^{2 0 2 6 0 5 1 7} No escape hatch.

Installable-asset evidence (`$ . . .`, User mandate - -)

For any user-distributable build artifact (package, bundle, installer, or container image), tests/challenges **MUST** open the artifact and verify each user-visible asset is **present** and **non-degenerate**. A PASS without artifact-opening verification is a `$ . . .` PASS-bluff. The failure mode: source file exists → build packages it → source-layer checks pass → artifact produced with asset stripped or misconfigured, and no gate opens the artifact to verify.

Required per consuming project: one challenge script per artifact type that opens the produced artifact and verifies every declared user-visible asset. The challenge **MUST** run as part of the standard QA gate. Classification: universal (per `$ . . .`). No escape hatch. See Constitution `$ . . .` for the full mandate.

Full-suite retest before release tag (§ . . , User mandate - -)

Release tag MUST NOT be created until a COMPLETE retest with ALL existing tests runs on a clean baseline AFTER every workable item in the batch is done, fixed, polished, individually verified. Spot-check retests FORBIDDEN – miss interaction defects. Comprises steps: () pre-build full sweep, () post-build full sweep, () on-device -phase cycle on EVERY owned device, () meta-test full mutation sweep, () Challenge bank full sweep, () Issues.md/Fixed.md state audit, () CONTINUATION.md sync check. Typically - h elapsed, NOT optional, NOT abbreviated. Composes with § . . (per-fix retest still required at fix granularity) + § . . (full-suite is authoritative baseline for closures) + § . . (per-feature on-device validation runs as step). Classification: universal (per § . .). No escape hatch. See Constitution § ^{1 1 4 4 1} . . for the full mandate.

Pre-force-push merge-first (§ . . , User mandate - -)

Forensic anchor – verbatim user mandate
(- -):

"make sure we bring everything from branches to our side before forc push is done! Afer everything is safely and fully merged and all potential conflicts (if any) resolved, then do force push! make sure nothing isnlost, broken or corrupted on bith sides!"

Any force-push authorised under CONST- MUST be preceded by a mechanical -step merge-first pipeline:

() fetch every remote (git fetch --all --prune --tags);
 () integrate every divergent commit locally (rebase /
 merge / operator-confirmed cherry-pick per appropriate
 strategy); () audit the integrated tree (no conflict
 markers anywhere in governance + source + test files; no
 file silently dropped; previously-passing tests still
 pass; captured-evidence artefacts still validate); ()
 only THEN execute git push --force-with-lease .

Two-gate composition with CONST- . \$. .
 does NOT relax CONST- – it adds a SECOND mechanical
 gate. CONST- alone authorises a push that loses
 remote work; \$. . alone risks pushing without
 operator awareness. Both required.

Three failure modes prevented: remote-side content loss
 (parallel sessions overwritten), stale-state acts (--
 force-with-lease reads stale local refs without prior
 fetch), conflict-driven corruption (markers committed
 verbatim – observed – – in helix_qa +
 containers governance files). Verification artefact: docs/
 changelogs/<tag>.md "Force-push merge-first audit" section
 captures fetch output, divergence log, integration
 strategy, conflict-marker scan, test delta, push output
 with lease SHA, + CONST- authorisation quote. Gate
 CM-FORCE-PUSH-MERGE-FIRST

- paired..mutation. Classification: universal
 (\$. . .). No escape hatch. Composes with \$. ,
 \$. . , \$. . , \$. . ,
 \$. . , \$. . , \$. . ,
 CONST- , CONST- . See Constitution
 \$. . for the full mandate.

Iteration-discipline mandate (§ 11.4.4, User mandate 11.4.9.3)

Work proceeds in priority-ordered cycles. Five mandatory steps per cycle: () select TOP + MIDDLE critical only (defer LOW until critical batch closed), () batch implementation per § 11.4.2 + § 11.4.5, () smoke gate (< 11.4.4 min), () ONLY if smoke GREEN and no new user/operator report → § 11.4.5 full retest (- h), () release-ready OR loop back. Composes with § 11.4.2. / § 11.4.5 / § 11.4.3 / § 11.4.5 / § 11.4.5 - § 11.4.5 is the meta-loop conductor binding them. Every PASS in smoke and full retest MUST carry positive captured evidence (§ 11.4.5 + § 11.4.5). Tests AND Challenges bound equally. No escape hatch. Classification: universal (per § 11.4.3). See Constitution § 11.4.4.3 for the full mandate.

TDD-Fix-Discipline (§ 11.4.5, User mandate 11.4.9.3)

Every fix follows -step workflow: **RED** (failing test FIRST, real product defect, captured evidence) → **LIVE-ADB-PROBE** (try on running device when feasible; INFEASIBLE for kernel / framework / HAL / vendor / init.rc / sepolicy / ro. / Android.bp - must rebuild; cite

LIVE_PROBE_INFEASIBLE: <reason> in commit) → **GREEN** (source patch, batched per § 11.4.5, four-layer coverage per § 11.4.5) → **VERIFY** (re-run RED test, PASSes under SAME conditions per § 11.4.5, -iteration reliability per § 11.4.5) → **DOCUMENT** (Issues.md → Fixed.md with type-aware closure § 11.4.5, CLAUDE.md Applied Fixes row, changelog, guides, HelixQA bank, CONTINUATION.md § 11.4.5 - all in SAME commit). Test-after-fix is a § 11.4.5 PASS-bluff: test agrees with fix but does not catch the bug. No escape hatch. Classification: universal

(\$. .). See Constitution \$. . for the full mandate.

Document revision header (\$. . , User mandate - -)

Every IN-scope tracked Markdown document (Issues.md / Issues_Summary.md / Fixed.md / Fixed_Summary.md / CONTINUATION.md / docs/guides/ / docs/research/ / docs/scripts/ / docs/changelogs/ / docs/superpowers/plans/ / docs/hardware/ / every other docs/ .md) carries **Revision:** N (monotonic positive integer)

- **Last modified:** YYYY-MM-DDTHH:MM:SSZ directly below the H title. CLAUDE.md / AGENTS.md / README / LICENSE / VERSION / rendered HTML / rendered PDF are OUT of scope. Agents MUST read the **Revision:** N literal from the document head when reporting document state, NEVER infer freshness from filename mtime or commit log – both can lie. Agents writing to an IN-scope doc MUST invoke scripts/doc_revision_bump.sh <file> as the final step before staging, OR rely on the pre-commit hook. Agents MUST NOT manually edit the revision number – only the bump script is authoritative.

Pre-build gates CM-DOC-REVISION-HEADER-PRESENT + CM-COVENANT-114-44-PROPAGATION + paired mutations. Composes with \$. (CONTINUATION.md header reuse), \$. , \$. , \$. , \$. . No escape hatch. Classification: universal (\$. .). See Constitution \$. . for the full mandate.

Integration-status-doc maintenance

(`$ . . , User mandate`
`- -`)

Every non-trivial domain integration MUST have a `docs/`
`<domain>/<integration>/Status.md` carrying the `$ . .`
revision header + auto-synced HTML+PDF + auto-colored
per `$ . .` + sync wrapper + captured-evidence
table (every claim cites the test log / recording / sink-
probe path) + closed status vocabulary (PASS / FAIL /
SKIP / PENDING_FORENSICS / OPERATOR-BLOCKED) + operator-
blocked items at the top + reference from CONTINUATION.md
`$` when non-terminal. `$ . .` is the generic form
of `$ . .` applied to every integration domain.

Pre-build gates `CM-COVENANT-114-45-PROPAGATION` + `CM-AF-`
`INTEGRATION-STATUS-DOCS` + paired mutations. Composes
with `$ . . / $ . . / $ . . /`
`$ . . / $ . . / $ . . /`
`$ . . / $ . .`. No escape hatch.
Classification: universal (`$ . .`). See
Constitution `$ . .` for the full mandate.

Validate-recent-work-before-post-flash-tests

(`$ . . , User mandate`
`- -`)

Every post-flash run MUST first run a recent-work
validation pass (targeted tests for Issues.md In-
progress / Ready-for-testing / Reopened + Fixed.md
last- -days + CONTINUATION.md `$` active work). Full
`test_all_fixes.sh` runs ONLY after % green.

Helper: `scripts/testing/recent_work_validate.sh --device`
`<serial>` writes `/data/local/tmp/.recent_work_validated`; full
suite refuses without it. Each recent fix needs paired

§ 11.4.42 RED-then-GREEN – naked GREEN is a bluff.

Composes with § 11.4.43 + § 11.4.44 + § 11.4.45

- § 11.4.46 + § 11.4.47 + § 11.4.48 +

§ 11.4.49. Pre-build gates

CM-COVENANT-114-46-PROPAGATION + CM-AF-RECENT-WORK-

VALIDATION-GATE + CM-AF-VALIDATION-ARTIFACT-FILE + paired

mutations. Classification: universal (§ 11.4.50).

No escape hatch. See Constitution § 11.4.51 for the full mandate.

Firestore data review (§ 11.4.52, User mandate – –)

Before every "bigger working round" (pre-build / pre-flash / pre-tag blocking; daily / post-deployment burn-in non-blocking) the operator/loop MUST execute scripts/firebase/review_round.sh . The pass queries Crashlytics (fatals + non-fatals + ANRs) + Analytics + Performance, classifies findings by severity, dedup- maps to existing Issues.md entries via a three-tier algorithm (exact Firestore Issue-ID match → stacktrace-similarity cluster hash → operator merge review), and drafts new Issue entries for unrecognised findings with full Firestore Console URL refs + § 11.4.53 (a) systematic-debugging output.

Five mandatory elements: -trigger cadence, -source query (all three sources), Issues.md output with Firestore metadata (Issue IDs + URL + Cluster Hash / KPI / Funnel), -tier dedup, and comprehensive root-cause analysis per Issue.

Pre-build gates CM-COVENANT-114-47-PROPAGATION + CM-AF-

FIREBASE-REVIEW-CADENCE + CM-AF-FIREBASE-ISSUE-XREF +

paired mutations. Composes with § 11.4.54 /

§ 11.4.55 (a) / § 11.4.56 / § 11.4.57 /

§ 11.4.58 / § 11.4.59 / § 11.4.60 /

§ 11.4.42. / § 11.4.47. / § 11.4.48. /
 § 11.4.49. / § 11.4.50. / § 11.4.51. /
 § 11.4.52. / § 11.4.53. No escape hatch – no --
 skip-firebase-review. Classification: universal
 (§ 11.4.54). See Constitution § 11.4.55 for the
 full mandate.

UI-driven video testing (§ 11.4.56, User mandate – –)

Every test that asserts video playback on a secondary display MUST traverse the user-equivalent UI path (launcher icon → app home → content list → tile tap → playback → in-app pause/resume → back button stop). NOT Intent/Broadcast shortcuts (am start -aVIEW , cmd media_session play). Real uiautomator dump element resolution

- input tap X Y . Coverage: every video-capable app in PRODUCT_PACKAGES + every stream type (progressive HTTP / HLS / DASH / RTMP / file-local / DRM) + every codec (H. 11.4.57 / VP / AV / MPEG- / MP V video; AC- / E-AC- / TrueHD/DTS/DTS-HD/MLP/Opus/AAC audio). Secondary display verified via ffprobe-on-captured-mp + VOM activeDecoder state. Arvus codec-state cross-check per § 11.4.58 + §CG screenshot.

Per § 11.4.59 four-layer: pre-build gate CM-AF-UI-DRIVEN-VIDEO- COVERAGE + propagation gate CM-COVENANT-114-48-PROPAGATION + on-device test framework at device/rockchip/rk3588/tests/ui_driven/ (Layer helper + Layer per-app drivers + Layer 11.4.60 scenarios) + Layer orchestrator . scripts/testing/run_ui_driven_video_suite.sh + paired meta-test mutations. No escape hatch – no --use-intent-shortcut , --skip-ui-traverse , --legacy-intent-mode flag. Classification: universal (§ 11.4.61). See Constitution § 11.4.62 for the full mandate.

Dual-approach testing (§ 11.4.43, User mandate 11.4.43)

Every feature test exercising a user-visible behaviour MUST ship in TWO variants: a UI-driven variant (uiautomator-based, § 11.4.43 surfaces A-E) AND an Intent/Broadcast-driven variant (`am start --es` / `am broadcast` -based). Either alone is a § 11.4.43 PASS-bluff for the OPPOSITE half of the stack – UI catches app-side bugs; Intent catches framework/system-server bugs (MediaCodec.configure hook, IVideoOutputManager bind timing, broadcast permission gating).

Shared assertion base `tests/lib/dual_approach_test_base.sh` – both variants share `dat_init` / `dat_start_capture` / `dat_assert_codec_state` / `dat_assert_video_frames` / `dat_assert_audio_channels` / `dat_arvus_dashboard_capture` / `dat_cleanup` / `dat_report_finding`. Both variants gather identical evidence into mirror dirs `qa-results/dual_approach/ <F>/<run-ts>/{ui,intent}/` so orchestrator `diffs` results and pinpoints which half of the stack contains a bug. Status mismatch between variants is itself a finding.

Kinopoisk 11.4.43 EAC 11.4.43 is the canonical first implementation. Both variants are RED per § 11.4.43 until the §CN decoder pipeline fix lands. Pre-build gates: `CM-COVENANT-114-49-PROPAGATION` .+. `CM-AF-DUAL-APPROACH-COVERAGE` + `CM-AF-KINOPOISK-5-1-DUAL- COVERAGE` . Three paired meta-test mutations. No escape hatch – no `--ui-only` / `--intent-only` / `--skip-dual` flag. Classification: universal (§ 11.4.43). See Constitution § 11.4.43 for the full mandate.

Deterministic consistency (§ 11.4.5.1, User mandate – –)

Every test that PASSES MUST have been executed N times (default N= normal tests, N= cycle-validation suites) against the same firmware MD + same device + same topology and produced IDENTICAL PASS in every iteration. A divergent N-iter run is auto-FAIL – there is no "first PASSED therefore X was a flake" path.

§ 11.4.5.1 's `intermittent` / `transient` / `flake` / `flaky` vocabulary is enforced MECHANICALLY by this mandate, not merely textually.

Coverage: every public API path (Activity / Service / Broadcast receiver / ContentProvider URI / IPC interface / JNI entry / sysprop write / sysfs node / init.rc trigger) MUST have ≥ dedicated test that drives it. Untested paths surface in a feature-coverage-matrix audit; the threshold ratchets → → → over phases.

Reliability mechanism: `ab_run_n_times` helper in the project anti-bluff library loops, captures evidence-hash per iter, asserts all N hashes + exit codes identical. NO operator-facing escape converts divergence to PASS.

Pre-build gates: `CM-COVENANT-114-50-PROPAGATION` + `CM-AF-RELIABILITY-CHECK-WIRED` + `CM-AF-FEATURE-COVERAGE-MATRIX`. Three paired meta-test mutations. No escape hatch – no `--allow-flake`, `--first-pass-suffices`, `--skip-n-iter`, `--skip-coverage-audit` flag. Classification: universal (§ 11.4.5.1). See Constitution § 11.4.5.1 for the full mandate.

Live-ADB-First maximization (§ 11.4.5.2, User mandate – –)

Every fix MUST be classified by rebuild-requirement before commit using the project's per-file-class decision matrix.

If `LIVE_ADB_TESTABLE` (on-device test scripts, host scripts, `atmosphere-.sh boot scripts`, `persist.` properties, markdown docs, test fixture assets, HelixQA YAML banks), the operator MUST first apply the fix to the running device via `adb push / setprop / pm install -r / mount -o remount,rw`, run the `$ . . . RED test live, capture PASS, THEN commit + rebuild + reflash`. Commit footer: `LIVE_ADB_VALIDATED: yes`. If `REQUIRES_REBUILD` (kernel, framework Java/AIDL, native C++ in APEX, sepolicy, `init.rc`, `ro.` properties, XML overlays, codec XML in APEX, `Android.bp/.mk`), the operator proceeds directly to source-side + rebuild. Commit footer: `REQUIRES_REBUILD: <reason>`. Mixed batches use partial.

`$ . . . REFINES $ . . . step` with mechanical enforcement. Helper: `classify_fix_rebuild_requirement.sh` walks `git diff --name-only`, looks up each file against the matrix, emits per-file classification + recommended commit-message footer. Unmatched paths classify as `REQUIRES_REBUILD: unmatched-path` (safe default per `$ . . .`).

Pre-build gates: `CM-COVENANT-114-51-PROPAGATION` + `CM-AF-CLASSIFY-FIX-HELPER-EXISTS` + `CM-AF-LIVE-ADB-FIRST-COMMIT-MARKER`. Three paired meta-test mutations. No escape hatch – no `--skip-classify` / `--assume-rebuild` / `--no-footer-required` flag.

Classification: universal (`$ . . .`). See Constitution `$ . . .` for the full mandate.

Autonomous-Validation (`$ . . .`, User mandate – –)

Forensic anchor – verbatim user mandate:

"Make sure we have full automation tests which will do all this work in full automation! IMPORTANT: Make sure that all existing tests and Challenges do work in anti-bluff manner – they MUST confirm that all tested codebase really works as expected! execution of tests and Challenges MUST guarantee the quality, the completion and full usability by end users of the product!"

11 4 5

Every user-facing feature MUST have at least one autonomous validation path: end-to-end via `adb shell` + scripted automation, captured runtime evidence per `$ . . .`, PASS/FAIL verdict WITHOUT human presence. Operator-attended tests are SUPPLEMENTARY, never PRIMARY. A feature whose ONLY path is operator-attended is a `$ . . .` violation: the path does not scale to CI, does not run on every commit, does not survive operator unavailability, and produces the exact "tests pass but feature doesn't work for users" failure mode `$ . . .` forbids.

Acceptable autonomous paths: instrumentation APK (SDK-API exercises + JSON result file), headless intent dispatch + state poll (`am start --es` + `dumpsys` / `/proc/<pid>/maps` polling), ADB-driven uiautomator (ONLY if hierarchy has `≥` clickable node – empty hierarchy demands fallback to APK/intent), network-side sink probe (`$ . . .`), HelixQA autonomous QA session (`$ . . .`).

Coverage ledger (`$ . . .`) classifies each feature as `AUTONOMOUS_VERIFIED` / `AUTONOMOUS_DESIGNED` / `OPERATOR_ATTENDED_ONLY` / `NOT_APPLICABLE`. `OPERATOR_ATTENDED_ONLY` blocks release until migrated; cite a tracked work item per `$ . . .` + `$ . . .`.

Pre-build gates `CM-COVENANT-114-52-PROPAGATION` + `CM-AF-AUTONOMOUS-PATH-PER-FEATURE` . Paired mutations. No escape hatch — no `--allow-operator-attended-only` / `--skip-autonomous-path` / `--manual-validation-suffices` flag.

Classification: universal (§ 1.1.4.5.3) . See Constitution § 2.0.2.6.0.5.1.8 for the full mandate.

Fixed_Summary parity (§ 2.1.2.6.0.5.1.8, User mandate — —)

Forensic anchor — verbatim user mandate (— — T : Z):

"Note: Just like for Issues we have Issues_Summary, for Fixed we MUST HAVE Fixed_Summary — like all other docs: ALWAYS in sync and up to date and ALWAYS exported into the PDF and HTML!"

`docs/Fixed_Summary.md` is the symmetric short-form summary of `docs/Fixed.md` . MUST be regenerated whenever `Fixed.md` changes. HTML + PDF exports MUST travel with the markdown. Stale exports are § 1.1.4.5.3 violations regardless of whether the underlying `.md` is correct. Same discipline as § 1.1.4.5.3 Issues_Summary applied to Fixed.md.

Generator: `scripts/testing/generate_fixed_summary.sh` (canonical). Auto-sync wrapper: `scripts/testing/sync_issues_docs.sh` regenerates BOTH Issues_Summary AND Fixed_Summary in one shot, exports HTML + PDF, colorizes per § 1.1.4.5.3, re-renders PDFs. MUST be invoked after any edit to `Fixed.md` . No `--issues-only` flag exists.

Sort order: closure date DESC (most-recent-Fixed first), §-letter / Fix- secondary. Documented at top of generated file.

Composes with \$ (Issues_Summary sibling),
\$ (atomic migration trigger), \$
(colorizer), \$ (type-aware closure terminal
values), \$ (revision header), \$
(CONTINUATION.md resumption).

Pre-build gates CM-FIXED-SUMMARY-SYNC + CM-COVENANT-114-53-
PROPAGATION . Paired mutations. No escape hatch – no --skip-
fixed-summary-sync , --issues-only , --summary-not-applicable
flag.

Classification: universal (\$). See
Constitution \$ for the full mandate.

**ATM-NNN ticket identifier (\$,
User mandate – –)**

**Forensic anchor – verbatim user mandate
(– –):**

"Every workable item in Issues / Issues_Summary /
Fixed / Fixed_Summary MUST carry a stable, unique,
auto-incremental ATM-NNN ticket identifier."

Every workable item heading in docs/Issues.md AND docs/
Fixed.md MUST carry [ATM-NNN] (form ## \$X.Y. [ATM-NNN]
<title> , zero-padded ≥ digits). Allocated by scripts/
testing/assign_atm_ticket_ids.sh and persisted to scripts/
testing/.atm_ticket_state.jsonl (jsonl). Identifiers are
monotonic, never renumbered, never reused, no gaps.
Issues_Summary.md and Fixed_Summary.md MUST carry an ATM
ID column as the leftmost data column.

Composes with \$ + \$ +
\$ + \$ + \$ +
\$ + \$ + \$.

Pre-build gates `CM-ATM-TICKET-IDS-COMPLETE` + `CM-ATM-TICKET-IDS-UNIQUE` + `CM-ATM-TICKET-IDS-MONOTONIC` + `CM-COVENANT-114-54-PROPAGATION`. Paired mutations. No escape hatch.

Classification: universal (§ 1.1.4.5.5). See Constitution § 2.0.2.6.0.5.1.9 for the full mandate.

Reopens-history tracking + per-item Reopens.md (§ 2.1.2.4.0.5.1.9, User mandate - -)

Forensic anchor – verbatim user mandate (- -):

"Add a Reopens-count column. For any item whose reopens-count > , create docs/issues/ATM-NNN/Reopens.md (+ HTML + PDF) with comprehensive reopen history + Fixed cycles."

Every workable item with `reopens_count > 0` MUST have `docs/issues/<ATM-NNN>/Reopens.md` (+ HTML + PDF). Required content: § 1.1.4.5.5 revision header, item identification, cycle counters, chronological timeline (By AI/User per § 1.1.4.5.5, On ISO date, Event, Reason, Evidence, Outcome), reasoning chain for each closure, most-recent state-change pointer.

Issues_Summary.md and Fixed_Summary.md MUST carry a `Reopens` column; count > hyperlinks to per-item Reopens.md.

Composes with § 1.1.4.5.5 (per-event capture), § 1.1.4.5.5 (ATM-NNN path), § 1.1.4.5.5 (revision header), § 1.1.4.5.5 (Status.md analogue), § 1.1.4.5.5 (Fixed_Summary parity – Reopens column symmetric).

Pre-build gates `CM-REOPENS-DOC-EXISTS-WHEN-COUNT-GT-ZERO` + `CM-REOPENS-DOC-REVISION-HEADER` + `CM-REOPENS-COL-IN-SUMMARIES` + `CM-COVENANT-114-55-PROPAGATION`. Paired mutations. No escape hatch.

Classification: universal (§ 11.4.56). See Constitution § 2.0.2.6 for the full mandate.

Status_Summary parity + two-audience format (§ 2.0.2.6, User mandate – –)

Forensic anchor – verbatim user mandate

(– –):

"Every Status.md doc gets a Status_Summary parity companion ALWAYS in sync, exported to HTML + PDF. Two-page format: page = non-developer audience (team-specific), page = software engineer summary."

For every `docs/<domain>/<integration>/Status.md` a companion `Status_Summary.md` MUST exist with: § 11.4.45 revision header; **Page** – **For the** (audience-specific – audio team for `docs/dolby/*`, video team for `docs/video/*`, etc.) plain-language summary, What works, What's broken or pending, Operator actions – NO \$-letter jargon, NO captured-evidence file paths; **Page** – **For software engineers** – \$-letter refs, gate names, commit hashes, captured-evidence paths, ATM-NNN cross-refs. HTML + PDF exports travel with the markdown.

Generator: `scripts/testing/generate_status_summary.sh <Status.md path>`. Invoked from `sync_integration_status.sh` (§ 11.4.12 wrapper) on every Status.md update.

Composes with § 11.4.45 (Status.md – Status_Summary COMPLEMENTS), § 11.4.56 + § 11.4.55 (parity

discipline), \$. . (revision header),
\$. . (colorizer), \$.
(CONTINUATION.md).

Pre-build gates CM-STATUS-SUMMARY-EXISTS-FOR-EVERY-STATUS +
CM-STATUS-SUMMARY-TWO-AUDIENCE + CM-STATUS-SUMMARY-REVISION-
HEADER + CM-COVENANT-114-56-PROPAGATION . Paired mutations.
No escape hatch.

Classification: universal (\$. .). See
Constitution \$. . for the full mandate.

README.md doc-link section + revision metadata

(\$. . , User mandate
- -)

Forensic anchor – verbatim user mandate

(- -):

"Add a doc-link section to README.md – links to Issues
+ Issues_Summary + Fixed + Fixed_Summary +
CONTINUATION + ALL Status docs + their exports. Each
link shows revision + last-modified."

Every project's top-level README.md MUST contain a section
titled Tracked-Items + Status Documents (heading MUST contain
literal Tracked-Items). The section is a markdown table
with columns: Document , Last modified , Revision ,
Markdown , HTML , PDF . Links to: Issues.md +
Issues_Summary.md, Fixed.md + Fixed_Summary.md,
CONTINUATION.md, every docs/**/Status.md + its
Status_Summary.md pair. Status docs auto-discovered by the
generator via find docs -name 'Status.md' .

Generator: scripts/testing/update_readme_doc_links.sh extracts
each doc's revision + last-modified from its \$. .
header, renders the table, replaces the section between

explicit markers (<!-- doc-link-section:begin --> / <!-- doc-link-section:end -->). Invoked from sync_issues_docs.sh AND sync_integration_status.sh .

Composes with \$. . + \$. . +
\$. . (Issues / Fixed / summaries),
\$. . (revision-header data source),
\$. . + \$. . (Status pairs),
\$. (CONTINUATION.md).

Pre-build gates CM-README-DOC-LINK-SECTION-PRESENT + CM-README-DOC-LINK-ROWS-COMPLETE + CM-README-DOC-LINK-FRESHNESS + CM-COVENANT-114-57-PROPAGATION . Paired mutations strip markers, remove rows, backdate Last modified , strip anchor literal. No escape hatch.

Classification: universal (\$. .). See Constitution \$. . for the full mandate.

Parallel-development methodology (\$. . , User mandate - -)

Forensic anchor – verbatim user mandate
(- - T~ : Z MSK):

"We MUST CREATE adjusted improved version of working methodology where multiple workable items could be done in parallel, all come into the central (main) branch as they are done, then we at particular moment rebuild and reflash the System and in background full testing with validation and verification is done. Each parallel work on one workable (or more) item(s) must use parallel agents as much as possible! ... Writing in depth tests (all supported types of the tests) with Challenges and full HelixQA use is MANDATORY! Every test we execute besides executed with success MUST

RESULT in proof that actual functionality being tested
REALLY DOES WORK with NO BLUFF of any kind!"

Project work proceeds through the **Parallel Work Unit (PWU) pipeline**. Each PWU is a self-contained workable item with: ATM-NNN identifier per \$. . , Issues.md entry per \$. . . +\$. . , file- scope manifest, \$. . RED test, source patch, pre-build gate, post-flash test, paired meta-test mutation per \$. , HelixQA Challenge bank entry, captured-evidence directory per \$. . +\$

-stage pipeline: Stage DEVELOP (parallel PWU agents in isolated worktrees) → Stage MERGE (serial via `commit_all.sh flock + $ . . -step merge-first`) → Stage REBUILD+FLASH (parallel where hardware allows) → Stage VALIDATE (parallel D +D +meta-test+coverage) → Stage SWEEP (parallel HelixQA + Fixed.md migration + README refresh). Stage of round N+ overlaps with Stages - of round N – throughput multiplier.

Synchronization: -layer lock hierarchy (parent flock / per- submodule git / contention-path advisory locks / per- PWU worktree). Disjoint-scope PWUs fully parallel.

Anti-bluff merge-time enforcement (mandatory, all four): RED-test captured (\$. .), paired meta-test mutation (\$. .), -iter deterministic-consistency (\$. .), captured-evidence per \$ Metadata-only / configuration-only / absence-of-error PASS REJECTED. HelixQA Challenge MANDATORY for every user-visible PWU.

Pre-build gates `CM-PWU-LOCK-HIERARCHY` + `CM-PWU-ANTI-BLUFF-COVERAGE`

- `CM-PWU-MERGE-QUEUE-DISCIPLINE` + `CM-PWU-PARALLEL-AGENT-LIMIT` + `CM-COVENANT-114-58-PROPAGATION`. Paired mutations cover each gate. No escape hatch.

Classification: universal (\$. .). See Constitution \$. . for the full mandate. Project-specific implementation reference in consumer-side `docs/guides/PARALLEL_DEVELOPMENT_METHODOLOGY.md`.

README always-sync (\$. . , User mandate - -)

`README.md` is a \$. . -class always-sync document. HTML + PDF exports refresh on every update via `scripts/testing/sync_readme_export.sh` (pandoc + weasyprint); auto-invoked by `sync_issues_docs.sh` so a single doc-sync run refreshes Issues / Issues_Summary / Fixed / Fixed_Summary / CONTINUATION / README (md + html + pdf). README carries a \$. . revision header and a Documentation Map section linking to every Status.md + Status_Summary.md + spec + plan + guide + script-companion doc + changelog + the constitution submodule, plus per-audience navigation (end user / developer / QA / agent). Pre-build gate `CM-README-EXPORT-SYNC` enforces mtime parity (`README.html` + `README.pdf` ≥ `README.md`). Paired meta-test mutation backdates HTML+PDF → gate FAILs. No escape hatch – no `--skip-readme-sync`, `--no-readme-export`, `--readme-stale-OK` flag. Composes with \$. .

- \$. . + \$. . + \$. . +
\$. . + \$. . + \$. . +
\$. .

Classification: universal (\$. .). See Constitution \$. . for the full mandate.

Documentation always-sync composite covenant (§ 1.1.4.6.0, User mandate – –)

Eight doc classes – Issues, Issues_Summary, Fixed, Fixed_Summary, CONTINUATION, README, every Status.md (domain-scoped), every Status_Summary.md – MUST be in sync across .md + .html + .pdf. Composite pre-build gate CM-DOCS-COMPOSITE-SYNC walks all classes (Status fleet via recursive docs/** find), FAILS the build if ANY .html or .pdf mtime is older than its .md. Per-class anchors § 1.1.4.6.0 / § 1.1.4.6.0 / § 1.1.4.6.0 / § 1.1.4.6.0 / § 1.1.4.6.0 / govern individually; the composite catches what they miss when one is silently bypassed. Paired mutation backdates docs/Issues.html → gate FAILs. No escape hatch.

Classification: universal (§ 1.1.4.1.0). See Constitution § 1.1.4.1.0 for the full mandate.

Credentials-handling (§ 1.1.4.1.0)

Forensic anchor – verbatim user mandate
(– –):

"Credentials or any secret and sensitive data MUST NOT leak!"

.env patterns git-ignored project-wide. Runtime-load-only – tests load credentials at execution time from operator-populated files under scripts/testing/secrets/ (or per-project equivalent). Test scripts MUST NEVER print or log credentials.

Host-session safety (§)

Project procedures MUST NOT use more than % of total system RAM. Heavy work wrapped in bounded execution scopes so the kernel OOM-kills only the scope – `user@<uid>.service` stays alive. Hibernating / suspending / logging out the user is FORBIDDEN (directly OR indirectly).

Continuation document (§ .)

`docs/CONTINUATION.md` MUST exist at the project root and reflect the live state of the work. Every non-trivial state change updates it in the same commit. Any agent must be able to resume work exactly where the previous session left off by reading this single file.

Data safety (§)

Hardlinked backup before any destructive operation. Post-op gate MUST pass. Force-push requires explicit per-session human authorization. Every history rewrite gets a `docs/changelogs/` audit entry.

CLI workflow expectations

- **Read before write.** Use file-reading tools to inspect the current state before editing. Edits in the dark are how bugs sneak in.
- **Use the project's commit wrapper.** Direct `git add` / `git commit` is forbidden unless explicitly carved out.
- **Commit messages cite forensic evidence** for any non-trivial fix (logs, captures, /sys readings, dropbox entries, strace). Speculative narratives are § . . violations.
- **Update CONTINUATION.md** in the same commit as the work (§ .).

- . **Run pre-build / post-build / runtime gates** before declaring a change complete. NEVER report "done" without running the relevant gate.
- . **Cite external research sources** for non-trivial fixes (\$.). Either a citation OR the literal "NO external solution found – original work".
- . **Refuse to bluff.** If a step requires information you don't have, say so and STOP. Don't guess.

Hierarchy of project authority

When a project files declares conflicting rules with this base AGENTS.md, resolve in this order:

- . Project's `docs/guides/<PROJECT>_CONSTITUTION.md` – most specific, project-tailored.
- . `constitution/Constitution.md` (this submodule) – universal.
- . Project's root `CLAUDE.md` / `AGENTS.md` – agent-facing summaries; lossy compared to the Constitution.
- . This `constitution/CLAUDE.md` / `AGENTS.md` – universal agent-facing summary; lossy compared to the Constitution.

If a project's own files contradict the Constitution, that is itself a Constitution violation and MUST be reported. Do NOT silently follow the contradicting project file.

When stuck

- Read `constitution/Constitution.md` .
- Cross-reference the project's `CLAUDE.md` / `AGENTS.md`.
- If still unclear, ask the operator. Do NOT guess. Do NOT bluff.

§ . . – OpenDesign UI design system mandate (User mandate, – –). Compact summary: every project producing user-facing interfaces (web, desktop, mobile, TUI) MUST use OpenDesign (<https://github.com/nexu-io/open-design>) as the mandatory UI design-and-refinement system – NOT ad-hoc CSS or one-off design tools; install as a project dependency and use its design tokens/themes system for: (a) all color palette definitions supporting BOTH light and dark themes from project brand assets (§ . .), (b) typography scale and spacing, (c) component-level design tokens; if a desired UI pattern is not supported, extend OpenDesign upstream per § . . (extend-don't-reimplement); every UI component MUST ship light + dark theme variants; elements MUST NOT overlap, fonts MUST NOT collide, labels MUST NOT overlay labels – any layout regression is a § violation; all UI changes MUST be covered by the project's standard test types including visual regression tests (before/after screenshots with per-pixel or perceptual-diff PASS/FAIL). Honest boundary (§ . .): OpenDesign governs design tokens and themes – it does NOT replace functional testing (§ . .), WCAG accessibility assertions (§ . .), nor the § /§ UI-driven and dual-approach testing methodology. Classification: universal § (§). Composes § /§ /§ (b)/§ /§ /§ /§ /§ Propagation gate CM-COVENANT-114-162-PROPAGATION + recommended gate CM-OPENDSIGN-UI-SYSTEM + paired § mutation.

§ – Conflict-free multi-track work-division + exactly-once claim registry + capability-aware deadlock-proof device-lock (User mandate, – –). Two mandatory multi-track arbitration layers, conflict-free by

construction: (A) exactly-once work-item/logical-group claim registry (extends § 11.4.11.1 / § 11.4.11.2); disjoint file-scope § 11.4.11.3 L + worktree/CoW § 11.4.11.4; no codebase loss § 11.4.11.5 / § 11.4.11.6 / § 11.4.11.7 / § 11.4.11.8); (B) capability-aware deadlock-proof device-lock (§ 11.4.11.9 single-owner; all-or-nothing + non-blocking + TTL-reap breaks all Coffman conditions; append-only JSONL + atomic snapshot § 11.4.11.10); (C) universal/decoupled (§ 11.4.11.11 / § 11.4.11.12 / § 11.4.11.13) + evidence-based resource-tuning (§ 11.4.11.14 / § 11.4.11.15, never guessed). Classification: universal (§ 11.4.11.16). Gates CM-WORK-DIVISION-EXCLUSIVE-CLAIM + CM-DEVICE-LOCK-DEADLOCK-FREE + CM-COVENANT-11.4.11.17 - PROPAGATION + paired § 11.4.11.18 mutation. Composes § 11.4.11.19 § 11.4.11.20 § 11.4.11.21 § 11.4.11.22 § 11.4.11.23 § 11.4.11.24 § 11.4.11.25 § 11.4.11.26 § 11.4.11.27 § 11.4.11.28 § 11.4.11.29 § 11.4.11.30 § 11.4.11.31 § 11.4.11.32 § 11.4.11.33 § 11.4.11.34 § 11.4.11.35 § 11.4.11.36 § 11.4.11.37 § 11.4.11.38 § 11.4.11.39 § 11.4.11.40 § 11.4.11.41 § 11.4.11.42 § 11.4.11.43 § 11.4.11.44 § 11.4.11.45 § 11.4.11.46 § 11.4.11.47 § 11.4.11.48 § 11.4.11.49 § 11.4.11.50 § 11.4.11.51 § 11.4.11.52 § 11.4.11.53 § 11.4.11.54 § 11.4.11.55 § 11.4.11.56 § 11.4.11.57 § 11.4.11.58 § 11.4.11.59 § 11.4.11.60 § 11.4.11.61 § 11.4.11.62 § 11.4.11.63 § 11.4.11.64 § 11.4.11.65 § 11.4.11.66 § 11.4.11.67 § 11.4.11.68 § 11.4.11.69 § 11.4.11.70 § 11.4.11.71 § 11.4.11.72 § 11.4.11.73 § 11.4.11.74 § 11.4.11.75 § 11.4.11.76 § 11.4.11.77 § 11.4.11.78 § 11.4.11.79 § 11.4.11.80 § 11.4.11.81 § 11.4.11.82 § 11.4.11.83 § 11.4.11.84 § 11.4.11.85 § 11.4.11.86 § 11.4.11.87 § 11.4.11.88 § 11.4.11.89 § 11.4.11.90 § 11.4.11.91 § 11.4.11.92 § 11.4.11.93 § 11.4.11.94 § 11.4.11.95 § 11.4.11.96 § 11.4.11.97 § 11.4.11.98 § 11.4.11.99 § 11.4.11.100 § 11.4.11.101 § 11.4.11.102 § 11.4.11.103 § 11.4.11.104 § 11.4.11.105 § 11.4.11.106 § 11.4.11.107 § 11.4.11.108 § 11.4.11.109 § 11.4.11.110 § 11.4.11.111 § 11.4.11.112 § 11.4.11.113 § 11.4.11.114 § 11.4.11.115 § 11.4.11.116 § 11.4.11.117 § 11.4.11.118 § 11.4.11.119 § 11.4.11.120 § 11.4.11.121 § 11.4.11.122 § 11.4.11.123 § 11.4.11.124 § 11.4.11.125 § 11.4.11.126 § 11.4.11.127 § 11.4.11.128 § 11.4.11.129 § 11.4.11.130 § 11.4.11.131 § 11.4.11.132 § 11.4.11.133 § 11.4.11.134 § 11.4.11.135 § 11.4.11.136 § 11.4.11.137 § 11.4.11.138 § 11.4.11.139 § 11.4.11.140 § 11.4.11.141 § 11.4.11.142 § 11.4.11.143 § 11.4.11.144 § 11.4.11.145 § 11.4.11.146 § 11.4.11.147 § 11.4.11.148 § 11.4.11.149 § 11.4.11.150 § 11.4.11.151 § 11.4.11.152 § 11.4.11.153 § 11.4.11.154 § 11.4.11.155 § 11.4.11.156 § 11.4.11.157 § 11.4.11.158 § 11.4.11.159 § 11.4.11.160 § 11.4.11.161 § 11.4.11.162 § 11.4.11.163 § 11.4.11.164 § 11.4.11.165 § 11.4.11.166 § 11.4.11.167 § 11.4.11.168 § 11.4.11.169 § 11.4.11.170 § 11.4.11.171 § 11.4.11.172 § 11.4.11.173 § 11.4.11.174 § 11.4.11.175 § 11.4.11.176 § 11.4.11.177 § 11.4.11.178 § 11.4.11.179 § 11.4.11.180 § 11.4.11.181 § 11.4.11.182 § 11.4.11.183 § 11.4.11.184 § 11.4.11.185 § 11.4.11.186 § 11.4.11.187 § 11.4.11.188 § 11.4.11.189 § 11.4.11.190 § 11.4.11.191 § 11.4.11.192 § 11.4.11.193 § 11.4.11.194 § 11.4.11.195 § 11.4.11.196 § 11.4.11.197 § 11.4.11.198 § 11.4.11.199 § 11.4.11.200 § 11.4.11.201 § 11.4.11.202 § 11.4.11.203 § 11.4.11.204 § 11.4.11.205 § 11.4.11.206 § 11.4.11.207 § 11.4.11.208 § 11.4.11.209 § 11.4.11.210 § 11.4.11.211 § 11.4.11.212 § 11.4.11.213 § 11.4.11.214 § 11.4.11.215 § 11.4.11.216 § 11.4.11.217 § 11.4.11.218 § 11.4.11.219 § 11.4.11.220 § 11.4.11.221 § 11.4.11.222 § 11.4.11.223 § 11.4.11.224 § 11.4.11.225 § 11.4.11.226 § 11.4.11.227 § 11.4.11.228 § 11.4.11.229 § 11.4.11.230 § 11.4.11.231 § 11.4.11.232 § 11.4.11.233 § 11.4.11.234 § 11.4.11.235 § 11.4.11.236 § 11.4.11.237 § 11.4.11.238 § 11.4.11.239 § 11.4.11.240 § 11.4.11.241 § 11.4.11.242 § 11.4.11.243 § 11.4.11.244 § 11.4.11.245 § 11.4.11.246 § 11.4.11.247 § 11.4.11.248 § 11.4.11.249 § 11.4.11.250 § 11.4.11.251 § 11.4.11.252 § 11.4.11.253 § 11.4.11.254 § 11.4.11.255 § 11.4.11.256 § 11.4.11.257 § 11.4.11.258 § 11.4.11.259 § 11.4.11.260 § 11.4.11.261 § 11.4.11.262 § 11.4.11.263 § 11.4.11.264 § 11.4.11.265 § 11.4.11.266 § 11.4.11.267 § 11.4.11.268 § 11.4.11.269 § 11.4.11.270 § 11.4.11.271 § 11.4.11.272 § 11.4.11.273 § 11.4.11.274 § 11.4.11.275 § 11.4.11.276 § 11.4.11.277 § 11.4.11.278 § 11.4.11.279 § 11.4.11.280 § 11.4.11.281 § 11.4.11.282 § 11.4.11.283 § 11.4.11.284 § 11.4.11.285 § 11.4.11.286 § 11.4.11.287 § 11.4.11.288 § 11.4.11.289 § 11.4.11.290 § 11.4.11.291 § 11.4.11.292 § 11.4.11.293 § 11.4.11.294 § 11.4.11.295 § 11.4.11.296 § 11.4.11.297 § 11.4.11.298 § 11.4.11.299 § 11.4.11.300 § 11.4.11.301 § 11.4.11.302 § 11.4.11.303 § 11.4.11.304 § 11.4.11.305 § 11.4.11.306 § 11.4.11.307 § 11.4.11.308 § 11.4.11.309 § 11.4.11.310 § 11.4.11.311 § 11.4.11.312 § 11.

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$ . $ . . $ 2026 07 04 $ . .
2026 07 09 $ . . $ . .

```

\$. . – Automatic multi-track ruler

orchestration (User mandate, – – ; landed

– –). Compact summary: every project incorporating this constitution MUST ship its multi-track parallel-development orchestration as a UNIVERSAL, AUTOMATIC, OUT-OF-THE-BOX, INHERITED capability – not a per-project reimplementation – via a single conductor session that programmatically SPAWNS, DRIVES, RESUMES, MONITORS, and CONTROLS per-track headless workers. ()

Headless spawn primitive – workers launch via native headless `claude -p --output-format stream-json --verbose` ; the worker's `session_id` is captured from the FIRST `init` event (never invented); success/failure is read from the terminal event's `result.is_error` field, NEVER the process exit code (a CLI can exit non-zero on a successful `result` , or zero on a failed one – exit-code-as-success-signal is a \$. . guess); continuation of an existing worker uses `--resume <session_id>` invoked from the SAME cwd + env the spawn used (the storage/resume-collision gotcha). () **Per-subscription auth + rate-limit fallback** – every spawn `unset` s any ambient

`ANTHROPIC_API_KEY` (the precedence trap that silently routes a "subscription" worker onto pay-per-token API billing) and sets the selected alias's `CLAUDE_CODE_OAUTH_TOKEN` / `CLAUDE_CONFIG_DIR` ; on a captured rate-limit signature (`isApiErrorMessage + apiErrorStatus:429` , quota-exhausted vs transient distinguished by evidence, never guessed) the ruler REBINDS the affected track onto the next healthy alias (cross-track reuse permitted), falls back to a provider alias where configured, and bounded-parks (`$ . .`) rather than hanging when every alias is cooled. () **Crash-resilient ruler self-supervisor** – the ruler's OWN state (per-track alias/session_id/worktree/

last-verdict) is persisted durably (write-temp-then-rename, NEVER tmpfs) so an external watchdog detects a dead ruler PID and re-hydrates every track's binding from the durable state + the \$. . . event stream – no work is silently lost to a ruler crash (\$. . . at the ruler layer). () **Idempotent bootstrap + auto-install + conductor-stays-home** – one idempotent bootstrap command wires the whole system out of the box (cwd-hook install, config validation, reconciliation), auto-invoked on every constitution pull via the \$. . .

`post_update_hook.sh` `skill-register seam`; the `conductor: -designated alias/session` is NEVER bound into a track worktree (the resolver returns nothing for it) so the conductor always runs from the main checkout. () **Project-agnostic engine, consumer-owned data** – the entire mechanism (config loader, alias→worktree resolver, cwd-hook, bind/fallback/cooldown orchestrator, session launcher, rate-limit monitor, supervisor) lives at `constitution/scripts/multitrack/`, inherited BY REFERENCE (\$. . . (B)/\$. . .), carrying NO project literal; every consuming project supplies its own per-host `config/multitrack/<hostname>.yaml` (tracks/mounts/serials/aliases/conductor) as DATA, never as an engine edit. Anti-bluff (\$. . . /\$. . .): every piece carries four-layer coverage (pre-build gate + on-host test + paired \$. . . mutation + captured evidence per \$. . .), and each fix declares a runtime signature verified on a clean host – never a claim the mechanism "should work." Honest boundary (\$. . .): this anchor mandates the MECHANISM; genuine per-subscription quota-isolation proof requires live tokens exercising real rate-limit boundaries (an operator-scheduled live acceptance window) – a mock-driven GREEN proves the spawn/resume/fallback CONTRACT, never live-quota behaviour, and the two are never conflated or substituted for each other. Classification: universal (\$. . .). Composes

11 4 24
 11 4 58
 11 4 70

```

$ . . . (subagent-driven-by-default) /
$ . . . (submodule decoupling + dependency
layout) / $ . . . (parallel-development PWU
pipeline) / $ . . . (subagent-driven execution
default) / $ . . . (autonomous-decision-over-
blocking – the bounded-park-on-all-cooled behaviour) /
$ . . . (continuous parallel-stream routine) /
$ . . . (resolve-by-stable-name – config by
serial, never by index) / $ . . . (real-time
conductor framework sync channel) / $ . . .
(single-resource-owner partitioning) / $ . . .
(default autonomous-loop mode) / $ . . . (always-
on device-recording – the analogous host-safety
discipline) / $ . . . (standing session-
resumption file) / $ . . . (crashed-agent
respawn-until-complete – the ruler-self-supervisor is its
ruler-layer instantiation) / $ . . .
(constitution auto-propagation hook seam) /
$ . . . (feature work-stream lifecycle) /
$ . . . (multi-track work-division + exactly-once
claim + device-lock) / $ . . . /$ . . . (host-safety
memory + concurrency ceilings – the spawn/launch pool is
budget-gated). Propagation gate CM-COVENANT-114-187-
PROPAGATION (literal 11.4.187) + recommended gates CM-
MULTITRACK-ENGINE-IN-CONSTITUTION (the generic ruler scripts
present under constitution/scripts/multitrack/, carrying no
project literal, bash -n + sh -n clean per
$ . . . ) / CM-MULTITRACK-AUTOINSTALL-WIRED / CM-
MULTITRACK-BOOTSTRAP-IDEMPOTENT / CM-MULTITRACK-BIND-ON-START /
CM-MULTITRACK-FALLBACK-MONITOR / CM-MULTITRACK-HEADLESS-
SPAWN / CM-MULTITRACK-RULER-SUPERVISOR + paired $ . . .
mutation (strip the literal → propagation gate FAILs; re-
introduce a project literal into the engine, or downgrade
a rate-limit rebind to exit-code-based success detection →
the corresponding mechanism gate FAILs). Canonical
authority: constitution submodule Constitution.md

```

\$. . . Non-compliance is a release blocker. No escape hatch – no `--skip-multitrack` , `--manual-multitrack-only` , `--no-auto-install` , `--per-project-reimpl-OK` , `--exit-code-is-success-signal` flag.

\$. . . – Website engineering-quality mandate: every project website MUST be fully responsive + completely SEO-optimized + uniquely OpenDesign-authored + bleeding-edge enterprise-quality, each PROVEN with captured evidence (User mandate, – –). Verbatim operator mandate: "every website we create inside our projects MUST BE fully responsive – working on all browsers, all operating systems, all platforms and all devices and screen sizes! Every Website MUST BE completely SEO optimized! Layouts and templates done MUST BE unique and everything fully designed from OpenDesign! Make sure this applies to our project Website as well! Everything must be bleeding-edge enterprise quality, stunning visual look and feel, innovative, creative and unique! There MUST BE no false or faulty result or bluff of any kind anywhere!" Every website / web-UI surface a project ships – INCLUDING this project's OWN website – MUST satisfy AND PROVE with captured physical evidence (\$. . . / \$. . . / \$. . . – never a bare claim, \$. . .) ALL of: (A) **FULL RESPONSIVENESS** – correct, usable rendering across ALL major browser engines (Chromium/Firefox/WebKit), all operating systems, all device classes (desktop/laptop/tablet/phone/large-display) and the full range of screen sizes/orientations/pixel-densities; NO broken layout / overflow / clipping / overlap / off-screen or unusable control at ANY breakpoint. (B) **COMPLETE SEO OPTIMIZATION** – semantic HTML + correct document outline; per-page unique `<title>` + meta description; Open Graph + Twitter cards; canonical URLs; structured data (schema.org / JSON-LD); `robots` + XML sitemap; crawlable / indexable markup; WCAG AA accessibility + Core Web Vitals performance (both feed

ranking). (C) **UNIQUE OpenDesign-AUTHORED templates/layouts** – EVERY layout + template designed FROM the OpenDesign design-token system (§ 11.4.190), NOT generic / off-the-shelf / templated; visually unique, innovative, creative, brand-distinct. (D) **BLEEDING-EDGE ENTERPRISE VISUAL QUALITY** – stunning, polished, professional look-and-feel in BOTH light AND dark themes (§ 11.4.191 tokens). (E) **ANTI-BLUFF PROOF** – NONE of (A)–(D) may be CLAIMED without captured physical evidence: responsiveness PROVEN by device-independent host-rendered screenshots across the real breakpoint × engine matrix + an OCR/vision layout oracle (no overlap / label-over-label / clip / overflow / off-screen); SEO PROVEN by an automated audit (Lighthouse SEO + structured-data validation + meta / OG / sitemap / robots presence) meeting a defined score floor; visual quality PROVEN by the § 11.4.192 device-independent host-rendered pixel proof per screen × state × {light,dark}; uniqueness PROVEN by OpenDesign-token provenance. A website reported responsive / SEO-optimized / unique / enterprise-quality WITHOUT these captured proofs is a § 11.4.193 PASS-bluff at the web-UI layer. § 11.4.194 is the strict WEB-quality expansion of § 11.4.195 (OpenDesign – it applies § 11.4.196's design-token mandate to the full responsive + SEO + enterprise-quality web surface) and binds § 11.4.197's device-independent host-rendered pixel proof as the correctness oracle; sibling of § 11.4.198 (exported-document independent visual validation) at the web-UI layer. Classification: universal (§ 11.4.199) – references no project-specific hardware; the consuming project supplies its concrete breakpoint × engine matrix, SEO score-floor, and OpenDesign token set per § 11.4.200. Composes § 11.4.201 / § 11.4.202 / § 11.4.203 / § 11.4.204 / § 11.4.205. Propagation gate CM-COVENANT-114-190-PROPAGATION (literal 11.4.190) +

recommended gates CM-WEBSITE-RESPONSIVE-PROVEN (host-rendered screenshots across the breakpoint × engine matrix + layout oracle – no overlap/clip/overflow/off-screen) / CM-WEBSITE-SEO-OPTIMIZED (automated SEO audit meets the score floor + structured-data / meta / OG / sitemap / robots present) / CM-WEBSITE-OPENDSIGN-UNIQUE (every layout/template's OpenDesign-token provenance, no generic/off-the-shelf template) + paired \$. mutation (report a website responsive/SEO/unique/enterprise-quality with NO captured proof, OR ship a generic non-OpenDesign template, OR skip the light/dark host-rendered pixel proof → the recommended gate FAILs; strip the literal → propagation gate FAILs; gate-code = separate work item). **Canonical authority:** constitution submodule Constitution.md

\$. . . Non-compliance is a release blocker. No escape hatch – no --skip-responsive-proof , --seo-optional , --generic-template-OK , --visual-quality-later , --claim-without-render flag.

\$. . . – Work-to-track/branch binding enforcement: a logic-group's work can NEVER be committed OR dispatched onto the wrong track/branch (research-derived, – –). Compact summary: every logical group of workable items binds to EXACTLY ONE canonical (track, branch) recorded ONCE in the single source of truth (the \$. . . workable-items DB logic_groups.destination = authoritative branch + a canonical_track column + a group_paths file-scope manifest), and NO change belonging to a group may LAND on, nor be DISPATCHED to, a track/branch other than that group's canonical one – the file-scope generalisation of \$. . . /\$. . . from branch-NAME consistency to work-PLACEMENT consistency (forensic FACT – – : srcpy/Remote-app work – group mistiq-vader , canonical feature/mistiq-vader /T – was committed onto main /T ; \$. . . 's guard-branch-consistency.sh never fired because committing onto an

existing branch is not a feature-branch CREATE, and no gate checked which FILES land on which BRANCH). ()

Single source of truth – the .logic_group → (branch, track, path-globs) ..binding is recorded ONCE in the DB and LOOKED UP, never re-invented (a commit/dispatch placing a group's file-scope on a non-canonical branch/track is a \$..no-guessing violation). ()

Preventive hook (\$.. /\$..) – a PreToolUse guard guard-work-track-binding.sh (inherited BY REFERENCE, never copied) BLOCKS (exit) a git-commit-class Bash call whose to-be-committed file set contains a path owned by a group whose canonical branch/track ≠ the current checkout, AND BLOCKS..an Agent/Task dispatch whose \$.. label track/branch ≠ the canonical (track, branch) of the ticket's group; fail-closed on unreadable registry, # guardrails:allow <reason> audited escape. ()

Detective gate (\$.. /\$..) – CM-WORK-TRACK-BINDING-ENFORCED asserts, on the current branch's own added commits (merge-base(HEAD,main)..HEAD), that every group_paths -owned file's group destination == current branch (and canonical_track == current track when pinned) – the actual enforcement boundary regardless of how the commit was made, the runtime signature being FAIL-on-wrong-placement / PASS-on-canonical-placement on a clean checkout. ()

Risk-free reconciliation (\$.. / \$.. /\$..) – a mis-placed change is reconciled by MERGING its commits onto the group's canonical branch (union, no commit lost, no -s ours / reset), ff-only push to all upstreams (NEVER force-push), then removing it from the wrong branch only after the merge..is..confirmed on all mirrors.

Classification:

universal (\$..) – references no project hardware; the consuming project supplies its (track, branch) map, group_paths seed, and commit-wrapper wiring per \$.. Composes

\$.. /. /. /. /. /. /. /. /. /. /.

§ . . Propagation gate CM-COVENANT-114-191-PROPAGATION
 (literal 11.4.191) + recommended gate CM-WORK-TRACK-BINDING-
 ENFORCED + paired § . mutation. **Canonical authority:**
 constitution submodule Constitution.md § . . .
 Non-compliance is a release blocker. No escape hatch – no
 --allow-cross-track-commit , --skip-work-binding , --commit-any-
 branch-OK , --dispatch-any-track-OK , --no-file-scope-check
 flag.

§. 07 11 . . – Continuous multi-track auto-backfill: a
 FREE track MUST be IMMEDIATELY re-assigned its next-
 highest-priority domain work-set, never left idle while
 its domain has actionable items (User mandate,
 – –). Verbatim operator mandate: "once a
 Track is free, we shall assign next set of workable items
 to it ... full continuous multi-track development ... no
 stopping ... this is MANDATORY". Compact summary: the
 moment any track's worker COMPLETES its current work-set
 (its item(s) closed with anti-bluff evidence, or merged),
 that FREE track MUST be IMMEDIATELY assigned the NEXT-
 highest-priority actionable work-set from its OWN domain
 (its § . . logic_group / canonical branch) – a
 track is NEVER left idle while its domain still holds
 actionable workable items, and the conductor/ruler
 CONTINUOUSLY re-assigns with NO stopping (full continuous
 multi-track development); PROMOTES the § . . (B)
 ≥ -parallel-background-streams-with-auto-backfill
 invariant into the multi-track (§ . . /
 § . . /§ /§ . .) layer
 (§ . . (B) backfills a STREAM the moment it is
 done; § . . backfills a TRACK the moment its
 worker is done, keyed to that track's canonical (track,
 branch) domain per § . .). () **WHAT** – on any
 track-worker completion the conductor/ruler
 (§ . .) MUST, per that track's domain, look up
 the NEXT actionable work-set, CLAIM it exactly-once
 (§ . . exactly-once claim registry, no two

tracks claiming one item), and DISPATCH it as the track's new § 11.4.192 -labelled work stream – without waiting for a scheduled wake or an operator prompt. () **PRIORITY** – the next work-set is the track domain's highest-priority actionable item (§ 11.4.192 / § 11.4.192 priority order, audio-first where in scope), never an arbitrary pick. () **SOURCE-SIDE ADVANCES EVEN WHEN VALIDATION IS GATED** – when the next item's on-device VALIDATION is blocked on a pending build / device availability (§ 11.4.192 clean-target / § 11.4.192 manual-QA / § 11.4.192 operator-gated flash), the track MUST still advance the item's SOURCE-SIDE work (investigation, RED-test § 11.4.192 , source patch, pre-build gate + paired § 11.4.192 mutation, docs) and honestly § 11.4.192 / § 11.4.192 SKIP-with-reason ONLY the genuinely device-gated validation portion – a build/device gate on the validation step is NOT a reason to idle the whole track. () **HONEST BOUNDARY** (§ 11.4.192) – a track idles ONLY when its domain GENUINELY has zero remaining actionable workable items OR every remaining item is externally blocked with no source-side advancement possible (§ 11.4.192 (A) / § 11.4.192 / § 11.4.192 idle-only-when-blocked closed set); "the merges are done" / "the current item is done" / "nothing visible right now" is NEVER a valid idle reason while the domain still holds progressable items (a § 11.4.192 + § 11.4.192 violation); an unavoidable per-item block PARKS one item, it does NOT stop the track (§ 11.4.192 – the track backfills its next non-blocked domain item). Classification: universal (§ 11.4.192 11.4.192 11.4.192) – references no project hardware; the consuming project supplies its track→domain map + priority source per § 11.4.192 . Composes § 11.4.192 / § 11.4.192 / § 11.4.192 / § 11.4.192 / § 11.4.192 / § 11.4.192 / § 11.4.192 . Propagation gate CM-COVENANT-114-192-PROPAGATION (literal 11.4.192) + recommended gate CM-CONTINUOUS-MULTITRACK-

BACKFILL (on a track-worker completion the ruler re-assigns that track's next-highest-priority domain work-set; a track with a completed worker AND remaining actionable domain items but no new dispatch → the gate FAILs) + paired \$. mutation (let a free track sit idle while its domain holds an actionable item, OR treat "merges done" as a valid idle reason → the gate FAILs; strip the literal → propagation gate FAILs; gate-code = separate work item). **Canonical authority:** constitution submodule Constitution.md \$. . Non-compliance is a release blocker. No escape hatch – no --allow-idle-track , --skip-backfill , --stop-multitrack , --merges-done-is-idle , --defer-track-reassign flag.

\$. . – Anti-blind-typing mandate: every UI interaction MUST be "seen" and "understood" via OCR/vision/screenshot proof, NEVER blind-typed (User mandate, – –). Verbatim operator mandate: "No blind typing of any kind anywhere!!!"; "Login to apps and services MUST BE done with understanding and seeing the whole UI ALWAYS!!!"; "blind typing is STRICTLY FORBIDDEN, and instead everything MUST BE seen and understood in coordination with proper models and means we have incorporated in HelixQA!!!". Blind typing – sending keystrokes/text/input events to a UI surface WITHOUT first capturing AND understanding what is on screen (OCR / vision / uiautomator / accessibility-tree / screenshot) AND confirming the result after each interaction – is STRICTLY FORBIDDEN everywhere and always, for every agent/tool/automation/platform/context. It is LITERAL bluff: claiming "credentials entered"/"login completed" without visual proof the keystrokes landed on the intended fields/screen producing the intended result is a \$. PASS-bluff at the UI-interaction layer (a credential typed into the wrong field, or a WebView that silently swallowed input, = end-user feature broken while the agent reported success). () **SEE-BEFORE-YOU-TYPE** – before ANY input

text / . input.keyevent / adb shell input / uiautomator
 gesture, capture the current screen state (screencap +
 Tesseract OCR ≥ DPI, OR uiautomator dump grep, OR
 accessibility-tree dumphsys, OR HelixQA VisionEngine frame
 per \$. . . / \$. . .) AND confirm the
 intended target element is visible AND focused – the
 capture IS the \$. . . / \$. . . touch_input
 evidence. (. . .) **VERIFY-AFTER-YOU-TYPE** – after EVERY
 interaction step, capture a NEW screen state + confirm the
 intended change (input text + final screenshot =
 blind typing). () **OCR/VISION ORACLE MANDATORY** –
 Tesseract ≥ DPI / uiautomator-dump grep / HelixQA
 VisionEngine / accessibility-tree dumphsys. () **WEBVIEW** –
 near-empty uiautomator hierarchy → OCR fallback per
 \$. . . ; if even OCR is infeasible (secure/DRM-
 blanked surface captures black) → the login is
 operator_attended-only, SKIP-with-reason per \$. . . ,
 NEVER blind-type-and-claim-success. () **CREDENTIAL-SAFE**
OCR – redact credential characters from OCR output + logs
 per \$. . . ; evidence = the redacted screenshot /
 OCR text. () **ANTI-BLUFF** – every login/interaction PASS
 cites a captured-evidence artefact containing pre-
 interaction state + post-interaction state + OCR/vision
 verification of the expected change; self-validated per
 \$ (. . .) with a golden-good + golden-bad
 fixture pair. Classification: universal (\$. . .).
 Composes \$. . . / \$. . . / \$. . . /
 \$. . . / \$. . . / \$. . . /
 \$. . . / \$. . . / \$. . . /
 \$. . . / \$ Propagation gate CM-
 COVENANT-114-193-PROPAGATION (literal 11.4.193) +
 recommended gate CM-NO-BLIND-TYPING + paired \$.
 mutation (strip OCR verification before an input text →
 gate FAILs; strip the literal → propagation gate FAILs;
 gate-code = separate work item). **Canonical authority:**
 constitution submodule Constitution.md \$

Non-compliance is a release blocker. No escape hatch – no
--blind-typing-ok , --skip-ocr-verification , --assume-input-
landed , --login-without-proof , --type-without-seeing flag.

\$. . . – **Branch-structure governance: taxonomy + merge-after-live-QA + flavor/product non-merge (User mandate, - -)**. Verbatim operator rules:
branch taxonomy feat/<NAME_OF_FEATURE> / product/
<NAME_OF_PRODUCT> / flavor/<NAME_OF_FLAVOR> ; tracks on
FEATURE branches or main MUST – after full confirmation +
validation + verification + full LIVE MANUAL TESTING
(\$. . .) – MERGE to main COMPLETELY (all
submodules + main repo), ff-only, NEVER force-push
(\$. . .), no commit lost (\$. . .); tracks with an
entirely-new FLAVOR/PRODUCT do ALL work on a flavor/ AND/
OR product/ branch (both may exist) which does NOT merge
to main the same way; commit + push ALL branches to ALL
upstreams (\$. . .), ff-only. Every controlled branch
belongs to one of THREE canonical taxonomy classes, each
with a distinct integration lifecycle: (A) **Taxonomy**
(extends \$. . . /\$. . . (D)) –
feat/<slug> = FEATURE (merges to main after live-manual-
QA, clause B); product/<slug> = PRODUCT line + flavor/
<slug> = FLAVOR line (do NOT merge to main the standard
way, clause C); <slug> lowercase snake/kebab
(\$. . .); the canonical branch NAME is minted ONCE
+ used IDENTICALLY across main repo + every touched owned
submodule (\$. . .), recorded once in the
\$. . . claim registry / \$. . .
logic_group → branch map + LOOKED UP never re-invented
(\$. . .); the prefix STRINGS are operator-canonical,
a project already on a different feature-prefix
(feature/...) is compliant so long as it is the ONE
canonical name per \$. . . , reconciled by the
\$. . . risk-free MERGE-onto-canonical path (ff-
only, no commit lost, never a rename that drops commits).
(B) **FEATURE/main → merge-after-live-QA** – before merging to

main satisfy IN ORDER automated four-layer
 (\$. . (b)) + \$. . full-suite retest THEN
 full LIVE MANUAL TESTING by the QA team (\$. .
 FINAL...human gate, automated GREEN necessary NOT
 sufficient); only after \$. . may the merge
 proceed + it MUST be COMPLETE (main repo + every touched
 owned submodule) ff-only onto latest.. main
 (\$), NEVER force-push, NO commit lost / NO
 history rewrite (\$. /\$. . step), pushed
 to ALL upstreams (\$.); binds the \$. . gate
 as an explicit precondition of the \$. . (I)/
 \$. . approval-gated back-merge. (C) **FLAVOR/
 PRODUCT lines → do-not-merge-to-main-the-same-way** – an
 entirely-new flavor/product line does ALL work on a
 product/<NAME> AND/OR flavor/<NAME> branch (both may
 coexist),...a divergent line with its own release lifecycle
 that does NOT flow back to main via clause B; STILL obeys
 \$. . (no force-push) / \$. (no commit
 lost) / \$. (push all branches all upstreams ff-
 only) / \$. . (MAY merge main INTO itself to
 stay close – trunk-into-line permitted; only line-into-
 trunk is gated/withheld). (D) **Push discipline** (\$. /
 \$) – commit + push ALL branches (feature/
 product/flavor) to ALL upstreams ff-only, NEVER force-
 push; new-branch creation is inherently ff (\$.) + non-
 disruptive from an existing HEAD, permitted in a dirty
 tree.. under \$ provided no commit is in-flight.
 Honest boundary \$. . . – governs branch STRUCTURE +
 integration DIRECTION,...does NOT prove any branch's work
 correct.. (still..\$. . . runtime-signature +
 \$ retest + \$. . . manual-QA).
 Classification:..universal (\$. .) – strict
 generalisation of \$. . + \$. . (D)/
 (I) + \$. . + \$. . . Composes
 \$. /\$ /\$. /\$. . /\$. . /
 \$. . /\$. . /\$. . /\$. . /

§ . . /\$. . /\$. . /
 § . . (D)/(I)/\$. . /\$. . /
 § . . /\$. . /\$. . /
 § . . . Propagation gate CM-COVENANT-114-195-
 PROPAGATION (literal 11.4.195) + recommended gate
 CM-BRANCH-TAXONOMY-CLASS (every controlled branch prefix ∈
 feat/ | product/ | flavor/ |reserved main / master ; a
 product/ | flavor/ branch is NOT the source of a merge onto
 main ; a feat/ | main back-merge cites a \$. .
 manual-QA confirmation) + paired \$. . mutation (a
 product/* merged onto main , OR a feat/* back-merge with
 no \$. . citation → gate FAILs; strip the
 literal → propagation gate FAILs; gate-code = separate
 work item). **Canonical authority:** constitution submodule
 Constitution.md § . . . Non-compliance is a
 release blocker. No escape hatch – no
 --any-branch-prefix-OK , --merge-product-to-main , --skip-live-
 qa-before-merge , --force-push-branch , --partial-submodule-
 merge flag.

§ . . – Native-alias-first priority + per-alias
 real-signal limit/subscription tracking + auto-rebind-on-
 recovery + resource-detection-by-real-identity-not-
 substring-match (User mandate, – –).
 Verbatim operator mandate: use ALL operational claude-
 NATIVE aliases FIRST – as long as ≥ native alias is
 operational (NOT session-rate-limited, NOT weekly-limit-
 reached, NOT expired-subscription) – before ANY provider
 alias; track per alias the session rate-limit / weekly-
 limit / subscription expiry+renewal from REAL signals
 (never faked, \$. .) + record when each becomes
 operational AGAIN; the moment a higher-priority native
 recovers, rebind tracks to it; and fix the host-budget
 guard footgun where a pgrep -f substring-match false-
 REFUSED every worker spawn. The multi-track alias-binding
 + rate-limit/subscription-tracking layer of the
 § . . ruler orchestration MUST implement ALL

of: (A) **NATIVE-ALIAS-FIRST PRIORITY** – every OPERATIONAL native (claudeN) alias is selected before ANY provider, UNCONDITIONALLY, while ≥ native is operational; the guarantee is STRUCTURAL – a native/provider CLASS partition scanned native-class-completely-before-provider-class (\$. . . resolve-by-CLASS-not-by-file-position), so a provider NEVER wins while a non-cooled native exists regardless of roster order; within a class the operator-mandated order (native equal-capability, then providers deepseek→xiaomi→opencode→kimi→...) is authoritative decision-DATA (\$. . .) never re-invented (\$. . .). (B) **PER-ALIAS REAL-SIGNAL LIMIT + SUBSCRIPTION TRACKING** – "operational?" per alias from REAL captured signals, NEVER faked/guessed (\$. . .): the \$. . . captured signature (isApiErrorMessage + apiErrorStatus:429 / api_error_status:429) sub-classified into reason-CLASS closed set {session | weekly | subscription} from real markers (reset/weekly-limit/session-limit); each limited alias records its CLASS + an operational-again EPOCH with a DIFFERENT window per class – session (self-heals at reset, short), weekly (~until the weekly reset, long), subscription (INDEFINITE – operational-again UNKNOWN until a REAL renewal signal, parked with a far-future sentinel; \$. . . NEVER guesses a renewal date, an operator/config supplies it). A limited alias is NEVER auto-selected until its epoch passes (session/weekly) or a real renewal/recovery clears it (subscription). (C) **AUTO-USE-ON-RECOVERY** – the moment a higher-priority native recovers (window elapsed / renewal marked), rebind tracks UPWARD to it (a promote -class op keyed off a comparable priority-rank: native class before provider, then config order), PRESERVING each track's worktree AND device leases (leases key on the STABLE track id – the alias changes, not the track; \$. . . / \$. . .); idempotent. (D) **RESOURCE-DETECTION BY REAL IDENTITY, NOT SUBSTRING-MATCH** (RB- guard fix) –

a "heavy work in flight" guard MUST identify a real resource by its ACTUAL process command line read from the OS (/proc/<pid>/cmdline), NEVER a bare pgrep -f REGEX substring match that false-matches a process merely MENTIONING a token – a pattern-CARRIER (a process quoting the whole alternation pattern; a claude /worker whose prompt embeds the token; a launcher/monitor/test), a same-tool ENGINE process, or the guard's own process/parent (the \$. self/sibling pgrep footgun – forensic FACT - - : the guard false-REFUSED EVERY \$. . spawn on a host with ZERO real builds because a live claude worker's prompt quoted the pgrep pattern); the guard re-reads each matched PID's real cmdline + excludes those carrier/engine/worker/self classes; an UNREADABLE/vanished cmdline is conservatively counted as the real resource (REFUSE – safe reversible default, \$. . /\$. . .), so a genuine build is still caught. (E) ANTI-BLUFF (\$. . /\$. . .) – four-layer coverage each (pre-build gate + on-host test + paired \$. mutation + captured evidence \$. .) with \$. . RED-polarity + \$. . determinism; honest boundary \$. . – mandates the MECHANISM + REAL-signal derivation of "operational", genuine live per-subscription quota-isolation proof still needs live tokens (operator-scheduled acceptance window), never conflated with a mock/hermetic-contract GREEN. Classification: universal (\$. .). Composes \$. . /. /. /. /. /. /. /. /. \$. /\$. /\$. . Propagation gate CM-COVENANT-114-196-PROPAGATION (literal 11.4.196) + recommended gate CM-ALIAS-PRIORITY-LIMIT-TRACKING (native-first CLASS partition; per-alias reason-class + operational-again tracking; auto-rebind-on-recovery; build-guard resolves by real cmdline not bare substring) + paired \$. mutation (swap native/provider priority bases so a provider outranks a native → recommended gate

FAILs; flip the guard's strict-cmdline-filter to the prefix bare `pgrep` → its RB- test FAILs; strip the literal → propagation gate FAILs; gate-code = separate work item). **Canonical authority:** constitution submodule `Constitution.md` § . . . Non-compliance is a release blocker. No escape hatch – no `--provider-before-operational-native`, `--fake-limit-state`, `--guess-renewal-date`, `--no-rebind-on-recovery`, `--substring-match-build-guard`, `--skip-real-signal-classification` flag.

§ . . . – **Research / kicked-off-work completion mandate:** every research effort **MUST** be driven to full, wired, verified completion or explicitly closed – never left un-wired in the backlog (User mandate, – –). Verbatim operator mandate: "all research work we start **MUST BE** fully finished, completely confirmed, validated and verified! It **MUST NOT** happen for it to stay at the backlog uncompleted and un-wired! Loss of requirements, failure to incorporate them is **FORBIDDEN!!!** Everything **MUST BE** fully completed! No exceptions!" Every research effort / kicked-off improvement / design doc / spike / prototype / PoC / partially-landed feature a project **STARTS** **MUST** be driven to **FULL** completion OR explicitly evidence-backed **CLOSED** – there is no third "sitting un-wired in the backlog" state; "we researched/designed/started it" is **NEVER** terminal, and a `docs/research/*` doc / design doc / spike / half-wired feature that is neither **COMPLETED**-and-wired nor explicitly-closed is a § . . . violation of § . **PASS-bluff** severity at the requirements-integrity layer (the requirement was accepted, work began, then it silently evaporated); loss of requirements + failure to incorporate started work are **FORBIDDEN**. **COMPLETED** = () implemented + () **WIRED** (integrated AND used by default, NOT merely present behind a dead flag / uncalled function / unreferenced module / never-selected path – mere-presence-without-wiring is the § . . .

dead-code + \$. . layer- /
SOURCE→ARTIFACT→RUNTIME gap) + () confirmed/validated/
verified with captured physical evidence \$. . /
\$. . /\$. . on a clean deployment
\$. . runtime-signature (never metadata-only/
config-only/absence-of-error/grep-without-runtime
\$. . /\$. .) + () tracked to a terminal Status
\$. . /\$. . . **Explicitly CLOSED** =
deliberately terminated with a documented evidence-backed
reason from the closed vocabulary – \$. .
Obsolete / \$. . structurally-impossible /
\$. . operator-confirmed drop – NEVER silent
abandonment ("no time"/"lower priority" = \$. .
Operator-blocked or a still-open tracked item, NOT a
close). **Mechanical tracker binding** (\$. .): every
research effort / design doc / kicked-off improvement MUST
map to ≥ tracked workable item (ATM-NNN \$. .)
whose completion the loop enforces; a research doc with NO
implementing item, or a tracked item stalled un-wired past
the cadence, is a violation; the autonomous loop
(\$. . / . . / . . / . . / . .) MUST
treat every un-wired research effort as an actionable
queue item – driving it to COMPLETED or explicitly CLOSED
is exactly the "keep working until the queue is genuinely
empty" contract. Honest boundary \$. . – "fully
completed" is bounded by wired+verified OR evidence-
backed-closed, NOT "every speculative idea ships"; every
STARTED effort reaches a documented terminal state, proven
never assumed. Classification: universal (\$. .).
Composes
\$. . / . . / . . / . . / . . / . . / . . / . .
Propagation gate CM-COVENANT-114-197-PROPAGATION (literal
11.4.197) + recommended gate CM-RESEARCH-COMPLETION-TRACKED
+ paired \$. . mutation. **Canonical authority:**
constitution submodule Constitution.md \$. .
Non-compliance is a release blocker. No escape hatch – no

11 4 118
--research-may-stay-unwired , --skip-completion-tracking , --
abandon-without-close , --present-but-unwired-is-done , --
backlog-research-OK , --lose-requirement-OK flag.
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\$. . – Default working mechanisms: multi-alias
native-first orchestration AND heavy token-optimization
are ALWAYS-ON defaults for single- and multi-track work
(User mandate, – –). Operator mandate:
multiple claude-toolkit aliases (native-first) AND heavy
token-optimization MUST be the DEFAULT mechanisms, always
used with single- and multi-track development and with
single or multiple working agents. Two proven mechanisms
promoted to STANDING always-on defaults, engaged
automatically for EVERY configuration (single- OR
multiple-agent, single- OR multi-track), never a per-
request opt-in: (A) **MULTI-ALIAS NATIVE-FIRST ORCHESTRATION**
– the claude-toolkit multi-alias mechanism (every
OPERATIONAL claude-native alias before ANY provider per
\$. . native-alias-first priority + per-alias
real-signal limit/subscription tracking + auto-rebind-on-
recovery, driven by the \$. . automatic multi-
track ruler) MUST be the default execution substrate for
ALL work, NOT only the multi-track case; a SINGLE-track/
SINGLE-agent session STILL binds its worker through the
same native-first alias-priority mechanism (best
operational native + rebind-on-recovery), a MULTI-track/
multiple-agent config uses it across every track
(\$. . exactly-once claim + \$. .
track+branch+alias label); running on an arbitrary/
hardcoded/provider-first alias when an operational native
exists, or bypassing the mechanism because "it is just one
agent", is a violation. (B) **HEAVY TOKEN-OPTIMIZATION** – the
\$. . token-efficiency discipline (compact-
anchor layout with full-one-hop-away, targeted reads over
whole-file dumps, subagent context isolation
\$. . / \$. . , evidence-hash references not
evidence dumps) MUST apply ALWAYS – every read/dispatch/

driven with the TEST'S EXACT sequence the mechanism provably ENGAGED. Every investigation / debugging pass / validation that reproduces a defect or exercises a code path MUST: () **use the existing reproduction's EXACT sequence** when a working reproduction already exists (a test/harness/recorded sequence that demonstrably reaches the defect) – same inputs, same order, same timing, same preconditions, same topology (§ . .); a hand-rolled approximation is NOT a substitute; () **a deviating variant MUST FIRST prove it reaches the precondition** with captured evidence before ANY conclusion is drawn from it; () **a repro that misses the precondition proves NOTHING** – it is NOT evidence of absence (§ . .), and concluding "defect present"/"defect absent"/"the mechanism never engages" from it is a § . /§ . . bluff at the INVESTIGATION layer (a FAIL-bluff when it wrongly condemns working code, a PASS-bluff when it wrongly clears broken code); () **record the exact driving sequence as captured evidence** so it is replayable + auditable, every deviation explicit + justified + precondition-proven.

STRENGTHENS § . . (Phase- reproduce) / § . . (reproduce-first) / § . . (RED on the broken artifact) – they mandate THAT you reproduce, § . . mandates HOW: with the sequence PROVEN to reach the defect. Classification: universal (§ . .). Composes

§ . . /. /. /. /. ./. /. /. /. /. /.

Propagation gate CM-COVENANT-114-199-PROPAGATION (literal 11.4.199) + recommended gate CM-EXACT-REPRO-SEQUENCE + paired § . mutation. **Canonical authority:** constitution submodule Constitution.md § . . . Non-compliance is a release blocker. No escape hatch – no --approximate-repro-OK , --skip-precondition-proof , --hand-rolled-repro-suffices , --absence-proves-absence , --deviating-sequence-conclusion-OK flag.

§ . . – Non-targetable deploy/flash tooling
MUST isolate exactly ONE eligible target and **VERIFY-AFTER-WRITE** on the **INTENDED** target (research-derived,
 – –). Forensic FACT (– –):
 a flashing tool with no device-selection argument **AUTO-SELECTED** a device from the bus, silently wrote the freshly-built image to the **WRONG** board (a sibling still attached), printed **SUCCESS**, and the **INTENDED** board kept its **OLD** build – every downstream test then validated a target that had **NEVER** received the artifact (the
 § . . **ARTIFACT→RUNTIME** layer silently broken while every tool reported green). Whenever a deploy / flash / install / write tool **CANNOT** address a specific target, the procedure **MUST**: () **targetability is a PRECONDITION** – EITHER pass an explicit **STABLE** selector (serial/UUID/stable id per § . . , never an enumeration ordinal) OR **GUARANTEE** exactly ONE eligible target is present at the moment of the write (siblings disconnected/powered-down/excluded), the enumerated eligible-target set captured as evidence (size ==);
 () **the tool's own success message is NOT proof** – an "ok"/exit- proves only that bytes reached the device the **TOOL** selected, **NEVER** the device the **OPERATOR** intended (§ . .); () **VERIFY-AFTER-WRITE on the intended target** – read back the deployed artifact's **IDENTITY** (build id / version / checksum / fingerprint) **FROM** the intended target and assert it equals the artifact just written; a mismatch or the **OLD** identity is a **FAIL**, never a warning (the § . . **ARTIFACT→RUNTIME** assertion applied to the deployment step); () **no validation on an unverified target** – any test run against a target whose deployed-artifact identity was **NOT** verified is **INVALID** and any **PASS** is a § . **PASS-bluff**. Honest boundary
 § . . – isolation + read-back prove the **INTENDED** target holds the **INTENDED** artifact, **NOT** that the artifact is correct (§ . . / . / .).

Classification: universal (\$. .). Composes
\$. . /. /. /. /. . /. .

Propagation gate CM-COVENANT-114-200-PROPAGATION (literal
11.4.200) + recommended gate CM-DEPLOY-TARGET-ISOLATED-AND-

VERIFIED + paired \$. mutation. **Canonical authority:**
constitution submodule Constitution.md \$. . .

Non-compliance is a release blocker. No escape hatch – no

--auto-select-target-OK, --trust-tool-success-message, --skip-
post-write-verify, --validate-unverified-target, --multiple-
targets-on-bus-OK flag.

\$. . – Every guard/gate **MUST** assert the **REAL**
condition: a false-positive refusal is a FAIL-bluff, a
false-negative pass is a PASS-bluff (research-derived,
– –). Forensic FACT (– – ,
two independent instances in one cycle): (a) a host-budget
guard used a bare `pgrep -f '<pattern>'` and REFUSED every
worker spawn on a host with ZERO real builds because a
live worker's own command line merely QUOTED the pattern
(the \$. . (D)/\$. . carrier footgun); (b)
a host-safety pre-flight refused a build on a "swap full"
reading that did not reflect a real memory-pressure
condition, clearing only after the operator manually
cycled swap – BOTH guards refused on a PROXY signal that
was NOT the real condition. Every GUARD / GATE / PRE-
FLIGHT (host-safety, resource-budget, readiness, lock,
capability, availability) **MUST** assert the **REAL** condition
it CLAIMS, resolved from the **AUTHORITATIVE** source (the
real `/proc/<pid>/cmdline`, the real cgroup/meminfo pressure
metric, the real lock-holder liveness via `kill -0`, the
real device identity), **NEVER** from a proxy signal that
something which is NOT the condition can satisfy: () a
FALSE-POSITIVE REFUSAL is a FAIL-bluff (\$. . .) –
blocking work when the condition is ABSENT is exactly as
forbidden as passing while a defect is present (it halts
real work AND teaches operators/agents to bypass guards);
() a **FALSE-NEGATIVE PASS is a PASS-bluff (\$. .);**

() **SELF-VALIDATED** (§ . . ()) – every guard ships a golden-TRUE fixture (condition present → guard FIRES) AND a golden-FALSE fixture INCLUDING the known CARRIER/decoy case (condition absent but proxy signal present → guard MUST NOT fire), both wired into the meta-test; a guard firing on its golden-FALSE fixture is ITSELF the defect; () **conservative-safe default on an unresolvable signal** (§ . . /\$) – refuse the destructive/high-blast-radius action and say so HONESTLY ("could not resolve the condition" is NOT a false positive; claiming a condition never verified IS one, § . .); () **every refusal reports its RESOLVED evidence** (real pid+cmdline, real metric+threshold, real lock holder) so a false positive is diagnosable in one step. GENERALISES § . . (D) (resource-detection-by-real-identity – to EVERY guard) + § . . (liveness PROVEN before reaping – to every guard's condition). Classification: universal (§ . .).

Composes

§ . . . / . / . . . () / . . . / . . . / . . .

§ /\$. /\$. /\$ Propagation gate CM-

COVENANT-114-201-PROPAGATION (literal 11.4.201) +

recommended gate CM-GUARD-ASSERTS-REAL-CONDITION + paired

§ . mutation. **Canonical authority:** constitution

submodule Constitution.md § . . . Non-compliance

is a release blocker. No escape hatch – no

--proxy-signal-OK , --substring-match-guard , --false-positive-refusal-acceptable , --unvalidated-guard , --refuse-without-evidence flag.

EXTENSIONS landed – – (existing anchors strengthened – full text in Constitution.md).

§ . . (G) subagent long-op discipline + conductor-owns-inherently-long-ops: a SUBAGENT MUST NOT run a single synchronous op exceeding the runtime's no-progress stream watchdog (~ min) – it backgrounds + polls (each poll IS the progress the watchdog observes); inherently-long ops

(full builds / full suites ./ device flashes) are the CONDUCTOR's to run (no such watchdog); a stall-killed subagent is a CRASH per § . . . (respawned, never absorbed as a completion); watchdog budget read from the runtime, never guessed (§ . . .). § . . . (F) write-seam hook enforcement:...the doc/DB sync MUST be enforced at the COMMIT seam (hook refuses a commit whose staged set...touches a chain source while its exports/DB are stale) + the BUILD seam (pre-build gate) .+ the CONSTITUTION-PULL seam (§ . . . post-update hook) – a stale CONTINUATION (§ . . .) / session-resumption file (§ . . .) / memory artefact while live state moved is a violation; honest boundary § . . . – seam hooks give eventual-consistency-at-every-write, NOT literal real-time (that needs a watch daemon, carried as a tracked § . . . upgrade path, never claimed as shipped). § . . . (e) API-QUOTA / RATE-LIMIT EXHAUSTION is a FIRST-CLASS CRASH CLASS: an agent killed mid-work by a weekly/session/subscription quota boundary is **crashed** (never **complete**) – record the REAL limit signal + reason-class per § . . . (B), REBIND to the next OPERATIONAL alias per § . . . (A)/(C) and RESPAWN there (bounded-PARK per § . . . only when EVERY alias is cooled), RESUME from the killed agent's captured transcript + partial state at its EXACT last point, and the § . . . /. /. /. loop done-condition MUST NOT read satisfied while any quota-killed entry is non-**complete** . § . . . (F) CONFIGURED ≠ IN USE: forensic FACT – the alias orchestrator was installed with a valid config yet its binding registry was EMPTY while THREE live workers ran outside it, so **promote** rebound ZERO tracks; EVERY spawn path MUST bind THROUGH the orchestrator, and the mechanism's readiness is its RUNTIME SIGNATURE (§ . . .) – **the binding-registry rows MUST EQUAL the live-worker set** – an empty/partial registry alongside

live workers is a FALSE-READY state and claiming "ready/in use" on config alone is a \$. /\$. . orchestration-layer bluff.

\$. . – Reporting directives: a report MUST auto-create a fully-populated, fully-synced workable item – never a prose acknowledgement (User mandate, – –). Verbatim operator mandate: introduce reporting directives `ISSUE` / `BUG` / `TASK` that create a proper workable item filled with all details, with the WHOLE workable-items system automatically up to date and fully in sync – the main DB, all documents, and all external tracking systems; universal, reusable on every project, fully decoupled, recognized immediately by every project that pulls the constitution. Every governed project MUST expose REPORTING DIRECTIVES – the \$. . registered actions `BUG` (Type=Bug), `TASK` (Type=Task), and `ISSUE` (generic report, CLASSIFIED into the \$. . closed set) – such that a plain-language report from the operator is TURNED INTO a real, tracked, fully-synced workable item, automatically, every time. A report that is discussed, acknowledged, answered in prose, or "noted" WITHOUT landing a tracked item is a \$. . violation of \$. . PASS-bluff severity at the requirements-intake layer: the requirement was accepted and then silently evaporated (the \$. . loss-of-requirements failure, at its entry point). () GRAMMAR (\$. .). The three directives are ordinary registry rows, so EVERY \$. . form works out of the box (`BUG :: <text>` , `DEFAULT::BUG :: <text>` , `/DEFAULT::BUG <text>` , `BUG --->` `<text>`), PLUS a SIXTH form this anchor adds – the **single-colon form** `NAME: <text>` (`ISSUE`: subtitles are late), the shape operators actually type. The single-colon form is **REGISTERED-ACTION-ONLY by construction**: a single-colon token that is NOT a registered action is a NO-OP (an ordinary prompt), NEVER an ASK and NEVER a sub-system

shortcut – because ordinary English prose (`TODO:` , `WARNING:` , `FIXME:`) shares that shape (`NOTE:` is now a registered \$. . severity marker), and routing it to the \$. . clarify path would question every such sentence (\$. . honest boundary: the other five forms keep their ASK-on-unknown behaviour – their shapes do not occur in prose). Host-command collisions are handled by the EXISTING \$. . conflict mechanism (`slash_bare: auto` + `slash_conflicts: [..]`), which already satisfies the operator's explicit-prefix requirement: a bare `/NAME` that collides with a built-in (e.g. the documented Claude Code built-in `/bug`) is NOT honored, while the namespaced `/DEFAULT::NAME` form is ALWAYS unambiguous – and the `NAME:` / `NAME ::` / `NAME --->` forms are never affected by any host command. () **THE ITEM.** The created item MUST carry, on ALL surfaces: a valid Status (\$. . , `Queued` at intake) + a Type from the CLOSED set {Bug | Feature | Task} (\$. . – `ISSUE` is a REPORTING CHANNEL, never a fourth type; inventing one is a violation) + a stable auto-incremented id (\$. .), and a COMPREHENSIVE structured description (\$. . D / \$. .): what / affected-scope / reproduction / acceptance. An undetermined section is recorded as an explicit `UNKNOWN:` gap – NEVER invented (\$. .). For `ISSUE` , the type is classified from the report's content and stated as FACT; if the content does not determine it, the operator is ASKED (\$. . / \$. .) BEFORE creation; ONLY when running autonomously where asking is impossible (\$. .) may the \$. . lowest-stakes ambiguity default `Task` be used – and then the defaulted classification MUST be recorded verbatim in the item and surfaced for reclassification, never silently asserted. () **THE FULL SYNC (the load-bearing half).** Creating the item is NOT sufficient: the directive MUST drive the WHOLE workable-items system into sync, in one

shot – (a) the item lands in the SQLite single-source-of-truth (\$₁₁ 4 65 . 11 4 106 / \$. .); (b) EVERY derived document is regenerated FROM the DB – the open + closed trackers, their summaries, and their .md / .html / .pdf / .docx siblings (\$. . / \$. . / \$. .); (c) the item is pushed to EVERY configured external tracker (\$. . D). () **NEVER A FAKED PUSH** (\$. . / \$. . / \$. .). A tracker whose credentials or whose client are absent is SKIPPED with an honest machine-readable reason (credentials_absent – names of the unset variables ONLY, never a value; tracker_client_absent – PENDING-OPERATOR-INPUT). A tracker PASS may be recorded ONLY after a real push command really exited . Fabricating, assuming, or silently omitting a push is a \$₁₁ 177 . bluff at the integration layer; a missing tracker NEVER blocks the item from landing in the DB + docs (the report is never lost because a mirror is unreachable). () **DECOUPLED ENGINE, CONSUMER-OWNED DATA** (\$. . / \$. .). The item-creation + full-sync engine lives in the constitution submodule (constitution/scripts/reporting/report_item.sh), is inherited BY REFERENCE (never copied), and carries ZERO project literals; every project-specific value – DB path, id prefix, sync command, tracker commands + their required env .vars – comes from a CONSUMER-OWNED config file (DATA, never an engine edit). The engine FAILS CLOSED with an actionable message when that config is absent – it never guesses a project's paths (\$. .). () **ANTI-BLUFF** (\$. . / \$. . ()). The mechanism ships four-layer coverage: a pre-build gate (CM-REPORTING-DIRECTIVES) whose FUNCTIONAL invariant SOURCES and RUNS the parser (never a grep-only assertion, \$. .), a paired \$. mutation test that proves each invariant is genuinely load-bearing, and an end-to-end suite run against a REAL SQLite DB with REAL binaries (no mocks,

§ . . .) that asserts a real row lands with the right Type/Status/id, that the docs are regenerated FROM the DB, and that an absent tracker SKIPS honestly – self-validated with a golden-good + golden-bad + negative-control fixture set (an oracle that passes its golden-bad fixture is itself the bluff). Classification: universal (§ . . .) – the consuming project supplies its DB path, id prefix, sync command, and tracker bindings per § Composes § (grammar) / § (DB SSoT) / § / § (status + type + id) / § / § (comprehensive description) / § / § / § (doc sync) / § (credentials) / § / § (clarify, never guess) / § (autonomous decision) / § (a started requirement never evaporates) / § / § (decoupling) / § (auto-registration on constitution pull) / § () / § Propagation gate CM-COVENANT-114-202-PROPAGATION (literal 11.4.202) + recommended gate CM-REPORTING-DIRECTIVES (registry declares ISSUE/BUG/TASK + the single-colon form is registered-only; the engine exists, is parse-clean, and is project-literal-free; the three directives really expand at runtime while prose does not; the honest-skip reasons exist and a tracker PASS is gated on a real exit) + paired § mutation (strip an action row, flip the single-colon form to non-registered-only, strip the parser branch, remove the prose NO-OP branch, make the engine fake a tracker push, or couple the engine to one project → the gate FAILs; strip the literal → the propagation gate FAILs).

Canonical authority: constitution submodule Constitution.md

§

Non-compliance is a release blocker regardless of context. No escape hatch – no `--report-without-item`, `--prose-acknowledgement-OK`, `--skip-item-creation`, `--skip-tracker-sync`, `--fake-tracker-push`, `--invent-a-fourth-type`, `--hardcode-project-in-engine` flag.

\$ `0.7.15` `.` `.` `.` – Instant multi-stream resume engine: whole-fleet continuation state is a durable, content-addressed, atomically-committed snapshot resumed in $O(\text{changed})$ reads, not an $O(\text{history})$ re-read (User mandate, `-` `-` `-`). Compact summary: the continuation mechanism (`$` `.` `.` `.` / `$` `.` `.` `.` / `$` `.` `.` `.` / `$` `.` `.` `.`) is EXTENDED – never replaced – by a mechanical engine so a fresh session resumes ALL work (every Track, main stream, and agent) at once, INSTANTLY and at token cost $O(\text{changed streams})$, by reading ONE snapshot manifest + one compact blob per stream, NEVER re-reading the full handoff history. () **Content-addressed Merkle store** – each stream's canonical state bytes are named by their sha (dedup: unchanged streams share a hash → a global fleet snapshot costs $O(\text{changed})$); a snapshot is ONE manifest referencing every stream's content-hash (`$` `.` `.` `.` `.` ()). () **$O(\text{streams})$ resume, $O(\text{history})$ NEVER on the resume path** – `RestoreAll` reads the manifest + one blob per stream; the append-only event ledger is audit + lost-update recovery only, never read to resume (`$` `.` `.` `.`). () **Tamper-evident + deterministic** – every blob is integrity-checked on read (bytes MUST hash to their id) and a deterministic round-trip (re-serialize must byte-match), no wall-clock in the hashed content, verified by a SELF-VALIDATING oracle: golden-good PASS + golden-bad detected FAIL + negative-control (a legitimately-older-but-valid snapshot) PASS – the false-positive guard proving the verifier distinguishes a LAGGING copy from a TAMPERED one (`$` `.` `.` `.` / `$` `.` `.` `.` () / `$` `.` `.` `.` ()). () **Durable + atomic + single-**

writer + crash-safe – atomic ref write
 (temp→fsync→rename→dir-fsync, \$. . ()),
 append-only ledger (\$. . ()), single-writer-
 per-stream refusal (\$. .), advisory lock with
 provably-stale reap (kill -0 liveness, never steals a live
 lock, \$. . /\$. .). () **Project-agnostic**
engine, consumer-owned data – the engine (github.com/vasic-
 digital/continuum , GitLab mirror) carries ZERO project
 literals, fails closed rather than guess a store path
 (\$. . /\$. .), and is inherited BY
 REFERENCE as a depth- . . submodule under the
 \$. . (C) carve-out (constitution/submodules/
 continuum/ , helix-deps.yaml , zero own-org deps); every
 consumer supplies its store path / actor / stream naming
 as DATA. Anti-bluff (\$. . /\$. .): the whole
 engine ships four-layer coverage (unit + integration +
 e . e . + stress + chaos \$. . , race-clean) with a
 measured resume Metric (blobs_read == 1 + streams is the
 O(streams) proof) and the self-validating oracle as
 captured evidence (\$. . /\$. . /
 \$. .); honest boundary (\$. .): it
 snapshots the state an agent REPORTS, not its execution
 image (not a CRIU/durable-execution runtime,
 \$. .), and est_tokens is a documented bytes/4
 heuristic, not a tokenizer count. Classification:
 universal (\$. .). Composes \$. . /
 \$. . / \$. . / \$. . /
 \$. . / \$. . / \$. . /
 \$. . / \$. . /
 \$. . () / \$. . /
 \$. . / \$. . / \$. . . Propagation
 gate CM-COVENANT-114-207-PROPAGATION (literal 11.4.207) +
 recommended gate CM-CONTINUUM-RESUME-ENGINE-PRESENT (the
 engine present under constitution/submodules/continuum/ , go
 test -race ./... green, continuum selfcheck = good=PASS/
 bad=FAIL/negctrl=PASS, zero project literals) + paired

\$. mutation (strip the literal → propagation gate FAILs; downgrade the resume path to read the ledger, or the oracle to pass its golden-bad → the mechanism gate FAILs; gate-code = separate work item). **Canonical**

authority: constitution submodule Constitution.md

\$. . . Non-compliance is a release blocker. No escape hatch – no --resume-reads-history , --skip-integrity-check , --unvalidated-oracle-OK , --hand-typed-resume-state , --per-project-resume-reimpl flag.

\$. . . – **Operator-request-history document:** every project maintains a project-local, always-in-sync ledger of every operator request/prompt (User mandate, – –). Verbatim operator mandate: introduce a request-history document capturing per operator request/prompt – full content, accepted-timestamp with explicit timezone, the track that processed it, the alias that took the workable item, and the model + effort used – always in sync, no false data, no bluff; the operator's load-bearing correction (\$. . . /\$. . .): the RULE is universal (→ constitution submodule), the request-history DOCUMENT itself is PROJECT-LOCAL (never placed into the constitution).. Every governed project MUST maintain a single project-local, append-only, newest-first **operator-request-history document** at a fixed project-declared path (e.g. docs/requests/history.md ; \$. . . , never moved without a \$. . . operator decision), each entry carrying EXACTLY:.. () **Request content** (verbatim where archived, else a faithful complete summary, marked which); (..) **Accepted (when)** – date + time + TIMEZONE stated EXPLICITLY on every entry (consuming project declares its default zone.. \$. . . ; date-only ⇒ time UNKNOWN , never fabricated); () **Track** (\$. . . /. . .); () **Alias** (\$. . . /. . .); () **Model + effort**. (A) **NO INVENTION** (\$. . .) – an unrecoverable field is recorded literally UNKNOWN , never guessed; a fabricated

timestamp/alias/model is a \$. /\$. . bluff at the requirements-record layer. (B) **HONEST RECONSTRUCTION BOUNDARY** – pre-mandate sessions are best-effort reconstructed from the durable dated record with unrecoverable fields marked UNKNOWN and the boundary stated explicitly, never silently implied complete. (C) **ALWAYS IN SYNC + EXPORTED** – \$. . revision header, \$. append-on-every-new-prompt, \$. . /\$. . four-format export (.md + .html + .pdf + .docx where applicable, siblings ≥ .md). (D) **KEEP-APPLYING MECHANISM** – a helper script and/or a UserPromptSubmit -class hook appends a .newest-first row per NEW prompt at capture time (track/alias derived deterministically \$. . , honest ? / UNKNOWN); a helper landed WITHOUT the auto-capture hook is honestly a partial mechanism (hook wiring = tracked \$. . / \$. . follow-up, never claimed as auto-capture it does not perform). (E) **COMPLEMENTS, NEVER REPLACES** any per-session no-loss ledger (\$. . / .), the workable-items DB SSoT (\$. . / .), or the \$. . /\$. . resumption artefacts; requirement loss at intake is FORBIDDEN (\$. .). (F) **RULE-UNIVERSAL / DOCUMENT-PROJECT-LOCAL** (\$. .) – the RULE lives in the constitution submodule; the document + its path + default timezone are consumer-owned project DATA (\$. . decoupling), NEVER placed into the constitution. Classification: universal (\$. .). Composes \$. . / . / . / . / . / . / . / . / . \$. . . Propagation gate CM-COVENANT-114-208-PROPAGATION (literal 11.4.208) + recommended gate CM-REQUEST-HISTORY-DOC-PRESENT + paired \$. . mutation. **Canonical authority:** constitution submodule Constitution.md \$. . . Non-compliance is a release blocker. No escape hatch – no --request-history-optional , --skip-request-log , --no-timezone-

stamp , --invent-missing-field , --request-history-in-constitution , --prompt-without-history-entry flag.

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§ 2.09.15.2.4.07.15 – Code-review MUST run on the Opus model at xhigh effort as PRIMARY substrate, with a genuine-unavailability Sonnet fallback – no Fable, no escalation tier (User mandate, – – , supersedes – – and – – ; amended – – to add the Sonnet-fallback clause).

Verbatim operator mandate (– – , CRITICAL): "Main model to use is Sonet, reviews and other heavy stuff have to be done by Opus instead of Fable!!! Fix this permanently on all levels!" Verbatim operator mandate (– – , CRITICAL), amending clause (A) – adds a genuine-unavailability fallback, never changes the primary substrate: "If we do not have access to Opus model for some reason but only to Sonnet, we MUST USE what we have! Reviews MUST BE done. Ideally with Opus model, however Sonet MUST BE used if Opus for any reason cannot be used! Add this / extend into main constiotution Submodule and apply the rule asap so nothing is blocked with...review which is not performed yet! We MUST BE maximally efficient!...Work MUST BE performed with all means we have at particular moment!" Every mandatory independent code-review the constitution requires – the

§ 1.10.4.12.4 code-review-agent gate before pre-build + main build, the § 1.10.4.12.4 universal every-change-review (no exception), the § 1.10.4.12.4 exhaustive all-scenario/all-angle review, iterated to a zero-finding GO per § 2026 11 18.18.18 – MUST be dispatched on the Opus model at xhigh effort as PRIMARY substrate, whatever the size or complexity of the reviewed change. (A) PRIMARY Opus-xhigh + genuine-unavailability Sonnet fallback (amended – –) – no escalation tier, no Fable – Fable is NOT permitted in any role, ever (the – – Fable-escalation clause and the – – Fable-first mandate remain WITHDRAWN,

kept verbatim as lineage in Constitution.md); no model other than Opus or the sanctioned Sonnet fallback below substitutes, and no lower effort for Opus itself is acceptable; when Opus-xhigh is genuinely unavailable (a captured FACT § . . , never a guessed unavailability) the review MUST be dispatched on **Sonnet** (at the highest effort the dispatch path can set) rather than blocked – the sole sanctioned fallback, per the verbatim

- - mandate above: reviews MUST be done with whatever capable means are actually available, Opus ideally, Sonnet when Opus genuinely cannot be used; the Opus-unavailability FACT and the Sonnet substitution MUST both be recorded honestly in the review's captured evidence (§ . . / .) – a Sonnet-fallback review is NEVER silently presented as if it ran on Opus; a review is NEVER left undone / blocked solely because Opus is unavailable while Sonnet is reachable (§ . . maximal multi-agent utilization). ONLY when BOTH Opus and Sonnet are genuinely unavailable (each established as its own captured FACT per § . . , never guessed) is the review BLOCKED and deferred – the fact recorded, the owed review tracked (§ . . / § . .); xhigh is REQUIRED on whichever substrate actually runs WHERE THE DISPATCH PATH CAN SET IT (the § . . all-angle depth is unreachable lower) – where it cannot (the pre-existing § . . (F.) harness gap), clause (C)'s honesty rule governs: the gap is recorded, never silently claimed as xhigh unverified. (B) **Independence preserved** (§ . . / .) – the reviewing model (Opus or the sanctioned Sonnet fallback) is the structurally-separated reviewer, never the author; § . . sets the review's MODEL + EFFORT, not its independence / iterate-until-GO (§ . .) / captured-evidence (§ . . / . / .). (C) **Honest boundary** (§ . .) – a review claimed done that ran on Fable, Haiku, or any model other than Opus or the sanctioned

Sonnet fallback, or below `xhigh` , or on Sonnet with no recorded Opus-unavailability fact, is a § . bluff at the review-substrate layer; the review's captured evidence records which model + effort actually performed it (where the dispatch path cannot set or report effort, § . . (F.), that gap is recorded instead of a claimed `xhigh`). (D) **Values already burned in a public repository** (§ . .) – any plaintext credential value that has appeared in documentation, a spec, or commit history is compromised from the moment of commit; an illustration never embeds a real-looking username/password pair without stating it is compromised-by-disclosure. STRENGTHENS

§ / . / . / . – does NOT replace them; it pins the review's execution substrate, now a PRIMARY-plus-sanctioned-fallback pin rather than a single-model pin. Classification: universal

(§ . . .). Composes

§ / . / . / . / . / . / . / . / . / . / . / .

– the `sonnet` rung this fallback draws on). Propagation gate `CM-COVENANT-114-209-PROPAGATION` (literal `11.4.209`) + recommended gate `CM-CODE-REVIEW-OPUS-XHIGH` (amended

`--review-any-model` : now accepts `model=Opus+xhigh` OR `model=Sonnet` carrying a captured Opus-unavailability fact; FAILS on Fable/Haiku/any other model, or on Sonnet with no recorded fact; gate-code = separate work item, NOT claimed shipped § . .) + paired § . mutation. **Canonical authority:** constitution submodule `Constitution.md`

§ Non-compliance is a release blocker. No escape hatch beyond the sanctioned (A) Sonnet fallback (itself requiring a captured Opus-unavailability FACT, never a guess) – no `--review-any-model` , `--review-lower-effort-OK` , `--skip-opus` , `--non-xhigh-review` , `--escalate-to-fable-or-haiku` , `--sonnet-fallback-without-recorded-fact` , `--leave-review-blocked-while-sonnet-reachable` flag.

§ 11.4.192 – Zero-loss request/prompt intake: every operator request/prompt MUST be mechanically CAPTURED, TRACKED, and PROCESSED – never skipped, ignored, avoided, or lost (User mandate, 11.4.192). Verbatim intent: "every request/prompt we issue MUST ALWAYS be respected, taken into account, executed and processed! No avoiding, ignoring, skipping or any form of loss!" Every operator request/prompt MUST be (a) CAPTURED in the 11.4.192 request-history ledger MECHANICALLY at intake via a WIRED verified-runtime-state UserPromptSubmit auto-capture hook (NOT a manual step, NOT a "tracked follow-up" – § 11.4.192 PROMOTES § 11.4.192 (D)'s hook from partial-follow-up to MANDATORY per the § 11.4.192 enforced-not-advisory pattern; forensic FACT 11.4.192: an entire release-drive session's prompts went un-recorded because the hook was left un-wired); (b) TRACKED → a § 11.4.192 workable item / § 11.4.192 completion, or an explicit evidence-backed no-op-with-reason; (c) PROCESSED – respected/executed, never dropped. Loss at ANY of the three stages is a § 11.4.192 PASS-bluff at the requirements-intake layer of § 11.4.192's severity class. Docs-chain-bound (§ 11.4.192), gated (§ 11.4.192 – hook verified-installed in the live hook path, + content-matched, ledger fresh); the autonomous loop (§ 11.4.192 /. 11.4.192 /. 11.4.192 /. 11.4.192) processes every captured request (done-condition never satisfied while any captured request is un-processed, § 11.4.192 (e)); honest § 11.4.192 (UNKNOWN never invented). Classification: universal (§ 11.4.192). Composes § 11.4.192 /. 11.4.192 /. 11.4.192 /. 11.4.192 /. 11.4.192 /. Propagation gate CM-COVENANT-114-210-PROPAGATION (literal 11.4.210) + recommended gate CM-REQUEST-CAPTURE-HOOK-WIRED + paired § 11.4.192 mutation. Canonical authority: constitution submodule Constitution.md § 11.4.192. Non-compliance

is a release blocker. No escape hatch – no `--skip-request-capture`, `--manual-ledger-OK`, `--hook-optional`, `--drop-unprocessed-request`, `--ledger-may-be-stale` flag.

\$ `main → feature/product/flavor` – Merge-conflict resolution during merges MUST run on the Opus model at xhigh effort as PRIMARY substrate, with a genuine-unavailability Sonnet fallback – no Fable (User mandate, `--sonnet-fallback`; amended `--no-fable`; amended again `--no-fable` to add the Sonnet-fallback clause). Verbatim operator mandate (`--no-fable`, CRITICAL), superseding the model choice of the `--no-fable` mandate: "Main model to use is Sonet, reviews and other heavy stuff have to be done by Opus instead of Fable!!! Fix this permanently on all levels!" Verbatim operator mandate (`--no-fable`, CRITICAL), amending clause (A) – adds a genuine-unavailability fallback, never changes the primary substrate: "If we do not have access to Opus model for some reason but only to Sonnet, we MUST USE what we have! Reviews MUST BE done. Ideally with Opus model, however Sonet MUST BE used if Opus for any reason cannot be used! Add this / extend into main.constitution Submodule and apply the rule asap so nothing is blocked with review which is not performed yet! We MUST BE maximally efficient! Work MUST BE performed with all means we have at particular moment!" The `--no-fable` mandate ("During the merge of main branch to all feature/product/flavor branches of tracks `--no-fable`, if conflicts happen they MUST BE resolved using Fable model with xhigh effort! If Fable is not available, then with Opus with xhigh!") remains WITHDRAWN as to its model choice and kept as lineage; its scope and its xhigh requirement stand. Any merge-CONFLICT resolution performed while integrating the canonical trunk (main / master) into a feature / product / flavor branch – the \$ `main → feature` regular main→feature merge cadence, the \$ `main → feature` branch-

taxonomy integration, the § . . . merge-first pipeline, the § . . . (D) trunk-into-stream sync AND the § . . . (I) approval-gated back-merge – MUST be carried out on the **Opus model at xhigh effort** as PRIMARY substrate; the merge-resolution analogue of § . . . , pinning it for the DECISION-MAKING of conflict resolution (which side to keep, how to UNION both sides without loss, marker removal, semantic reconciliation of divergent edits). (A) **PRIMARY Opus-xhigh + genuine-unavailability Sonnet fallback (amended** § . . .) – **no Fable:** Fable is NOT permitted in any role, ever, and no lower effort for Opus itself is acceptable; when Opus-xhigh is genuinely unavailable (not configured / rate-limited / unreachable, established as FACT. § . . . – never a guessed unavailability) the conflict resolution MUST be carried out on **Sonnet** (at the highest effort the dispatch path can set) rather than blocked – the sole sanctioned fallback, per the verbatim § . . . mandate above; the Opus-unavailability FACT and the Sonnet substitution MUST both be recorded honestly in the merge's captured evidence

(§ . . . / . . .) – a Sonnet-resolved merge is NEVER silently presented as if Opus resolved it. ONLY when BOTH Opus and Sonnet are genuinely unavailable (each established as its own captured FACT per § . . . , never guessed) is the resolution BLOCKED and deferred – the fact recorded, the merge NOT completed, the branch left quiescent, with no conflict marker committed; **xhigh REQUIRED** on whichever substrate actually runs WHERE THE DISPATCH PATH CAN SET IT (a careless resolution silently drops a commit or commits a marker) – where it cannot (the pre-existing § (F.) harness gap), clause (D)'s honesty rule governs: the gap is recorded, never silently claimed as xhigh unverified. (B) **Scope:** the operator named tracks – (**feature / product / flavor**); GENERALISES per § . . . to EVERY controlled

main→(feature|product|flavor) merge on EVERY track that hits a conflict... (C) **Safety preserved:** sets the MODEL + EFFORT of the conflict-resolution work only, BOUNDED BY \$. . . (NEVER force-push) / \$ / \$. . . (no commit lost, hardlinked pre-op backup before any large/risky merge) ./ \$. . . step (ZERO conflict markers committed, ZERO file silently dropped, union of both sides preserved) / \$. . . () anti-bluff (post-merge smoke GREEN + marker-scan empty + no-lost-commit) / \$. . . (quiescent-only) / \$. . . / \$. . . (fetch-first). (D) **Honest boundary** (\$. . .): a merge whose conflicts were resolved on Fable or any model other than Opus or the sanctioned Sonnet fallback, or below xhigh , or on Sonnet with no recorded Opus-unavailability fact, is a \$. bluff at the merge-resolution-substrate layer – the merge's captured evidence records which model + effort resolved the conflicts (and, when blocked, the FACT that both Opus and Sonnet were unavailable). Where the dispatch path cannot set or report the effort (the \$ 11 4 188 • 11 • 115 11 4 (F.) harness gap), the evidence records that gap honestly and the merge is never claimed to have run at xhigh on an unverified basis. STRICT EXTENSION of \$ 11 4 289 • 183 • 188 115 41 to merge-CONFLICT resolution; STRENGTHENS \$ 9 9 2 • 6 • 167 / \$. . . / \$. . . by pinning the substrate of their conflict-resolution step. Classification: universal (\$. . .). Composes \$. . . / . / . / . / . / . / \$ / \$. / . / . (D) (I) / . / . / . / . / . (A – the sonnet rung this fallback draws on). Propagation gate CM-COVENANT-114-211-PROPAGATION (literal 11.4.211) + recommended gate CM-MERGE-CONFLICT-OPUS-XHIGH (amended – – : now accepts model=Opus+xhigh OR model=Sonnet carrying a captured Opus-unavailability fact; FAILS on Fable/any other model, or on Sonnet with no

recorded fact; gate-code = separate work item, NOT claimed shipped \$. .) + paired \$. mutation. **Canonical authority:** constitution submodule Constitution.md

\$. . . Non-compliance is a release blocker. No escape hatch beyond the sanctioned (A) Sonnet fallback (itself requiring a captured Opus-unavailability FACT, never a guess) – no --merge-conflict-any-model , --skip-opus-merge-resolution , --non-xhigh-merge-conflict , --resolve-conflict-lower-effort-OK , --escalate-to-fable , --sonnet-fallback-without-recorded-fact , --leave-merge-blocked-while-sonnet-reachable flag.

\$. . – Main README is the canonical starting-point / entry point for ALL project documentation – no doc may be an orphan unreachable from README (User mandate, – –). Verbatim operator mandate: "Manual tests catalog must have linking all through the main README as the starting point of this and any other Project documentation! This fact and mandatory rule MUST be added into the constitution." The main README (README.md at the repository root) is the canonical STARTING POINT / entry point for ALL project documentation, for THIS project and EVERY project incorporating this constitution. Every project document in the \$. . . export scope – INCLUDING the manual-tests catalog (the \$. . manual-QA testing catalog), the \$. . per-feature Status / Status_Summary docs, the Issues / Fixed trackers + their summaries (\$. . / \$. . / \$. .), and every guide / research note / changelog / script companion doc – MUST be reachable / linked (DIRECTLY or TRANSITIVELY) from the main README. No project document may be an ORPHAN – a doc that no link-path from README reaches is a \$. . violation. STRICT GENERALISATION of \$. . (which mandated a Tracked-Items + Status Documents doc-link section covering the tracker / status set) to EVERY \$. . – scope document, so the README is the single navigable root

of the project's documentation tree. (A) **Reachability** = a **link...path**. README → guide → sub-doc is fine (transitive); the requirement is that SOME path from README reaches every in-scope doc, not that every doc is linked directly at top level. (B) **Always-sync** (\$. . . / \$. . . docs-chain). A newly-added \$. . . -scope.doc that no README-reachable path links is a \$. . . violation the moment it lands; the manual-tests catalog specifically MUST be linked from the README (the operator's load-bearing example). (C) **Honest boundary** (\$. . .). Reachability is a link-graph property (every doc FINDABLE from README), NOT a claim about each doc's CORRECTNESS or freshness (those stay \$. . . / \$. . . / the doc's own gates). (D) **Composition with** \$ \$ 11.4.212 . . . 's **Tracked-Items** section remains the mandated README landing block for the tracker / status set; \$. . . adds that EVERY OTHER in-scope doc is likewise reachable — \$. . . 11.4.22 is the specialisation, \$. . . 11.4.22 the generalisation. Classification: universal (\$ 11.4.193 . . .) — the consuming project supplies its README path, its manual-tests-catalog path, and its \$ 11.4.22 . . . doc-set enumeration per \$ Composes \$. . . / \$. . . / \$. . . / \$. . . / \$. . . / \$. . . / \$. . . / \$. . . / \$. . . / \$ Propagation gate **CM-COVENANT-114-212-PROPAGATION** (literal **11.4.212**) + recommended gate **CM-README-DOC-ENTRYPOINT-COMPLETE** + paired \$. . . mutation. **Canonical authority:** constitution submodule **Constitution.md** \$ 11.4.213 . . . Non-compliance is a release blocker. No escape hatch — no **--orphan-doc-OK**, **--readme-not-entrypoint**, **--skip-readme-link**, **--doc-unreachable-OK** flag. \$. . . — **FEATURE** research-scheduling directive: recognized via all \$. . . forms, **SCHEDULES** (never synchronously executes) a deep, enterprise-grade

research + implementation-planning effort as a tracked workable item (operator-directed mandate, - -). Compact summary: extends the \$. . reporting-directive family with a fourth registered action FEATURE – recognized via ALL SIX \$. . grammar forms (FEATURE :: x / DEFAULT::FEATURE :: x / /FEATURE x / /DEFAULT::FEATURE x / FEATURE ---> x / the single-colon FEATURE: x , registered-action-only so lowercase prose like "feature: ..." never expands and is never asked about, \$. .) – that SCHEDULES rather than synchronously executes a deep research + implementation-planning effort: the directive creates a Type=Task / Status=Queued workable item (\$. . /. /.) by INVOKING the SAME \$. . report_item.sh engine (never a duplicate implementation of the item-creation/DB-sync/tracker-push machinery – a second divergent tracker-push path would itself be a bluff-surface regression) and appends it to a durable docs/requests/feature_queue.md queue (mirroring BACKGROUND 's docs/requests/background_queue.md pattern) so a scheduled request is never dropped; the item's comprehensive description (\$. . /.) embeds the FULL research mandate as its acceptance-defining work program – deep web research on best incorporation (\$. . /. /.), systematic-debug of every enumerated weak-spot/gap/danger-zone with a risk-free rock-solid design per gap (\$. .), always enterprise-grade/bleeding-edge/innovative, exhaustive documentation down to lines-of-code + micro-POCs + diagrams + SQL + templates (\$. . /.), reuse-first investigation of vasic-digital/HelixDevelopment components kept decoupled (\$. . /. /.), full test-type + Challenges + HelixQA planning (\$. . /.), fine-grained phased/task/subtask breakdown, enterprise-scalability + max-performance planning, OpenDesign UI/UX wireframes where applicable

(`$ . . /.`), full CodeGraph integration
(`$ . . /. /.`), and fully-synced workable
items across the SQLite SSoT + every connected external
tracker (`$ . . /. /. D`), honestly
SKIPPING any absent tracker (`$ . .`) – NEVER a
faked push. The multi-day research itself is EXECUTED by
the standing autonomous loop
(`$ . . /. /. /. /.`) when it
claims the item, driven to a genuinely COMPLETED-and-wired
or explicitly evidence-backed CLOSED terminal state under
`$ . .` – never left un-wired in the backlog;
"scheduled" is never reported as "done" (`$ . .`).
Anti-bluff four-layer coverage: pre-build gate `CM-FEATURE-DIRECTIVE`
(FUNCTIONAL – sources + RUNS the `$ . .`
parser, never grep-only, `$ . . . . .`) + paired `$ . .`
mutation + a real-DB end-to-end evidence trail (temp
SQLite DB, no mocks, `$ . .`). Classification:
universal (`$ . .`). Composes
`$ . . /. /. /. /. /. /. /.`
Propagation gate `CM-COVENANT-114-213-PROPAGATION` (literal
`11.4.213`) + recommended gate `CM-FEATURE-DIRECTIVE` + paired
`$ .` mutation. **Canonical authority:** constitution
submodule `Constitution.md` `$ . .` . Non-compliance
is a release blocker. No escape hatch – no `--feature-runs-`
`synchronously`, `--skip-feature-queue`, `--feature-without-`
`research-mandate`, `--reimplement-tracker-push`, `--hardcode-`
`project-in-engine`, `--skip-item-creation` flag.

EXTENSIONS landed – – (existing anchors
strengthened – full text in `Constitution.md`; research-
derived from a consuming project's status-without-custody
forensics). Forensic FACTs (genericised, all measured
– –): tracked items,
claiming a done/ready status, (%) with NO
registered regression guard – an operator manually sampled
done-claiming items and found / broken, the
expected draw of that arithmetic, not bad luck; recorded

reopen-per-recorded-fix % (published elite < %),
itself an UNDERCOUNT because recurrences re-entered the
tracker as NEW ids and statuses moved with ZERO history
rows (one item: Reopened events, terminal
events); the standing guard suite's exit code read ONLY
its FAIL count – measured live on a release-candidate
target: **registered guards → PASS / FAIL /**
PENDING → exit "not blocked" – and the
release-tag tooling consulted the guard suite **ZERO times**,
so to the release gate "never verified" and "verified
good" were the same colour; the project's own history
selected the discriminator – every fix whose done-claim
was preceded by an on-device RED→GREEN polarity flip that
HELD recorded zero reopens, and every fix confirmed at
source-green bounced. **No new anchor number was minted:**
§ . . . and § . . . already said what
the operator mandates and did not bind because they are
prose consulted by agents, not conditions checked by seams
– the deliverable is the MECHANICAL BINDING (the
§ . . . test applied: "if I documented this and
nobody implemented the hook, would anything catch it?").
§ . . . (F) – **machine-written verdict pairs +**
guard-viability: RED and GREEN are harness-written verdict
files, never prose – carrying the item id, guard identity,
polarity, exit code, the target's artifact fingerprint
READ FROM THE TARGET AT RUN TIME, iterations (≥ ,
§ . . .), and evidence files whose CLASS matches the
defect's layer (a source-grep transcript can never satisfy
a user-visible/pixel/audio class – the wrong-layer oracle
dies by construction); GREEN requires a prior RED with
exit ≠ on the PRE-fix artifact and a DIFFERENT
fingerprint (identical fingerprints prove the fix was
never deployed); a guard never observed FAILing on the
genuinely-broken artifact is unvalidated instrumentation
and mints no verdicts (a guard structurally unable to FAIL
– or to PASS – satisfies nothing); the canonical § .

mutation for a landed fix is the fix-commit's REVERT, and a mutation whose diff only deletes the literal strings the gate greps for is a refused tautology. Recommended gate `CM-GUARD-VERDICT-MACHINE-WRITTEN` . § . . (verdict-coverage extension) – **ABSENCE of a verdict blocks exactly as a FAIL does:** the suite's exit semantics MUST incorporate COVERAGE, never only the FAIL count; the release seam refuses the tag while `uncovered = registered ^ topology-present ^ no-verdict-for-the-candidate-fingerprint` is non-empty OR any FAIL exists; the release-tag tooling MUST consult the suite/verdict store mechanically; REQUIRES-REBUILD PENDING is honest on intermediate artifacts and self-clearing on the candidate (re-run on the candidate, or the item leaves the release's done-set – both forced, neither a permanent exemption); topology-absent guards are exempt ONLY via a checked-in target-pool topology map and MUST be enumerated in the release changelog as honest gaps (`$ . . /$ . .`) – never silent; a naive "PENDING always blocks" is REJECTED as a `$ . .` FAIL-bluff; pre-existing unguarded backlogs adopt via a ONE-TIME monotone-decrease ratchet snapshot so day one is green and the disease cannot spread. Recommended gate `CM-RELEASE-VERDICT-COVERAGE` (self-tested with golden-good / golden-bad / negative-control per `$ . . / $ . . ( )`). § . . (D) – **status-custody mechanical binding:** a done/ready status write is REFUSED without the file chain registry row keyed by the item's EXACT stable id (checked-in alias map for legacy keys – a guard keyed to a non-item id fails the join and satisfies nothing) → executable registered guard → §11.4.115(F). RED+GREEN verdict pair → class-matched evidence – prose appears nowhere in the chain; Seam A = the workable-items engine (`$ . .`) refuses the write (project-agnostic per `$ . .` , custody paths are consumer DATA per `$ . .`); Seam B = a FULL-TABLE sweep gate on every pre-build run (not a diff check – catches raw-DB writes that bypass Seam A);

Seam C = the § . . . release seam;
 § . . . construction order – the executable sweep
 + its golden-bad meta-fixture (a temp SSoT overlay with a
 chain-less terminal item; the meta-test RUNS the sweep and
 requires FAIL, and asserts the sweep is WIRED) land in the
 SAME commit as any governance text citing them.
 Recommended gate CM-STATUS-CUSTODY . Honest boundary
 (§ . . .), carried explicitly: the binding proves no
 status outruns its evidence and the REPORTED defect cannot
 silently return under the guard's conditions – it does NOT
 prove the feature bug-free (family dedup § . . .
 remains), does NOT make a miscalibrated oracle honest
 (§ . . . () goldens are necessary, not
 sufficient), does NOT replace the § . . . manual-
 QA sufficiency gate, and makes target availability a HARD
 dependency of status progress BY DESIGN (a visible block
 replaces a silent pass). All Classification: universal
 (§ . . .) – no hardware/vendor/project literal;
 consumers supply their registry, verdict-store, sweep,
 ratchet, and topology-map paths as DATA per §
 Propagation gates for the literals 11.4.115 / 11.4.135 /
 11.4.146 are unchanged and stay GREEN; the three
 recommended mechanism gates + paired § . . . mutations are
 gate-code = separate work item.

EXTENSIONS + NEW ANCHORS landed – –
 (round – the carrier/measurement-integrity harvest).
 Research-derived from one cycle's forensics in a consuming
 project, where EVERY false finding had ONE shape: a token
 that MENTIONS X was matched as X, or an instrument
 returned silence that was read as data. Genericised FACTs:
 a removal's completeness was measured by counting
 references to the removed component and the count matched
 the AUDIT-TRAIL COMMENTS documenting the removal (a
 completed removal read as failed); a pattern's escapes
 were eaten by an intervening shell so the far side
 searched a LITERAL and matched nothing (a false "the

component is missing" that would have implied the target could not boot – it had booted); `\|` under an extended-regex dialect is a LITERAL pipe so a mandated-fields check searched a string nobody wrote and nearly reported the fields absent (all were present); a `check | head` pipeline returned the PAGER's exit status so a parse-check reported "clean ✓" three times without checking anything; a line-start-anchored pattern missed an indented, colour-escaped line and read a present verdict as no-verdict; a binary extractor split a multi-byte character and reported a present string as absent. `$ . . (D)'s process-carrier` and `$ . 's self-match` had ALREADY anchored this shape TWICE and it kept recurring – that recurrence is the argument for the general rule. The self-demonstrating instance: a mutation-residue scan run over the governance repo flagged the very anchor that DEFINES the mutation markers, because that anchor quotes them in its rule text (proven a carrier, not residue, by an identical HEAD-vs-worktree count). **The fix, proven in-session: the CONTROL NEEDLE** – an artifact check returned hits; instead of reporting absence it ran a string KNOWN present through the SAME path, got hit, and only THEN reported the zero as real.

EXTENSION – `$ . . ( )( )`: scope widened to EVERY load-bearing measurement + carrier-vs-thing + control needle + metric-validation. `( )` clauses `( )-( )` bind guards (decision seams that refuse/allow) with a refusal-shaped taxonomy; `( )-( )` bind **every measurement whose result is load-bearing** (grep/count/parse/exit-code/scan/metric/artifact-inspection) with a MEASUREMENT-shaped taxonomy – a **FALSE-NULL** (a zero read as absence when the instrument could not see) and a **FALSE-MATCH** (a hit on a carrier) are `$ . /$ . .` bluffs at the measurement layer; the null is more dangerous because a broken instrument and a clean artifact return the identical quiet zero. `( )`(a) MATCH STRUCTURE, NOT

SUBSTRING – distinguish the THING from a CARRIER that MENTIONS it. () (b) **A NULL IS NOT EVIDENCE until a CONTROL NEEDLE proves the instrument can see** – same instrument, same path, same artifact, a known-present needle MUST return non-null; needle absent ⇒ the instrument is blind and the zero says NOTHING (report the blindness, never the absence § . .). **The needle MUST share the certified query's LOAD-BEARING FEATURES** (same dialect constructs / anchoring / quoting / encoding): a bare-literal needle certifies only the layers a bare literal crosses – the () (c) dialect FACT refutes the naive form directly, since a literal needle sails through the same tool+path and would 'certify' a zero from a query still searching a string nobody wrote, whereas a needle carrying the SAME alternation dies with it. Earned inference is therefore BOUNDED: needle found ⇒ the path can see the needle's QUERY CLASS ⇒ a zero from that same class is evidence; a zero from a query using features the needle never exercised needs a needle of its own class. Generalises § . . (F)'s validate-the-detector from guards to every measurement, PER-QUERY – **golden fixtures cannot substitute** (§ . . (D): goldens necessary-not-sufficient), an instrument can pass every fixture on the authoring host and be blind on the real path. () (c) **THE PATH IS PART OF THE INSTRUMENT** – the tool PLUS every layer the query crosses (wrapper-shell quoting, regex dialect, pipeline exit status, anchoring, encoding); each returns a CLEAN CONFIDENT WRONG answer without crashing, so § . . 's script-bug clause (which covers crashes-into-FAIL) never fires and § . . 's parse-check (parse-time only) cannot see it – the control needle is the ONLY mechanism that tests the path. () **A METRIC MUST BE VALIDATED AGAINST THE DEFINITION OF DONE** – construct the correct end-state and evaluate the metric there; if the correct end-state does not move it to target, the metric is INVALID and MUST NOT

gate work (a metric the correct outcome REFUTES will, if enforced, drive work AWAY from correctness); honest boundary § . . . – this constitution already ships one such gating metric (§ . . . 's adoption ratchet, blocking at the RELEASE-TAG seam) whose validation against the definition of done is OWED, not claimed, and which every consuming project operating a ratchet MUST TRACK as a § . . . item (stated as the obligation it is – this anchor asserts NO project has already filed it; asserting an unverified tracker state would be the § . . . violation the clause forbids). **Interim precedence:** until that validation lands the § . . . () ratchet KEEPS GATING – clause () does NOT disarm it (an unvalidated ratchet blocking a release is bounded + visible + operator-resolvable; disarming the only adoption mechanism would silently reopen the unguarded-backlog hole § . . . closes – the § . . . reversible-safe choice); clause () binds every NEW metric from this anchor forward, and the ratchet's validation as owed work – never as a licence to switch it off. Gate contract landed in the canonical file: **CM-GUARD-ASSERTS-REAL-CONDITION** EXTENDED to the measurement layer ((i) a reported absence cites a class-matched control-needle result; (ii) the needle's class matches the certified query's load-bearing features) + NEW recommended sibling **CM-METRIC-VALIDATED-AGAINST-DONE** (every gating/ranking metric carries a recorded definition-of-done evaluation reaching target) + paired § . . . mutations (bare-literal needle on a dialect-dependent query → (ii) FAILS; needle stripped → (i) FAILS; metric whose correct end-state moves AWAY from target → FAILS). Gate-code = separate work item – the contract is landed, the code is NOT claimed shipped (§ . . .).

§ . . . (NEW) – **Recurrence-links-not-mints:** a defect that returns MUST reopen its existing item, never enter as a new id. FACT: recurrences of already-"fixed"

defects re-entered as BRAND-NEW ids ($\geq$ chains, one three deep), so the ORIGINAL's reopen counter never incremented – and the $\$$. . . `reopens_count`, the exact signal $\$$. . . (d)/ $\$$. . . use to rank the most-fragile work for the deepest scrutiny, **stays quiet precisely on the items that keep breaking**; the recorded % reopen rate was a KNOWN UNDERCOUNT for this reason; ~ items mapped to ~ real root causes, so "fix all " would have forked duplicate/conflicting fixes on one cause. Before minting a NEW id from ANY intake path, check for an existing item describing the SAME defect $\rightarrow$ RESOLVE the match THROUGH its `duplicate-of` links to the CANONICAL CHAIN HEAD ($\$$. . . 's closure vocabulary includes `duplicate-of`, and clause () MANUFACTURES such items – reopening one would resurrect a non-canonical copy beside its live canonical item, forking one defect into two and crediting the $\$$. . . signal to the WRONG item), then LINK it, and REOPEN ($\$$. . .) **iff the resolved head is in a TERMINAL state**; multi-match $\Rightarrow$ resolve every match to its head and act on the SINGLE survivor, > distinct head or a broken/circular link $\Rightarrow$ UNDECIDED (ask / mint-with-link, never guess $\$$. . .). **Open-original case:** a matched item still OPEN gets LINK-ONLY – 'reopen an open item' is undefined in the $\$$. . . status machine and $\$$. . . 's `Reopened` presupposes a closure to demote FROM ($\$$. . .); the reopen path exists to correct the specific lie 'closed as fixed and it is not', and an open item is not telling it. $\$$. . . **precedence (explicit, never left to a consumer):** $\$$. . . 's mandate is that no report is ever LOST – SATISFIED IN FULL by a SAME-DEFECT report landing as link + (terminal-only) reopen; it forbids the report evaporating into prose, it does NOT require a fresh id per report. $\$$. . . refines WHICH item a report lands on; $\$$. . . owns THAT it lands; DISTINCT/UNDECIDED verdicts run

§ 11.4.24. . . 's mint path unchanged. () EVERY intake path binds (reporting directives § . . . , AI-detected test/gate failures, regression-guard verdicts, cycle re-discovery, operator manual testing – § . . . 's own reason vocabulary enumerates them; wiring dedup ONLY into the reporting channel under-covers exactly the machine-driven paths that recur most). () Key on ticket else normalised (subject, scope) – NEVER a bare subject substring (§ . . .). () **A wrong merge silently DELETES a real defect – so distinct-but-similar is a FIRST-CLASS verdict:** SAME-DEFECT ⇒ link+reopen, DISTINCT ⇒ mint (correct + expected), UNDECIDED ⇒ ask (§ . . . / . . .) or mint-WITH-a-candidate-duplicate-link; autonomous default (§ . . .) is mint-with-link over merge – a spurious id is cheap + recoverable, a wrongly-merged defect is LOST (the asymmetry decides). () Self-validated oracle (§ () / §): golden-good + golden-bad + a **negative-control** (two genuinely DISTINCT, same-subject items that MUST NOT merge) – an oracle without it is a false-merge engine. () Id stability preserved (§ – reopen the original, never renumber/reuse/retire; an already-minted duplicate is LINKED via § duplicate-of, never silently deleted). () The restored signal IS the point – an undercounting reopen counter is a § . . . / § . . . bluff at the PRIORITISATION layer: it reports the fragile set as stable and aims the deepest scrutiny at the wrong items. Honest boundary § – dedup makes the signal TRUE, not the item correct, and a same-symptom recurrence with a different root cause is real (§ decides; the linked+reopened item is where that is recorded, never a reason to fork). Gate **CM-RECURRENCE-LINKS-NOT-MINTS** (every CONFIRMED-duplicate link whose target was TERMINAL at link time has a reopen event on that target) – **its own scope is § -bound:**

it MUST NOT fire on an OPEN-target link (LINK-only is correct there) nor on a clause-() UNDECIDED candidate-duplicate link (un-reopened BY DESIGN, and the mandated autonomous default), or it becomes exactly the
§ . . . () false-positive refusal this same round harvests against; its golden-FALSE set MUST include both + paired § . . . mutation. Why NEW: § . . . CONSUMES the count, § . . . presupposes the status is already Reopened, § . . . forbids only one-id-two-items, § . . . 's DEDUP needs DIVERGENT metadata and gates export/sync/commit not INTAKE, and
§ . . . 's operative text, read WITHOUT this anchor, reads as mint-every-time – the reading the precedence paragraph above corrects (they are NOT in conflict: § . . . owns THAT a report lands, § . . . refines WHICH item it lands on).

§ . . . (NEW) – A doc that BINDS work MUST live, tracked, in the repository where that work happens. FACT: the document DEFINING a binding acceptance gate was UNTRACKED and lived in an out-of-tree scratch directory belonging to a different checkout – so the gate could not be honoured (unreadable), reviewed (no diff/history/authorship), version-controlled (no revision/rollback), or reconciled when it contradicted the source of truth. A binding artifact that lives nowhere is a rumour with an imperative mood. () **BINDING ⇒ TRACKED, and un-tracked ⇒ NOT BINDING** – a binding claim sourced from an untracked/out-of-tree doc MUST NOT be enforced; land the doc or decline the gate, never obey a citation of it. () IN THE REPOSITORY THE WORK HAPPENS IN – a sibling checkout / scratch dir / chat message / issue comment is NOT tracked for this purpose (the test: can someone who clones ONLY that repo read the binding text at its declared path?); multi-repo binding docs are inherited BY REFERENCE from a tracked shared dependency (§ . . . / § . . .), never copied, never out-of-tree. ()

The prior question § . . cannot ask –
 § ENUMERATES the in-repo scope and checks reachability, so a doc in no checkout is never enumerated and its gate cannot FAIL; § answers "is it findable?", § answers "is it here at all?" (once in-repo, § /. /. /. /. all attach as normal – this is their precondition, not their competitor). () The generalisation of § (whose entire content is "authoritative ⇒ tracked, NOT a build artefact, IS authoritative source data" for ONE named artifact), exactly as § was minted as the generalisation of § § is the counterweight and stays intact – logs/forensics/scratch remain out-of-tree and untracked by DEFAULT; **binding is what pulls a document into the repository**, and a doc nobody cites as binding is governed by § , not this anchor. Honest boundary § – in-repo presence proves AVAILABLE + ATTRIBUTABLE + VERSION-CONTROLLED, never CORRECT/current (§ /. /.), and never well-seated (§ 's seam clause – whose forensic case was carried by this SAME out-of-tree document: two defects, one artifact, which is itself the argument that an unreviewable binding doc is where bad gates hide). Gate CM-BINDING-DOC-IN-REPO + paired § mutation.

All Classification: universal (§) – no hardware/vendor/project literal; consumers supply their intake wiring, id scheme, normalisation function, and binding-doc paths as DATA per § Propagation gates CM-COVENANT-114-214-PROPAGATION + CM-COVENANT-114-215-PROPAGATION (literals 11.4.214 / 11.4.215); the literals 11.4.201 / 11.4.112 / 11.4.120 / 11.4.69 are unchanged and their propagation gates stay GREEN. All recommended mechanism gates + paired § mutations = gate-code, a separate work item.

DESIGN-METHODOLOGY MANDATES landed

(§ 11.4.214 - § 11.4.223) – full text in `Constitution.md` ; proving project: the Helix Thready MVP design package at `docs/public/research/mvp/design/`). The complete design methodology built and verified end-to-end in Helix Thready is codified as eight universal anchors so every governed project inherits it out of the box. All Classification: universal (§ 11.4.214) – consumers supply brand values, platform lists, paths, and submodule pointers as DATA per § 11.4.214 . Propagation gates CM-COVENANT-114-216-PROPAGATION ... CM-COVENANT-114-223-PROPAGATION (literals 11.4.216 – 11.4.223) + recommended mechanism gates + paired § 11.4.223 mutations = gate-code, a separate work item.

§ 11.4.223 – Canonical machine-readable design-token source (User mandate, § 11.4.223). Compact summary: every UI-shipping project maintains its design tokens as ONE canonical machine-readable source – CSS custom properties, light in `:root {}` , dark via the `[data-theme="dark"]` override (composing the sanctioned dark mechanisms) – and EVERY surface binds via `var(--...)` or a GENERATED binding produced by codegen FROM that file, never a hand-kept copy (hand-copies drift silently); NO INVENTED VALUES (§ 11.4.223 applied to design): every color/size/duration traceable to captured evidence – brand colors to a recorded eyedrop/measurement of the canonical brand asset (proving project: brand `#B6E376` = mean of `light` , `dark` , logo pixels), accent pairs to MEASURED WCAG ratios against their PAIRED background (`light` : `light` / `dark` : `dark`), structural values to the documented shared scale; a hex resolving to no token is a violation – use the nearest token, never mint one; the token source is the white-label seam (§ 11.4.223 closed override set); not-yet-built codegen bridges are [OPEN]-tracked (§ 11.4.223), never claimed shipped. Composes

§ . . /. /. 11 / 216 /. /. .
 Propagation gate CM-COVENANT-114-216-PROPAGATION +
 recommended gate CM-DESIGN-TOKEN-SINGLE-SOURCE + paired
 § . mutation. **Canonical authority:** constitution
 submodule Constitution.md § . . . Non-compliance
 11 4 217 is a release blocker. No escape hatch – no --hand-copied-
 palette-OK , --inline-hex-OK , --invent-token-value , --light-
 only-tokens , --platform-keeps-own-palette flag.
 1026 07 22

§ . . – OpenDesign brand contract: –section
 DESIGN.md + tokens.css twin, disjoint color roles, AA-
 pinned accents, locked attribution footer (User mandate,
 – –). Compact summary: every UI-shipping
 product carries under design/opendesign/ (or its declared
 equivalent § . .) a DESIGN.md in the VERIFIED
 OpenDesign –section schema (H + > Category:
 blockquote immediately after the H ; number-prefix-parsed
 sections; a "Font labels for catalog extraction" block;
 real hex; CSS variables wrapped in :root {} with dark via
 [data-theme="dark"] ; real component CSS with
 prefers-reduced-motion targeting specific elements, never a
 global * ; :focus-visible on every interactive component –
 format verified against the tool's CURRENT source per
 § . . , never from memory) + the machine-readable
 tokens.css twin (the § . . canonical file or a
 compiled view; prose and tokens never diverge); THREE
 DISJOINT color roles by contract – brand DECORATIVE only
 (never body text), accent INTERACTIVE + AA-PINNED
 (measured ≥ . : per mode), semantic STATE never
 overridden by any brand (a brand color can never mask an
 error signal); white-label = a CLOSED override set (brand
 tokens, accent pair, logo, slogan) with accent overrides
 SERVER-VALIDATED to WCAG AA or REJECTED carrying the
 measured ratio + an AA-passing suggestion; structural/
 neutral/semantic tokens NOT account-overridable; the
 organization attribution footer is LOCKED – present on
 every surface, never removed even fully white-labeled.

Composes

```
$ . . /. /. /. 114 217 /. /. .  
Propagation gate CM-COVENANT-114-217-PROPAGATION +  
recommended gate CM-OPENDESIGN-BRAND-CONTRACT + paired  
$ . mutation. Canonical authority: constitution  
submodule Constitution.md $ . . . Non-compliance  
is a release blocker. No escape hatch – no --skip-brand-  
contract , --prose-only-design-md , --brand-as-text-OK , --  
unvalidated-accent-override , --remove-attribution-footer , --  
semantic-tokens-brandable flag.
```

```
$ . . – Living design-library catalogue + per-  
cell-provenance reusability matrix (User mandate,  
– – ). Compact summary: every UI-shipping  
project maintains a LIVING component catalogue rendering  
EVERY component × declared state × light/dark from SELF-  
CONTAINED artifacts (zero external requests – no CDNs,  
vendored libs inline, honest font fallbacks), token-bound  
($ . . ), with REAL interactive behaviors (real  
dialog/sort/toggle; prefers-reduced-motion honored per  
$ . . ) – proving project: groups, ~  
components, ~ state cells, ~ rendered  
states, both themes auditable via a real persisted toggle;  
PLUS a components × platforms reusability matrix where  
EVERY cell carries closed-set provenance – VERIFIED (re-  
checked at source, dated) / PROPOSED ([DEFAULT –  
adjustable]) / ASSUMED (labeled inference); an unmarked  
cell reads ASSUMED; a scaffold cell claiming VERIFIED is a  
$ . . . PASS-bluff at the design-planning layer; the  
matrix is read BEFORE implementing any component on any  
platform. Spec'd-but-unrendered components/states are  
tracked gaps ($ . . ). Composes  
$ . . /. /. 114 218 /. /. .  
Propagation gate CM-COVENANT-114-218-PROPAGATION +  
recommended gate CM-DESIGN-LIBRARY-LIVING-CATALOGUE + paired  
$ . mutation. Canonical authority: constitution  
submodule Constitution.md $ . . . Non-compliance
```

is a release blocker. No escape hatch – no `--cdn-in-catalogue`, `--single-theme-catalogue`, `--mocked-behaviors-OK`, `--unmarked-matrix-cell`, `--scaffold-claims-verified` flag.

\$. . – Per-app screen catalogues: full-IA coverage, per-platform chrome, honesty markers (User mandate, – –). Compact summary: every shipped surface (web / mobile / desktop / TUI / marketing / ...) has a screen catalogue covering EVERY node of its information architecture (an IA node with no catalogue screen is a tracked gap, never silently skipped); per-platform CHROME VARIANTS rendered where one design serves several hosts (mobile OS chrome toggle, per-OS window chrome, terminal-width constraints); each screen SELF-CONTAINED, token-bound (\$. .), light + dark, populated with REALISTIC product data and a per-screen state contract (loading/empty/error/populated); UNBUILT or UNDECIDED behavior is HONESTLY marked – [GAP: id] / [OPEN: id] (\$. .), labelled placeholders, inline notes, disabled controls – NEVER a designed-as-if-real fake flow (a fake flow presented as settled is a \$. PASS-bluff at the product-design layer). Proving..project:.. web + mobile (Android/iOS chrome toggle) + desktop shell + TUI + marketing screens, every unbacked flow marked.

Composes

\$. . /. . . /. . . /. . . /. . . /. . .

Propagation gate `CM-COVENANT-114-219-PROPAGATION` + recommended gate `CM-SCREEN-CATALOGUE-FULL-IA` + paired \$. mutation. **Canonical authority:** constitution submodule

`Constitution.md` \$. . . Non-compliance is a release blocker. No escape hatch – no `--partial-ia-coverage`, `--one-chrome-fits-all`, `--fake-flow-OK`, `--lorem-ipsum-content`, `--single-theme-screens` flag.

\$. . – Open-first design tooling: self-hosted PenPot primary via the PenPot submodule; proprietary tools

import/export only; in-repo open-format sources (User mandate, - -). Compact summary: the collaborative design platform is an open-source self-hostable tool – PenPot as the standing default – run SELF-HOSTED and consumed ONLY via the dedicated reusable PenPot submodule (working name `vasic-digital/PenPot`, final repository name pending – the generic referent is normative meanwhile) carrying ALL PenPot mechanisms/utilities (deployment recipe, RPC client, access-token minting, import tooling, browser-hydration automation, MCP wiring), itself consuming the Containers submodule (`$ . . . , rootless $ . . .`) and DECLARING that dependency in its `$ . . . helix-deps.yaml`; projects MUST NOT hand-roll PenPot deployment/client code (`$ . . . /$ . . .` – extend the submodule upstream); the submodule carries mechanisms ONLY, never project design content. Proprietary tools (Figma or equivalent) are OPTIONAL import/export/review bridges, NEVER the source of truth – no design value may exist only inside a proprietary tool/cloud; design sources live IN-REPO in open, diffable formats (SVG masters, CSS tokens, JSON manifests/IR/variables payloads, Lottie JSON, self-contained HTML – `$ . . .` applied to design). Tooling capability claims are RUNTIME-PROBED (`$ . . . /$ . . .`) with a headless-vs-operator-interactive matrix + an honest-limits section – a cloud push claimed without a real credentialed run is a fabrication. Operational rules (research-verified): the HEADLESS write path is the token-authenticated RPC API – access tokens (minted under the `enable-access-tokens` backend flag, handled per `$ . . .`) are the MANDATED automation auth mechanism, never interactive OAuth relays; API-created PenPot files REQUIRE a browser hydration pass before handoff (thumbnails/text layout materialize in the browser runtime, not the write API) – an unhydrated API-created file handed off as render-ready is a `$ . . .`.

PASS-bluff at the design-handoff layer. Composes
\$. . /. /. /. 11 4 20 /. /. /. /.
Propagation gate CM-COVENANT-114-220-PROPAGATION +
recommended gate CM-OPEN-DESIGN-TOOLING-PRIMARY + paired
\$. mutation. **Canonical authority:** constitution
submodule Constitution.md \$. . . Non-compliance
is a release blocker. No escape hatch – no --figma-is-
source-of-truth , --cloud-only-design , --proprietary-format-
source , --hand-rolled-penpot-stack , --oauth-relay-automation ,
--skip-hydration-pass , --unprobed-tooling-claim , --rootful-
design-stack flag.

\$. . . – **Motion discipline: token-bound timing + machine-readable manifest + reduced-motion static fallbacks** (User mandate, – –). Compact summary: every duration/easing binds to the
\$. . . motion tokens (an ad-hoc timing literal bypassing the token layer violates \$. . .); every playable animation ships a STATIC reduced-motion fallback DERIVED FROM ITS OWN KEYFRAMES (poster frame for loopers, final frame for one-shots) in an open format, named per animation in a versioned MACHINE-READABLE motion manifest (id, loop mode, poster, reduced-motion mode, reducedMotionFallback) – a deliberate instant-cut is an EXPLICIT null-with-reason, never a silent absence; prefers-reduced-motion: reduce honored on EVERY surface via targeted component gates (never a global *), swapping in the manifest-named fallback; NO animation is claimed working without a REAL render in a real runtime (parse + structural schema + rendered geometry with progression assertions – the \$. . . () analyzer discipline applied to motion assets), unexercised runtimes [OPEN]-tracked. Proving project: hand-authored Lottie animations + keyframe-derived fallback SVGs + manifest v , lottie-web/jsdom + headless-Chromium validated, mobile runtimes honestly [OPEN]. Composes
\$. . . ()/. /. /. /. .

Propagation gate CM-COVENANT-114-221-PROPAGATION +
 recommended gate CM-MOTION-TOKEN-BOUND-REDUCED-FALLBACK +
 paired \$. mutation. **Canonical authority:** constitution
 submodule Constitution.md \$. . Non-compliance
 is a release blocker. No escape hatch – no
 --ad-hoc-duration , --no-reduced-motion-fallback , --global-
 star-motion-kill , --manifest-optional , --unrendered-animation-
 claimed-working flag.

\$. . – **Design-completion export wave: a design is not done until the exports land, honestly (User mandate, – –)**. Compact summary: a design (system / brand / screen set / change wave) is NOT done until its EXPORT WAVE runs with verified outputs – () per-platform token codegen regenerated from the canonical source (or [OPEN]-tracked); () the .FULL icon/asset raster matrix, documented per OS surface and locked : to its generator, produced by an HONEST-OUTPUT pipeline: every candidate renderer PROBE-TESTED with a content-verified probe render before use (\$. . – presence ≠ correctness), fall-through on probe failure, absent tools SKIP-with-note (\$. . /\$. . .), NEVER a placeholder/blank/faked output, every raster verified non-degenerate (size floor + non-blank; multi-resolution bundles opened + verified per \$. .) – proving project: real rasters, zero faked; () both-theme screen renders (deterministic theme forcing, every render verified non-blank) + the bound design book; () hand-off bundles with per-leg HONEST statuses (produced / deferred-operator-only / impossible-with-reason); () the \$. . /\$. . doc twins regenerated. A design wave reported complete without its export wave – or with any faked/unverified export output – is a \$. PASS-bluff at the design-delivery layer.

Composes

\$. . /. /. /. /. /. /. /. /.
 Propagation gate CM-COVENANT-114-222-PROPAGATION +

recommended gate CM-DESIGN-EXPORT-WAVE-COMPLETE + paired
§ . mutation. **Canonical authority:** constitution
submodule Constitution.md § . . . Non-compliance
is a release blocker. No escape hatch – no --skip-export-
wave , --unprobed-renderer , --placeholder-raster-OK ,
--single-theme-renders , --fake-bundle-status , --stale-doc-
twins flag.

§ . . – Provenance-marker discipline for design
docs + central open-item registry (User mandate,

– –). Compact summary: every design
document carries inline provenance markers from a closed,
project-declared vocabulary, whose MINIMUM CORE is
[VERIFIED] (with evidence/check date) / [OPERATOR] /
[DEFAULT – adjustable] / [OPEN: id] (extensions:
[RESEARCH] , [ASSUMED] , [BUILD-NEW] , [GAP: id] , [IN-HOUSE:
...] , [CONSTITUTION \$x] per § . .); an unmarked
load-bearing claim reads UNVERIFIED, and an assumption
presented as verified is a § . . violation of
§ . PASS-bluff severity at the design-documentation
layer; every unresolved decision is an [OPEN: id] with a
stable, never-renumbered id (§ . . discipline)
owned by a named file; the design area index maintains the
SINGLE CANONICAL open-item registry, TREE-VERIFIED in both
directions (every tree id in the registry, no registry
ghosts – the § . . consistency discipline
applied to design opens), closures citing the closing
revision, narrowed items recording exactly what closed and
what remains; registry items map to tracked workable items
(§ . . /§ . . /§ . .) so the
§ . . completion mandate applies – no design
open is silently abandoned. Proving project:
registered ids across nine families, tree-verified,
closed with citations. Composes

§ . . /. /. /. /. /. /. /.
Propagation gate CM-COVENANT-114-223-PROPAGATION +
recommended gate CM-DESIGN-DOC-PROVENANCE-MARKERS + paired

\$. mutation. **Canonical authority:** constitution
 submodule Constitution.md \$. . Non-compliance
 is a release blocker. No escape hatch – no --unmarked-
 claims-OK , --assumption-as-verified , --unregistered-open-
 item , --registry-optional , --silently-abandon-open flag.

\$. . – Test-first (TDD) for ALL work, not only
 fixes + a minimum code-coverage floor that is NECESSARY-
 never-sufficient (User mandate, – –).

Verbatim operator mandate: "Any work we do MUST START by
 writing the test! We MUST FULLY comply and follow the TDD
 – The test driven development! It MUST BE applied for any
 change, implementation, wiring, tests, challenges and any
 other form of codebase we write! Bash scripts as well!

Coverage with TDD MUST BE close to %, not less than

% !!!" **The gap closed:** \$. . 's operative text
 binds FIXES only ("Every fix MUST follow the –step TDD-
 fix workflow"), so a NEW feature / wiring / gate /
 Challenge / BASH SCRIPT inherited no test-first
 obligation; and \$. . /\$. . 's " %"

is test-TYPE breadth (which KINDS of test exist per
 feature × platform), stating no CODE-coverage figure
 anywhere – \$. . closes exactly those two holes
 and claims nothing else as new. **(A) TEST-FIRST BINDS ALL**

WORK – every work product with an executable surface
 (change, feature, implementation, wiring, refactor, gate,
 \$. . mutation, Challenge, test-harness code, AND bash/
 shell scripts): the test is written FIRST, run FIRST, and
 OBSERVED TO FAIL for the right reason before the

implementation exists (RED on the broken artifact per
 \$. . where one exists; RED against the absent
 behaviour where the work is new); a test authored AFTER
 its implementation proves only that it AGREES with the
 code, never that it CATCHES the defect (the \$. .

PASS-bluff generalised). Per-work-class test named
 explicitly (\$. .): bash/shell script → an executing
 test through its REAL invocation path asserting exit

status + output + state delta, NEVER a `bash -n parse-check` alone (parse-cleanliness is § 11.4.110. . . 's separate obligation and proves nothing about behaviour); a pre-build GATE → its paired § . . . mutation IS the test-first artifact, authored and observed to make the gate FAIL BEFORE the gate is trusted (§ . . . (F) – observation-before-trust, NOT authorship order); a Challenge → its scored assertion against captured evidence (§ . . . /§ . . .); a behaviour-bearing config/data change (any committed data an executing consumer reads to decide behaviour) → an executing test that drives that CONSUMER through the changed value and asserts the changed observable behaviour. (B) **CODE-COVERAGE FLOOR** ≥ %, ~ % **TARGET**, applying to the executable codebase INCLUDING bash/shell (the operator's "Bash scripts as well!"). (C) **NECESSARY, NEVER SUFFICIENT – anti-gaming** (§ . . . ()/\$. . . /§ . . .): MEASURED FACT (in-session, - -) – a line-granular instrument reported an `if/else` line COVERED while the `else` branch was NEVER taken, and an assertion-free test raises the number identically to a proving one; therefore the REAL bar remains catch-its-own-negation proven by a paired mutation (a line executed by an assertion-free test is NOT covered in the sense that matters), the floor is a MINIMUM ON A PROXY, and citing "≥ % covered" as proof a change works is a § . . . PASS-bluff at the metric layer; the floor NEVER stands alone (composes § . . . + § . . . 2024 . 12 /§ . . . / § . . . captured evidence + § . . . runtime signature). (D) **METRIC VALIDATION** (§ . . . ()) – **SPLIT VERDICT, honestly stated:** directional validation PERFORMED - - (at the correct end-state – every behaviour test-driven, every test catching its negation – the implementing lines ARE executed by their tests, so the metric DOES reach target; the floor is VALID as a NECESSARY condition); the CONVERSE

is REFUTED by the clause-(C) measured gameability fact;
and the EMPIRICAL per-corpus calibration of the %
threshold is explicitly OWED, NOT claimed – every
consuming project enforcing the floor MUST track it as a
§ item (this anchor asserts no project has
already filed it – asserting an unverified tracker state
would be the § violation the clause prevents);
INTERIM PRECEDENCE mirroring § ()'s own
§ -ratchet precedent – the floor KEEPS GATING
while the calibration is owed (an uncalibrated floor
blocking a merge is bounded, visible, operator-resolvable;
disarming it silently re-opens the untested-code hole –
the § reversible-safe choice), never a
licence to switch it off. (E) HOW MEASURED, per language,
honestly (§ /§): the consuming
project supplies its per-language instrument as DATA; the
universal rule is that an instrument MUST exist and MUST
actually be RUN – a floor nobody can measure is an
unmeasurable gate and itself a § bluff, and
a percentage nobody measured is a §
fabrication. MEASURED host reality (- -):
Go built-in coverage (go test -coverprofile + go tool
cover) + gcov PRESENT, but kcov / bashcov / bats /
shellspec / shunit2 ALL PROBED ABSENT for bash (shellcheck
IS present but is a LINTER, not a coverage instrument, and
satisfies nothing here). PROVEN-FEASIBLE zero-tooling bash
mechanism, measured in-session never asserted: PS4='+COV:\$
{BASH_SOURCE##*/}:\${LINENO}:' bash -x <script> emits executed
line numbers, and executed ÷ executable (non-blank, non-
comment) source lines yields line coverage – a probe
correctly identified an uncovered function body, and
subshell bodies ARE traced. HONEST LIMITS (measured,
stated as fact): LINE coverage NOT branch coverage (the
clause-(C) same-line if/else counted covered with one
branch never taken); set +x regions and traps unaccounted
– a FLOOR-measuring instrument only, so branch-level

assurance requires a real tool (e.g. `kcov`) or an honestly-recorded shortfall. Where NO instrument exists for a language: INSTALL one, or record an § . . / § . . . SKIP-with-reason + a tracked § . . item – NEVER an invented percentage, NEVER a silently-dropped floor. EXCLUSION-LIST FENCE (amendment, remediation round of – – – the § . . . exemption pattern applied to the coverage corpus): a corpus exclusion is legal ONLY via a CHECKED-IN exclusion list, every entry justified from the closed class set {generated-code (output of a committed generator per § . .) | vendored-third-party | non-shipping-fixtures-and-golden-assets} and enumerated as honest gaps never silent; excluding FIRST-PARTY executable code additionally REQUIRES a tracked § . . item naming the plan to bring it into scope – an unlisted, unjustified, or un-tracked-first-party exclusion VOIDS the measured figure (a consumer that excludes most of its first-party executable code and truthfully measures ≥ % on the remainder has satisfied the floor in letter and voided it in substance – the silent-evasion channel this fence closes). (F) **HONEST BOUNDARY / SCOPE** (§ . . .): "all work" = work products with an EXECUTABLE surface; a pure prose/governance/documentation edit has no executable surface and is NOT coverage-scoped (its discipline is § . . / § . . / § . . / § . . + the § . . universal review – demanding a coverage percentage there would be a § . . () false-positive refusal), BUT a governance change that SHIPS an executable artifact (gate, hook, script, mutation) is FULLY in scope for (A) and (B); test-first PRECEDES § . . four-layer verification / § . . full-suite retest / § . . manual-QA, never replaces them; and the CODE-coverage axis is DISTINCT from § . . / § . . 's TEST-TYPE breadth – both apply, neither

substitutes, satisfying one is never evidence for the other.

Classification: universal (§) – the consuming project supplies its per-language instruments, corpus scope, CHECKED-IN clause-(E)-fenced exclusion list, and threshold calibration per § . Composes

§ /./././././././././.
§ . Propagation gate CM-COVENANT-114-224-PROPAGATION (literal 11.4.224) + recommended gates CM-TEST-FIRST-ALL-WORK + CM-COVERAGE-FLOOR (contract amended, remediation round of - : the measured numerator counts ONLY RED-CAPABLE tests – every contributing test carries a recorded failing-first run OR a paired mutation observed to make it FAIL, zero-assertion contributors rejected by construction, sampling-based mutation verification an acceptable stated implementation – so ONE real RED test plus assertion-free padding to % FAILS the gate; and the gate FAILS on any exclusion that is unlisted, outside the clause-(E) closed class set, or first-party without its tracked \$ item;

brownfield adoption is DELIBERATELY UNDEFINED – the owed gate-code item MUST surface the adoption question to the OPERATOR per § with concrete options {immediate hard floor | one-time \$ -style monotone-decrease ratchet | per-corpus phase-in | changed-code-only with a scheduled full-corpus deadline} BEFORE first enforcement, recording the answer as consumer DATA – never an invented ratchet, a \$ -class decision the operator owns) + paired \$ mutations;

gate-code = a separate work item – the CONTRACT is landed, the gate CODE is NOT claimed shipped (\$).

Canonical authority: constitution submodule `Constitution.md`

\$. . . Non-compliance is a release blocker. No escape hatch — no `--test-after-OK` , `--skip-red-first` , `--bash-exempt-from-tests` , `--coverage-not-applicable` , `--below-floor-OK` , `--coverage-proves-correct` , `--unmeasured-percentage-OK` , `--assertion-free-test-counts` flag.

\$. . - Scheduler-quota burst-throttling
 telemetry + interactive-scope isolation from bursty fleets
 (research-derived, - -). Compact summary:
 progressive / bursty / load-correlated interactive
 degradation MUST be diagnosed from the scheduling scope's
 THROTTLING ACCOUNTING - on cgroup v , cpu.stat
 nr_throttled / throttled_usec as DELTAS over a measured
 window under load, as a fraction of enforcement periods,
 WITH a quiet-phase control - NEVER from averages: quota
 enforcement is per-period, so a workload far below quota
 ON AVERAGE can still throttle a large fraction of periods
 and stall every co-resident process, while CPU average /
 load average / RSS / thread counts all read HEALTHY (their
 quiet zeros are \$. . () FALSE-NULLS; forensic
 FACT - - : a session scope at quota .
 CPU/ ms on a -CPU host, average demand .
 - UNDER quota - yet . % of periods throttled during
 active phases, % quiet, worsening with session age, the
 co-located terminal-multiplexer server freezing ≥ ms
 per throttled period into multi-second interactive
 stalls); an "averages look healthy" verdict without
 throttle-accounting deltas is a \$. . guess, and
 the same discipline binds every quota'd scheduler
 (container CPU limits, VM caps/steal). DESIGN half: an
 interactive-latency-sensitive process (terminal server /
 TUI / editor / operator shell) MUST NOT share a fixed-
 quota NO-BURST scope with a bursty agent/build fleet -
 grant a measured burst allowance, isolate the interactive
 process in its own scope, or both; quota values are HOST-
 ADAPTIVE (computed from detected capacity per \$. ,
 \$. . never hardcode - sibling FACT: a FIXED -
 CPU default quota on a -core host → . % of
 periods throttled, s forced idle, fixed by host-
 adaptive computation proven with a four-quadrant polarity
 test reading the LIVE cgroup back per \$. .);
 the THIRD orthogonal host-exhaustion axis beside \$.

memory + \$. threads – no axis's instrument
substitutes for another's. Honest boundary \$. . –
throttle telemetry proves WHERE periods were lost, not
that throttling is the only degrader (RTT floors / GC load
ranked separately in the founding investigation), and the
design half never licenses removing a quota host policy
requires (\$). Classification: universal

(\$. .). Composes

\$. . /. /. /. /. (.)/. /
\$. /\$. /\$. /\$. . Propagation gate

CM-COVENANT-114-225-PROPAGATION (literal 11.4.225) +

recommended gates CM-QUOTA-THROTTLE-TELEMETRY + CM-
INTERACTIVE-SCOPE-NOT-QUOTA-STARVED + paired \$.

mutations (averages-only verdict on a progressive-
degradation issue → telemetry gate FAILs; fixed low quota
literal on a detected many-core host, or terminal server
co-located with the fleet in a no-burst quota scope →
isolation gate FAILs; strip the literal → propagation gate
FAILs; gate-code = separate work item). **Canonical**

authority: constitution submodule Constitution.md

\$. . . Non-compliance is a release blocker. No
escape hatch – no ^{2016 17 23} --averages-prove-healthy , --skip-throttle-
telemetry , --fixed-quota-default-OK , --interactive-in-fleet-
scope-OK , --no-burst-needed , --skip-quiet-phase-control flag.

EXTENSIONS landed ^{11 4 6} – – ^{11 4 115} (the systematic-
debugging / measurement-semantics harvest – full text in
Constitution.md ; every lesson verified against captured

evidence or re-demonstrated under control conditions
before codifying, \$. .). \$. . (G) –
defect-identity binding for CONSTRUCTED reproductions: a
RED test's PRECONDITION must be TRACEABLE to the reported
defect's real evidence (the reporter's sequence

\$. . , a captured failure trace, or a probe
proving the precondition occurs in the reported
environment), recorded with the RED verdict as a
precondition-provenance field { observed | constructed }; when

NO working repro exists and the precondition is CONSTRUCTED, the fix mints at most DEFENSIVE HARDENING – labelled as such, NEVER closing the reported item (it stays OPEN § . . .); systematic-debugging refutation of every mechanism a pending fix addresses REVOKES its done-claim on the spot (forensic FACT – : an exit-hang fix nearly shipped on a hand-made-lock-holder RED→GREEN; real-PTY § . . . – shape re-driving refuted every addressed mechanism – / clean exits – and the change was honestly re-labelled in-source "defensive hardening ... NOT a fix for the reported exit-hang"); strengthens (A) from artifact-synthetic to MECHANISM-synthetic; gate CM-RED-PRECONDITION-DEFECT-TRACEABLE (a constructed provenance on a defect-CLOSING claim FAILs; the same provenance on an honest hardening item PASSes – the § . . . () false-positive guard) + paired § . . . mutation.

§ . . . ()()() – measurement-semantics quartet: () FIELD-IDENTITY + UNITS + SEMANTIC CLASS {capacity|occupancy|rate|cumulative|high-water} verified against the authoritative source definition before ANY summary-read metric gates or propagates – CAPACITY≠OCCUPANCY is the named canonical instance (forensic FACT: rcv_wnd , a window-CAPACITY ceiling, misread as a "~ . s standing send queue"; real Send-Q a few hundred bytes app_limited ; propagated into THREE agent briefs + operator reports before the raw field was re-read, then publicly withdrawn – a downstream consumer of an unverified metric INHERITS the bluff);

() OBSERVER DECONTAMINATION – an instrument contributing to its own measured population (probe sockets in a socket census, sampler load in a load profile) identifies + EXCLUDES its footprint AND re-proves the decontaminated instrument still sees via a ()(b) control needle (FACT: TIME-WAIT – raw → – decontaminated, ~ × observer-induced, while the same

scan still caught a real + MB/ → -socket step);
 composes \$. . () observer-effect budget +
 \$. . single-resource-owner for measurement
 arms; () ARTIFACT-USABILITY-NOT-PREREQUISITE-PRESENCE
 – a readiness/install/health gate probes the ARTIFACT
 through its real invocation path (\$. . (A)),
 never a build prerequisite's presence (FACT: a command -v
 <compiler> gate turned a host holding a WORKING prebuilt
 into a false "cannot launch" exit- – "[ok] verified"
 and "[FAILED] cannot launch" in ONE transcript, review-
 graded BLOCKING; the proxy fails BOTH directions – clause
 () and () at once); () the SHELL-INSTRUMENT
 FOOTGUN CHECKLIST – NEW standing reference docs/guides/
 shell_instrument_footguns.md (inherited by reference
 \$. . /\$. .): exec-wrapper classes
 (timeout / nice / env) cannot run shell functions/aliases
 (rc= false refusal), pgrep -f self/carrier match
 (\$. . (D)/\$. .), delimiter-keyed
 extractors TRUNCATING the unit under test into a FALSE
 PASS, bash -x xtrace BLINDED by the code under test's
 exec ... 2>/dev/null (countermeasure BASH_XTRACEFD), and a
 background watchdog subshell spawned INSIDE a \$(...)
 command-substitution – it inherits the substitution's pipe
 WRITE-END, and an early disarm (kill) ORPHANS its sleep
 grandchild which keeps that write-end OPEN, so the \$(...)
 BLOCKS for the FULL timeout budget after the real work
 completed instantly, every verdict and exit code CORRECT
 (countermeasure: redirect the watchdog subshell's fds away
 from the pipe – (...) >/dev/null 2>&1 & – or have the probe
 write to a file) – all five DEMONSTRATED under control
 conditions – – ; every shell-touching
 \$. . /\$. . review consults it,
 every shell-instrument zero stays bound by the ()(b)
 class-matched needle; CM-GUARD-ASSERTS-REAL-CONDITION
 EXTENDED with invariants (iv) metric-semantics-verified /
 (v) census-decontaminated / (vi) artifact-probed-not-

prerequisite / (vii) checklist-consulted + paired \$. mutations (gate-code = separate work item).

\$. . () – sourced-function `exec` redirection-scoping: a command-less `exec` applies EVERY redirection on its line to the CURRENT shell PERMANENTLY, so in sourced/library functions an fd-manipulation `exec` MUST brace-scope its side redirections ({ `exec 9>>"$f";` } 2>/dev/null); a bare `exec N>file ... 2>/dev/null` in sourced code is a review defect + greppable gate class (forensic FACT

– – : a sourced lock helper's bare `exec 9>>lock 2>/dev/null` silenced the operator's interactive-shell stderr for the shell's LIFETIME – one provider launch left the terminal unable to print any error – while the paired unlock had used the correct brace form all along; marker-probe proven pre/post-fix); gate `CM-SHELL-EXEC-REDIRECTION-SCOPED` (brace-scoped forms and `exec -with-command exempt` – the \$. . () false-positive guard) + paired \$. . mutation. \$. . – teaching forensic FACT added (– –): a transport-hardening change was committed AND pushed to `main` BEFORE any independent review (honestly flagged); the post-hoc \$. . -substrate review returned NO-GO with ONE BLOCKING finding – an unsafe security-relevant default, already public on every mirror – plus Important + Minor, and remediation consumed a full fix-forward cycle a pre-push review would have folded into the original change; the rule needed NO change (\$. . /. /. /. already bind it with no carve-out) – the incident is cited as its newest forensic anchor; remediation fix-forward per \$. . , never a history rewrite. ALREADY-COVERED verdicts this round: measurement-arm single-resource-ownership + observer-effect budget (\$. . + \$. . () – the decontamination instance landed as \$. . (), not a new anchor); review-before-push as a rule (bound in full by \$. .

family – teaching FACT only). All Classification: universal (§ . . .) – every toolchain literal genericised; consumers supply scopes, quotas, instruments, and paths as DATA per § The literals 11.4.115 / 11.4.201 / 11.4.67 / 11.4.142 are unchanged – their propagation gates stay GREEN.

§ . . . – Evidence-class-at-closure + standing detection pressure: machinery presence does NOT predict whether a fix holds – the EVIDENCE CLASS at closure (runtime vs source) under real DETECTION PRESSURE does; prose does not bind, seams do (research-derived,). Compact summary: the decisive finding of the reopen/first-touch root-cause forensics, confirmed across four independent evidence lines (internal forensics; a-source external literature pass; one cross-project corpus WITHOUT tracking machinery; one WITH exemplary machinery): **every fix whose done-claim was preceded by an on-target RED→GREEN polarity flip that held recorded ZERO reopens; every fix confirmed only at source-green bounced** (runtime-evidence corpus: median attempts-to-stick . . . , ~ . . . % corrected reopen – a floor under weak detection pressure; source-green corpus: . . . % recorded, itself an undercount; the healthy corpus's ONLY bouncing family was a wrong-layer echo-assertion; PENDING-BUILD claim-moves; the last audited release entered manual QA with . . . / . . . items user-layer-validated – the operator was the FIRST user-layer oracle most claims ever met, so first-touch rediscovery was the deterministic pipeline output, not bad luck). The mandate – the STRICT FORMALIZATION of § (F)/ § (D)/§ : () closed evidence-class RANKING {runtime > artifact > source} with a per-defect-layer floor at every closure seam – a user-visible/runtime defect can NEVER close on artifact/source-class evidence (source-on-source stays legal, the § () false-refusal guard); () class

proven by MACHINE FIELDS never a label (runtime $\Rightarrow$ target fingerprint read FROM the target + runtime observable; artifact $\Rightarrow$ path + content hash; source $\Rightarrow$ ref; label-without-fields = SHAPE-INCOMPLETE, unreadable = BLIND);
 () ANTI-ECHO – a "runtime observable" that IS a grep/source-scan transcript is WRONG-LAYER, refused (a grep can never observe a pixel; the pixel-defect-by-five-greps closure dies at this seam); () per-status evidence PRECONDITIONS – every \$. . . /\$. . . status carries a checked precondition, not only the terminal write (FACT: a penultimate "ready"-class status parked items – % of the done-claiming population – as design-complete-but-UNVERIFIED; the operator's / - broken sample was drawn from exactly that set); () FIX-KIND honesty – a state-only repair/mitigation cannot close a defect item as fixed (sibling-corpus FACT: recurrence chains, median . / max attempts, dominated by state-only "fixes"); () STANDING DETECTION PRESSURE – pressure is a CAUSAL INPUT to every measured quality rate (a rate without its pressure regime is incomparable, \$. . . ; the %-vs-~ . % split is explained by pressure + closure class, not machinery; the healthy corpus's one recurrence surfaced at days = the next SWEEP not the next use): every registered, topology-present guard carries a FRESHNESS CONTRACT (verdict on the CURRENT artifact fingerprint within a declared staleness budget); the never-executed/stale/over-budget set feeds a STANDING risk-ordered re-run queue (most-reopened-first \$. . . , then stalest-first) drained by the zero-idle loop (\$. . . / \$. . .) as regular development-cycle work – the \$. . . release seam is the blocking BACKSTOP, never the only executor (FACT: two standing guards, finally executed during the forensics, immediately emitted the latent FAILs – one reproduced the operator's loudest defect / ; registration was treated as coverage while

execution was nobody's job); topology-absent guards
 enumerated never queued, an empty registry is BLIND never
 "all fresh". Binding: applies during EVERY regular
 development cycle; CHECKED through every code-review per
 § . . . (); on every constitution pull the
 § . . . sweep + § . . . hook analyze +
 apply the additions and violations are tracked + cleared
 (§ . . .) – no parallel mechanism. Honest boundary
 § . . . : a measured correlation with a stated
 mechanism and zero counter-cases – NOT a randomized trial
 (partially-censored exposure carried openly); fabricated
 machine fields stay closed by § . . . (F) harness-
 written verdicts + the § . . . candidate-
 fingerprint join (layered, not eliminated); oracle
 calibration stays § . . . (); the layer
 taxonomy is authored data (mislabelling is the
 § . . . ()(a) review class's detective target);
 re-running a weak guard proves little – oracle strength
 stays § . / § . . . (F), orthogonal discovery
 stays § . . . / § Reference mechanisms
 (proven, NOT wired § . . .):
 SOL- / / / RED-first POCs with golden-good/
 golden-bad/negative-control fixtures under docs/research/
 quality/solutions/ (/ GREEN, deterministic second
 iteration) – no force until their seams are wired by
 tracked § . . . work items. Classification:
 universal (§ . . .) – consumers supply layer
 bindings, evidence stores, verdict registries, and
 staleness budgets as DATA per § Propagation
 gate CM-COVENANT-114-226-PROPAGATION (literal 11.4.226) +
 recommended gates CM-EVIDENCE-CLASS-AT-CLOSURE + CM-GUARD-
 FRESHNESS-SCHEDULER + paired § . . . mutations (close a
 runtime-layer item on source evidence / label-without-
 fields / echo-observable / stale guard with no queue entry
 → the gates FAIL; strip the literal → propagation FAILs;
 gate-code = separate work item, NOT claimed shipped

\$. .). **Canonical authority:** constitution submodule
 Constitution.md \$. . . Non-compliance is a
 release blocker. No escape hatch – no --source-green-closes-
 runtime-defect , --label-is-class , --echo-observable-OK , --
 registration-is-coverage , --rate-without-pressure-regime , --
 mitigation-closes-defect , --skip-freshness-queue flag.

\$. . – **Governance-corpus self-custody:** the
 rule corpus is bound by the same custody it imposes –
 every named gate implemented-or-registered-deferral under
 a monotone ratchet, and propagation gates count anchor
 BLOCKS (exactly-once + lockstep-identical), never bare
 literals (research-derived, . . . – . . .). Compact
 summary: measured on THIS corpus – . . .
 (commands + control needles preserved in the root-cause
 analysis): . . . , . . . canonical lines,
 distinct anchors; . . . named CM- . . . gates –
 (. . . %) with no implementation in either canonical gate
 site – plus . . . literal "gate-code = separate work
 item" deferrals; the custody seam designed
 . . . – . . . was still prose on
 . . . – . . . (the corpus's highest-value
 remediation fell into its own gap within five days); the
 fix-confirmation anchors predate the failures they failed
 to prevent by five-plus weeks – while **every fully-**
mechanised rule in the record either prevented or exposed
its failure mode (the PENDING-semantics + release verdict-
 coverage seams landed within a day of diagnosis and now
 block): the gap is not rule quality, it is the prose-to-
 seam conversion rate. The SHAPE defect: the propagation-
 gate family – the only family that runs everywhere –
 asserts literal-presence ≥ . . . , mechanically rewarding
 restatement over implementation ("the corpus measures its
 own health in words present, and so it grows words");
 consequences measured: two anchors duplicated verbatim-
 divergently in ALL FOUR mirrors while every presence-gate
 stayed green (the F . . . incident); one anchor NUMBER

heading two different mandates (the § . . . /
 § . . . collision, independently re-detected by
 the block-integrity instrument – resolution stays an
 operator-owned re-mint); two canonical entry-point
 mandates contradicting (the mandated resumption file
 unreachable from the mandated README); external
 corroboration: mandated-checklist adoption without
 workflow embedding moved nothing across . . . hospitals
 (the Ontario null). The mandate: **(A) NAMED-GATE LEDGER +
 RATCHET** – every CM-* -class token named in the corpus is,
 at every build, either IMPLEMENTED (token in an EXECUTABLE
 gate site; prose carriers never count § . . . ()
 (a)) or covered by a REGISTERED DEFERRAL row pointing at a
 tracked § . . . item; silent gate debt refused;
 the unimplemented count is MONOTONE-DECREASING (validated
 against the definition of done per § . . . () –
 at the correct end-state it is); a vanished gate NAME
 without an explicit removal citation FAILs (the delete-
 names-to-lower-the-count gaming channel closed; repeals
 stay legal + visible, the § . . . pattern); an
 anchor's "done" state is its SEAM landing, not its TEXT
 landing – a rule authored without its seam is in
 § . . . 's worse-than-nothing state, and the ledger
 makes that state visible, counted, and shrinking;
 brownfield baseline = operator § . . . decision
 (the § . . . (E) adoption-fence pattern), never an
 invented ratchet. **(B) ANCHOR-BLOCK INTEGRITY** – every
 propagation-class gate counts BLOCK-STARTS (structural
 line-anchored heading patterns), never bare literals:
 exactly ONE block per anchor per governance file (=
 absence, > = DUPLICATION, FAIL naming anchor + file);
 content-hash EQUALITY across the declared § . . .
 lockstep mirror set (divergent copies FAIL naming both
 files; compact-vs-full variance ACROSS layers stays
 legitimate – canonical-vs-mirror drift is the
 § . . . family's job); mid-body citations are

CARRIERS and never count; ZERO blocks extracted = BLIND-OR-EMPTY never clean (§ . . . ()); an anchor NUMBER heading two mandates is a COLLISION and FAILS (§ . . . never-reuse applied to anchor ids); the extractor captures full dotted ids so a sub-anchor never prefix-matches its parent (a measured § . . . () false positive in the founding instrument, fixed RED-first). (C) semantic contradiction between two DIFFERENT anchors stays undecidable-by-hashing – only the decidable projections are mechanized (fenced scopes § . . . (), seam declarations § . . .), the rest is § . . . /§ . . . review territory, stated never claimed; this anchor adds the enforcement-ratio INSTRUMENT, not a growth ban. Binding: the ledger regenerates + diffs on every governance edit + every build (regular development cycles, never an on-demand audit); review-checked per § . . . ()(c); on every constitution pull the § . . . sweep + § . . . hook run the ledger + block-integrity checks and violations are tracked + cleared. Reference mechanisms (proven, NOT wired § . . .): SOL- + SOL- RED-first POCs under docs/research/quality/solutions/ (SOL- 's live run reproduced the - gate census exactly; SOL- 's live run confirmed the four mirrors clean post-F + re-detected the / collision) – no force until wired by tracked work items. Classification: universal (§ . . .) – consumers (and this governance repo itself) supply gate-site lists, deferral registries, and lockstep mirror sets as DATA per §

Propagation gate CM-COVENANT-114-227-PROPAGATION (literal 11.4.227 – itself REQUIRED block-start-shaped per (B)) + recommended gates CM-GATE-LEDGER-RATCHET + CM-ANCHOR-BLOCK-INTEGRITY + paired § . . . mutations (name a gate with no implementation + no deferral / delete a name without citation / duplicate a block / diverge two lockstep

copies / re-mint a number → the gates FAIL; strip the literal → propagation FAILs; gate-code = separate work item, NOT claimed shipped \$. .). **Canonical authority:** constitution submodule Constitution.md \$. . . Non-compliance is a release blocker. No escape hatch — no --name-gate-without-seam , --silent-gate-debt-OK , --delete-name-to-lower-count , --presence-gte-1-suffices , --duplicate-anchor-OK , --mirror-drift-OK , --reuse-anchor-number flag.

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EXTENSION landed (round — the reopen/first-touch root-cause harvest; full text in Constitution.md). \$. . () — the quality-harvest review classes: every code-review (\$. . universal / \$. . gate / \$. . Opus-xhigh substrate, iterated to G0 per \$. .) MUST additionally CHECK, per reviewed change: (a) **closure-evidence class vs defect layer** (\$. .) — a terminal/ready-class status moving on evidence below the defect layer's floor, a user-visible property "verified" by file-existence + greps, an echo-assertion, or a state-only repair closing a defect as fixed is a FINDING; (b) **needed measurements** — every load-bearing measurement in the change/its tests satisfies \$. . ()-(), and NO count was converted into a finding without reading the underlying lines — a COUNT IS A LEAD, THE LINES ARE THE FINDING (forensic FACT: twice in one measured cycle a count was read as state — a "mixed half-written doc" that was actually coherent, and an inflated FAIL total that counted its own summary line); (c) **governance-and-gate integrity** (\$. .) — a change NAMING a gate lands its implementation or registered deferral in the same change; a governance edit preserves anchor-block integrity; (d) **reviewer-authored adversarial mutations** — the author's self-authored mutation set is NECESSARY but NOT SUFFICIENT (forensic FACT, from the dated forensic corpus cited in the — — root-cause

analysis: an author's six self-authored mutations were ALL caught while an independent reviewer's FIRST THREE different mutations ALL sailed through, each restoring a user-visible defect); the reviewer MUST attempt ≥ mutation the author did not write (or record honest infeasibility per § 11.4.228) – a guard whose reviewer-authored mutation survives is unvalidated instrumentation per § 11.4.194 (F) and a FINDING. The literal 11.4.194 is unchanged – its propagation gate stays GREEN.

§ 11.4.228 – **Cross-agent extension lifecycle**
mandate: per-platform compatibility, source tracking, documentation with compatibility matrix, and auto-wiring on constitution pull (research-derived from Extensions Catalog forensics, 11.4.228 – 11.4.229). Compact summary: every governed extension (Skill, MCP, ACP, LSP, plugin) MUST (A) ship a per-platform compatibility declaration (Claude Code, OpenCode, Gemini CLI, Qwen Code minimum) with closed vocabulary {SUPPORTED|UNTESTED|UNSUPPORTED|CONFIG-ONLY}, NEVER cause loops/broken behavior in any platform, and carry a per-platform liveness test § 11.4.229; (B) be traceable to its exact canonical origin (repo + commit SHA + path, recorded in EXTENSION_SOURCE.yaml – symlinks preferred over copies, blind copies without provenance are § 11.4.229 dead-code-in-waiting), with a per-project extensions catalog (docs/extensions/ EXTENSIONS_CATALOG.md) as the single source of truth, § 11.4.229 roster/corpus-backed fingerprint; (C) the extensions catalog MUST carry a per-platform compatibility matrix + liveness-testing evidence per extension × platform per § 11.4.229 feature_class=extension_loading , four-format export per § 11.4.229 , auto-synced via § 11.4.229 docs-chain; (D) the § 11.4.229 post_update_hook.sh is EXTENDED to auto-detect all governed agent platforms, symlink constitution skills into each (.claude/skills/ , .opencode/skills/ , platform equivalents), register MCP servers into each platform's config, self-

validate, and be idempotent – closing the gap where constitution skills exist in source but are un-wired to any platform (the § . . . source-present-runtime-absent gap). Honest boundary § . . . – wiring guarantees REACHABILITY, never runtime acceptance. Classification: universal (§ . . .). Propagation gate CM-COVENANT-114-228-PROPAGATION + recommended gates CM-EXTENSION-PER-PLATFORM-COMPAT / CM-EXTENSION-SOURCE-TRACKED / CM-EXTENSION-CATALOG-DOCUMENTED / CM-CONSTITUTION-EXTENSIONS-AUTO-WIRED + paired § . . . mutations. Full text in Constitution.md §

§ . . . – Live in-session task/todo tracker MUST always be up to date and fully in sync with the real work state – never stale, never completed-shown-as-pending nor pending-shown-as-done (User mandate, – –). Verbatim operator mandate: "make sure it is ALWAYS up to date fully in sync! CRITICAL: This MUST BE one of mandatory rules / constraints! Add it with all relevant details into the constitution Submodule and relevant related files." The agent's LIVE IN-SESSION TASK/TODO TRACKER – the harness-provided task list / `TodoWrite` -class todo tool / any equivalent moment-to-moment execution-state surface the runtime exposes DURING a session – MUST ALWAYS be kept up to date and FULLY IN SYNC with the real live work state: never stale, never showing completed work as pending, never showing pending / blocked / abandoned work as done or in-progress; updated the MOMENT any tracked task's state changes (started → in-progress, completed → done, blocked, abandoned, split, re-scoped, newly discovered), never batched to a later checkpoint, never reconstructed from memory at the end, never left reflecting a superseded plan – a live tracker that lags the real work is a § . . . PASS-bluff at the execution-transparency layer (an operator or resuming conductor per § . . . makes decisions on a false picture of what is done / in-flight / owed). (A) every state-change

pending-may-be-done , --completed-may-stay-pending flag. Full text in Constitution.md § . . .

§ . . . – Parallelized-pipeline methodology:
 build validate overlap + affected-stages-only re-run +
 parallel multi-device testing/recording + fan-out-
 subagents-always + incremental delivery granularity with a
 declared QA-build cadence (research-derived,
 - - ; clause (D) added
 - -).

Forensic motivation (genericised): the sequential pipeline shape – build FULLY completes → THEN validation begins → on ANY discovery, restart the WHOLE cycle from scratch → all devices tested one-at-a-time – serialises the operator's wall-clock onto build latency + one-device-at-a-time evidence gathering, so a multi-hour containerised build followed by serial per-device validation and a from-scratch restart-on-discovery makes time-to-a-manual-QA-capable-release a multiple of what the genuinely-parallel work requires. § . . . 's PWU pipeline already overlaps Stage of round N+ with Stages – of round N; § . . . already backgrounds long tests; § . . . already mandates ≥ auto-backfilled parallel streams; § . . . already partitions shared hardware by single-owner; § . . . / § . . . already mandate always-on recording + availability-following. § . . . BINDS those into one operating discipline and adds the four invariants they did not each state on their own: (A) the build validate CONCURRENCY seam + affected-stages-only re-run-on-discovery, (B) all-devices-concurrent + worker-pool recording/processing, (C) fan-out-subagents as the standing default, (D) incremental delivery granularity + a declared QA-build cadence (added - -). It is the parallelization capstone the way § . . .

is the continuous-parallel-stream capstone – a strengthening bind, never a weakening.

(A) PIPELINE-OVERLAP PARALLELIZATION. Validation/verification work MUST run CONCURRENTLY WITH the build wherever a validation stage does NOT depend on the build's own not-yet-produced artifact – never deferred until the build finishes when it could have run alongside it. Deploy the moment the build is done (build-completion is the deploy trigger, per \$. . . no-commit-while-build-writes-tracked-artifacts + \$. . . artifact-must-exist ordering – the deploy waits for the artifact, the deploy-independent validation does not). Concretely, WHILE a build runs, the following run in parallel per the \$. . . SAFE-during-build catalogue: pre-build source-layer gates + meta-test mutation sweeps + coverage/clash checks (\$. . .), the \$. . . /\$. . . /\$. . . independent code-review on the batch's SOURCE, doc/DB sync (\$. . .), and on-device validation of the PRIOR build (device-independent of the in-flight build, \$. . .). On any NON-huge-blocker discovery, the cycle MUST: (. . .) bump the version code (a new, distinctly-identifiable candidate artifact – never re-use the failing artifact's id, \$. . . stable-name + \$. . . target-identity); () fix at the proven root cause (\$. . . systematic-debugging first, \$. . . RED-on-the-broken-artifact); () RE-RUN ONLY THE AFFECTED STAGES, PARALLELIZED – the stages whose inputs the fix changed (its own targeted validation FIRST per \$. . . , then the affected regression set), NOT a serial restart-from-scratch of the entire cycle. The explicit efficiency target is a $\times$ reduction in time-to-a-manual-QA-capable-release versus the fully-sequential shape – a measured target, tracked as build-and-validate wall-clock deltas (\$. . . build-resource stats), never an assumed one (\$. . .). **HONEST BOUNDARY**

(`$ . . / $ . . / $ . .`): the CURRENT build's ARTIFACT-layer (`$ . . . . layer`) and RUNTIME-ON-CLEAN-TARGET-layer (`layer`) validation CANNOT precede its own artifact – a gate whose precondition is the build's output belongs at a post-build seam, never overlapped onto authorship (`$ . . seam-placement; $ . . artifact_not_yet_built`). "Concurrent with the build" therefore means the deploy-INDEPENDENT stages overlap; the CURRENT-artifact runtime validation begins the moment that artifact exists, in parallel across devices per (B). **"Affected-stages-only re-run" is bounded to NON-huge-blocker discovery:** a HUGE BLOCKER (release-blocking-severity defect / regression invalidating the cycle / blast radius reaching the batch's other fixes) still triggers the `$ . . . . STOP` → `fix-all` → full-RESTART protocol (restart, never resume) – `$ . . (A)` does NOT weaken `$ . .`, `$ . . full-suite-retest-before-tag`, `$ . . validate-the-fix-first`, `$ . . risk-ordered validation`, or `$ . . manual-QA-final`; it narrows the from-scratch restart to exactly the cases those anchors require it, and parallelises everything else.

(B) PARALLEL MULTI-DEVICE TESTING + RECORDING/PROCESSING.

Every available, reachable test device MUST be exercised CONCURRENTLY – one validation stream per device, never one-device-at-a-time when the devices are independent – each stream the SINGLE EXCLUSIVE OWNER of that device's exclusive resources per `$ . . . .` (the owner drives; any concurrent stream targeting the same device is read-only), so per-device evidence is never cross-contaminated (`$ . . evidence-integrity`). Every device is ALWAYS-RECORDED per `$ . .` (non-invasive, side-effect-free, deterministic layout) and AVAILABILITY-FOLLOWED per `$ . .` (a dropped device logs an honest offline event + reconnect-and-

resume, never a silent corpus hole, never a faked continuity). Recording CAPTURE + ANALYSIS/PROCESSING MUST be fanned out through a BOUNDED WORKER-POOL (a Go worker-pool or platform equivalent) so N devices' recordings are captured + analysed in parallel and per-device evidence never serialises onto a single processor – the pool bounded by \$. memory + \$. thread headroom. Per-device PASS cites its own captured-evidence artefact (\$. . . /\$. . .) on a CLEAN/fresh deployment (\$. . . runtime-signature, \$. . . verify-deployed-artifact-identity-per-device); the highest-risk / most-reopened cases (\$. . . /\$. . .) get the deepest scrutiny FIRST on every device. **HONEST BOUNDARY** (\$. . .): genuinely-shared exclusive resources (one HDMI/audio sink, one media-playback path) serialise per \$. . . single-owner-over-time even across devices; "all devices concurrent" means independent devices, never two owners of one exclusive resource.

(C) FAN-OUT-SUBAGENTS-ALWAYS. Subagent fan-out (\$. . . /\$. . . subagent-driven; \$. . . multi-track ruler; \$. . . auto-backfill) is the **STANDING DEFAULT** for EVERY genuinely-parallelizable effort – batch investigation, test-authoring, gate-authoring, per-device validation (B), per-PWU development (\$. . .), doc/research work – not a per-request opt-in and not reserved for "big" tasks. A parallelizable effort executed monolithically in the foreground when a fan-out path exists is a \$. . . (C) violation (the \$. . . / \$. . . /\$. . . zero-idle discipline applied to HOW work is dispatched, not only WHETHER it runs). **HONEST BOUNDARY** (\$. . . / \$. . .): "always fan out" is BOUNDED by () genuine parallelizability (contending/redundant/serially-dependent subagents are NOT fan-out – they are waste, and spawning

them is NOT maximization per § . . . maximum-USEFUL-throughput); () § . . . % memory ceiling; () § . . . RLIMIT_NPROC thread headroom; () § . . . ≤ concurrent-agent cap; () § . . . single-resource-owner (two subagents may not both drive one exclusive device); () critical-state sequencing (commits/pushes/tags stay serial per § . . . /§ . . .). The bar is maximum USEFUL parallel throughput, never agent count; fan-out that exceeds a host-safety ceiling or contends a single resource is a violation, not compliance.

(D) INCREMENTAL DELIVERY GRANULARITY + DECLARED QA-BUILD CADENCE – a long-running spec/feature ships every already-proven-correct increment as it clears its own bar, never withholding delivery until the **WHOLE** spec/feature is done (operator mandate, – – , **CRITICAL** ; research-derived; added – –). Verbatim operator mandate (– –): "All work we do, especially SpecKit specs, gets incorporated right away. Fully deterministic validation/verification tests of all supported test types relying on machine-produced evidence get incorporated too, aiming to cover everything critical. All intensive testing (gates, checks, fixing, polishing) keeps running in BACKGROUND continuously until fully done. IN PARALLEL we keep building specs one by one, phase by phase, story by story, task by task. We get fast delivery results and AS SOON AS POSSIBLE builds that can be handed to QA teams for manual testing RIGHT AWAY, while in parallel all testing/fixing/polishing/improvement continues exactly as it would anyway. We do NOT wait for all of that to finish first – that can take days [...]. QA reports discovered issues; background agents keep working until everything is absolutely flawless. As background work progresses and fixes/improvements land in the main codebase, we keep producing REGULAR builds periodically delivered to QA teams for ongoing full manual testing. It

is critical that background agents keep ALL workable items, documentation, and other materials in sync so nothing ever goes out of date." Forensic anchor (genericised, FACT): in one consuming project a multi-phase spec had RED tests drafted for every phase and many source slices independently reviewed to a source-correctness GO, yet nothing reached QA for days, because delivery was implicitly held at the WHOLE-SPEC boundary while ONE slice's iterate-to-GO review loop ran past forty rounds – the finished, proven work of every other slice sat undelivered. The GAP this clause closes, stated precisely so it is not restated elsewhere

(§ . . .): clause (A) overlaps validation with the build of ONE candidate; § . . . (A) makes a batch build-eligible on source-correctness alone;

§ . . . (D) makes that eligibility PER SLICE inside a batched review – but none of them obliges an ELIGIBLE increment to actually be DELIVERED before the rest of its spec completes, none sets a delivery CADENCE, none says how an incomplete spec stays safe to ship, and none binds the hand-off to tracker truth. Build-eligibility that is never exercised is the same withheld delivery under a different name. ALL of the following hold:

(D.) THE DELIVERY UNIT IS THE INCREMENT, NOT THE SPEC. A spec / feature / epic decomposed into phases / stories / tasks (vertical, independently testable slices per the § . . . SLICING extension – Small + Testable) MUST integrate each increment into its INTEGRATION BRANCH – the trunk for main -targeted work, or the feat/ / product/ / flavor/ branch (§ . . .) the increment actually lives on for branch-scoped work – and make it part of the next QA deliverable (D.) cut from that branch, the moment THAT increment holds its OWN source-correctness GO per § . . . (A)/(D). When an increment exists on MULTIPLE coexisting integration branches – for example both a product/* AND a flavor/*

branch per § 4.1.1 (A) – EACH branch is independently obligated under (D. 1)'s own per-branch cadence: the increment's delivery obligation is evaluated separately per branch it lives on, never as a single ambiguous "the branch". A QA build MAY be cut from a `feat/` / `product/` / `flavor/` branch. For a `feat/` branch that QA build IS the live-manual-testing vehicle § 4.1.1.2 (B) requires BEFORE the branch may merge to trunk, so the feature-branch QA build is the PRECURSOR to trunk-landing, never a consequence of it; a `product/` / `flavor/` line, which never merges to trunk the standard way (§ 4.1.1.3 (C)), receives its cadence builds from its own branch. Clause (D) relaxes neither § 4.1.1.2 (B) (no `feat/` merge to trunk before live manual QA) nor § 4.1.1.3 (C), and a later merge of the branch to trunk still follows § 4.1.1.4 / § 4.1.1.5. Withholding an already-build-eligible increment solely because OTHER increments of the same spec are unfinished, or because its own or another increment's `test-instrumentation` / `process-doc` loops (§ 4.1.1.6 (D)) are still open, is a § 4.1.1.7 (D) violation. An increment WITHOUT its own source-correctness GO is not deliverable – incremental delivery is never a licence to ship unreviewed or unproven work.

(D. 1) **INCOMPLETE-SPEC, SAFE DELIVERABLE.** Shipping an increment while its spec is incomplete MUST leave the deliverable correct on every path a user or QA tester can reach: EITHER the increment is a complete, independently valuable vertical slice, OR the unfinished surface is LATENT – unreachable by construction, behind a release toggle (P. Hodgson, "Feature Toggles", martinfowler.com, 2010: "Release Toggles allow incomplete and un-tested codepaths to be shipped to production as latent code which may never be turned on"), or behind branch-by-abstraction (trunkbaseddevelopment.com: a technique "to effect a 'longer to complete' change in the trunk"). A toggle is a

PERMITTED mechanism, never a mandated one (consistent with § 4.1.1's own statement that it mandates none), and it is inventory with a carrying cost (Hodgson: "Savvy teams view their Feature Toggles as inventory which comes with a carrying cost"): every release toggle carries a tracked § 4.1.1 removal item and a consumer-declared expiry (§ 4.1.1), and a toggle past its expiry is a § 4.1.1 finding. A toggle that needs to outlive its expiry – for example because its latent path's (D. 1) hardening loop legitimately runs many rounds – is extended ONLY through a § 4.1.1 waiver (rostered non-producer authoriser, a new unelapsed expiry, a named tracked item), never by a silent expiry edit, by enabling the still-unproven path, or by deleting finished work. A latent path is NOT an advertised capability (§ 4.1.1) until it is enabled and proven.

(D. 1) A DECLARED QA-BUILD CADENCE, NOT A ONE-SHOT. Every project declares as DATA (§ 4.1.1) its QA-build trigger: `per-increment` (each build-eligible increment landing on its integration branch, D. 1), `periodic` (a fixed interval – the release-train pattern; SAFE's "Develop on Cadence. Release on Demand" separates the development rhythm from the release decision), or `whichever-first`; a project that declares nothing gets `per-increment`, the safe default (§ 4.1.1). A `periodic` or `whichever-first` interval MUST be recorded as DATA together with its written justification (§ 4.1.1) and stated in every QA hand-off record (D. 1), so an interval long enough to defeat this clause is visible rather than silent; this clause sets no numeric maximum – any such cap is an operator decision (§ 4.1.1). Whenever an integration branch (D. 1) holds build-eligible content not yet in a QA build cut from that branch and the declared trigger has fired, a QA deliverable MUST be cut from that branch and handed to QA for ongoing manual testing (§ 4.1.1). Builds COALESCE: a QA build

already in flight covers every increment that had landed on its branch before that build started, and increments landing WHILE it is in flight roll into the NEXT build rather than each demanding a fresh one – the obligation is that eligible content never waits past the next build, not one build per increment. A build that FAILS, or is REFUSED at § 4.2.1, covers NOTHING – its increments roll into the next build. "Completed" in this clause and in its two amended gate contracts (D's recommended gates, below) means HANDED TO QA: a build provides coverage only once it has genuinely been handed to QA for manual testing, never merely once the build process itself exits successfully. Each such deliverable is a § 4.2.2 (B) cycle boundary with its own monotonic version increment (§ 4.2.2), passing § 4.2.2 readiness on its own candidate fingerprint, and built once then promoted (§ 4.2.2). The § 4.2.2 "mandated validation" for a cadence build is AT MINIMUM the source-correctness GO plus the pre/post-build gates of § 4.2.2 (A), PLUS every operator-declared blocker bound to the QA-deploy seam per § 4.2.2 and any § 4.2.2 critical-invariant scenarios the increment touches – NOT the open background-hardening backlog, which by § 4.2.2 (A) never gates a build. "Deliverable" is project-defined DATA (§ 4.2.2): a binary, a container, a device image – or, for a dev-tooling / governance feature, the feature's own demonstrable, testable state at a fingerprinted commit; no artifact form is implied by this clause.

(D.) CONTINUOUS BACKGROUND HARDENING THAT NEVER BLOCKS THE NEXT INCREMENT. Test-instrumentation hardening, gate / check / mutation authoring, polishing, and the fixes for QA-reported findings on ALREADY-delivered increments run as standing background streams (§ 4.2.2 / § 4.2.2 / clause (C)), each a tracked item (§ 4.2.2) driven to its own zero-finding GO

(§ . . . – not relaxed), landing on its integration branch (D.) as it reaches G0 and riding the NEXT cadence build cut from that branch. They never gate a LATER increment's build-eligibility, with two exceptions that follow from existing anchors rather than relax them: an open source-defect finding (§ . . . (D)) in code the later increment touches or depends on is a source-defect of that later increment too; and a huge blocker still triggers § . . . STOP → fix-all → full restart.

(D.) NEVER-STALE TRACKING IS A PRECONDITION OF EVERY HAND-OFF, NOT AN AFTERTHOUGHT. With many increments and many background streams in flight at once, the existing sync machinery – § . . . (F) write-seam doc/DB sync, § . . . live tracker, § . . . / § . . . / § . . . intake and request history, § . . . item integrity, § . . . completion – applies at EVERY increment landing and EVERY cadence build (cited, not restated). Each QA hand-off records the deliverable's fingerprint and the exact set of tracked items it contains; a hand-off whose tracker or documents disagree with the deliverable (an item marked done whose fix is not in that build, or a fix in the build that no item records) is a § . . . PASS-bluff at the delivery layer.

(D.) QA FINDINGS ON EARLY INCREMENTS ARE STILL COVERAGE ESCAPES. Earlier, more frequent QA builds surface defects earlier; they do NOT make manual QA the discovery mechanism. Every defect manual QA finds on an increment build triggers the § . . . (C) coverage-escape audit and a new automated check, exactly as for a release candidate.

(D) honest boundary, research grounding, and gates. Honest boundary (§ . . .): clause (D) changes WHEN and HOW OFTEN proven-correct work reaches QA; it does NOT change

what gates a release candidate. It does not weaken § . . . (huge-blocker full restart), § . . . (full-suite retest before a release TAG – a cadence QA build is not a tag), § . . . (manual-QA-final terminal gate for a release), § . . . (readiness verdict per candidate), § . . . / § . . . (build once and promote; progressive delivery), § (every finding of every layer still driven to GO), or § . . . (D)'s conservative-safe default (an unclassifiable finding is a source-defect and does block). It does not prove any increment correct – that stays § . . . / § Research grounding, stated honestly: DORA's "Working in small batches" capability (dora.dev) – "break them down into work units that can be completed independently and in short timeframes", with dark launching / feature toggles named as the way to deploy incomplete features hidden from users; Hodgson's toggle taxonomy and toggle-as-inventory discipline (quoted in D.); branch-by-abstraction (quoted in D.); SAFE's "Develop on Cadence. Release on Demand" (D.); and GitHub Next's "Continuous AI" / Agentic Workflows technical preview, which its principal researcher calls "the agentic evolution of continuous integration" – scheduled or event-triggered agent loops for recurring work such as CI-failure investigation and documentation-drift fixing, cited here only as industry direction for (D.)/(D.), not as an adopted mechanism. No single published methodology was found that combines per-increment QA delivery with continuously running agent hardening streams; that composition is this clause's own synthesis, not a claim of prior art. Classification: universal (§) – consumers supply the cadence trigger, interval, toggle mechanism and expiry policy, and the deliverable form as DATA per § Composes § . . . / § . . . / § . . . (F) / § . . . /

§ 11.4.230 • • / § 231 • • / § • • /
 § 11.4.230 • • 11.4.231 / § • • / § • • /
 § 11.4.244 • • 11.4.245 / § • • / § • • /
 § 11.4.271 • • / § • • / § • • /
 § • • (A)(B)(D) / § • • /
 § • • / § • • / § • • /
 § • • / § • • / § • • /
 § • • • Recommended gates CM-INCREMENT-DELIVERED-

NOT-SPEC-WITHHELD (FAIL: an increment holding its source-correctness GO is absent from every completed-and-handed-to-QA or in-flight QA build from its integration branch (D. 231) after the declared trigger fired while no build from that branch is in flight – coalescing per (D.) so an increment waiting on an in-flight build's successor is never a FAIL, and a build that FAILED or was REFUSED at § • • counts as NEITHER completed NOR in-flight so it provides no coverage and the increment rolls to the next build; FAIL, the § • • () dual guard: an increment WITHOUT that GO appears in a QA build) + CM-QA-BUILD-CADENCE-DECLARED (FAIL: an integration branch (D.) holds build-eligible content that no completed-and-handed-to-QA or in-flight QA build from that branch covers, the declared trigger has fired, and no QA build from that branch is in flight; coalescing per (D.) – a build in flight covers every 231 increment that landed before it started and later increments roll into the NEXT build, so an in-flight build is never a FAIL and the gate never demands one build per increment; a build that FAILED or was REFUSED at § • • covers NOTHING and does not satisfy this gate; FAIL also: a periodic / whichever-first interval with no recorded justification; an absent declaration resolves to per-increment , never to "no cadence") + CM-RELEASE-TOGGLE-TRACKED-EXPIRY (FAIL: a release toggle with no tracked removal item or no expiry, or past its expiry) + CM-QA-HANDOFF-TRACKER-IN-SYNC (FAIL: a hand-off record whose item set disagrees with the deliverable's

contents) + paired \$. mutations (withhold an eligible increment until spec completion; ship an increment lacking its GO; delete the cadence declaration and assert a "no cadence" reading; add a toggle with no expiry; mark an item done whose fix is absent from the build → each gate MUST₂₂₇ FAIL; golden-FALSE: an increment shipped on the first trigger after its GO, behind a tracked unexpired toggle, with a matching hand-off record → none fires). Gate-code = separate work item, NOT claimed shipped (\$. . / \$. .); the four gates are registered as tracked deferrals in constitution/scripts/gates/gate_ledger_deferrals.tsv . No escape hatch — no --withhold-until-spec-complete , --hardening-gates-next-increment , --no-qa-cadence , --untracked-release-toggle , --stale-tracker-at-handoff , --manual-qa-as-discovery flag.

(Process note for future extenders: \$. . 's next free clause letter is (E); never reuse (A)–(D).)

Classification: universal (\$. .) — a platform-neutral parallelization discipline reusable by ANY project that builds an artifact, validates it across multiple targets, and records device evidence; the consuming project supplies its concrete device set + serials, build/flash tooling, recording layout, and worker-pool implementation per. \$. . . Composes \$_{11 4 94 . 11 4 95} / \$. . (wall-clock stats prove the — × target) / \$₁₀₂ . . / \$. . (the PWU₁₁ pipeline this binds + extends) / \$. . / \$. . / \$_{11 4 115 . 11 4 116} / \$_{11 4 115 . 11 4 119} / \$. . (SAFE-during-build catalogue) / \$. . / \$_{11 4 120} / \$. . / \$. . (artifact/runtime layer ordering — the honest boundary) / \$. . / \$. . (single-resource-owner — B's load-bearing partition) / \$. . (seam-placement — A's honest boundary) /

§ 11.4.132 • 11.4.144 / § 11.4.130 / § 11.4.130 /
 § 11.4.130 (always-on recording – B's evidence
 source) / § 11.4.130 (huge-blocker full-restart –
 the un-weakened carve-out) / § 11.4.130 /
 § 11.4.130 / § 11.4.130 (availability-
 following – B's drop-handling) / § 11.4.130
 (maximum-USEFUL-parallelism – C's honest boundary) /
 § 11.4.130 (manual-QA-final – the un-weakened
 terminal gate) / § 11.4.130 / § 11.4.130 /
 § 11.4.130 (per-device deployed-artifact
 verification) / § 11.4.130 / § 11.4.130. Propagation gate
 CM-COVENANT-114-230-PROPAGATION (literal 11.4.230 across the
 consumer fleet) + recommended gates CM-PIPELINE-OVERLAP-
 VALIDATION (a build that runs with zero deploy-independent
 validation stages running concurrently when SAFE-catalogue
 work is queued → FAIL; a NON-huge-blocker discovery that
 restarts the whole cycle from scratch instead of re-
 running only the affected stages → FAIL) + CM-PARALLEL-
 DEVICE-WORKER-POOL (available independent devices validated
 serially instead of concurrently, OR recording capture/
 analysis serialised onto one processor when a worker-pool
 path exists → FAIL) + CM-FAN-OUT-SUBAGENTS-DEFAULT (a
 genuinely-parallelizable effort run monolithically in the
 foreground when a fan-out path exists AND all host-safety
 ceilings have headroom → FAIL) + paired § 11.4.230 meta-test
 mutations (per § 11.4.230 ; gate-code = separate work item, NOT
 claimed shipped § 11.4.230).

Canonical authority: constitution submodule Constitution.md
 § 11.4.230 .

Non-compliance is a release blocker regardless of context.
 No escape hatch – no `--sequential-build-then-validate` , `--`
`restart-whole-cycle-on-discovery` , `--serial-device-testing` , `--`
`serialise-recording-processing` , `--no-fan-out` , `--monolithic-`
`parallelizable-work` flag exists.

§ . . – Nano-precision model-tier selection: lightest-capable-model-first with dynamic mid-task escalation (User mandate, – , CRITICAL).

Verbatim operator mandate: every work effort MUST use the LIGHTEST CAPABLE model, escalating tier only as complexity rises – a nano-precision, per-task model-tier choice, NOT "everything on the heaviest model." Forensic motivation (genericised, § . .): dispatching every task – a trivial doc-sync, a one-line status read, a mechanical format fix – on the heaviest available model wastes capacity + token budget + wall-clock that a lighter tier would spend identically well, while the OPPOSITE failure (a hard multi-factor root-cause investigation dispatched on a tier too light to reason through it) produces exactly the shallow, guessed, or wrong-conclusion work

§ . . / § . . / § . . forbid. Neither "always heaviest" nor "always lightest" is compliant; the correct discipline is a PER-TASK, EVIDENCE-BASED tier choice that tracks the task's own complexity, re-evaluated as that complexity becomes known – the nano-precision selection this anchor mandates.

(A) **LIGHTEST-CAPABLE-FIRST + ESCALATE-ON-COMPLEXITY.** Every dispatched work effort – main-stream or subagent (§ . . / § . .), single-track or multitrack (§ . . / § . .) – is assigned the LIGHTEST model tier from the closed, capability-ordered ladder sonnet < opus genuinely sufficient to complete it correctly, with sonnet the DEFAULT working tier and opus for every review and other heavy work (operator mandate – , CRITICAL , verbatim: "Main model to use is Sonet, reviews and other heavy stuff have to be done by Opus instead of Fable!!! Fix this permanently on all levels!"). The review/merge-conflict-resolution substrate is a SEPARATE, already-pinned Opus-xhigh tier per clause (E), not an extra rung of this ladder; fable

is not a rung and is not a permitted dispatch tier; **haiku** is excluded per clause (D.). Tier selection is a DECIDABLE heuristic, not a guess (§ . .) – evaluated against, at minimum: (i) **task scope** (single mechanical edit / focused single-file change / multi-file-or-multi-component change); (ii) **reasoning depth** (a lookup-and-apply action vs. a multi-factor root-cause investigation per § . . / § . .); (iii) **blast radius** (a change touching one isolated file vs. one crossing shared state, hardware safety § . . , or governance/gate seams § . .); (iv) **novelty** (a previously-solved, pattern-matched task vs. a genuinely new problem with no established precedent). Closed-set tier bindings (illustrative, non-exhaustive – the consuming project supplies its own precise mapping per § . .): **sonnet** (the default) – trivial/mechanical AND moderate work (simple greps/searches, doc-sync regeneration, status/field reads, format/lint fixes, boilerplate scaffolding, a focused single-component debug, a bounded single-file implementation, straightforward research or audit reads); **opus** – complex and other heavy work, including every review (architecture/design decisions, multi-factor reasoning per § . . (), cross-feature blast-radius analysis per § . . . Pass / , hard root-cause investigation per § . . . that resists a first-pass hypothesis). A task whose complexity is genuinely UNDECIDABLE from available evidence starts at the NEXT tier UP from the best available estimate (the safe-reversible default per § . .) rather than guessed downward.

(B) DYNAMIC MID-FLIGHT SWITCHING. Tier assignment is NOT a one-time, start-of-task decision – it is RE-ASSESSED continuously as the task's real complexity becomes known during execution. A task DISPATCHED as simple that PROVES harder mid-flight (an unexpected multi-factor interaction surfaces, a root cause resists the first two hypotheses,

the blast radius turns out to cross a governance/hardware-safety seam) MUST ESCALATE to the next-heavier tier for its remainder – continuing on an under-powered tier past the point complexity is KNOWN is a clause-(D) false economy, not compliance. Conversely, a task DISPATCHED as heavy that DECOMPOSES into genuinely light, bounded, low-blast-radius subtasks (mechanical file edits following an already-designed pattern, doc regeneration, per-item status reads) dispatches THOSE subtasks on a LIGHTER tier via subagent fan-out (\$. . . /\$. . . / \$. . .) rather than running every subtask on the parent's heavier tier by default – capability-matching applies at EVERY dispatch granularity, not only the top-level task.

(C) ALWAYS-ON COMPOSITION, NEVER A PER-REQUEST OPT-IN.

Nano-precision tier selection is a STANDING DEFAULT mechanism, engaged automatically for EVERY work effort from the first prompt of a session onward (mirrors \$. . . default-autonomous-loop / \$. . . default-mechanisms-always-on) – applied TOGETHER WITH, never as a substitute for: \$. . . (heavy token-optimization – a lighter model dispatched with a bloated context is not the efficiency this anchor targets); \$. . . (maximum-USEFUL-parallelism – tier-matching each parallel stream, not merely running more streams on one fixed tier); \$. . . / \$. . . (the multi-track ruler orchestration + continuous auto-backfill – each track's worker is tier-assigned per this anchor, not defaulted to one blanket tier project-wide); and \$. . . /\$. . . (native-alias-first priority + always-on default mechanisms – this anchor governs WHICH TIER within whichever alias/account is selected, composing with, never replacing, \$. . . 's ALIAS-layer priority). None of these five composed anchors is weakened, duplicated, or

re-specified by § 1.1.1 – this anchor adds the TIER axis to their existing ALIAS/PARALLELISM/TOKEN axes.

(D) HONEST BOUNDARY (§ 1.1.1) – "LIGHTEST CAPABLE" MEANS CAPABLE, NOT CHEAPEST-REGARDLESS. "Lightest capable" is a CONJUNCTION, not a discount: a tier choice that fails to actually complete the task correctly – producing a shallow investigation, a guessed root cause, an under-verified fix, or ANY form of bluff per the § 1.1.1 anti-bluff covenant – is NOT "lightest capable," it is UNDER-POWERED, and under-powering a task that then fails, bluffs, or requires costly rework is a FALSE ECONOMY and a § 1.1.1 violation of equal severity to any other PASS-bluff (the token/time saved at dispatch is spent many times over in re-investigation, re-review, and the risk of a shipped defect). Capability determinations are made from the clause-(A) decidable heuristic + observed task outcomes – EVIDENCE-BASED, never guessed (§ 1.1.1): "this always works fine on sonnet" without a captured basis is exactly the guess § 1.1.1 forbids, and a tier's genuine capability for a task class is established by the same anti-bluff discipline (§ 1.1.1 / § 1.1.1.1 / § 1.1.1.2) that governs every other claim in this constitution. The ladder binds the DISPATCHER's tier ASSIGNMENT: a dispatcher MUST NOT assign any tier outside sonnet < opus, – never fable (– – mandate), and never haiku where clause (D.) applies. A serving model the platform selects for a turn WITHOUT a dispatcher assignment is recorded honestly in the § 1.1.1 label (whatever id the platform reports, never re-labelled); it is neither a dispatch-tier violation by the dispatcher nor a licence to assign that model, and review/merge-conflict scopes run on Opus-xhigh per clause (E) regardless.

(D.) haiku PROHIBITED WHERE IT CANNOT RECEIVE ITS GOVERNING CONTEXT – measured fact (– –). In

the reference consuming repository a `haiku` subagent dispatch failed before doing any work with "Prompt too long (~ k tokens vs k limit)" (request id `req_011CfSBXm7Kn6enyUT1UEaWd`, recorded in that repository's agent registry `docs/requests/agent_registry.jsonl`): the governance context every dispatched agent must receive there exceeds the `haiku` tier's context window, so a `haiku` dispatch cannot carry its binding context and `haiku` is PROHIBITED as a dispatch tier in that repository. A consuming project MAY re-admit `haiku` as a rung below `sonnet` ONLY on its OWN captured evidence (`$ . . / $ . .`) that a `haiku` dispatch in that project receives its full governing context and completes its task correctly – never by assumption, and never by trimming the governing context to make it fit. A `haiku` dispatch that fails this way is recorded honestly as a failed dispatch (`$ . .`), never silently retried on the same tier.

(E) HARD PINS OVERRIDE THIS DEFAULT FOR THEIR SCOPES.

`$ . .` sets the DEFAULT working tier for NON-review, non-merge-conflict-resolution work. It does NOT relax, override, or compete with the two existing hard model-substrate pins: `$ . .` (every mandatory independent code-review – `$ . . / $ . . / $ . .` iterated to G0 per `$ . .` – runs on **Opus at xhigh effort** as PRIMARY substrate, with a genuine-unavailability SONNET fallback, amended – – – no Fable, no Haiku, no other fallback model; BLOCKED only when BOTH Opus and Sonnet are genuinely unavailable, NEVER a lower tier or any other model) and `$ . .` (every main→feature/product/flavor merge-CONFLICT resolution runs on the SAME Opus-xhigh-primary-plus-Sonnet-fallback substrate, blocked only when both Opus and Sonnet are genuinely unavailable). Those two anchors are STRICTER, SCOPE-SPECIFIC pins that this anchor's general ladder does

not reach – review and merge-conflict-resolution are NOT eligible for sonnet/opus-by-complexity dispatch; they run on their pinned substrate ALWAYS, regardless of how "simple" the reviewed change or the conflict appears. A review or merge-conflict resolution dispatched on anything other than the \$. . . /\$. . . Opus-xhigh-primary-plus-sanctioned-Sonnet-fallback substrate (amended - -) is a violation of THOSE anchors, not a valid application of \$. . . 's general ladder.

(F) EFFORT-TIER SELECTION – the SAME nano-precision discipline applied to the reasoning-EFFORT axis (operator IMPORTANT mandate, - - ; verbatim: "Besides dynamic determining proper models to use, we MUST decide the effort as well and switch between the effort settings same as we do now for chosen model! Labeling MUST BE extended for this details as well. Example: (T /main - claude - opus - high)."). Every dispatched work effort is assigned not only a model tier per clauses (A)-(E) but ALSO the LIGHTEST reasoning-EFFORT setting from the closed ladder low < medium < high < xhigh < max genuinely sufficient to complete the task correctly – chosen by the SAME clause-(A) decidable heuristic (scope / reasoning-depth / blast-radius / novelty), NOT a guess (\$. . .). Illustrative bindings (non-exhaustive; the consuming project supplies its precise mapping per \$. . .): low / medium – trivial / mechanical work (doc-sync regeneration, status / field reads, format / lint fixes, boilerplate); high – moderate-to-complex reasoning (focused single-component debug, multi-factor analysis per \$. . . / \$. . .); xhigh / max – the HARDEST verification / merge-conflict / architecture work. (F.) DYNAMIC MID-TASK ESCALATION – effort, like model tier per clause (B), is RE-ASSESSED continuously as the task's real complexity becomes known; a task dispatched at a lower effort that PROVES harder mid-flight MUST ESCALATE

its effort for the remainder (continuing under-effort past the point complexity is KNOWN is the clause-(D) false economy), and a heavy task decomposing into light bounded subtasks dispatches THOSE at a lighter effort – effort-matching applies at EVERY dispatch granularity, not only the top-level task. (F.) HONEST HARNESS BOUNDARY (\$. . , load-bearing – do NOT overstate the mechanism) – effort-tiering is REAL where the execution path exposes an effort control and a DOCUMENTED CAPABILITY-GAP where it does not: the current Claude Code **Agent tool** (subagent dispatch) exposes a `model` parameter but **NO effort parameter**, so a subagent's effort is NOT presently settable via that path (the \$. . . `<effort>` label field degrades honestly to ? there); the **Workflow tool's agent()** DOES take an `effort` argument, so effort IS settable there – the label reports whichever path is live, never fakes one. This is the SAME aspirational-vs-harness state the clause-(E) \$. . / \$. . Opus- `xhigh` pins already occupy: the pins mandate the effort, and whether a dispatch path can SET, it is a harness capability reported honestly, never faked (\$. .). (F.) HARD-PIN FLOOR PRESERVED – clause (E)'s hard pins are ALSO effort pins: every \$ 11 4 20 11 4 211 mandatory independent code-review AND every \$ 2024 10 04 merge-conflict resolution runs at `xhigh` effort on Opus as PRIMARY substrate (or, on a captured genuine Opus-unavailability FACT, on the \$. . /\$. . sanctioned Sonnet fallback, amended – – – Fable and Haiku remain forbidden in every role), and this general effort ladder NEVER tiers those two scopes below `xhigh` where the dispatch path can set it, regardless of how "simple" the reviewed change or the conflict appears – a review / merge-conflict resolution run below `xhigh` on a path that COULD set it, or run on a model other than Opus or the sanctioned Sonnet fallback, is a \$. . /

§ . . violation, not a valid application of this ladder. (F. .) HONEST BOUNDARY (§ . .) – "lightest capable effort" is a CONJUNCTION, not a discount: an effort setting that yields a shallow / guessed / under-verified result is UNDER-POWERED, a FALSE ECONOMY, and a § . violation of PASS-bluff severity, exactly as an under-powered MODEL tier is per clause (D); capability determinations are evidence-based, never guessed (§ . .).

Classification: universal (§ . .) – a platform-neutral model-capacity-matching discipline reusable by any project dispatching work across a tiered model roster; the consuming project supplies its own precise tier task-class mapping and its own model-name ladder per

§ . . (the sonnet / opus ladder with Sonnet as the default, the clause-(D.) haiku prohibition, and the Opus-xhigh pins are this project's current concrete instantiation, not a universally-fixed vocabulary – a consumer with a differently-named tier roster maps its own ladder onto the SAME lightest-capable-first + dynamic-escalation + always-on-composition + honest-boundary + hard-pin-override structure, re-admitting a lighter rung only on its own captured evidence per clause (D.)).

Composes § . . (no-guessing – capability determinations are evidence-based) / § . . / § . . (subagent-driven dispatch – the granularity at which tier-matching applies) / § . . (blast-radius analysis feeds the clause-(A) heuristic) /

§ . . (safe-reversible default on undecidable complexity – escalate up, never guess down) /

§ . . / § . . (reasoning-depth signals for the heuristic) / § . . /

§ . . / § . . / § . . /

§ . . / § . . / § . .

(the always-on composition set, clause (C)) /

§ . . / § . . (the hard-pinned

review/merge-conflict substrate this anchor does NOT touch, clause (E)). Propagation gate `CM-COVENANT-114-231-PROPAGATION` (literal `11.4.231` present as a block-start, exactly-once per governance file, lockstep content-hash equality across the consumer fleet per § . . . (B)) + recommended gate `CM-MODEL-TIER-LIGHTEST-CAPABLE` (a dispatched effort whose logged/declared model tier exceeds the clause-(A) heuristic's indicated tier for a task later confirmed trivial-and-successful-at-the-lighter-tier is a WARN/candidate finding – never a hard FAIL on tier choice alone, since tier-adequacy is only provable in retrospect per clause (D); the gate instead HARD-FAILS the two decidable violations: (i) a review or merge-conflict-resolution effort NOT run on the § . . . / § . . . pinned substrate, and (ii) a task whose complexity was PROVEN mid-flight [an escalation event was logged, or a downstream failure/bluff was traced back to it] that did NOT escalate tier before continuing, and (iii) a dispatch assigned `fable`, or assigned `haiku` where clause (D.) applies) + paired § . . . mutation (gate-code = separate work item, NOT claimed shipped – § . . . ; mutation plan: () dispatch a code-review labeled as running on `sonnet` instead of `Opus-xhigh` → gate MUST FAIL; () inject a logged mid-task complexity-escalation event with NO corresponding tier-escalation record → gate MUST FAIL; (b) record a dispatch assigned `fable`, or assigned `haiku` where clause (D.) applies → gate MUST FAIL; () golden-FALSE fixture: a task that stays on its initially-assigned tier throughout, with no escalation event logged → gate MUST NOT fire, per § . . . () the false-positive guard).

Canonical authority: constitution submodule `Constitution.md`
 §

Non-compliance is a release blocker regardless of context.
 No escape hatch – no `--always-heaviest-model`, `--always-`

lightest-model , --skip-complexity-reassessment , --fixed-tier-per-session , --relax-review-pin , --relax-merge-conflict-pin , --per-request-tier-opt-in flag exists.

\$. . . – Anti-mess long-op orchestration: every long-op is registered, single-owned-per-purpose, liveness-PROVEN, handed-off-on-stop, safely-reaped, consistency-checked, and escape-bounded – the single source of truth for what is running, never inferred (research-derived, - -).

Forensic anchor (genericised): in one cycle a verification sweep produced five distinct mess states – () multiple concurrent same-purpose sweeps ran unintentionally (no single-owner-per-purpose claim); () a sweep HUNG on a consumer-blocked-on-an-unclosed-producer pipeline and was undetected because liveness was inferred from process-alive (a \$. . . () FALSE=NULL: a wedged op and a progressing op returned the identical "process alive" signal); () stopping the owning subagent SILENTLY KILLED its child long-op because ownership was implicit (no handoff-on-stop); () a stale/non-growing log was read as live progress (the same false-null, reader-side); () an un-completable tooling gate blocked a release for hours (no escape to a terminal state). Per \$. . .

Phase . . . these are ONE architectural state-management flaw, not five bugs: **there was no single source of truth for what long-op is running, who owns it, its state, and its PROVEN liveness** – the \$. . . (F) configured-in-use gap generalised to project long-ops, so every mess state was un-observable, and an un-observable bad state is a \$. . . /\$. . . bluff at the orchestration layer.

The mandate – a **long-op orchestration layer** binding every long-op (any op expected to exceed the runtime background-execution threshold per \$. . . – verification

sweep, meta-test sweep, build, flash/deploy, test/QA suite, dispatched subagent), ALL of which hold:

(A) SINGLE SOURCE OF TRUTH – the long-op lifecycle registry. Every long-op MUST register BEFORE it runs and reach a terminal state (complete|failed|reaped|handoff|blocked-escape) when it ends, in a durable (never-tmpfs, write-temp-then-rename, \$. . . append-only event stream + atomic snapshot) registry recording {op_id, purpose_key, owner, pid|pgid, cwd, state, last_heartbeat, heartbeat_seq, progress_offset, log_path, verdict, evidence_path} . Success is read from the op's terminal verdict/ result , NEVER its process exit code. A long-op running with no registry record – or a record with no live op – is a FALSE-READY drift state.

(B) SINGLE-OWNER-PER-PURPOSE. No two same- purpose_key long-ops run at once; acquisition is a probe-race-safe claim keyed on purpose_key (\$. . .). A live claim forces exactly ONE of REUSE (attach) or explicit SUPERSEDE (graceful handoff of the running owner, logged), never a silent second run.

(C) LIVENESS TRUTH-SOURCE. Progress is PROVEN, never inferred from process-alive: every long-op emits a monotonic heartbeat and/or an advancing progress offset; a watchdog reads the DELTA over a window and flags HUNG on no-advance past the op's declared no-progress budget (the \$. . . ()-()/\$. . . discipline – kill -0 is a necessary-not-sufficient pre-filter, a stale heartbeat/non-growing log is NEVER "progressing"). A consumer blocked on an unclosed producer write-end (\$. . . ()) is caught by the flat heartbeat regardless of internal state.

(D) OWNERSHIP-HANDOFF-ON-STOP. Stopping/killing an owner MUST hand its long-ops to the registry (handoff , re-adoptable per \$. . .) or deliberately reap them

(reaped, reason logged) – NEVER silently orphan or accidentally cascade-kill; long-ops launch detached (own process-group) so an owner stop does not cascade-signal them, and the stop path records each owned op's disposition. The loop done-condition is not satisfied while a handoff op is un-adopted.

(E) **SAFE REAPING.** An orphaned/hung op is reaped ONLY on PROVEN staleness – holder PID dead (kill -0 fails) OR no-progress-budget exceeded per (C) – resolved by the process's real /proc/<pid>/cmdline (\$. . (D)/ \$.), never a bare pgrep /substring that could match a carrier or the reaper itself; a LIVE, ADVANCING op is NEVER reaped (\$.); every REAPED/KEPT decision + resolved evidence is logged (\$. .).

(F) **CONSISTENCY INVARIANT SWEEP.** A fast invariant check runs before every gated transition (build/flash/tag) and on a standing cadence, asserting: no duplicate owner per purpose, no orphan, no stale lock (\$. .), no hung op (C), no registry-vs-reality drift (live-long-op set == registry live rows – the \$. . (F) check generalised). Any violation is surfaced (named, actionable) and the gated transition is REFUSED until resolved. The sweep is itself a guard (\$. .): golden-TRUE + golden-FALSE-with-carrier fixtures, conservative-safe REFUSE on an unresolvable signal, resolved evidence printed on every refusal.

(G) **UN-COMPLETABLE-BLOCKER ESCAPE.** A gate that cannot reach a terminal verdict MUST degrade, after a bounded budget, to an explicit evidence-backed decision – NEVER an indefinite silent block: \$. . first (hung/(C) vs duplicate/(B) vs wrong-seam/\$. . re-seat-at-activation); NOT-YET-RUNNABLE → \$. . artifact_not_yet_built (non-blocking on an intermediate artifact, blocking only on the release candidate per \$. .); genuinely un-completable →

```

$      . .      (autonomous reversible SKIP-with-reason +
tracked $      . .      item, ELSE a bounded
$      . .      operator decision) – the loop keeps
progressing every other non-blocked item, it does not
idle, and the escape NEVER fabricates a PASS.

```

Honest boundary (§ 11.4.1). This layer reduces orchestration mess; it does NOT make product bugs impossible, does NOT make a gate correct (§ 11.4.2 / § 11.4.3 still bind each gate), and does NOT remove the operator from genuine blocks – it converts an indefinite silent stall into a bounded, evidence-backed, operator-visible decision. A heartbeat proves ADVANCE not CORRECTNESS; the registry proves what CLAIMS to run, (F) proves registry==reality. **Reuse-not-rewrite:** it is the multitrack per-track agent registry (§ 11.4.4 / § 11.4.5 / § 11.4.6) generalised from per-track workers to any long-op and bound through, per § 11.4.7 (F) – no parallel mechanism is invented.

Classification: universal (\$. .) – no hardware/
vendor/project literal; consumers supply their long-op
set, ¹⁴⁷ purpose keys, ¹⁴⁸ no-progress budgets, and stop-hook
wiring as DATA per \$ ₆ . ₁₂ . . Composes

$\$_{11 \ 4 \ 232 \ \bullet \ 1 \ \bullet \ 4 \ 63 \ 12 \ / \ \$ \ \bullet \ \bullet \ / \ \$ \ \bullet \ \bullet \ /$
 $\$_{12 \ 12 \ \bullet \ 9 \ 2 \ \bullet \ / \ \$ \ \bullet \ \bullet \ / \ \$ \ \bullet \ \bullet \ /$
 $\$ \ \bullet \ \bullet \ / \ \$ \ \bullet \ \bullet \ / \ \$ \ \bullet \ \bullet \ /$
 $\$ \ \bullet \ \bullet \ (D)(F) \ / \ \$ \ \bullet \ \bullet \ (\) (\) - (\) \ /$
 $\$ \ \bullet \ \bullet \ / \ \$ \ \bullet \ \bullet \ / \ \$ \ \bullet \ \bullet \ /$
 $\$ \ \bullet \ \bullet \ / \ \$ \ \bullet \ \bullet \ .$ Propagation gate CM-COVENANT-114-232-
 PROPAGATION (literal 11.4.232) + recommended mechanism
 gates CM-LONGOP-REGISTRY-SINGLE-SOURCE (A) / CM-LONGOP-SINGLE-
 OWNER-PER-PURPOSE (B) / CM-LONGOP-LIVENESS-HEARTBEAT (C) /
 CM-LONGOP-OWNERSHIP-HANDOFF-ON-STOP (D) / CM-LONGOP-SAFE-REAP-
 PROVEN-STALE (E) / CM-ORCHESTRATION-CONSISTENCY-SWEEP (F) /
 CM-UNCOMPLETABLE-BLOCKER-ESCAPE (G) + paired \$.
 mutations (run two same-purpose sweeps with no reuse/

supersede → (B) FAILS; infer liveness from `kill -0` alone / drop the heartbeat-delta check → (C) FAILS; a stop that leaves an owned op with no terminal/handoff record → (D) FAILS; reap a live-advancing op OR reap by bare- `pgrep` substring → (E) FAILS; a live long-op with no registry row passes the consistency sweep → (F) FAILS; an un-completable gate blocks with no escape/decision → (G) FAILS; strip the literal → propagation FAILS; gate-code = separate work item, NOT claimed shipped \$. .). No escape hatch – no `--allow-duplicate-longop` , `--infer-liveness-from-pid` , `--silent-orphan-on-stop` , `--reap-live-op` , `--skip-consistency-sweep` , `--block-without-escape` , `--longop-without-registry` flag.

EXTENSION landed – – – \$. .
model-in-label + flawless alias/provider model
consistency + real-time always-in-sync (User mandate,
 – – .). Verbatim operator mandates
 (– –): "Extend our logging to present
 models used since we will now have dynamic models choice
 and switching! From (example) (T /main – claude) into:
 (T /main – claude – model_used), model MUST be
 dynamically obtained so we present name equivalents –
 short names – like: haiku, opus, sonnet, fable. However we
 MUST fully support all Claude Toolkit providers aliases,
 so real model used names MUST BE obtained in this case(s)
 as well! Examples: (T /main – claude – haiku), (T /
 main – claude – fable), (T /main – claude – opus),
 (T /main – claude – deepseek), (T /main – claude –
 xiaomi) ... Labels MUST BE everywhere ALWAYS up to date in
 real time and in sync!" and "Adding model names into
 labels considers that providers names / aliases will be up
 to date too and in relation to model used since models
 belong to providers / aliases !!! Consistency MUST BE
 flawless!" This EXTENDS the \$. . label
 discipline (it does NOT re-mint the anchor – the literal
 11.4.182 and its propagation gate CM-COVENANT-114-182-

PROPAGATION are UNCHANGED and stay GREEN; this carrier block cites § . . . , it is not a second § . . . block-start per § . . . (B)).

() **LABEL FORM EXTENDED** – the mandated § . . . label is extended from (T<N>/<branch> - <alias>) to (T<N>/<branch> - <alias> - <model> - <effort>) (the trailing <effort> field added - - per § . . . clause (F) – see the EFFORT-FIELD clause () below), <model> recording which model is CURRENTLY answering for that alias: for a native Claude alias, the SHORT name (opus / sonnet / haiku / fable , case-insensitive-substring-derived from the model id the agent platform reports for the most recent turn – DYNAMIC, obtained live at label-emission time, never a fixed/assumed value since native aliases have no persistent per-alias model and the serving model can legitimately switch mid-session per § . . .); for a claude-toolkit PROVIDER alias (deepseek , xiaomi , kimi-for-coding , opencode , ...) the REAL provider model id as declared in that provider's host-config registry (the SAME value the alias's own launch mechanism uses – a single source of truth, never re-derived). () **MODEL-FIELD DERIVATION** (§ . . . , never guessed) – a closed two-case split on the SHAPE of the already-derived <alias>: (a) native-alias → read the model id from the CURRENT session's own transcript (the platform's per-project per-session transcript; LAST assistant-turn's reported model → short name); (b) provider-alias → read the model id from the host multi-account-toolkit provider registry (CMA_PROVIDER_MODEL). Either lookup that cannot resolve yields the honest ? – NEVER a fabricated or last-known value; an id obtained but matching none of the four native short-names is printed VERBATIM (real-but-unrecognised is more honest than a false ?). The -field legacy form (no <model>) remains a VALID, ACCEPTED label during migration – the enforcing guard-hook LABEL_RE admits both

forms, and NEVER treats an honest ? model, or an absent model field, as a block-cause. () **FLAWLESS ALIAS/ PROVIDER MODEL CONSISTENCY INVARIANT** (operator NOTE, **load-bearing**) – the <alias> /provider field and the <model> field MUST be MUTUALLY CONSISTENT AT ALL TIMES: the model shown MUST be a model that genuinely BELONGS to the alias/provider shown (models belong to providers/ aliases), and the alias/provider name MUST itself be current – never a stale alias beside a fresh model, never a model that does not belong to the shown alias. This is guaranteed BY CONSTRUCTION, not by coincidence: the native model is read from THAT alias's own live session transcript, and the provider model is read from THAT provider's own registry entry – so the derived <model> is definitionally a model of the shown <alias> ; a derivation that ever emits a model not belonging to the shown alias is a \$. . / \$. . consistency violation, never a valid label. () **REAL-TIME, ALWAYS-IN-SYNC** (\$. . /\$. . (F)) – the label is STATELESS and RE-COMPUTED at EVERY emission (every human-invoked labeler run AND every synchronous PreToolUse guard-hook firing on every Agent/Task/TaskCreate dispatch) against the LIVE environment (current CLAUDE_CONFIG_DIR , current session id, current transcript tail, current provider .env) – so a mid-session model switch (/model , an autonomous \$. . escalation) is reflected in the VERY NEXT emission with no cache to go stale: real-time BY CONSTRUCTION, not by a sync daemon. Honest boundary (\$. .): the live label has NO persisted artifact that could drift, so the Docs-Chain (\$. .) sync engine – which propagates PERSISTED documents/DBs – does not apply to the stateless label itself; the operator's "use/extend Docs Chain for real-time responsiveness" ask attaches to any PERSISTED label-history / model-switch LOG surface (a \$. . follow-up to be tracked), which, if adopted, IS

\$. . -Docs-Chain-synced and \$. . -
 request-history-adjacent – never faked as already-shipped.
 () **HONEST SUBAGENT-INTROSPECTION LIMIT** (\$. . /
 \$. .) – the native-alias transcript lookup is
 proven reliable when invoked from the DISPATCHING
 process's own shell (the guard-hook's actual execution
 context at dispatch time); it is NOT proven reliable when
 a background-dispatched subagent introspects its OWN model
 from inside its own subprocess on hosts where the subagent
 shares its parent's session-transcript with no subagent-
 vs-conductor marker – recorded as an open limitation to be
 tracked, never silently claimed solved. () **DIVERGENT-
 LABELER RECONCILIATION** (\$. . 11 4 137) – the LIVE
 labeler wired via `.claude/settings.json` is `constitution/
 scripts/multitrack/track_branch_label.sh` ; the top-level
`scripts/multitrack/` path is a thin pass-through WRAPPER
 that `exec s` that canonical copy (\$. .
 inherited-by-reference), so it mirrors nothing, cannot
 drift, and emits byte-identical output by construction..
 The reconciliation LANDED in commit `a48e919f52d` and is
 gate-enforced at `pre_build_verification.sh:41911-41915` (the
 gate detects the `exec` wrapper and follows it to the
 canonical implementation), never a silent divergence. ()
EFFORT FIELD (added - - per the operator
 IMPORTANT mandate – verbatim: "Besides dynamic determining
 proper models to use, we MUST decide the effort as well
 and switch between the effort settings same as we do now
 for chosen model! Labeling MUST BE extended for this
 details as well. Example: (T /main - claude - opus -
 high)."). The `..-field` label appends `<effort>` after
`<model>` , recording which reasoning-EFFORT setting is
 CURRENTLY in force for that work-stream, drawn from the
 closed set `{low | medium | high | xhigh | max}` (the SAME
 effort ladder \$. . clause (F) selects). (a)
 DERIVATION (\$. . , never guessed) – `<effort>` is
 read from whatever LIVE effort signal the execution path

exposes (the Workflow-tool `agent()` `effort` argument, a future Agent-tool `effort` parameter, an env/config `effort` setting); a path exposing NO `effort` signal, OR any value outside the closed set, yields the honest `?` – EXACTLY like the `<model>` field's `?` fallback, NEVER a fabricated / assumed / last-known value. (b) HONEST HARNESS BOUNDARY (`$ . . , load-bearing` – do NOT overstate the mechanism) – `effort`-tiering is REAL where the execution path exposes it and a DOCUMENTED CAPABILITY-GAP where it does not: the current Claude Code **Agent tool** (subagent dispatch) exposes a `model` parameter but NO `effort` parameter, so a subagent's `effort` is NOT currently settable via that path and its `<effort>` field degrades honestly to `?`; the Workflow tool's `agent()` DOES take an `effort` argument, so `effort` IS settable + derivable there – the label reports whichever path is live, never invents one. This is the SAME aspirational-vs-harness state the `$ . . / $ . . Opus-xhigh` review/merge pins already occupy: the pins mandate the `effort`, and whether a given dispatch path can SET it is a harness capability the label reports honestly and NEVER fakes. (c) LEGACY-FORM ACCEPTANCE (migration) – the `-field (T<N>/<branch> - <alias>)` and `-field (T<N>/<branch> - <alias> - <model>)` label forms remain VALID and ACCEPTED; the guard `LABEL_RE` admits all three shapes and NEVER treats an absent `<effort>` field, or an honest `?` `effort`, as a block-cause (`$ . . – mirroring the $ . . track/alias/model honest-? rule`). (d) CONSISTENCY – the `effort` field is selected TOGETHER WITH the `model` by `$ . . (both tiers chosen for one task)` and never contradicts it; a stale `effort` beside a fresh `model`, or an `effort` outside the closed set, is a `$ . . / $ . . consistency violation`, never a valid label. (e) REAL-TIME, ALWAYS-IN-SYNC (`$ . . )` – like `<model>`, `<effort>` is STATELESS and RE-COMPUTED at EVERY label emission against the LIVE environment, so a

mid-task § . . effort escalation is reflected in the VERY NEXT emission with no cache to go stale. Propagation: CM-TRACK-BRANCH-LABEL is EXTENDED to also assert the labeler emits a parseable <model> field AND a parseable <effort> field, the guard LABEL_RE accepts the -field / -field / -field label forms (legacy -field and -field ALWAYS remain accepted during migration; a ? effort is NEVER a block-cause), and the alias/provider model consistency holds; paired § . mutation: strip the <model> derivation OR emit a model not belonging to the shown alias → the extended gate FAILs (gate-code = separate work item, NOT claimed shipped § . .). No new escape hatch – the existing § . . "no escape hatch" list applies to the extended label unchanged.

§ . . – Standing anti-mess control plane: a level-triggered, idempotent, invariant-catalogue reconciliation loop over the WHOLE persistent System state, orchestrating every per-domain enforcer, refusing gated transitions on un-reconciled drift (research-derived, - -).

Forensic anchor (genericised): every existing consistency enforcer in a multi-track / multi-agent / multi-submodule / multi-upstream System is EDGE-TRIGGERED – a PreToolUse hook fires on an action, a check runs at a seam (lock-acquisition / merge-time / build-time / commit-time), a gate runs when a build runs – so a mess that arises WITHOUT an edge firing is un-observed until it causes harm (two live examples captured § . . .

- - : a large uncommitted-batch pile-up sat unreconciled through a session, and multiple provably-stale VCS/wrapper locks were held by dead holders with no acquisition attempt to reap them – the § . . "a hook never installed / a check that runs only when asked" forensic generalised to the whole persistent state). Per

§ 1.1. Phase 1. the ~ persistent-state mess classes (Plane R repository: uncommitted pile-up, stale/duplicate locks, orphaned worktrees, half-applied cross-repo commits, drifted submodule pointers, partial merges, divergent mirrors, mis-placed work; Plane G governance: divergent mirrors, pointer drift, gate-ledger drift, anchor-number collision; Plane S tracking: doc/DB drift, cross-track SSoT divergence, ledger drift, handoff-doc drift; Plane P runtime: the § 1.2. long-op set + un-reaped streams + dangling review state) are ONE architectural flaw – there is no LEVEL-TRIGGERED control plane that re-derives the actual persistent state and reconciles it against a declared clean state; an unobserved drift is a § 1.3. /§ 1.4. bluff at the organization layer ("the System is clean" while a mess silently accumulates).

The mandate – a standing **anti-mess control plane** (the desired-state/actual-state reconciliation-loop + control-plane/data-plane pattern of distributed systems^{11 4 23}), ALL of which hold^{11 4 31}:

(A) **DECLARED INVARIANT CATALOGUE.** A single machine-readable consumer-owned catalogue (DATA per § 1.5. / § 1.6.) enumerating every consistency invariant across the persistent planes, each carrying {id, plane, desired-condition, detector, reconcile-class ∈ {auto-safe | operator-gated | owned-by <anchor>}, evidence-shape} ; the catalogue IS the desired clean state; an un-catalogued mess class is an honest tracked GAP (§ 1.7. / § 1.8.), never silently claimed covered.

(B) **LEVEL-TRIGGERED LOOP.** The loop re-derives the ACTUAL state (not an event) and diffs it against the catalogue BEFORE every gated transition (commit-batch/build/flash/merge-to-trunk/tag/constitution-pull) AND on a standing cadence (§ 1.9. / § 1.10.) – catching drift regardless of how it arose (the Kubernetes level-triggered

property)); actual-vs-desired uses a \$. . fingerprint diff drilling to the offending record (anti-entropy), never a whole-scan where a fingerprint suffices.

(C) IDEMPOTENT RECONCILE-OR-REFUSE. Each drift takes exactly one dedup-keyed action: RECONCILE-AUTO-SAFE (reversible + evidence-determinable + bounded-blast per \$. . . - reap a provably-dead lock \$. . , regen a stale derived doc \$. . , scope-group-commit a placement-clean batch \$. . , retire a merged worktree \$. .), idempotent (a no-op on a clean state); OR REFUSE-THE-GATE + a bounded operator decision (irreversible/high-blast/indeterminate - id-colliding SSoT \$. . , anchor collision \$. . , partial cross-repo merge, divergent mirror) per \$. . /\$. . ; a gated transition NEVER proceeds while a catalogued invariant is violated and un-reconciled; the loop keeps progressing every other non-blocked item (park-one-not-the-loop).

(D) SAGA FORWARD-RECONCILE (no-loss). A partially-applied cross-repo operation is reconciled complete-forward under a \$. pre-op backup (finish the remaining legs / bump the pointer / complete the merge zero-markers-zero-drop \$. . step / ff the lagging mirror) - NEVER force-push/reset/ -s ours (\$. . /\$. .); compensation is \$. backup-restore only; the saga reaches fully-completed, never rolled-back-with-loss.

(E) THE LOOP IS A GUARD (\$. .). Every detector ships golden-TRUE + golden-FALSE-with-carrier fixtures (\$. . ()); a false-positive REFUSAL of a clean state is a \$. . () FAIL-bluff exactly as a false-negative PASS is a \$. . PASS-bluff; every "no drift" null is control-needle-proven (\$. . ()-

(
)); conservative-safe REFUSE on an unresolvable
signal, resolved evidence printed on every refusal.

(F) **CONTROL-PLANE / DATA-PLANE SEPARATION.** The control
plane_DETECTS + RECONCILES state and NEVER performs the
data-plane_work (the actual build/commit/merge/long-op –
\$. . . + the build/flash/commit wrappers); the
two never merge; the control plane's own state is durable
(\$. . . () temp→fsync→rename, never tmpfs, may
reuse \$. . .).

Honest boundary (\$. . .). Converges the CATALOGUED
invariants; does NOT₁₀₁ make an un-catalogued mess class
impossible (extensible catalogue, never "provably mess-
free"), does NOT₁₀₁ make a gate correct (\$. . . /
\$. . .), does NOT auto-fix irreversible drift
(operator-gated \$_{11 4 100} . . . /\$_{11 4 100} . . .), and is
eventual-consistency-at-every-gate-and-cadence NOT a real-
time_daemon (\$_{11 4 147} . . . ()). **Reuse-not-rewrite:** it
is the level-triggered COMPOSER over the existing per-
domain enforcers (\$_{11 4 220} . . . /\$. . . /
\$_{11 4 232} . . . /\$. . . /\$. . . /
\$. . . /\$_{11 4 233} . . . /\$. . . /
\$. . .), inheriting each's detector – no parallel
enforcer is invented (\$. . .). It composes
\$. . . (the RUNTIME_{11 4 17} data-plane long-op
orchestrator – \$. . . clause (F) is Plane P of
this catalogue) as the plane it does not touch.

Classification: universal (\$. . .) – no hardware/
vendor/project literal; consumers supply their invariant
catalogue, plane detectors, gated-transition list,
cadence, and reconcile-class bindings as DATA per

\$_{11 4 147} Composes \$. . . / \$. . . /
\$. . . / \$. . . / \$. . . /
\$. . . / \$. . . / \$. . . /
\$. . . / \$. . . / \$. . . /
\$. . . / \$. . . / \$. . . /

```

$ ./... / $ . . / $ . . /
$ ./... / $ . . / $ . . /
$ . . (F) / $ . . /
$ . . / $ . . / $ . . /
$ . . / $ . . / $ . . /
$ . / $ . . Propagation gate CM-COVENANT-114-233-
PROPAGATION (literal 11.4.233 ) + recommended mechanism
gates CM-ANTIMESS-INVARIANT-CATALOGUE (A) / CM-ANTIMESS-LEVEL-
TRIGGERED-LOOP (B) / CM-ANTIMESS-RECONCILE-OR-REFUSE (C) /
CM-ANTIMESS-SAGA-FORWARD-NO-LOSS (D) / CM-ANTIMESS-LOOP-IS-A-
GUARD (E) / CM-ANTIMESS-CONTROL-DATA-PLANE-SPLIT (F) + paired
$ . mutations (drop the standing-cadence trigger so
the loop is edge-only → (B) FAILs on the pile-up/stale-
lock fixture; let a gated transition proceed on a
catalogued un-reconciled drift → (C) FAILs; a saga that
force-pushes/resets to reconcile a mirror → (D) FAILs; a
detector that REFUSES a clean golden-TRUE fixture, or
PASSes a golden-FALSE-with-carrier → (E) FAILs; the
control plane invokes a data-plane build/commit → (F)
FAILs; strip the literal → propagation FAILs; gate-code =
separate work item, NOT claimed shipped $ . . /
$ . . ). No escape hatch – no
--edge-triggered-only , --skip-consistency-catalogue , --
proceed-on-drift , --reconcile-with-force-push , --unvalidated-
detector , --control-plane-does-work , --assume-clean-without-
reconcile flag.

```

\$. . – Dedicated hook-validation script: hook checks run as an explicit stage of a dedicated commit/push script, and the commit/push mechanism is ALWAYS unblocked (operator mandate, – –).

Forensic anchor (genericised): a consumer project wired its full local CI gate as an automatic git pre-push hook (core.hooksPath → hook → multi-minute build/test/sweep stage on EVERY push). The result, captured – – : routine pushes took many minutes each,

and when one gate stage inside the hook failed or stalled, the push was rejected outright – the commit/push mechanism itself became blocked, with pending commits stranded locally and no operator-usable path forward short of a forbidden `--no-verify` bypass. The gate logic was sound; wiring it as an uninterruptible implicit hook on the commit/push code path was the architectural flaw – an edge-triggered enforcer (`$ . . .`) placed directly on the operator's only forward path, violating the `$` single-entripoint discipline by making the entripoint unusable.

The mandate, ALL of which hold:

11 4 35

(A) DEDICATED SCRIPT AS THE SINGLE COMMIT/PUSH ENTRYPPOINT.

Every consumer MUST ship a dedicated, executable, idempotent script (Lava binding: `scripts/commit-push-all.sh`; consumers bind their own path as DATA per `$ . . .`) that runs, in order: () submodule/workspace sync, () commit of all pending state, () the hook validations as an EXPLICIT stage – the unmodified hook logic fed the real push range, Layer-cheap checks always, long gates stage-separated, () push to ALL configured upstreams (`$ . . .`), () a closing green/clean verification (clean status + zero unpushed commits, recursive over submodules). The script prints per-stage evidence and is safe to re-run; direct `git commit / git push` on the main repo ceases to be the routine workflow (`$ . . .`).

(B) HOOKS MUST NOT BLOCK THE MECHANISM.

Automatic git hooks wired via `core.hooksPath` MUST NOT gate routine commit/push in a way that can leave the repository uncommittable or unpushable. Disconnecting means: `core.hooksPath` unset, the hook file itself preserved UNMODIFIED in the repo, and `--no-verify` NEVER used as the routine path. Validation relocates from the implicit hook

to the explicit script stage (A) and to tag/release-time gates – it is never deleted.

(C) NO GATE IS LOST. Every check the disconnected hook performed MUST remain executed – by the dedicated script's validation stage, by the tag/release gate, or by a documented scheduled run. Dropping a check silently because the hook no longer fires is a `$ . . . PASS-bluff` at the process layer ("the gate still exists" while nothing invokes it); the script's stage list MUST name every check it runs and every check it defers, with the deferral target recorded.

(D) ALWAYS-UNBLOCKED INVARIANT. The commit/push mechanism MUST always be able to complete. A failing validation yields a clear per-check report plus a documented remediation path – never an opaque hung or rejected push. Long gates (multi-minute build/test/sweep stages) MUST be separable from the cheap checks and skippable via an explicit, documented flag (Lava binding:

`LAVA_SYNC_SKIP_CI=1`), with every skip recorded in the commit message or evidence pack so the deferred gate is tracked, never forgotten. The escape hatch is a recorded deferral, not a bypass: the skipped gate remains owed and is caught at the next full run or tag gate.

Honest boundary (`$ . . .`). This anchor relocates WHERE validation executes (implicit hook → explicit script stage + release gates); it does NOT weaken any substantive gate – the Sixth/Seventh-Law-equivalent anti-bluff covenant, the tag-time full gate, and `$ . . .` 's sweep-after-constitution-pull all still bind at their own seams. It does NOT mandate deleting hooks (the hook file is preserved), does NOT authorise `--no-verify` as routine practice, and does NOT make a failing gate pass – it makes the mechanism reachable while the failure is remediated. Composes `$ / $ . . .` (single entrypoint + all-upstreams push – this anchor keeps that entrypoint usable),

§ . . . (sweep still runs, now as a named stage),
 § . . . (no force-push history-rewrite – the
 unblocked path never rewrites), § . . . (the
 script is a guard: a false GREEN verification is a FAIL-
 bluff), § . . . (amendment discipline),
 § . . . (the commit/push path is a gated
 transition the anti-mess control plane reconciles).

Classification: universal (§ . . .) – no hardware/
 vendor/project literal; consumers supply their script
 path, upstream-remote list, cheap-vs-long gate split, and
 skip-flag name as DATA per § Propagation gate
 CM-COVENANT-114-234-PROPAGATION (literal 11.4.234) +
 recommended mechanism gates CM-DEDICATED-HOOK-VALIDATION-
 SCRIPT (A) / CM-HOOKS-NEVER-BLOCK-PUSH (B) / CM-NO-GATE-LOST-
 ON-HOOK-DISCONNECT (C) / CM-COMMIT-PUSH-ALWAYS-UNBLOCKED (D) +
 paired § . . . mutations (wire the full gate back as an
 automatic blocking pre-push hook → (B) FAILS; disconnect
 the hook AND drop a check from every stage → (C) FAILS; a
 validation failure that strands commits with no
 remediation report → (D) FAILS; strip the literal →
 propagation FAILS; gate-code = separate work item, NOT
 claimed shipped § . . . /§ . . .). No escape
 hatch beyond the recorded (D) deferral – no --no-verify-
 routine , --skip-all-gates-silently , --delete-blocking-hook ,
 --push-without-any-validation flag.

§ . . . – Build-and-deploy the moment source
 fixes are proven-correct (test-side hardening is a
 parallel stage, never a build gate) + post-deploy version
 increment (operator mandate, – –).

(A) BUILD-AND-DEPLOY THE MOMENT THE SOURCE FIXES ARE
 PROVEN-CORRECT – TEST-SIDE WORK IS A PARALLEL STAGE, NEVER
 A BUILD GATE. The moment the SOURCE fixes of a batch are
 done AND proven-correct – proven-correct = the mandatory
 independent SOURCE-correctness review has returned a clean
 GO (§ . . . /§ . . . /§ . . . on

the source diff, on the \$. . . Opus- xhigh substrate, iterated to zero-finding/zero-warning GO per \$. . .) – the build MUST be triggered, pre-build + post-build tests EXECUTED (\$. . . layer-before, layer- after), and the artifact deployed. IN PARALLEL WITH the build AND the manual-QA session (\$. . .), NEVER gating the build: () gate-instrumentation hardening + new-gate authoring (\$. . .); () paired \$. . . mutations + four-layer \$. . . (b) coverage + HelixQA Challenges; () "gates and checks of all kinds" (\$. . .); () the re-review of THAT test-instrumentation work. The build's GO-SIGNAL is source-correctness alone; test-guard hardening (future-regression coverage) + its re-review are a deploy-independent parallel stage per \$. . . (A). Holding the build sequentially – waiting for the test-instrumentation buildout + its re-review – when the source is already proven-correct is a \$. . . (A) violation.

(B) POST-DEPLOY VERSION INCREMENT + MANUAL-QA-DEPLOY CYCLE BOUNDARY (extended – – – operator mandate, verbatim: "Increase upcoming release to . . . since new cycle starts every time we deploy release for manual QA testing! This is how we will be working this ALWAYS! Extend the constitution!"). The moment image(s)/artifact(s) deploy, the version code/version is INCREMENTED (next build targets the next version; every deployed artifact carries a distinct, monotonic, greppable id – \$. . . /\$. . . /\$. . .). Two deployments sharing one version id defeats per-deploy artifact identity → \$. . . (B) violation. EVERY deployment of a release FOR MANUAL QA TESTING (\$. . .) is a DEVELOPMENT-CYCLE BOUNDARY: the manual-QA deploy CLOSES the outgoing cycle and STARTS a NEW development cycle, and the version code/version is incremented AT THAT POINT – the new cycle's work targets

the NEXT monotonic version (§ 11.4.126 project-prefixed naming..preserved) – ALWAYS, the STANDING working model per clause (C), never a per-request opt-in. The manual-QA deploy is the canonical clause-(B) increment trigger (a deploy-for-manual-QA IS a deploy), and it SHAPES the § 11.4.125 release-cycle framing: fixes for QA findings against the deployed artifact land in the NEW cycle on the NEW version id, never patched onto the already-QA-deployed version id. Starting a new cycle's work still targeting the QA-deployed version id is a § 11.4.125 (B) violation.

(C) STANDING DEFAULT (§ 11.4.125 / § 11.4.142), never a per-request opt-in.

(D) FINDING-LAYER CLASSIFIER – the decidable split clause (A) relies...on, mandatory at every § 11.4.125 / § 11.4.142 review verdict (added – – ; research-derived from the § 11.4.125 four-agent audit of this project's own Speckit-T remediation saga). Every finding a code-review verdict records (§ 11.4.125 (F) machine-written) MUST carry exactly one `finding_layer` from the closed set `{source-defect | test-instrumentation | process-doc}` (a field DISTINCT from any other per-finding classification the review record already carries – e.g. this project's own `review_record.py` pre-existing "class" field `{mechanical | judgment | false-positive}` is a different axis...and is unaffected): `source-defect` – the finding shows the SHIPPED artifact's own behaviour wrong, unsafe or exploitable as it will actually run, OR invalidates the evidence that proves THIS batch's fixes correct (their § 11.4.125 RED→GREEN guard, § 11.4.125 runtime signature, or covering test per § 11.4.125 () – a proving guard that cannot fail or whose reviewer-authored mutation survives per § 11.4.125 () (d) means the source is NOT proven-

correct); `test-instrumentation` – the finding concerns only future-regression coverage, mutation-guard robustness or detection completeness of a guard/gate/meta-test whose protected behaviour was ALREADY proven, with no production decision path shown wrong and no proof-of-this-batch evidence invalidated; `process-doc` – labels, comments, citations or prose with no behavioural effect. The layer is recorded by the REVIEWER (§ . . . / § . . . producer#verifier) – the producer MAY contest it only via a fresh review round and MUST NOT lower it unilaterally. A finding that cannot be decidably classified defaults to `source-defect` (the conservative-safe default per § . . . ()); secondarily § . . . safe-reversible). § . . . is NOT relaxed: every finding of every layer is still driven to a zero-finding, zero-warning GO in its own loop; clause (D) decides only which GO is the build GO-signal – a batch/slice is BUILD-ELIGIBLE when its `source-defect` findings reach that GO, while outstanding `test-instrumentation` / `process-doc` findings continue as their own tracked § . . . item on a § . . . (A)/ § . . . parallel stream. A BATCHED review (one review spanning many slices/files) MUST record its verdict PER SLICE/FILE, never one undifferentiated batch verdict, so one slice's unresolved `test-instrumentation` loop cannot withhold build-eligibility from the other slices. Clause (D) applies prospectively; recorded verdicts are never rewritten or reclassified (§ . . . / § . . .). Honest boundary (§ . . .): this clause does not make classification easy or infallible – it makes it a required, recorded, reviewer-owned decision instead of an implicit one.

11 4 125

Honest boundary (§ . . .): relocates the TEST-INSTRUMENTATION re-review + gate/check hardening OFF the build critical path; does NOT weaken any substantive gate. The SOURCE-correctness review (§ . . . /

§ 11.4.135 . . . /§ . . . Opus- xhigh
 § 11.4.129 . . . 11 . . . 41 → GO § . . .) STILL PRECEDES +
 gates the build (it IS "source proven-correct"); pre/post-
 build tests (§ . . . L -) run at their seams;
 § . . . manual-QA-final un-weakened;
 § . . . /§ 11.4.136 1.2 un-weakened. "Test-side
 work is parallel, never a build gate" is bounded to
 future-regression instrumentation – a check that proves
 the SHIPPED artifact wrong is a source/artifact-
 correctness gate (§ . . . L -), stays on the
 critical path. Clause (D) supplies the mandatory
 classification mechanism this boundary's own test
 presupposes – a review that never classifies its findings
 has not actually applied this honest boundary, it has
 merely stated it.

Classification: universal (§ . . .). Propagation
 gate CM-COVENANT-114-235-PROPAGATION (literal 11.4.235) +
 recommended CM-BUILD-ON-SOURCE-PROVEN-NOT-TEST-SIDE (FAIL:
 build held past source-GO w/ unfinished test-
 instrumentation; FAIL: build started with NO source-review
 GO – the § . . . () dual-assertion guard) + CM-
 VERSION-INCREMENT-ON-DEPLOY (FAIL: new-artifact deploy shares
 a version id; FAIL: work of the cycle opened by a manual-
 QA deploy still targets the QA-deployed version id – the
 clause-(B) cycle-boundary increment skipped) + paired
 § . . . mutations. CM-BUILD-ON-SOURCE-PROVEN-NOT-TEST-SIDE is
 IMPLEMENTED (constitution/scripts/gates/
 cm_build_on_source_proven_not_test_side.sh + its paired
 _mutation_test.sh), as is CM-VERSION-INCREMENT-ON-DEPLOY
 (constitution/scripts/gates/cm_version_increment_on_deploy.sh +
 its paired mutation test) – the prior "gate-code =
 separate work item" disclaimer for those two was stale and
 is corrected here. NEW recommended gate CM-REVIEW-FINDING-
 CLASS-RECORDED for clause (D) (FAIL: a batched-review
 verdict covering more than one slice/file with no per-
 slice verdict or no per-finding finding_layer recorded;

FAIL: a slice marked build-eligible while any
 source-defect finding in it has not reached GO; FAIL: a
 finding_layer lowered by the producer without a fresh
 review round) + paired \$. mutation – gate-code =
 separate work item, NOT claimed shipped (\$. . /
 \$. .), registered as a tracked deferral in
 constitution/scripts/gates/gate_ledger_deferrals.tsv ; it binds
 ADDITIVELY by computing whether the existing
 source_review_go marker may be written, never by modifying
 cm_build_on_source_proven_not_test_side.sh 's own tested logic.

Non-compliance is a release blocker. No escape hatch – no
 --build-behind-test-hardening , --test-side-gates-the-build , --
 skip-source-correctness-review , --deploy-without-version-
 increment , --not-default-speed-mode , --serialise-build-and-
 validate . \$. . – QA-deploy-readiness gate: no
 manual-QA hand-off until the mandated validation produced
 a candidate-fingerprinted PASS verdict; a blocker MUST
 bind to a seam, never prose (research-derived,
 – –). Compact summary: a release artifact
 MUST NOT be deployed / handed off for manual QA
 (\$. .) until the mandated autonomous
 validation(s) for THAT artifact produced a machine-written
 PASS verdict whose artifact fingerprint == the candidate
 (\$. . (F), read from the target at run time);
 ABSENCE of the verdict BLOCKS the deploy exactly as a FAIL
 does (\$. .); the refusal lives in the deploy
 TOOL (the seam), never in prose (\$. . /
 \$. .) – a source-gate asserting the harness
 merely EXISTS is necessary-not-sufficient; any operator-
 declared blocker (a \$. . CRITICAL / "release-
 blocker") MUST bind to a mechanical seam at the point it
 gates, never satisfied by prose nor DOWNGRADED to a
 follow-up, and when its instrument bugs the response is
 \$. . /\$. . (fix the instrument, never
 route around it); override only via an explicit, recorded
 operator flag (\$. . /\$. .), never a

silent bypass; the gate is self-validated golden-good/
golden-bad (\$. . ()/\$. . , cannot
false-allow) and the verdict-emitter reads the fingerprint
from the target at run time, never a hardcoded per-release
value (\$). Honest boundary \$. . –
the seam proves a candidate-matching PASS verdict EXISTS,
it does NOT replace the \$ manual-QA
sufficiency gate nor make the underlying validation honest
(that stays \$. . /. /.). Classification:
universal (\$). Composes

\$ (. . .)/. /. (F)/. /.
Propagation gate CM-COVENANT-114-236-PROPAGATION (literal
11.4.236) + recommended gate CM-QA-DEPLOY-READINESS-GATE +
paired \$. . mutation (gate-code = separate work item,
NOT claimed shipped \$ /\$).

Canonical authority: constitution submodule Constitution.md
\$ Non-compliance is a release blocker. No
escape hatch – no --deploy-without-verdict , --prose-blocker-
suffices , --downgrade-blocker-to-followup , --skip-readiness-
gate , --verdict-fingerprint-optional , --silent-override flag.

\$ – Mandatory exhaustive context-and-
spirit-aware translation review: every localized artifact
independently reviewed for CONTEXT (source) + SPIRIT
(target-language idiom/register) + technical/non-
translatable fidelity – accuracy necessary, never
sufficient (operator mandate, –).

Forensic anchor (operator mandate, –): a
translated artifact that is merely ACCURATE – every source
token faithfully rendered – can still read as stilted,
foreign, machine-literal, register-wrong, or idiom-broken
to a native speaker of the target language, and a
localization pass that silently mangles a technical term
or "translates" a non-translatable product/proper name
ships a defect worse than untranslated text; the recurring
failure mode is a translation that scores green on an

accuracy check while a native reader immediately perceives it as non-native. Accuracy is NECESSARY; it is NOT SUFFICIENT for a slick, polished, professional localization.

The mandate (ALL hold):

(A) EVERY LOCALIZED ARTIFACT IS INDEPENDENTLY REVIEWED – NEVER SELF-REVIEWED. Every translation and every translated artifact – UI strings, pages, documents, PDFs, manuals, marketing copy, captions/subtitles, help text, any localized content – MUST undergo a HEAVY, EXHAUSTIVE, independent review by a reviewer (agent or human) STRUCTURALLY SEPARATE from the translator that produced it (§ . . . /§ . . . independence). The translator's own pass (§ . . . multi-pass self-review) PRECEDES this review and NEVER satisfies it (§ . . . 's no-self-review-substitution discipline applied to localization; § . . . precedes, never satisfies). A localized artifact accepted without an independent context-and-spirit review is a § . . . PASS-bluff at the localization layer.

(B) THE REVIEWER HOLDS AND APPLIES THREE DIMENSIONS. (a) FULL CONTEXT OF THE SOURCE MATERIAL – the original ideas, the subject/domain, the purpose, and the intended audience; the reviewer reads the source MEANING, not merely the source STRING, so intent, nuance, and tone are carried across, never lost. (b) THE SPIRIT OF THE TARGET LANGUAGE – its tradition, idiom, register, tone, and cultural conventions – so the result reads as NATIVE, natural, fluent, and MAXIMALLY POLISHED, never literal / word-for-word / calqued; a construction correct in the source that reads foreign in the target is REWRITTEN into the natural target-language form. (c) CORRECT TECHNICAL VOCABULARY AND VERBATIM PRESERVATION OF NON-TRANSLATABLE TERMS – domain/technical terms rendered with their established target-language equivalents, and NON-

TRANSLATABLE tokens (product names, brand names, proper nouns, technology/library/API names, code identifiers, trademarks) preserved VERBATIM – never "translated", never accidentally transliterated, never localized away.

(C) THE GOAL IS MAXIMALLY POLISHED LOCALIZATION – ACCURACY NECESSARY, NEVER SUFFICIENT. The bar is a slick, polished, professional, native-reading localization; fidelity to the source is the FLOOR, not the ceiling. A rendering that is accurate but stilted / foreign / register-wrong / idiom-broken is a FINDING, not a pass.

(D) ANTI-BLUFF – CAPTURED MACHINE-READABLE POSITIVE EVIDENCE + FAIL-LOOPS-TO-CLEAN. The review emits captured, machine-readable POSITIVE evidence per § 11.4.139 / § 11.4.140 (`feature_class = translation_review`): a per-artifact verdict carrying a PER-DIMENSION rationale across the closed dimension set {fidelity, fluency/naturalness, spirit/register, terminology, completeness, non-translatable-preservation} – a PASS with no captured per-dimension rationale is a § 11.4.140 / § 11.4.141 bluff. A FAIL on ANY dimension LOOPS BACK to re-translation/polish and re-review, iterating to a zero-finding clean verdict per § 11.4.142 – a single pass that "looked fine" never closes the artifact.

Honest boundary (§ 11.4.143). This anchor REFINES the existing localization discipline from PASS/FAIL accuracy checking to CONTEXT-AND-SPIRIT native-polish review; it does NOT replace § 11.4.144 's canonical translation pipeline (the HelixTranslate-canonical-translation-pipeline sense of that anchor number) nor § 11.4.145 's independent per-language review (the independent-per-language-translation-review sense of that anchor number – a KNOWN operator-owned § 11.4.146 anchor-number collision on 11.4.140 / 11.4.141 , re-mint still operator-owned; cited here strictly in those senses, introducing no new collision), it STRENGTHENS them

(accuracy → native polish). It does not make a mistranslation impossible (a miscalibrated reviewer stays § . . . () golden-good/golden-bad fixture territory), and where a target language genuinely has no established term for a source concept, that gap is an honestly recorded finding (§ . . . SKIP-with-reason), never a silently invented word.

Classification: universal (§ . . .) – no vendor/project/language literal; the consuming project supplies its language set, term glossary, non-translatable token list, and reviewer substrate as DATA per § Composes § 92 (canonical translation pipeline) / § (independent per-language translation review – this REFINES it from PASS/FAIL accuracy to context+spirit polish) / § / § / § / § Propagation gate CM-COVENANT-114-237-PROPAGATION (literal 11.4.237) + recommended gate CM-TRANSLATION-CONTEXT-SPIRIT-REVIEW (every localized artifact carries an independent per-dimension context-and-spirit review verdict with captured rationale across {fidelity, fluency/naturalness, spirit/register, terminology, completeness, non-translatable-preservation}, FAIL looping to re-review per §) + paired § mutation (a translation accepted on an accuracy-only check, OR self-reviewed by its own translator, OR carrying a golden-bad-seeded mistranslated idiom / a "translated" non-translatable product name → the gate FAILs; strip the literal → the propagation gate FAILs; gate-code = separate work item, NOT claimed shipped § / §).

Canonical authority: constitution submodule Constitution.md §

Non-compliance is a release blocker regardless of context. No escape hatch – no --skip-spirit-review ,

11 4 218
--accuracy-only-OK , --self-review-translation-OK , --literal-translation-OK , --translate-non-translatable-OK , --skip-per-dimension-rationale flag exists. 1020 00 00

§ . . – Automated QA must be the **DISCOVERER**: manual QA finds nothing new, and anything it finds is a coverage escape (User mandate, – –).

Verbatim operator mandate: *"Helix QA full websites testing MUST find all such issues, not us! Once we do manual QA we shall not be able to find anything! THIS IS MANDATORY RULE / CONSTRAIN / GOAL to achieve with this and any other project and MUST BE part of the root constitution!"*

Forensic case study (FACT, – –): in one cycle...five real defects were found – a health endpoint returning {"status":"healthy","components":[]} that was structurally incapable of ever reporting unhealthy; a gateway routing every completion at a port nothing served (HTTP on the product's primary capability); two submodule pins on main that upstream answers with not our ref , so no fresh clone could check them out; and two hardware-profiler defects reading a key from a file where it does not exist. **Every one was found by an agent reading source or probing by hand; not one by the automated QA system**, which reported green throughout. The mandate (ALL hold): (A) **the automated QA system is the DISCOVERER, not the confirmer** – discovery by a human, an operator's manual pass, an agent reading code, or an end-user report is a FAILURE of the QA system; the finding is still fixed, but the QA gap it exposes is a defect of equal standing and closing only the reported defect is the violation. (B) **manual QA must find NOTHING NEW** – by the time manual QA runs (§ . .) the automated regime must already have surfaced everything, so manual QA is a CONFIRMATION and sufficiency gate, never the discovery mechanism; the target is explicit and measurable – a manual pass discovering zero previously-unknown defects. (C) **every escape triggers the § . . loop, widened to**

EVERY out-of-band channel: systematic-debug the defect (§ 11.4.102), produce a **coverage-escape audit** naming WHY the regime missed, it cited to the specific missing check / unexercised path / absent assertion (never "we didn't think of it"), register a new or strengthened automated check that WOULD have caught it into the standing regime (§ 11.4.102) with a § 11.4.102 RED capturing the escaped defect, and commit the audit as captured evidence. (D) **coverage is measured by what the regime can CATCH, not what it runs** – every surface falsifiable per § 11.4.102 (a check whose paired mutation does not make it fail is decoration), exercising real user-reachable paths end-to-end (§ 11.4.102 / § 11.4.102 / § 11.4.102 / § 11.4.102 / § 11.4.102). (E) **directional and measured** – track the discovery-channel split (automated vs out-of-band) and drive out-of-band toward zero; honest boundary (§ 11.4.102): this does NOT claim a regime can be provably complete (§ 11.4.102) and does NOT license closing an escape by narrowing the definition of "defect". **EXTENSION – Discovery-channel record schema + escape-ratchet no-data-point rule (spec-derived, speckit -anti-slop-enforcement, - -)**.

§ 11.4.102 (C) states the CONCEPT (a coverage-escape audit citing the specific missing check) but fixes no record schema, and (E)'s "toward zero over successive cycles" names neither an input condition nor an observable pass/fail outcome – the ratchet is the decidable form. (11.4.102). **Discovery-channel record schema:** every recorded defect MUST carry a discovery channel from the CLOSED SET {automated_seam, manual_qa, operator, end_user, agent_inspection} (or the consuming project's declared equivalent per § 11.4.102) AND a should_have_been_caught_by value naming a specific seam or gate, OR the literal none with a WRITTEN justification; a record missing either field is REFUSED. The record MUST be WRITTEN BY THE VERIFIER SEAM,

never by the producer of the defect's fix (the
 § . . . producer≠verifier separation applied to
 the record OF the escape, not only the record of the fix).
 () **Monotone-decreasing escape count:** a cycle whose
 escape count EXCEEDS the recorded baseline is REFUSED at
 the release seam; a cycle whose count is lower moves the
 baseline DOWN. () **A manual-QA-not-run cycle contributes
 NO DATA POINT and MUST NOT lower the baseline** – the
 § . . . () false-null applied to this ratchet: an
 unmeasured cycle and a genuinely-clean cycle both read
 zero, and only the `manual_qa_ran` flag distinguishes them.
 () **should_have_been_caught_by: none** records are counted and
 reported SEPARATELY, so narrowing the definition of an
 escape is visible rather than silent. Honest boundary
 (§ . . .): at the correct end-state – every escape
 caught by a seam – the count reaches , so the ratchet
 is a valid NECESSARY condition; it is NOT sufficient,
 since zero also results from nobody looking, which the no-
 data-point rule and the separate `none` tally exist to
 expose. Composes § . . . ()-() /
 § . . . () / § Recommended
 mechanism gates `CM-DISCOVERY-CHANNEL-RECORD-COMPLETE` + `CM-
 ESCAPE-RATCHET-NO-DATA-POINT-HONEST` + paired § .
 mutations (a record with no channel, or a `none` with no
 justification, or a producer-authored record → the schema
 gate MUST FAIL; a `manual_qa_ran` false cycle lowering the
 baseline → the ratchet gate MUST FAIL; golden-FALSE per
 § 11 4 138 . . . (): a genuine manual-QA cycle with a real
 zero-escape result MUST be allowed to lower the baseline);
 gate=code = separate work item, NOT claimed shipped
 (§ 4 91 . 11 . 134 / § 11 4 140 . . .). Composes § . . . /
 § 11 4 138 . . . 10 4 159 11 4 (trigger set widened) / § . . .
 (manual QA becomes confirmation, never discovery) /
 § . . . / § . . . / § . . . /
 § . . . / § . . . / § . . . /
 § . . . / § . . . / § . . . /

11 4 17

11 4 35

\$. . / \$. . / \$. . Classification:
universal (\$. .) – the consuming project supplies
its QA system, surface inventory, and discovery-channel
ledger per \$. . . Propagation gate. CM-
COVENANT-114-238-PROPAGATION (literal 11.4.238) +
recommended gates CM-QA-IS-THE-DISCOVERER / CM-MANUAL-QA-
FINDS-NOTHING-NEW + paired \$. . mutation (gate-code =
11 4 238
separate work item, NOT claimed shipped \$. . /
\$. .).

Canonical authority: constitution submodule Constitution.md
\$. . . Non-compliance is a release blocker. No
escape hatch – no --manual-qa-may-discover , --skip-coverage-
11 4 238
escape-audit , --found-by-human-is-fine ,
--close-escape-without-new-check , --qa-green-is-enough , --
discovery-channel-not-tracked flag.

\$ 00 15 . . – Critical-invariant work-class

**Definition of Done mandate – money/safety/availability/
integrity code MUST include failure-path scenarios as
gates, not merely happy-path passes (research-derived,
– –). Compact summary: every project's DoD
MUST identify a closed CRITICAL-INVARIANT WORK CLASS
(money movement, safety-of-life/property outputs,
availability of the primary user-facing capability,
integrity of the persistent record) and MUST require that
every change touching this class carry EXPLICIT FAILURE-
PATH SCENARIOS as passing tests BEFORE closure – a happy-
path-only DoD on money/safety/availability/integrity code
is the exact "green means done" bluff \$. . /\$. .
forbids (extraction line [un-resolvable offset per
the \$. . CITATION NOTE below; verified locator
module_27:30]: "a Definition of Done that stops at 'the
code compiles and the happy-path test is green' is a
machine for generating reopened tickets on the paths that
move money"). (A) The four critical-invariant categories
(money/safety/availability/integrity) with consumer-owned**

domain extensions per § . . ; (B) domain-appropriate MINIMUM failure-path scenario sets rendered as Given/When/Then acceptance criteria with "Then" observing real₂₃₉ state deltas (§ . .), never "no error" – the canonical MONEY set: fresh / idempotent retry / same-key-different-payload₂₄₀ / crash-in-the-middle (extraction lines – [un-resolvable offsets per the § . . CITATION NOTE below; verified locator module₂₇:312]); (C) mechanical precondition binds through the existing § 11 4 10. . (D) status-custody chain (the₂₄₁ status-write seam refuses a closure without the scenario map cited, each scenario mapped to a § . . (F) machine-written verdict pair); (D) composes with § . . four-layer / § . . huge-blocker₂₄₂ / § . . manual-QA-sufficiency / § . . automated-QA-discovers – automation NECESSARY, manual QA still the terminal human oracle; (E) honest boundary – directional (forbids happy-path₂₄₃-only), not exhaustive (§ . . discovery-pressure stays in force). **CITATION NOTE** (– – citation audit, § . .): the "extraction file" offsets cited throughout § . . -§ . . are absolute offsets into a consolidated extraction artifact that is NOT tracked in this repository and whose numbering does NOT resolve against the tracked corpus at docs/research/ai_curriculum_2026_08/source_corpus/ – the highest cited offset () exceeds the -line total of ALL NINE module files concatenated. Each extraction offset is therefore retained ONLY as an un-resolvable provenance pointer, NEVER as a resolvable location, and is accompanied by a VERIFIED module_{NN}:LINE locator measured against the tracked corpus with a control-needle-proven instrument per § . . ()(b). The corpus landed – – , AFTER these anchors landed, so it corroborates them; it is not their derivation source. Full

note: constitution/Constitution.md § . . .
 Classification: universal (§ . . .). Composes
 § . . . /. /. /. 239 /. /. /. (F)/.
 Propagation gate CM-COVENANT-114-239-PROPAGATION +
 recommended gate CM-CRITICAL-INVARIANT-FAILURE-PATH-DOD +
 paired § . mutation. **Canonical authority:** constitution
 submodule Constitution.md § . . . Non-compliance
 is a release blocker. No escape hatch – no --happy-path-
 suffices-for-money , --skip-failure-scenarios , --defer-failure-
 path-tests , --criticality-not-applicable-without-reason , --
 closure-without-scenario-map flag.

2024 04 15

§ . . – **Producer ≠ Verifier scope separation:**
 the actor that produces a change/artifact/verdict cannot
 be the one that gates it – least privilege over cleverness
 (research-derived, – –). Compact summary:
 every project MUST enforce STRUCTURAL SEPARATION between
 the actor that PRODUCES a change/artifact/verdict and the
 actor that GATES that same work – the producer cannot
 write, weaken, delete, or edit the acceptance tests, the
 oracle, the gate configuration, the verdict store, the
 paired § . mutation, or any other check gating its own
 work; enforcement is by CAPABILITY (structurally cannot
 reach), never by INSTRUCTION (told not to reach),
 following the "least privilege over cleverness" principle
 (extraction line [un-resolvable offset per the
 § . . CITATION NOTE; MODULE CORRECTED
 – – – the quoted passage resolves to
 module , not module ; verified locator
 module_28:393]: "the durable fix is least privilege over
 cleverness ... the load-bearing move is that the agent
 chasing the green gate cannot reach the gate, the oracle,
 or the evidence store"). (A) Four scopes MUST resolve to
 distinct actors – PRODUCER / ORACLE / GATE / VERIFIER; a
 two-scope-or-more collapse involving the producer is a
 § . . finding; (B) enforcement channels are
 filesystem scope / branch protection (§ . .

PROTECTED-SPEC + CODEOWNERS) / verdict-store isolation / clean-checkout execution isolation – NEVER prompts, policies, or social contracts (extraction line [un-resolvable offset per the § . . . CITATION NOTE; module attribution VERIFIED CORRECT – – , verified locator `module_34:87`] catalogues "we told it in the prompt not to hardcode" as the shallow / red-flag answer); (C) three forbidden collapses named: PRODUCER=ORACLE (tautological testing), PRODUCER=GATE (weakened thresholds), PRODUCER=VERIFIER (self-certification); (D) composes with the existing verifier layer – § . . . (F) machine-written verdicts, § . . . /. /. independent review, § . . . (D) status custody, § . . . release verdict coverage, § . . . independent verifier agent, § . . . oracle independence; (E) honest boundary – separation forbids the CLASS of self-certification, does NOT prove verifier correctness (§ . . . () still binds). **(F) Independence-tier vocabulary + Tier-C capability-separation triggers, degrading honestly on a single-uid host (extension, spec-derived, speckit – anti-slop-enforcement, – –)**. Clauses (A)-(E) state the SEPARATION PRINCIPLE in the abstract; this clause supplies the CLOSED VOCABULARY for recording HOW MUCH separation was ACHIEVED, plus the two triggers that raise the required strength – because claiming "capability separation" where only "instance separation" is achievable is itself the theatre this anchor exists to prevent. () **Closed tier set:** every accepted verdict's recorded basis of independence MUST name its tier from the CLOSED SET {instance, model, capability} – instance (a separate agent/process instance sharing no in-flight state with the producer), model (a DIFFERENT model), capability (a genuine host-level access-control boundary separating the producer's write scope from the verifier's). () **Model**

diversity is PERMITTED, NEVER REQUIRED – the false-positive guard: a verdict from an independent INSTANCE running the SAME model as the producer MUST be ACCEPTED on that basis alone; a check that refuses same-model independent verdicts is the exact class of false-positive refusal § 1.1.1 forbids. This does NOT weaken the constitution's OTHER, narrower model-substrate pins – § 1.1.2 (translation review runs on a DIFFERENT model/provider than the translator) and § 1.1.3 (every mandatory independent code-review runs on Opus at xhigh as PRIMARY substrate, with a genuine-unavailability Sonnet fallback, amended – – ; Fable and Haiku remain forbidden in every role) both remain in force unmodified for their own domains. () **Tier-C trigger – producer holds write access to the evidence/verdict store:** the verdict MUST be written through the MOST RESTRICTIVE separation genuinely available on the host, and the ACHIEVED separation (not the aspired-to one) MUST be reported. () **Tier-C trigger – the three named high-blast-radius seams:** at release-tag , qa-deploy or manual-qa-handoff (or the consumer's declared equivalents per § 1.1.4) a producer-authored verdict is REFUSED regardless of the write-access analysis. () **No demand for capability outside the two triggers:** at a seam matching NEITHER trigger a separate agent INSTANCE MUST SUFFICE and capability separation MUST NOT be demanded – over-applying Tier-C strength to every seam is itself the same false-positive-refusal violation. **Honest boundary** (§ 1.1.5): on a SINGLE-UID host capability separation CANNOT be established by any mechanism an agent can set up – ownership defeats access-control lists, the append-only attribute requires privilege the agent lacks, and any key the producer can use it can also read – so such a host reports instance HONESTLY and never capability , until the operator takes the one host-boundary action agents are correctly forbidden from creating (§ 1.1.6 /§ 1.1.7);

the field records WHAT WAS ACHIEVED, never what was ASPIRED to, and honest degradation IS the compliant path. Recommended mechanism gate CM-INDEPENDENCE-TIER-HONESTLY-RECORDED + paired \$. mutation (record capability where no verified host boundary exists → MUST FAIL; refuse a same-model independent verdict at an ordinary seam → MUST FAIL; accept a producer-authored verdict at release-tag / qa-deploy / manual-qa-handoff → MUST FAIL; golden-FALSE per \$. . (): an honest instance -tier verdict at an ordinary seam MUST NOT fire it); gate-code = separate work item, NOT claimed shipped (\$. . / \$. .). Classification: universal (\$. .). Composes \$. . / . / . / . () / . (F) / . / . (mutual composes with \$. . four-role + flight-recorder architecture per \$. . reviewer MINOR- - \$. . the separation principle, \$. . the four-role taxonomy that instantiates it). Propagation gate CM-COVENANT-114-240-PROPAGATION + recommended gates CM-PRODUCER-NOT-VERIFIER-SCOPE-SPLIT + CM-LEAST-PRIVILEGE-CAPABILITY-NOT-INSTRUCTION + paired \$. mutations. **Canonical authority:** constitution submodule Constitution.md \$. . . Non-compliance is a release blocker. No escape hatch – no --producer-may-write-tests , --self-certification-OK , --exhortation-suffices-as-enforcement , --producer-authored-verdict-accepted , --capability-limit-optional , --policy-document-is-enforcement , --single-actor-for-all-scopes flag.

\$. . – Illegal-state-unrepresentability preference (types > API-shape > lint > property-test > runtime-assert) (research-derived, - -). Compact summary: every load-bearing invariant a change relies on MUST land at the STRONGEST enforcement rung the target language/toolchain supports, walking a closed ordered ladder – () TYPE SYSTEM (invalid state cannot be constructed: sum types / newtype wrappers / NonEmpty /

opaque handles), () API SHAPE (only exposed constructor path enforces it: private constructors + smart factories, sealed hierarchies, capability tokens), () LINT (invalid pattern refused by tree before build), () PROPERTY-BASED TEST (invariant asserted across generator-driven input space), () RUNTIME ASSERTION (last-line guard raises when invalid state actually forms). Defect PREVENTABLE at rung N caught only at rung > N is a finding. Operator honest boundary - - - rungs - conditional on toolchain support (mypy present, TypeScript strict, Pydantic used), rungs - unconditional (every language has a linter + runtime). \$. mutation: drop the type constraint / relax `PositiveInt` to `int` / make private constructor public → compiler/type-checker/API gate...FAILs to build; if it still builds, the rung was decoration. Composes \$. . / \$. . . (pushes enforcement LEFT of Layer) / \$. . / \$. . / \$. . / \$. . . Classification: universal (\$. .). Propagation gate `CM-COVENANT-114-241-PROPAGATION` + recommended gate `CM-ILLEGAL-STATE-LADDER-PREFERENCE` + paired \$. mutation. **Canonical authority:** constitution submodule `Constitution.md` \$. . . Non-compliance is a release blocker. No escape hatch – no `--runtime-assert-suffices`, `--skip-type-preference`, `--dynamic-language-exempts-all-rungs`, `--property-test-in-lieu-of-type`, `--decorative-type-annotation-OK` flag.

\$. . – **Bisection-not-blame regression cause-finding + failure modes** (research-derived, - -). Compact summary: regression localisation FIRST-choice mechanism is `BISECTION` (`git bisect`, $O(\log N)$) over the range from the last known-good tag (\$. .) to current HEAD, NOT blame-log walking ($O(N)$ linear) NOR speculative root-cause hunting (unbounded). Bisection produces the introducing commit as CAPTURED EVIDENCE (\$. .) feeding \$. .

systematic-debugging's root-cause phase with the actual change in hand. MONOTONE-SEARCH INVARIANT: bisection valid only when observable is monotone across the range (broken stays broken); non-monotone (flaky, environment-dependent, timing-race) MUST stabilise first – bisection over flaky signal produces WRONG "guilty" commit with same confidence as correct one. FAILURE-MODE CATALOGUE (closed): () flaky/non-monotone observable, () build-broken commits (mistaken for "bad", use `bisect skip`), () merge-commit introductions (defect in conflict resolution not either parent), () external/environmental changes (upstream dep, OS update), () delayed-fuse introductions (latent commit + later trigger), () test-infra regressions masquerading as product regressions. `git bisect run script` MUST use `exit-code discipline` (`=good`, `-` `=bad`, `=skip`); mapping every non-zero to "bad" collapses the taxonomy. Composes `$ . . . / $ . . . / $ . . . / $ . . . / $ . . . / $ . . . / $ . . . / $ . . .`. Classification: universal (`$ . . .`). Propagation gate `CM-COVENANT-114-242-PROPAGATION` + recommended gate `CM-BISECTION-BEFORE-BLAME` + paired `$ . . .` mutation. **Canonical authority:** constitution submodule `Constitution.md`

`$ . . .`. Non-compliance is a release blocker. No escape hatch – no `--blame-log-suffices`, `--skip-bisection`, `--flaky-observable-bisect-anyway`, `--run-script-exit-code-collapse-OK`, `--regression-guess-without-localisation` flag.

`$ . . .` – **Characterization / golden-master safety net before behaviour-changing modification of legacy (research-derived, `-` `-`)**. Compact summary: before ANY behaviour-changing modification to code that qualifies as legacy (lacks tests asserting observable behaviour, or written spec diverges from actual, or predates the constitution's TDD discipline, or the modifier cannot describe its behaviour from reading source), the modifier MUST FIRST author CHARACTERIZATION

TESTS – tests asserting the CURRENT observable behaviour as GROUND TRUTH whether "correct" or not – and land them as a green baseline. Only AFTER the safety net exists may the modification proceed; characterization tests then act as the oracle catching unintended behaviour changes. Pattern originates in Feathers's WORKING EFFECTIVELY WITH LEGACY CODE + generalises GOLDEN-MASTER (record outputs across representative input space; assert future runs match). Characterization is DESCRIPTIVE not prescriptive – it captures WHAT IS not WHAT SHOULD BE; if current behaviour is subtly wrong, ASK operator per § 11.4.244 whether modification should preserve or correct it (never silently). Composes § 11.4.244 / § 11.4.244 / § 11.4.244 / § 11.4.244 (RED-first – characterization is legacy counterpart) / § 11.4.244 (sampled input space = honest gaps, not completeness) / § 11.4.244 (investigate-before-remove – parallel discipline) / § 11.4.244 / § 11.4.244 (test-first backfilled for legacy). Classification: universal (§ 11.4.244). Propagation gate CM-COVENANT-114-243-PROPAGATION + recommended gate CM-CHARACTERIZATION-BEFORE-LEGACY-MODIFICATION + paired § 11.4.244 mutation. **Canonical authority:** constitution submodule Constitution.md § 11.4.244. Non-compliance is a release blocker. No escape hatch – no --modify-untested-legacy-directly, --skip-characterization, --claim-tests-not-needed-because-obvious, --decorative-assertion-suffices, --refactor-without-safety-net flag.

§ 11.4.244 – **Cross-boundary contract tests + can-i-deploy gate** (research-derived, – –).

Compact summary: changes modifying a boundary between independently-deployable components (service consumer, library dependents, DB schema readers/writers, message topic subscribers) MUST be covered by CONSUMER-DRIVEN CONTRACT TESTS on both sides AND a can-i-deploy gate MUST verify proposed provider version compatibility against

every currently-deployed consumer version (and vice versa) BEFORE the change ships. CDC pattern: consumers publish assumptions they make about the provider (Pact / Spring Cloud Contract / gRPC schema-evolution); providers verify each consumer's contract against actual implementation; provider change breaking a live consumer's contract fails provider's build. CAN-I-DEPLOY: at deploy attempt, gate queries broker for compatibility matrix ($P \times C \times C \times C$); single red cell BLOCKS deploy naming incompatible pair, NEVER warning-with-continue; MISSING contract = first-class refusal reason (\$. . , not implicit "assume compatible"). SemVer becomes MEASURABLE: PATCH bump breaking contract = mis-labelled version. Backward-compatibility window declared per \$. . (support current + previous N minor versions, deprecate with M weeks notice); consumers outside window flagged for \$. . . operator decision. \$. . MERGES what the task brief labelled contract-tests + can-i-deploy – they are useless in isolation. Composes \$. . / \$. . . / \$. . . / \$. . . (per-side four-layer) / \$. . (contract tests as permanent regression guards) / \$. . (composed-system manual QA still needed) / \$. . (gate honesty). Classification: universal (\$. . .). Propagation gate CM-COVENANT-114-244-PROPAGATION + recommended gates CM-CONTRACT-TESTS-BOTH-SIDES + CM-CAN-I-DEPLOY-GATE-WIRED + paired \$. mutations. **Canonical authority:** constitution submodule Constitution.md \$. . . Non-compliance is a release blocker. No escape hatch – no --skip-contract-tests , --deploy-without-compatibility-check , --assume-compatible-when-broker-silent , --patch-may-break-contract , --contract-tests-optional-for-internal-boundaries flag.

\$. . – Oracle-problem-first test authoring: identify the oracle BEFORE authoring the test (research-derived, – –). Compact summary: every test

MUST FIRST identify its ORACLE – the mechanism deciding PASS vs FAIL for the input it drives – and the oracle MUST be DIFFERENT from the code under test AND from the test's own assertion mechanism. Authoring without explicit oracle identification produces the "test agrees with code" anti-pattern (test asserts what code happens to produce, so it passes as long as code produces something, never catches wrong-output). ORACLE STRATEGY CLOSED SET, each test's oracle NAMED in source (comment/docstring/tag): () SPECIFIED (spec/RFC/standard names expected output), () DERIVED (independent reference implementation), () METAMORPHIC (metamorphic relations between related inputs), () GOLDEN-MASTER / CHARACTERIZATION (past captured behavior – § . . .), () INVARIANT (universal property – sort output sorted + permutation of input; serialize-deserialize round-trip = original), () STATISTICAL (distribution properties for stochastic outputs, never specific value), () HUMAN (§ . . . manual QA). NO other classes. STRUCTURAL INDEPENDENCE: oracle's SOURCE of expected value must not depend on code being tested – different implementation / spec / previous run / mathematical property / inverse operation / human. Reviewer per § . . . verifies (i) NAMED strategy, (ii) structural independence, (iii) paired § . . . mutation flips the oracle. Composes § . . . / § . . . / § . . . () (self-validated analyzers – analyzer-as-oracle) / § . . . / § . . . / § . . . / § . . . / § Classification: universal (§ . . .). Propagation gate CM-COVENANT-114-245-PROPAGATION + recommended gate CM-ORACLE-STRATEGY-NAMED-AND-INDEPENDENT + paired § . . . mutation.

Canonical authority: constitution submodule Constitution.md § Non-compliance is a release blocker. No escape hatch – no --test-agrees-with-code , --skip-oracle-

identification , --oracle-is-the-code , --assertion-is-oracle ,
 --implicit-oracle-OK flag.

\$. . – Reproducible + hermetic builds +
 supply-chain integrity at SLSA Build Level as fleet-
 wide minimum (operator decision, – –).
 Compact summary: every consuming project's build pipeline
 MUST produce REPRODUCIBLE + HERMETIC artifacts and meet
 SLSA Build Level as fleet-wide MINIMUM (operator
 decision – – – L + tracked as backlog
 per \$. .). REPRODUCIBLE = same source at same
 time on any conformant host produces byte-identical
 artifacts (or diff-set fully accounted for by declared
 non-determinisms pinned to canonical values). HERMETIC =
 no undeclared inputs (every dep version-pinned + lockfile-
 hashed, every tool version declared, no unpinned public-
 network reach at build, no host-env leakage). SLSA-L =
 signed provenance describing source URI + commit hash +
 builder identity + invocation + artifact hash, on hosted
 tamper-resistant build platform (boba already satisfies
 ancillary: SSH-only \$. , rootless podman
 \$. . , fixed-tag pins \$. . /
 \$. .). Every consumer MAINTAINS docs/security/
 SLSA_LEVEL.md (or equivalent per \$. .) recording
 CURRENT L level with cited evidence + TARGET L + tracked
 \$. . upgrade items – CURRENT below L
 without tracked item is violation. SUPPLY-CHAIN EXTENSION:
 deps either vendored (checked in hash-verified), from
 provenance-attested source (npm/PyPI/crates.io as they add
 SLSA), OR mirrored through hash-pinning local registry –
 an unattested public source is integrity risk. Threat
 model NARROW: covers source→artifact chain, does NOT cover
 malicious commit landing in source (\$. .
 review) nor compromised dep source (supply-chain
 extension) nor bug-free artifact (\$. . four-
 layer + \$. . manual QA). Composes \$. /
 \$. . / \$. . / \$. . /

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$ . . / $ . . / $ . . /
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```

Classification: universal (\$. .). Propagation gate CM-COVENANT-114-246-PROPAGATION + recommended gates CM-BUILD-REPRODUCIBILITY + CM-BUILD-HERMETICITY + CM-SLSA-L2-PROVENANCE-VERIFIED + CM-SUPPLY-CHAIN-DEPS-PROVENANCE-ATTESTED + paired \$. mutations. **Canonical authority:**

constitution submodule Constitution.md \$. . . Non-compliance is a release blocker. No escape hatch – no `--skip-reproducibility`, `--allow-network-during-build`, `--no-provenance`, `--slsa-optional`, `--below-l2-OK`, `--unpinned-dependency-OK`, `--host-env-may-leak-into-build` flag.

\$. . – **Composite-output layer-move must move every layer (research-derived, – –)**.

Compact summary: whenever a change MOVES a system-boundary layer (service migrates hosts, endpoint URL changes, API version bumps, DB schema shifts, message topic renames, ingress path re-routes, storage bucket moves), the change MUST move EVERY LAYER of the composite output atomically or via explicit compatibility window – NEVER piecemeal.

Recurring failure: endpoint moves `/v1/foo` → `/v2/foo`; server handler + client call-site moved; load balancer + CDN + API gateway + OpenAPI spec + SDK stubs + integration tests + metrics + traces + logs + dashboards + alerts + docs still reference `/v1/foo`; operator declares victory; residuals silently break when `/v1/foo` shim retires. LAYER TAXONOMY per \$. . baseline: (i) server (handler + route + OpenAPI/proto/gRPC descriptor), (ii) client (call-sites + SDK stubs + integration tests), (iii) infrastructure (LB + ingress + API gateway + CDN + service-mesh + DNS), (iv) data stores (cached keys + DB URL rows + MQ subscriptions), (v) observability (metric labels + span names + log fields + dashboards + alerts + SLOs), (vi) documentation (user + dev + runbook + README + cookbook), (vii) tests (integration + e e + contract per

\$. . + load + smoke). Consumer provides scripts/audit/layer_move_completeness.sh reporting residual references delta; change proposing move MUST show ZERO residuals OR explicit compatibility-window plan per residual with tracked \$. . items.

COMPATIBILITY-WINDOW: old value keeps working alongside new until stated date/version/event, deprecation warnings named per caller, tracked deadline, clean removal at expiry – no "clean up later." Composes \$. . / \$. . / \$. . / \$. . / \$. . / \$. . (residual = coverage escape) / \$. . . Classification: universal (\$. .). Propagation gate CM-COVENANT-114-247-PROPAGATION + recommended gate CM-LAYER-MOVE-COMPLETENESS-CHECK + paired \$. mutation. **Canonical authority:** constitution submodule Constitution.md \$. . . Non-compliance is a release blocker. No escape hatch – no --partial-layer-move-OK , --skip-completeness-check , --residuals-cleaned-up-later , --observability-layer-optional , --doc-move-can-lag flag.

2024 08 15

\$. . – Flaky-test quarantine + protected-regression-spec gate via [PROTECTED-SPEC: ATM-NNN] tag + CODEOWNERS required reviewer (operator decision, – –). Compact summary: flaky tests are the SILENT ANTI-BLUFF DECAY vector (every flake conditions team to ignore red until real regression is dismissed as "probably flaky, re-run"). Two disciplines bound: (A) QUARANTINE – mechanical detection (test-history aggregator flags two-consecutive-run-different-verdict OR rerun-after-red pattern) moves flaky test to tests/quarantine/ (per \$. .), opens tracked \$. . stabilisation deadline, blocking suite runs WITHOUT quarantined test so green means green; deadline past = \$. . extension or violation; (B) PROTECTED-REGRESSION-SPEC – regression tests codifying known-broken

defect fixes (per § . . . permanent-guard suite) MAY be tagged # [PROTECTED-SPEC: ATM-NNN] (target-language comment equivalent) referencing the workable-item ATM-NNN (§ . . .) they guard, marking test as load-bearing regression guard whose modification requires elevated review. Tag applied deliberately per test (not blanket – that voids the discipline). CODEOWNERS-REQUIRED-REVIEWER GATE: consuming project's CODEOWNERS declares modifications to [PROTECTED-SPEC] -tagged tests OR regression/**/* path scope require designated required-reviewer approval BEFORE merge; composes with § . SSH + § . . . no-force-push (gate cannot be silently bypassed); audit trail in PR review thread + commit history. Gate CATCHES change so reviewer distinguishes STRENGTHEN (approve) vs FIX-DEFECT-IN-TEST (approve after verifying ATM-NNN's defect not silently re-introduced) vs WEAKEN/REMOVE (refuse or § . . . escalate) – makes modification VISIBLE not preventable. Composes § . / § . . . / § . . . / § . . . (ideal empty quarantine) / § . . . / § . . . / § . . . / § . . . / § . . . / § . . . / § Classification: universal (§ . . .). Propagation gate CM-COVENANT-114-248-PROPAGATION + recommended gates CM-FLAKY-TEST-QUARANTINE-WIRED + CM-PROTECTED-REGRESSION-SPEC-GATE + paired § . mutations. **Canonical authority:** constitution submodule Constitution.md § Non-compliance is a release blocker. No escape hatch – no --rerun-until-green , --quarantine-permanent , --protected-spec-review-optional , --codeowners-skip , --blanket-protect-everything , --tag-without-atm-id flag.

§ . . . – **Producer ≠ oracle ≠ gate ≠ verifier + flight recorder** (research-derived, – –). Compact summary: every load-bearing quality mechanism has FOUR distinct roles that MUST be structurally separate – () PRODUCER (code computing/emitting/building output),

() ORACLE (mechanism deciding correctness per § 11.4.249), () GATE (seamlessly consuming oracle verdict to allow/block/proceed), () VERIFIER (mechanism auditing gate's decision – paired § 11.4.201 mutation proving gate can refuse a bad output). Collapsing any two produces specific bluff class: producer=oracle (§ 11.4.249 oracle-agrees-with-code); oracle=gate (refuses to say "cannot decide" defaults to allow, § 11.4.249 () FALSE-NULL); gate=verifier (self-auditing gate – § 11.4.201 self-review anti-pattern applied). FLIGHT RECORDER: mechanism CAPTURING producer inputs + oracle verdict + gate decision + verifier audit as durable replayable artifact (structured JSONL event stream per § 11.4.249, content-addressed per § 11.4.249) so post-hoc "why did this go through" has evidence not shrug. Recorder is NOT the gate – a gate reading its own recorder to decide collapses gate + verifier + recorder. DETECTABLE COLLAPSES reviewer per § 11.4.249 scans for: () producer=oracle (test's assertion calls code being tested), () oracle=gate (no "cannot decide" branch, defaults PASS), () gate=verifier (meta-test IS the gate – mutation making gate always-pass ALSO makes meta-test always-pass; meta-test MUST live OUTSIDE gate), () recorder=decision-substrate (gate uses recorder contents as decision input – precedent-based gating not authorised). Composes § 11.4.249 / § 11.4.249 / § 11.4.249 (§ 11.4.249) / § 11.4.249 / § 11.4.249 / § 11.4.249 / § 11.4.249 (producer verifier separation principle – the WHY behind this four-role architecture) / § 11.4.249. Classification: universal (§ 11.4.249). Propagation gate CM-COVENANT-114-249-PROPAGATION + recommended gates CM-ROLE-SEPARATION-DECLARED + CM-FLIGHT-RECORDER-CAPTURED + paired § 11.4.201 mutations.

Canonical authority: constitution submodule Constitution.md § 11.4.249. Non-compliance is a release blocker. No

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escape hatch – no --collapse-producer-oracle , --gate-self-audits , --skip-flight-recorder , --oracle-defaults-pass , --decision-without-recorded-rationale flag.

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\$. . – Heuristic-tower signals a primitive defect: when N heuristic layers stack to compensate, the primitive is broken (research-derived, – –). Compact summary: codebase accumulating TOWER of heuristics compensating for same underlying issue (retry-with-backoff + connection-pool-warmup + DNS-cache-flush + health-check-with-startup-grace – all guarding one flaky DB connection) IS the diagnostic: PRIMITIVE is defective, every layer papers over a manifestation without fixing cause. Recognition signals: (i) each layer justification "we added because sometimes X happens" – reactive, (ii) removing any layer INCREASES failures BUT keeping all shows occasional residual, (iii) no single layer owns responsibility, (iv) layers added at different times by different authors per specific incidents, (v) monotone layer-count growth over time – debugging session for residual adds layer N+ rather than fixing ..N. DISTINCT from defense-in-depth: defense-in-depth is deliberate + each layer targets DIFFERENT failure class + designed together not accreted + removing layer produces SPECIFIC different failure not "same failure more often". SYSTEMATIC-DEBUGGING RESPONSE on tower depth ≥ : () STOP adding layers, () INVOKE \$. . systematic-debugging on PRIMITIVE, () proceed root-cause / pattern / hypothesis / implementation phases, () fix primitive, () REMOVE layers one at a time verifying with \$. . evidence, () commit each removal per \$. . with git-history evidence citing primitive fix. Composes \$. . / \$. . / \$. . / \$. . / \$. . (outer layer masks \$. . () false-null) / \$. . . Classification: universal (\$. .). Propagation

gate CM-COVENANT-114-250-PROPAGATION + recommended gate CM-HEURISTIC-TOWER-TRIGGERS-PRIMITIVE-DEBUG + paired \$.

mutation. **Canonical authority:** constitution submodule

Constitution.md \$. . . Non-compliance is a release blocker. No escape hatch – no --add-another-heuristic-layer , --skip-primitive-debug , --tower-is-defense-in-depth-without-justification , --reactive-mitigation-OK , --layer-per-incident flag.

\$. . . – **Byte-identical-fork prohibition + role-as-data-pack (research-derived, – –)**.

Compact summary: when project needs would be met by TWO

NEAR-IDENTICAL CODE COPIES differing only in

configuration / data / role-specific parameters (prod + staging with hardcoded hostnames; reader + writer with different flag values; web + mobile with different feature flags), copies MUST NOT be maintained as parallel forks.

Instead ONE code copy retained + role-specific values extracted into DATA PACK (role-config file / env var set / capability descriptor / feature-flag bundle) consumed by the one code to instantiate as either role. Anti-pattern: FORK-AND-DIVERGE (copies drift as one patched not other, until drift = defect surface no one owns). DETECTION:

reviewer or mechanical scan finds byte-identical fork when diff shows every difference is a config value (hostname/port/flag/label/env/credential-ref) NOT behavioral, common git-history ancestor, > % byte identity; per

\$. . . threshold-DATA. EXTRACTION: () IDENTIFY configuration surface (closed value set differing), () EXTRACT to data pack per role, () REFACTOR single code copy to consume at instantiation (constructor injection / factory / config-loader-at-startup), (.). DELETE one fork, () VERIFY per \$. . . both role instantiations produce same observable behavior their forks previously produced (\$. . . characterization before refactor), () commit extraction per \$. . . with git-history evidence. LEGITIMATE FORKS: temporary

migration (tracked \$. . . end date),
 experimental / A-B (bounded end date), truly divergent
 behavior (copies differ in algorithm/protocol/data-model
 not just config). Composes \$. . . / \$. . . /
 \$. . . (decoupling – role-configurable code shared
 across consumers) / \$. . . (data-pack location) /
 \$. . . / \$. . . / \$. . . /
 \$. . . / \$ Classification:
 universal (\$. . .). Propagation gate CM-
 COVENANT-114-251-PROPAGATION + recommended gate CM-NO-BYTE-
 IDENTICAL-FORK + paired \$. . . mutation. **Canonical**
authority: constitution submodule Constitution.md
 \$ Non-compliance is a release blocker. No
 escape hatch – no --fork-for-role-OK , --skip-data-pack-
 extraction , --copy-and-modify-config , --duplicate-code-per-
 environment , --near-identical-forks-are-fine flag.

\$. . . – Fail-closed-on-dangerous-combination
 for mutating/credential surfaces (research-derived,
 – –). Compact summary: every code path
 COMBINING ≥ dangerous capabilities MUST FAIL CLOSED –
 refuse operation unless every capability's precondition
 verifiably satisfied – NEVER fail open on ambiguous/
 missing/unresolvable input. Security-engineering
 generalisation of \$. . . guard-honesty applied
 to code paths where false-negative blast radius dwarfs
 operator inconvenience of false-positive. DANGEROUS-
 COMBINATION TAXONOMY: () MUTATION of shared resource
 (DB / filesystem / container state / config), ()
 UNTRUSTED INPUT (user string / network bytes / external
 env var / unverified file), () CREDENTIAL ACCESS (secret
 read / authenticated API call / signing key use), ()
 EXTERNAL SIDE EFFECT (email / webhook / MQ publish /
 payment endpoint), () SHELL / EXEC (subshell / binary
 with user-influenced args / template eval with user
 values), () IRREVERSIBLE (delete / drop / truncate –
 \$. . . force-push already absolutely forbidden).

Any path combining $\geq$ = dangerous-combination, MUST fail closed. INVARIANT: () VERIFY every precondition (authorization + input shape + credential presence + target identity), () REFUSE when ANY unverifiable with ERROR EXIT naming specific unresolved precondition, () EMIT captured evidence per § . . of refusal (log reason + resolved inputs + unresolved precondition + caller identity), () NEVER default to allow/proceed/retry-with-relaxed on ambiguity – § . . conservative-safe default = REFUSE. FAIL-OPEN ANTI-PATTERNS refused: `catch { /* ignore */ }, credential = credential || default, if (!valid) { log("warn"); return proceed(input); }`, `target = user_input || default_target`, unbounded `retry { dangerous(); }`. Distinct from graceful degradation (cache-miss returning cache-miss response is legitimate – NOT dangerous combination). Composes § . . / § . . ./ § . . (credentials specialisation) / § . . (operator escalation when refuse stuck) / § . . / § . . (max instantiation) / § . . / § . . . Classification: universal (§ . .). Propagation gate CM-COVENANT-114-252-PROPAGATION + recommended gate CM-DANGEROUS-COMBINATION-FAIL-CLOSED + paired § . mutation. **Canonical authority:** constitution submodule Constitution.md § . . . Non-compliance is a release blocker. No escape hatch – no `--fail-open-on-error`, `--swallow-exception`, `--default-missing-credential`, `--proceed-on-ambiguous-input`, `--retry-until-succeeds` flag.

§ . . – Idempotency under retry + DB-level durable uniqueness guard (research-derived, - -). Compact summary: every retryable operation (network calls with auto-retry, at-least-once MQ consumers, cron with overlapping schedules, HTTP endpoints with client-retry, DB transactions with retry-on-serialisation-failure, deploy pipelines with rerun capability) MUST BE IDEMPOTENT: N executions produce same

observable end state as one. Idempotency ENFORCED at LOWEST LAYER where "same end state" is durably provable – typically DATABASE UNIQUENESS CONSTRAINT on caller-supplied idempotency key, NEVER only application-layer check-then-insert (races). Prevents DOUBLE-CHARGE / DOUBLE-EMAIL / DOUBLE-DELETE / DOUBLE-EFFECT bugs.

IDEMPOTENCY-KEY DISCIPLINE: every retryable op accepts (or generates + passes forward) unique identifier naming THIS logical op; persistence layer has UNIQUE INDEX / CONSTRAINT on key column; second attempt REFUSED by DB at durable level not application logic reading-then-writing; refusal caught by caller as "duplicate – already succeeded" = SUCCESS (op happened, no further retry).

ANTI-PATTERNS: (i) check-then-insert without unique index races, (ii) app-level dedup table read without spanning-transaction, (iii) key scoped to app instance (in-memory not shared), (iv) effect persisted BEFORE key (crash between = double-effect on retry), (v) no key at all + "the effect is 'set X to V' so redoing is fine" (safe for pure sets, NOT for INSERTs / MQ emissions / external-system side effects), (vi) idempotency-key expires – converts to time-bounded-idempotent = § . . mislabelling if caller expects unbounded.

CHAOS VERIFICATION per § . . : () SIMULTANEOUS RETRIES (same key from concurrent callers), () MID-OPERATION CRASHES (kill receiver between "check exists" and "insert effect"), () RETRY AFTER APPARENT SUCCESS (caller believes fail, receiver persisted, retry MUST recognise duplicate). External-system side effects (Stripe/PayPal charging card + local DB write): idempotency key MUST pass through to external; if external does not support, op is not fully idempotent – label honestly per § . . .

Composes § . . / § . . / § . . / § . . / § . . . Classification: universal (§ . .).

Propagation gate CM- COVENANT-114-253-PROPAGATION + recommended gates CM-

IDEMPOTENCY-DB-LEVEL-UNIQUE-GUARD + CM-IDEMPOTENCY-CHAOS-VERIFIED + paired \$. mutation. **Canonical authority:** constitution submodule Constitution.md \$. . . Non-compliance is a release blocker. No escape hatch – no
 11 4 254 --check-then-insert-OK , --application-dedup-suffices , --skip-idempotency-chaos-test , 2016 08 15 --claim-idempotent-without-key , --expire-idempotency-marker-silently flag.

\$. . – **Boot-time invariant assertion + capability matrix** (research-derived, – –). Compact summary: every service / process / long-running executable MUST assert BOOT-TIME INVARIANTS at startup – required env vars present + valid; required config files parseable + declared schema version matches; required deps reachable; required credentials present + not-expired; required schema migrations applied; required feature-flag states declared; required host resources present (disk / memory / thread limits per \$. / port availability) – and FAIL FAST + LOUD if any violated, BEFORE accepting request / performing work. Prevents "service silently starts with broken invariant + serves requests that fail deep in request path hours after boot." INVARIANT CLASSES: (i) environment, (ii) configuration, (iii) dependencies, (iv) credentials, (v) schema, (vi) capabilities, (vii) host. FAIL-FAST-LOUD: () EMIT structured error to stderr + captured log naming specific failed invariant (env var name / config path / dep URL / credential file / schema version), () EXIT non-zero (traditionally EX_CONFIG for config, consumer supplies per \$. .), () DO NOT accept requests / begin main event loop / bind listening port. Service booting partially + serving requests-that-will-fail is worse than refusing to start – partial-boot masquerades healthy long enough for load balancer to route traffic. Dual to \$. . fail-closed on dangerous combinations: both refuse to proceed on unresolvable preconditions. CAPABILITY MATRIX: consumer's docs/services/

CAPABILITY_MATRIX.md (per § 11.4.5 . . .) declares per service (i) boot-time invariants, (ii) runtime capabilities OFFERED (endpoints / topics / callable functions), (iii) runtime capabilities REQUIRED from other services. Regenerated / verified on every commit per § 11.4.5 . . . docs-chain + § 11.4.6 . . . fingerprint. Undeclared endpoints emitted = violation; silent upstream deps not declared = violation (invisible to § 11.4.128 . . . contract testing). Composes § 11.4.128 . . . / § 11.4.252 . . . / § 11.4.128 . . . / § 11.4.17 . . . / § 11.4.17 . . . Layer (boot IS runtime-on-clean-target check) / § 11.4.128 . . . / § 11.4.128 . . . (contract tests consume matrix) / § 11.4.128 . . . / § 11.4.128 . . . Classification: universal (§ 11.4.128 . . .). Propagation gate CM-COVENANT-114-254-PROPAGATION + recommended gates CM-BOOT-TIME-INVARIANT-ASSERTED + CM-CAPABILITY-MATRIX-PRESENT-AND-IN-SYNC + paired § 11.4.128 . . . mutations. **Canonical authority:** constitution submodule Constitution.md § 11.4.128 . . . Non-compliance is a release blocker. No escape hatch – no § 11.4.257 . . . --skip-boot-time-assertion , --boot-partial-OK , --default-missing-env-silently , --capability-matrix-optional , --service-with-undeclared-dependency-OK flag.

§ 11.4.15 . . . – Comprehensive user-manual + guide + FAQ coverage: every component / service / feature ships an operator-usable manual, task-oriented guide, and FAQ (BACKGROUND :: REMINDER :: IMPORTANT operator mandate, – –). Compact summary: every consuming project MUST cover the whole project (every component, service, subsystem, binary, container, plugin, feature, user-visible workflow) with three cooperating documentation classes – (a) complete USER MANUAL per component/service/product, (b) task-oriented GUIDES for every user-facing workflow, (c) properly-created FAQs derived from real operator/QA/end-user questions – always in sync (§ 11.4.15 . . . / § 11.4.15 . . .), always exported

(§ . . .), always reachable from README
 (§ . . .), always tracked when incomplete
 (§ . . .); a capability that ships without its
 manual+guide+FAQ trio is a § . . . violation of
 § . . . PASS-bluff severity at the documentation layer.
 GENERALISES § . . . + § . . . to whole-
 project doc coverage; BINDS § . . . /
 § . . . / § . . . / § . . . /
 § . . . / § . . . / § . . . into
 the always-in-sync guarantee. Honest boundary § . . .
 – mandates existence + sync + reachability, not doc
 semantic-correctness (that stays § . . . /
 § . . . / § . . .). Classification:
 universal (§ . . .). Propagation gate CM-
 COVENANT-114-257-PROPAGATION + recommended gates CM-USER-
 MANUAL-PER-COMPONENT / CM-GUIDE-PER-USER-WORKFLOW / CM-FAQ-
 FROM-REAL-QUESTIONS + paired § . . . mutation. **Canonical
 authority:** constitution submodule Constitution.md
 § Non-compliance is a release blocker. No
 escape hatch – no --skip-user-manual , --guide-later , --faq-
 optional , --capability-without-docs-OK ,
 --fake-doc-placeholder-OK , --source-of-truth-is-code-comments
 flag.

§ . . . – **Architectural + data-flow + state-
 machine + sequence-diagram coverage:** every project ships
 accurate embedded diagrams across four canonical classes
 (BACKGROUND :: REMINDER :: IMPORTANT operator mandate,
 – –). Compact summary: every consuming
 project MUST cover the whole project with properly created
 graphs/diagrams/schemes across FOUR canonical classes –
 (a) SYSTEM ARCHITECTURE (services, containers, ports,
 deployment topology), (b) DATA FLOW (information movement
 between components/storage/external), (c) STATE MACHINE
 (lifecycle of every stateful entity: workable items
 § . . . , long-ops § . . . , sessions
 § . . . , agents § . . . , deploys

\$. . , builds), (d) SEQUENCE (per critical user workflow AND per critical cross-component protocol) – INCORPORATED at the point of use into the existing doc surface (README/manuals/guides/Status docs), NEVER an orphan gallery; every diagram open-format-diffable (Mermaid/PlantUML/D /diffable SVG per \$. .), rendered non-blank and non-degenerate (\$. . (. .) analyzer), reachable from README (\$. .), synced via \$. . (a code change invalidating a diagram REFUSES commit); prefer derived-from-SSoT diagrams over hand-drawn where possible. STRENGTHENS \$. . /\$. . (design-side) to whole-project doc coverage. Honest boundary \$. . – mandates existence + accuracy + embedding + honest render; does not require every class for a project that lacks the class's subject. Classification: universal (\$. .). Propagation gate CM-COVENANT-114-258-PROPAGATION + recommended gates CM-ARCHITECTURE-DIAGRAM-PRESENT-ACCURATE / CM-DATAFLOW-DIAGRAM-PER-SERVICE / CM-STATEMACHINE-DIAGRAM-PER-STATEFUL-ENTITY / CM-SEQUENCE-DIAGRAM-PER-CRITICAL-WORKFLOW / CM-DIAGRAM-EMBEDDED-NOT-ORPHAN + paired \$. . mutation. **Canonical authority:** constitution submodule Constitution.md \$. . . Non-compliance is a release blocker. No escape hatch – no --skip-11 4 259architecture-diagram , --diagram-in-orphan-directory , --proprietary-master-only , --stale-diagram-OK , --dataflow-implied-from-code , --sequence-later flag.

\$. . – README quality-status badge row: comprehensive badges with a closed green→amber→red vocabulary at the top of README, each machine-derived, production-readiness gauge summarising deploy-readiness (BACKGROUND :: REMINDER :: IMPORTANT operator mandate, – –). Compact summary: every consuming project's README MUST carry, at the TOP (immediately below the H , above the introduction – reader's first visual signal), a comprehensive BADGE ROW: closed color

vocabulary GREEN (ok) / AMBER (attention) / RED (not ok) / GRAY (N/A with reason), MINIMUM class floor covers build / test-breadth (\$. . . seven types) / code coverage (\$. . .) / security posture (\$. . . + supply-chain) / documentation completeness (\$. . .) / diagram completeness (\$. . .) / live health / open defects (\$. . . with most-reopened highlighted \$. . .) / supply-chain integrity level / zero-shortcomings verdict (\$. . .) / machine-evidence coverage (\$. . .) / PRODUCTION-READINESS GAUGE (composite; GREEN only when every clause and every \$. . . blocker clear; BINDING on release); every badge MACHINE-DERIVED (\$. . . no hand-typed colors), provenance embedded in tooltip or docs/BADGES.md (\$. . . -fingerprint-driven sync via \$. . .); badge-computer self-validated golden-good/golden-bad/negative-control per \$. . . ()/\$. . . ; stale = \$. . . violation. EXTENDS \$. . . with a symmetric badge section at the same first-visible-position privilege. Honest boundary \$. . . – mandates presence + machine-derivation + closed vocabulary + production gauge; does not claim green = bug-free (necessary-not-sufficient); GENUINELY N/A = GRAY with reason (silence-as-badge is \$. . . () FALSE-NULL). Classification: universal (\$. . .) – palette/placement/vocabulary/derivation universal; specific per-project badge set = consumer \$. . . DATA. Propagation gate CM-COVENANT-114-259-PROPAGATION + recommended gates CM-README-BADGE-ROW-AT-TOP / CM-BADGE-CLOSED-COLOR-VOCABULARY / CM-BADGE-MACHINE-DERIVED-SOURCE / CM-PRODUCTION-READINESS-GAUGE / CM-BADGE-SELF-VALIDATED + paired \$. . . mutation. **Canonical authority:** constitution submodule Constitution.md \$ Non-compliance is a release blocker. No escape hatch – no

```
--badges-optional , --hand-type-color , --skip-production-  
gauge , --badge-not-derived-from-source , --silent-omit-red-  
badge , --badge-in-footer-OK flag.
```

```
$ . . . - Cutting-edge enterprise quality +  
production-deployment readiness: every work product built  
for production + max stability + zero nasty surprises  
(BACKGROUND :: REMINDER :: IMPORTANT operator mandate,  
- - - ). Compact summary: every work product  
MUST be built for CUTTING-EDGE ENTERPRISE QUALITY targeted  
at PRODUCTION DEPLOYMENT + MAXIMAL STABILITY + ZERO NASTY  
SURPRISES - standing default from first prompt  
($ . . . /$ . . . /$ . . . /  
$ . . . ), never "good enough" / "will do for  
now" / "MVP-first". "Cutting-edge" sourced via  
$ . . . +$ . . . +$ . . . , never novelty-  
for-its-own-sake, genuine gaps honestly tracked  
($ . . . +$ . . . markers). Closed  
PRODUCTION-READINESS INVARIANT SET ( . . . clauses) every  
change satisfies: functional completeness  
($ . . . /$ . . . ), observability  
($ . . . /$ . . . ), resilience  
($ . . . chaos), security ($ . . . /  
$ . . . + fleet supply-chain minimum), performance  
($ . . . measured), maintainability  
($ . . . Opus-xhigh review → $ . . .  
GO), deployability ($ . . . /$ . . . /  
$ . . . ), documentability ($ . . . /  
$ . . . /$ . . . GREEN), testability  
($ . . . /$ . . . breadth + $ . . .  
floor + $ . . . mutations + $ . . . guards),  
operational runbook ($ . . . guide class). Zero  
nasty surprises: no known-broken ships ($ . . . ),  
no silenced gate ($ . . . /$ . . . ), no  
disabled test w/o tracked follow-up ($ . . . ), no  
workaround-as-fix ($ . . . ), no flakiness tolerance  
($ . . . /$ . . . ), no silent-failure try/
```

catch/swallow. docs/PRODUCTION_READINESS.md tracks per-component × invariant × evidence-source (\$. . sibling); is the SoT for the \$. . production gauge. STRENGTHENS \$. . (web-only) to universal. Honest boundary \$. . – mandates posture + invariants + tracker; does not claim provably-bug-free (\$. . discovery-pressure remains); unmet invariant honestly red-badged, blocks release; genuine research-frontier gaps honestly documented as \$. . original work. Classification: universal (\$. .). Propagation gate CM-COVENANT-114-260-PROPAGATION + recommended gates CM-PRODUCTION-READINESS-TRACKER-PRESENT / CM-CUTTING-EDGE-POSTURE-ALWAYS-ON / CM-ZERO-NASTY-SURPRISES-AUDIT + paired \$. . mutation. **Canonical authority:** constitution submodule Constitution.md \$. . . Non-compliance is a release blocker. No escape hatch – no --mvp-quality-OK , --good-enough , --will-fix-later , --skip-production-readiness-tracker , --production-readiness-optional , --acceptable-flakiness , --for-now-implementation flag.

\$. . . – Zero shortcomings / gaps / weak spots / danger zones invariant with mechanical audit ratchet: standing invariant, monotone-decreasing findings ledger (BACKGROUND :: REMINDER :: IMPORTANT operator mandate, – –). Compact summary: every consuming project MUST hold at ALL TIMES exactly ZERO shortcomings, gaps, weak spots, or danger zones across code / services / components / gates / docs / config – enforced by a MECHANICAL AUDIT SWEEP script inventorying findings from a CLOSED VOCABULARY (classes: shortcomings, gaps, weak spots, danger zones, TODOs/FIXMEs/"for now"/placeholders/dead-code \$. . , skipped tests w/o SKIP-OK, bluffs, unresolved \$. . items, divergent/stale/orphan artifacts per \$. . , un-catalogued anti-patterns w/o \$. . mutation), extensible via \$. . DATA never subtractable; runs at pre-build + release + constitution-pull; emits a machine-readable

ledger₁₀ {finding_id, class, file, line, description, status, tracker_ref, mitigation_evidence} (append-only + snapshot \$_{11 4 61 11 4 224}, git-tracked \$. .); each finding class ships golden-good/golden-bad fixtures per \$. . (). MONOTONE-DECREASING RATCHET (\$. . pattern) – brownfield snapshot per \$_{11 4 10} . /\$_{4 140} . (E), per-class + total, never increases, may decrease; every finding EITHER CLOSED (evidence at defect layer per \$. . , code repaired, disappears from sweep) OR TRACKED (\$. . item + \$. . description + \$. . marker + \$. . mitigation + \$. . regression guard).. Silent absorption is \$. PASS-bluff. Honest "zero" is against the closed vocabulary (\$_{11 4 124} . . . discovery-pressure remains); narrowing vocabulary to declare zero is \$. . fabrication + release-blocker; \$. . escape triggers vocabulary extension + ratchet snapshot down. GENERALISES \$. . + \$. . + \$. . to whole-project zero-finding invariant. Classification: universal (\$. .). Propagation gate₁₁₁ CM-COVENANT-114-261-PROPAGATION + recommended gates CM-ZERO-FINDINGS-AUDIT-SWEEP / CM-ZERO-FINDINGS-MONOTONE-RATCHET / CM-EVERY-FINDING-CLOSED-OR-TRACKED + paired \$. mutation.

Canonical authority: constitution submodule Constitution.md \$. . . Non-compliance is a release blocker. No escape hatch – no --skip-findings-audit , --allow-todo-without-tracker_{11 4 242} , --ratchet-may-increase , --silently-absorb-finding , --narrow-vocabulary-to-declare-zero , --for-now-in-production flag.

\$. . – Machine-created evidence at every gate: every claim of works/passes/verified cites captured machine-derived machine-verifiable evidence, no operator eyeballing, no narrative-only PASS (BACKGROUND :: REMINDER :: IMPORTANT operator mandate, – –). Compact summary: every claim by the

System that something WORKS / PASSES / IS VERIFIED / IS PRODUCTION-READY MUST cite MACHINE-CREATED + MACHINE-VERIFIABLE evidence produced by the gate itself, cited by path + sha + timestamp (\$. . content-addressed); operator eyeballing / narrative-only PASS / "looks fine" / "no error" / prediction / guessing / assumption are FORBIDDEN (\$. /\$. . / \$. .). STRENGTHENS the \$. covenant preamble from CAPTURED to CAPTURED-AND-MACHINE-VERIFIABLE evidence, applying \$. . () self-validating analyzers UNIVERSALLY (not only AV playback). THREE-GATE COVERAGE: () CONFIRMATION (source review + static + clash \$. .), () VALIDATION (\$. . four layers each with its own machine verdict + artifact), () VERIFICATION (\$. . human + \$. . machine sufficiency, each with captured evidence). MACHINE-CREATED four properties: produced by the gate not the tester, schema-parseable/machine-verifiable (JSON verdict, JSONL event stream, OCR/vision oracle per \$. . /\$. . , audio RMS/loudness), content-addressed, semantic-class-matched (\$. . wrong-layer echo forbidden). Every evidence analyzer golden-good/golden-bad/negative-control validated per \$. . (); analyzer that passes its golden-bad = bluff itself + release-blocker (\$. .). Four-layer binding with \$. . – every layer PASS cites machine evidence PRODUCED AT THAT LAYER; higher layer citing lower layer only = \$. . wrong-layer violation. Standing default from first prompt (\$. . /\$. .). Honest boundary \$. . – mandates evidence + analyzer validation; does not claim machine evidence proves human satisfaction (\$. . remains); does not license shipping when machine evidence disagrees with reality (\$. . escape). Classification: universal (\$. .). Propagation gate CM-COVENANT-114-262-PROPAGATION +

recommended gates CM-MACHINE-EVIDENCE-AT-EVERY-GATE / CM-EVIDENCE-ANALYZER-SELF-VALIDATED / CM-NO-NARRATIVE-ONLY-PASS / CM-EVIDENCE-LAYER-MATCH-108 + paired \$. mutation.

Canonical authority: constitution submodule Constitution.md
\$. . . Non-compliance is a release blocker. No escape hatch – no --narrative-pass-OK , --operator-eyeball , --should-pass-suffices , --skip-evidence-analyzer-validation , --evidence-optional , --pass-without-machine-artifact , --predict-success flag.

\$. . . – Process-group signal-safety mandate:
NEVER signal pgid ≤ , NEVER trust a mock-derived pid/pgid – validate as int > before every killpg / kill(-pid, sig) / pkill / killall call (BOB- forensic anchor, - -). Compact summary: SEVEN forced-logout incidents on the operator's workstation over ~ h (BOB- / / / / / /) traced by kernel audit trail to a single Python test – AsyncMock().pid unset → MagicMock.__int__ == 1 default → os.getpgid(1) == 1 → os.killpg(1, SIGKILL) → glibc → kill(-1, SIGKILL) = SIGKILL every UID- process (systemd user@ manager, gnome-shell, tmux, ssh, browsers, Claude Code) – with contextlib.suppress(Exception) swallowing nothing because the syscall SUCCEEDED.
Universal mandate: (A) NEVER signal pgid ≤ 1 in any language – killpg(1,sig) == kill(-1,sig) == the disaster syscall on Linux/BSD/macOS; (B) validate isinstance(pid,int) and pid > 1 AND isinstance(pgid,int) and pgid > 1 before every process-group signal call (Python os.killpg , Go syscall.Kill(-pgid,sig) , Rust nix killpg , Bash kill -<pgid> / pkill -g , C killpg / kill(-pid,sig)); (C) TESTS mocking subprocess/proc objects MUST explicitly set mock.pid = <int> (never rely on mock library defaults) AND patch the killpg call itself AND assert killpg was called with pgid > 1 ; (D) DEFENSE-IN-DEPTH four-layer coverage per \$. . (b) – pre-build grep gate, post-build

CodeGraph query, runtime kernel audit rule `syscall=kill`
`a0=0xffffffff a1=9`, paired `$ .` mutation regression
test; (E) honest boundary – hardens process-group signal
APIs; does NOT prohibit legitimate `start_new_session=True`
child-owned groups (guard still cheap); does NOT prevent
kernel OOM-killer / systemd-oomd (`$ .` +
`$ . .` apply); (F) standing default from first
prompt – every code review scans for `killpg` calls +
verifies guard, every subagent writing `subprocess-cleanup`
code includes the guard, every test-writing subagent
includes explicit `mock.pid`. Classification: universal
(`$ . .`). Composes `$ . . /$ . . (b)/`
`$ . . /$ . . /$ . . /`
`$ . . ( )/$ . . /$ . . /`
`$ . . /$ . . /$ . .`
Propagation gate `CM-COVENANT-114-263-PROPAGATION` +
recommended gates `CM-KILLPG-PGID-GUARD` + `CM-TEST-MOCK-PID-`
`EXPLICIT-INT` + paired `$ .` mutations. **Canonical**
authority: constitution submodule `Constitution.md`
`$ . .`. Non-compliance is a release blocker
regardless of context. No escape hatch – no `--skip-killpg-`
`guard`, `--mock-pid-optional`, `--killpg-1-is-ok`,
`--kill-minus-1-permitted`, `--suppress-catches-syscall`, `--test-`
`may-not-explicit-pid` flag.

`$ . .` – No-Fakes-Beyond-Unit-Tests +
`%-Test-Type-Coverage` Mandate (User
mandate, – –)

Forensic anchor – verbatim user mandate
(– –):

"Mocks, stubs, placeholders, TODOs or FIXMEs are
allowed to exist ONLY in Unit tests! All other test
types MUST interact with real fully implemented
System! No fakes, empty implementations or bluffing is

allowed of any kind! All codebase of the project MUST BE % covered with every supported test type: unit tests, integration tests, e e tests, full automation tests, security tests, ddos tests, scaling tests, chaos tests, stress tests, performance tests, benchmarking tests, ui tests, ux tests, Challenges (fully incorporating our Challenges Submodule). EVERYTHING MUST BE tested using HelixQA (fully incorporating HelixQA Submodule). HelixQA MUST BE used with all possible written tests suites (test banks) for every applications, service, platform, etc and execution of the full HelixQA QA autonomous sessions! All required dependency Submodules MUST BE added into the project as well (fully recursive!!!)."

Two cooperating invariants:

(A) No-fakes-beyond-unit-tests. Mocks, stubs, fakes, placeholders, `TODO`, `FIXME`, "for now", "in production this would", or empty-implementation patterns are PERMITTED only in unit-test sources. Every other test type – integration, E E, full automation, security, DDoS, scaling, chaos, stress, performance, benchmarking, UI, UX, Challenges, HelixQA – MUST exercise the real, fully implemented system against real infrastructure. Production code MUST NOT import mock paths. Gate `CM-NO-FAKES-BEYOND-UNIT-TESTS` + paired mutation.

(B) % test-type coverage. Codebase MUST be covered by every supported test type the domain warrants: unit, integration, E E, full-automation, security, DDoS, scaling, chaos, stress, performance, benchmarking, UI, UX, Challenges (vasic-digital/ Challenges submodule fully incorporated), HelixQA (HelixDevelopment/ HelixQA submodule fully incorporated). HelixQA autonomous sessions

drive end-to-end execution of every registered test bank with captured wire evidence per check.

Required dependency submodules (recursive per CONST-):

- Challenges – `git@github.com:vasic-digital/Challenges.git`
- HelixQA – `git@github.com:HelixDevelopment/HelixQA.git`
- Any other functionality submodules under `vasic-digital/` * / `HelixDevelopment/` orgs the project depends on.

Pointers bumped to upstream HEAD in same commit as cascade work (\$. . step); pointer drift = \$. . violation.

Classification: universal (\$. .). Severity-equivalent to a \$ PASS-bluff at the release-gate layer. No escape hatch. Composes with \$, \$. , \$. . - \$. . (esp. \$. . – this is its strict expansion into per-type-of-test territory). See Constitution \$. . for full mandate.

\$. . – **Mandatory HelixTranslate canonical translation pipeline** (User mandate, - -).

Compact summary: ALL translations of ANY content – UI strings, documents, articles, CVs, READMEs, marketing copy, in-app messages, release notes, legal text, ANY human-language string rendered in a non-source language – **MUST** be produced by **HelixTranslate**

(`git@github.com:HelixDevelopment/HelixTranslate.git`) driven through the canonical universal translation pipeline shipped in the constitution submodule at `scripts/translation/` (`translate-pipeline.sh` + `render-articles.sh`) – a hard, non-negotiable constraint. FORBIDDEN as

\$. . violations: (a) **hand-translation** – no human-authored, manually edited, or "good enough" hand-written translation of any content, ever; the engine is the single source of every translated string; (b) **other**

engines – no Google Translate, DeepL, raw ChatGPT/Claude prompting, OS translation APIs, library in auto-translators, or any engine other than HelixTranslate's `unified-translator`; there is exactly one engine; (c) **silent fallback** – no quiet substitution of the source string, no empty/partial output passed off as a translation, no swallowed engine error, no absence-of-error PASS (`$ . /$ . .`); on exhaustion of all configured providers the pipeline **MUST FAIL LOUD** (non-zero exit, destination NOT written) so the caller detects it, a missing translation being a hard error never a degraded-but-shipped string – provider-to-provider failover **WITHIN** HelixTranslate (primary → fallback, with logged, evidenced retries) is **NOT** a silent fallback and is permitted, while falling back to **ANY** non-HelixTranslate source is forbidden. **Mandatory validation** – every generated translation **MUST** be validated for accuracy with real (physical) captured evidence per `$ . . / $ . . /$ . .`, no metadata-only / config-only / grep-without-runtime PASS (`$ . /$ . .`), no false results, no bluff of any kind (`$ . .`); the per-file run log (provider used, byte counts, retries, success/failure) **IS** the evidence artifact, and an unvalidated or unevidenced translation is treated as **ABSENT**. **Explicit provider routing** – provider + model routing **MUST** be explicit and recorded, never implicit/defaulted-in-the-dark; the canonical routing is provider `groq` / model `llama-3.3-70b-versatile` with documented failover to provider `mistral` / model `mistral-large-latest`, the provider that actually produced each output **MUST** be recorded in the per-file log, and any deviation from the canonical routing **MUST** be stated explicitly in configuration and logged. **Canonical pipeline location** – `scripts/translation/translate-pipeline.sh` (universal Markdown / plain-text + article-frontmatter-preserving translator) + `scripts/translation/render-articles.sh`

(article-source → HTML fragment renderer); both reusable verbatim across projects, the engine binary supplied by the consuming project via `HELIX_TRANSLATE_BIN` and built from the HelixTranslate repository. **Rationale** – a single, evidenced, fail-loud translation engine guarantees every translated string in the fleet has a known provenance, a reproducible pipeline, and a captured accuracy proof, eliminating the silent drift, inconsistent quality, and untraceable "where did this string come from" failures that hand-translation and ad-hoc engines produce: one engine, one pipeline, one evidence trail. Classification: universal (§ 11.4.255) – a platform-neutral, language-neutral translation discipline reusable by ANY project; the consuming project supplies its concrete source files, target languages, and the HelixTranslate engine build per § 11.4.255. Composes § 11.4.255 / § 11.4.255 / § 11.4.255 / § 11.4.255 / § 11.4.255 / § 11.4.255. Propagation gate `CM-COVENANT-114-255-PROPAGATION` (literal `11.4.255`) + recommended gate `CM-MANDATORY-HELIXTRANSLATE-PIPELINE` + paired § 11.4.255 meta-test mutation (gate-code = separate work item). **Canonical authority:** constitution submodule `Constitution.md` § 11.4.255. Non-compliance is a release blocker regardless of context. No escape hatch – no `--allow-hand-translation`, `--other-engine-OK`, `--skip-translation-validation`, `--silent-fallback` flag exists.

§ 11.4.255 – **Mandatory independent per-language translation review** (User mandate, § 11.4.255). Compact summary: every translated artifact produced under § 11.4.255 MUST, BEFORE it is shipped / committed / deployed, be INDEPENDENTLY REVIEWED for quality by a per-language review agent – a hard, non-negotiable gate that composes with, and NEVER replaces, the § 11.4.255 generation mandate. Requirements: (a) **Independence** – the reviewer MUST be a different model/provider than the one that produced the translation (e.g. translator `mistral/`

mistral-large-latest → reviewer groq/llama-3.3-70b-versatile , or any distinct verified model); a model reviewing its own output is NOT an independent review and is forbidden. (b) **Per-language specialization** – the reviewer MUST act as a NATIVE speaker / linguistic reviewer of the SPECIFIC target language, evaluating that language's grammar, orthography, and correct script (Cyrillic for ru/be/kk, Arabic for ar/fa, Han/Kana/Hangul/Devanagari for zh/ja/ko/hi, Latin for sr/de/es/fr/tr, etc.). (c) **Full criteria** – each review MUST judge accuracy/fidelity, fluency/naturalness, completeness (nothing omitted or invented), correct script + orthography, and absence of untranslated source leftovers (proper nouns ./ URLs / code excepted). (d) **Strict machine-readable verdict + evidence** – each review MUST emit a structured PASS/FAIL verdict with per-criterion scores and an issues list, persisted as a physical evidence artifact (§ . . . / § . . . / § . . .); a FAIL – or an ERROR / unavailable-reviewer – BLOCKS the content and only a real PASS may ship, with no metadata-only / absence-of-error PASS (§ . . . / § . . .) and no bluff (§ . . .). (e) **Remediation loop** – a FAIL routes the artifact back through the § . . . pipeline (re-translate / multipass-polish) and is re-reviewed, iterating until PASS, NEVER hand-patched to pass (hand-translation remains forbidden under § . . .).

Canonical tooling – scripts/translation/review-translations.sh (per-language review driver) + scripts/translation/review_translation.py (independent reviewer engine emitting strict JSON verdicts); reusable verbatim across projects, the consuming project supplying its source/translation file pairs and a reviewer provider DISTINCT from its translator. **Rationale** – § . . . guarantees a translation's provenance and a fail-loud pipeline, but not its linguistic quality; an independent, native-level, per-language reviewer with a captured PASS/FAIL verdict is

what turns "a string was produced" into "a string was verified correct", making translation quality an auditable fact, rather than an assumption. Classification: universal (\$ 4.1.1.1.5.1.6) – a platform-neutral, language-neutral translation-quality-assurance discipline reusable by ANY project that ships translated content. Composes

$$\begin{array}{l} \$ \quad . \quad . \quad \quad \quad (\text{the translation it gates}) / \$ \quad . \quad / \\ \$ \quad . \quad . \quad \quad / \$ \quad . \quad . \quad \quad / \$ \quad . \quad . \quad \quad / \\ \$ \quad . \quad . \quad \quad / \$ \quad . \quad . \quad \quad / \$ \quad . \quad . \end{array}$$

(rock-solid evidence). Propagation gate `CM-COVENANT-114-256-PROPAGATION` (literal `11.4.256`) + recommended gate `CM-MANDATORY-TRANSLATION-REVIEW` + paired \$. meta-test mutation (gate-code = separate work item). **Canonical authority:** constitution submodule `Constitution.md`

\$. . . Non-compliance is a release blocker regardless of context. No escape hatch – no `--skip-translation-review`, `--self-review-OK`, `--accept-unreviewed-translation` flag exists.

§ 1.1.1 – Build once, promote ONE immutable content-addressed artifact through every environment; never rebuild per stage (research-derived, § 1.1.2). Compact summary: exactly ONE artifact is produced per release candidate and THAT artifact is promoted through every environment; rebuilding per stage silently BREAKS THE CHAIN OF EVIDENCE, because every gate that ran against the staging artifact measured a DIFFERENT binary than production runs – making each of those green verdicts a § 1.1.1 PASS-bluff about the shipped artifact (corpus module 1.1.1: *"The thing you tested must be the thing you ship ... Rebuilding per environment silently breaks the chain of evidence"*; red flag: *"We rebuild for production to be safe."*). (A) BUILD ONCE per release candidate under § 1.1.1 (reproducible + hermetic + SLSA-L1); determinism makes a rebuild detectably equivalent, it is never a licence to rebuild instead of promote. (B) IDENTITY IS A CONTENT ADDRESS – a digest,

never a mutable tag (*"tags get re-pushed; digests cannot change without changing the digest"*; the attack defended is *"the same version, different bytes"*), composing

\$. . . and \$. . . (identity read BACK from the intended target). (C) PROMOTION IS A VERIFICATION – *"verify this exact digest is authentic and approved for the next environment," not "make a new thing and hope it's equivalent"*; signature/SBOM/provenance travel WITH the bytes. (D) ENVIRONMENT CONFIG IS INJECTED AT DEPLOY, NEVER BAKED AT BUILD (baking per-environment config re-creates the per-stage rebuild; \$ 11 4 200 . . . still forbids baking a secret). (E) ROLLBACK IS DETERMINISTIC BY CONSEQUENCE – point at the previous known-good digest, *"not a rebuild-from-source scramble under incident pressure"* – the precondition that makes \$ 240 . . . 's auto-abort a real undo. (F) REBUILD-TO-DODGE IS GATE-EVASION – *"re-running until green, or rebuilding to get a different result, launders a real failure into a pass"*; retries are counted and surfaced (\$ 4 200 . . .). Honest boundary \$. . . – guarantees the tested bytes are the shipped bytes; does NOT prove them correct (\$. . . / \$. . . / \$. . .), that the deploy reached the intended target (\$. . .), or that injected config is right (\$ 11 111 . 100 . 200 .). Classification: universal (\$. . .); environment ladder, digest scheme and config-injection mechanism are consumer DATA per \$. . . 240 . 250. Composes

\$. . . /. /. /. /. /. /. /. /. (its DEPLOY-side complement: \$. . . governs how the artifact is BUILT, \$. . . that it is not built AGAIN)/. /. /. /\$. . . Propagation gate CM-COVENANT-114-264-PROPAGATION (literal 11.4.264) + recommended gates CM-BUILD-ONCE-PROMOTE-DIGEST + CM-DEPLOY-CONFIG-INJECTED-NOT-BAKED + paired \$. . . mutations (add a second per-environment build, or swap a promoted digest for a mutable tag, or bake an environment endpoint into

the build → the gates FAIL; golden-FALSE per
§ . . . (): one build + N digest-referencing
promotions MUST NOT fire either); gate-code = separate
work item, NOT claimed shipped (§ . . . /
§ . . .). **Canonical authority:** constitution
submodule Constitution.md § Non-compliance
is a release blocker. No escape hatch – no --rebuild-per-
environment , --promote-by-tag , --bake-env-config-at-build ,
--rebuild-until-green , --skip-digest-pin , --deterministic-
build-means-rebuild-is-safe flag.

§ . . . – **Progressive delivery: traffic-shifted
rollout gated by automated analysis on SLO and business
metrics, with automatic abort to stable** (research-derived,
– –). Compact summary: a release reaches
users by a bounded-blast-radius strategy – CANARY (gradual
weighted traffic shift, smallest blast radius, old+new run
concurrently so schema/compat must tolerate it) or BLUE-
GREEN (full parallel stand-up, one-shot cutover, rollback
window; *"at the switch, all users hit the new version
simultaneously"*) – never an ungated cutover; for
§ . . . critical-invariant surfaces the corpus
prefers canary with blue-green's instant-undo posture. (B)
THE PROMOTE/ABORT DECISION IS AUTOMATED, NOT A HUMAN
WATCHING A DASHBOARD – a controller compares the canary
against the stable baseline over successive analysis
windows and *"auto-promotes ... or auto-aborts and rolls back
all traffic to stable – no human needed"*; a manual
approval click is explicitly rejected (*"Manual LGTM is a
rubber stamp with no receipt; under agents it looks like
oversight while providing none"*), composing
§ . . . and § (C) THE LOAD-
BEARING CLAUSE – THE ANALYSIS MUST INCLUDE BUSINESS
METRICS, NOT ONLY INFRASTRUCTURE METRICS; the corpus names
infra-only measurement *"the classic mistake": "'HTTP error
rate is fine' does not tell you the checkout-completion
rate quietly dropped % ... the failures that matter most*

\$. .). **Canonical authority:** constitution submodule Constitution.md \$. . . Non-compliance is a release blocker. No escape hatch – no
 11 4 246 --full-cutover-ok , --infra-metrics-suffice , --manual-approval-is-the-gate , --single-window-analysis , --skip-smoke-before-traffic , --merge-implies-deploy flag.

\$ 00 20 . . – **Claim-vs-reality ledger:** every 15
ADVERTISED capability is a row, typed from the closed bluff-type vocabulary, and a capability with no passing challenge is a release blocker (research-derived,
 – –). Compact summary: corpus module names this the FIRST and cheapest of its portfolio gates –
"A shared claim-vs-reality ledger per repo ... every advertised capability is a row with a Bluff type and a severity, and a capability with no passing challenge is a release blocker. This makes stubbed-core, byte-identical-fork, and config-present-but-unwired visible and blocking rather than discovered later." THE DIRECTION OF ENUMERATION IS THE POINT and is what no existing anchor supplies: \$. . 's coverage ledger,
 \$. . 's autonomous-path column and
 \$ 11 4 239 . . 's test-type column are all keyed on FEATURES the project knows it has; a claim-vs-reality ledger is keyed on what the project ADVERTISES (README, docs, API/CLI surface, \$. . capability matrix, \$. . badges), so a claim with no feature row is invisible to a feature-keyed ledger by construction –
 \$. . adds a keying, not a restatement (\$. .). (A) ONE LEDGER PER REPOSITORY, claim-side; the advertised-surface set is consumer DATA (\$. .). (B) EACH ROW CARRIES A BLUFF TYPE FROM THE CLOSED SEVEN-MEMBER VOCABULARY – green-but-broken / coverage-theater / rubber-stamp-verified / stubbed-core / doc-vs-code-drift / config-present-but-unwired / byte-identical-fork – plus a severity (the corpus states NO severity scale; consumer DATA). (C) THE TYPE IS A ROUTING

KEY TO A. COUNTER-GATE – "it turns 'another bug' into 'another instance of type T,' which is what lets you install one gate that kills the whole type" – each type binding to the anchor that already owns its counter (\$. . . / . , \$. / \$. . . (C), \$. . . (D)) / . / . , \$. . . / . / . / . , \$. . . / . (F), \$. . .); \$. . . makes the ROUTING mandatory, it does not restate the counters. (D) THE RELEASE-BLOCKING RULE – a row whose challenge is absent, never executed, or executed only against a stale artifact fingerprint blocks exactly as a FAILING challenge does (\$. . . / \$. . . absence-blocks-as-FAIL, applied to the claim side). (E) THE ESCAPE HALF is \$. . . 's, cited not duplicated. **PROVENANCE CORRECTION** (\$. . .): secondary project documentation has called this material an "FP-1 ... FP-14" catalogue, an "L-1 ... L-16" list, and an "F-1 ... F-31" bluff taxonomy; measured against the tracked corpus, no F-n , FP-n or L-n identifier exists in any of the nine modules (control-needle proven – the same scan returns the real ATM-277 / ATM-343 / LVA-098 / ITE-6 / API-33 ids and for a negative control). The real structures are a -row numbered failure-pattern table, a -bullet portfolio-gate list, and this -member bluff-type vocabulary; those labels MUST NOT be cited as corpus wording. Honest boundary \$. . . – proves every ADVERTISED capability has a passing challenge; does NOT prove the challenge strong (\$. . . / \$. . . () / \$. . .), the capability bug-free (\$. . .), or enumerate capabilities nobody advertised (claim-keyed and feature-keyed ledgers are complements; running one leaves an honestly-stated blind side). Classification: universal (\$. . .). Composes \$. . . / . / . / . / . / . / . / . / . / . \$. . . Propagation gate CM-COVENANT-114-266-PROPAGATION

(literal 11.4.266) + recommended gates CM-CLAIM-REALITY-LEDGER-COMPLETE + CM-UNCHALLENGED-CAPABILITY-BLOCKS-RELEASE + CM-LEDGER-ROW-TYPED-FROM-CLOSED-VOCABULARY + paired \$. mutations (advertise a capability with no row, blank a row's challenge, or retype a row to an invented type → the gates FAIL; golden-FALSE: a fully-populated ledger with fresh candidate-fingerprinted PASS verdicts MUST NOT fire any); gate-code = separate work item, NOT claimed shipped (\$. . /\$. .). **Canonical authority:** constitution submodule Constitution.md \$. . . Non-compliance is a release blocker. No escape hatch – no --ledger-optional , --unchallenged-capability-ok , --untyped-ledger-row , --invent-bluff-type , --feature-ledger-suffices , --advertise-without-row flag.

\$. . – Shared attempt record + converge-on-evidence / escalate-on-stall: a failed approach is never silently retried, and a non-converging loop is itself a signal (research-derived, – –). Compact summary: corpus module – "No memory of what was tried → the loop repeats itself. Agents re-attempt discarded approaches. Fix: a shared record of attempts and outcomes (in the task state / PR thread), so a failed approach is not silently retried; and an escalation rule – after N failed iterations, stop and route to a human or a stronger model, rather than burning cycles." The premise that makes it structural: "The agent has no memory of prior attempts. It will re-implement the same wrong fix because it does not remember the last three" – so the record is externalised memory, not a convenience log. WHAT THIS IS NOT (the \$. . boundary, stated first): \$. . governs a CRASHED agent, \$. . (C) a HUNG long-op, \$. . the TERMINAL exhaustion audit – none governs a LIVE, HEALTHY, PROGRESSING loop re-walking a dead end its predecessor already proved dead; \$. . closes exactly that gap. (A) A SHARED, DURABLE ATTEMPT RECORD across every

actor that may take the next attempt (successor agent, § . . . respawn, § . . . other track); the § . . . DB and § . . . event stream are the substrates and MUST be reused, not duplicated. (B) CONSULTED BEFORE ACTING— the record is READ before the next attempt; a recorded-failed approach is not silently retried, and a re-attempt is permitted only on a recorded material difference (changed precondition, corrected § . . . sequence, new § . . . (F) RED); a written-but-never-read record is the § . . . (F) configured-≠-in-use failure applied to agent memory. (C) CONVERGE ON EVIDENCE – *"Each loop must monotonically reduce uncertainty"*, on a SINGLE SOURCE OF TRUTH (*"the verifier's build exit beats the producer's fluent paragraph every time"*) with CONCRETE FAILURE FEEDBACK (*"feed the exact captured error (argv, exit code, failing assertion, the offending line) back as the next input"*). (D) STALL IS A DETECTED SIGNAL WITH A BOUNDED RETRY AND AN ESCALATION PATH – two named signatures (*"the same criterion red after N tries"*, *"oscillating diffs"*), the stall itself tracked and trended, and the forbidden shape explicit: *"an unbounded retry loop with no ground-truth feedback and no escalation – that is how agents thrash expensively while appearing busy."* Escalation composes § . . . (re-tier), § . . . , § . . . /§ (E) N IS CONSUMER DATA, NEVER A CONSTITUTIONAL LITERAL – the corpus leaves N symbolic in every occurrence and states no numeric, wall-clock, token or cost bound; this anchor mandates that a bound EXISTS and is DECLARED, never a value. (F) THE LOOP'S DONE-CONDITION – not satisfied while an attempt is in-flight with no outcome, and every escalation reaches a terminal disposition (§ . . . /. /. /. /. /.). Honest boundary § . . . – prevents the RECORDED dead end from being re-walked; does NOT guarantee convergence

(an escalation is an honest outcome), prove any attempt competent, or substitute for §

Classification: universal (§). Composes § /. /. /. /. /. /. /. /. /. /. /.

§ Propagation gate CM-COVENANT-114-267-PROPAGATION (literal 11.4.267) + recommended gates CM-ATTEMPT-RECORD-SHARED-AND-CONSULTED + CM-STALL-BOUNDED-AND-ESCALATED + paired § mutations (dispatch blind without reading the record, remove the retry bound, or record an escalation with no terminal disposition → the gates FAIL; golden-FALSE: a loop converging on attempt with both attempts recorded MUST NOT fire either); gate-code = separate work item, NOT claimed shipped (§ / §). **Canonical authority:** constitution submodule Constitution.md § Non-compliance is a release blocker. No escape hatch – no --no-attempt-record, --record-without-reading, --unbounded-retry-loop, --retry-same-failed-approach, --stall-without-escalation, --invent-N-and-cite-corpus flag.

EXTENSIONS landed – – (existing anchors strengthened – full text in Constitution.md; research-derived from the tracked AI-curriculum corpus docs/research/ai_curriculum_2026_08/source_corpus/, modules / / / / / /). Five anchors gain clauses rather than five new numbers being minted (§ – a rule already covered is EXTENDED, never restated as a fresh anchor). § (G) – **dependency-pointer preflight fails closed on THREE classes, not one:** clause (A)'s Plane-R catalogue named only *drifted* submodule pointers; the catalogue is extended with **ABSENT/UNINITIALISED** (declared but not checked out – an empty directory read as a satisfied dependency) and **UNFETCHABLE** (the recorded pointer resolves on NO configured remote – the not our ref condition, invisible in the checkout that created it and failing only on a fresh clone). Corpus module row : "The fix is

real and merged in a submodule, but the top-level image ships the old pointer, so the bug returns from a clean checkout"; its prescription is a HARD gate – "A hard preflight gate that fails the build ... 'the fix is in the submodule but the image shipped old code' must be impossible, not caught after" – with four mechanical steps: "resolves every submodule pointer, verifies it matches the intended ref, fails closed on a missing checkout, and – for the truly load-bearing components – asserts a runtime signature on the flashed/deployed artifact." General principle: "fail closed when an input to your correctness is unverifiable ... a missing thing silently read as a present, correct thing." Gate CM-

DEPENDENCY-POINTER-PREFLIGHT-FAIL-CLOSED . § . . . –

delta debugging: bisection is not only over commits.

§ . . . 's own honest boundary limits it to "the INTRODUCING COMMIT of a MONOTONE regression", leaving defects with no known-good commit and defects whose discriminating variable is not a commit. Corpus module

supplies the generalisation and its containment direction – **git bisect** is delta debugging over commits, not the reverse – with the enumerated axes "Bisect a config ... Bisect the data ... Bisect the change set ... Bisect dependencies" (plus feature flags; STATE is NOT an enumerated axis and is not claimed as one). A non-monotone or non-commit-discriminated failure is narrowed to a MINIMAL failing case which becomes the § . . .

RED test's input, under a "stable, fast, unambiguous test predicate" run repeatedly where the failure is noisy

(§ . . . /§ . . .). Gate CM-DELTA-DEBUG-MINIMAL-CASE . § . . . (B)() – **observability: the**

mechanism, not only the property. The clause NAMED observability; three mechanical requirements were missing. Corpus module : the **three complementary signals** – structured logs ("queryable key-value events (JSON), not prose strings"), metrics, traces – plus **crash/ANR**

analytics; a **correlation id** threading all of them that MUST survive async boundaries (*"I make sure the correlation ID survives the queue ... or I lose the thread exactly where the hard bugs live"*), pivoting **metric** → **trace** → **log** under *"narrow before you theorize"*; and **crash-free users/sessions as a release gate** – *"the honest health signal I gate releases on ... I treat a new signature after a deploy as a release-blocker even at low volume"* (binding § . . . and § . . .). Thresholds are consumer DATA – the corpus states none and cautions that crash-free thresholds *"change over time and by version"*. CITATION DISCIPLINE (§ . . .): the corpus does NOT use "observability triad" or "three pillars", and contains NO claim of the form *"a system you cannot observe in production cannot be verified in production"* – what it licenses is narrower: *"Observability turns 'sometimes' into a query ... the system must be built to explain itself."* Gates **CM-CORRELATION-ID-THREADED-ACROSS-BOUNDARIES** + **CM-CRASH-FREE-RELEASE-GATE** . § . . . – **suite shape: the shape is a consequence, and the generating rule is push-tests-down.** § . . . freezes WHICH KINDS of test must exist but not WHERE a behaviour is tested. Corpus module REFUSES to prescribe a shape (*"the shape is a consequence, not a goal"*; the trophy-is-always-correct answer is explicitly marked WRONG; no numeric ratios are stated) and instead mandates the generating rule: *"push each test as far down as it will still catch the bug; if a high-level test fails with no lower-level test failing, add the lower-level one and consider deleting the redundant high-level one."* Two named anti-patterns bound it – the **ice-cream cone** (*"slow, flaky, still misses unit edges"*) at the top and **mock-heavy unit theater** (*"it never touched the system it claims to test", "can hit % coverage ... and still catch nothing"*) at the bottom, the latter already forbidden by § . . . (A) and § . . . (C). Gate **CM-SUITE-**

SHAPE-DECLARED-AND-PUSHED-DOWN . § . . – **slicing:**
an item too big to test is an item whose "done" cannot be proven. § . . governs an item's integrity once it exists and § . . (B) its acceptance criteria; neither governs SIZE AND SHAPE, the upstream cause of the reopen they then have to catch. Corpus module :
"INVEST (Bill Wake) ... Independent, Negotiable, Valuable, Estimable, Small, Testable" – and, stated by the corpus itself, only TWO letters are defined: *"the two load-bearing letters are Small and Testable, and they are linked: an item too big to test end-to-end is an item whose 'done' you cannot prove, so it closes on partial evidence and reopens on the untested part."* Vertical slices only – the horizontal slice...*"is never independently valuable or testable"*; three amigos and specification by example pin the criteria before code; and under agents *"a big fuzzy item is one an agent will report 'done' on with nothing to check it against"* (§ . . / § . . entering at the item-definition layer). CITATION DISCIPLINE (§ . .): the corpus names "SMART Tasks" once, in a source title only, never expands or mandates it – no SMART obligation may be grounded on it; and "small" is testability-in-isolation, NOT a point/hour/line count (the corpus states no size metric). Gate CM-ITEM-SLICED-SMALL-AND-TESTABLE . All five gates + their paired § . mutations = gate-code, a separate work item, NOT claimed shipped (§ . . / § . .). The literals 11.4.233 / 11.4.242 / 11.4.260 / 11.4.169 / 11.4.148 are unchanged – their propagation gates stay GREEN.

§ . . – Tamper-evident evidence chain + periodic anchor record: deletion, reordering, and tail-truncation of an accepted evidence record are DETECTED, not merely content-mutation (spec-derived, speckit -anti-slop-enforcement, -)

Compact summary: every accepted evidence record (the § . . (F) machine-written verdict store, the § . . event stream, or an equivalent evidence-bearing ledger a consuming project maintains) MUST be tamper-evident against THREE distinct attacks, not only the content-mutation attack § . . (F)/ § . . already cover: () deleting an entry, () reordering two entries, and () truncating the tail – and the mechanism that detects each MUST be named, because a forward-hash-chain alone detects () and () only when the chain is left internally inconsistent, and does NOT detect an attacker who deletes an entry and then RECOMPUTES the chain forward from that point (spec.md: , FR- 's own measured boundary: *"an actor who deletes an entry and then recomputes the whole chain is NOT detected by the chain alone (. s for k entries); that case is detected only against an anchor"*) – closing that gap requires a SEPARATE periodic anchor record the producer can append to but never rewrite (spec.md: – , FR- –). **The mandate (ALL hold): (A) EVERY STATE-CHANGE ENTRY WITHIN THE DECLARED COVERED-CALL SET, NO CITED-READ VOLUME CREEP.** Scope (§ . .): this clause binds the enforcement system's own tool-call ledger under § -anti-slop-enforcement (spec.md: , FR- is written against THAT ledger, not every write/exec/deploy on the host) – the consuming project declares its own concrete COVERED-CALL SET (which state-changing operations this evidence chain tracks) as DATA per § . . ; § . .

does NOT itself mandate a fleet-wide "every write/exec/ deploy anywhere produces a chain entry" obligation, only that within the declared set, coverage is complete and cited-vs-uncited is honoured. A state-changing call WITHIN the declared covered-call set (write, exec, deploy – spec.md: , FR-) MUST produce a matching chain entry naming it. An UNCITED read of the store – a read whose result backs no claim – MUST NOT be required to produce an entry (spec.md: , FR- : *"this guards against both volume creep and a gate that refuses correct behaviour"*); a read result CITED in support of a claim MUST have a chain entry (spec.md: , FR-). The distinction is CITED-vs-UNCITED, never present-vs-absent-in-storage. **(B) CHAIN VERIFICATION DETECTS DELETION + REORDER WITHOUT RECONSTRUCTION.** Deleting one entry, or reordering two entries, WITHOUT reconstructing the chain that follows, MUST be reported by the integrity check (spec.md: , FR-) – mutation-detection of a single entry's content alone does NOT satisfy this; the check walks the FULL forward chain, recomputing every link, not spot-checking. This clause's own honest limit is stated in (C): a reconstructed (recomputed) chain defeats (B) alone. **(C) THE ANCHOR IS THE ONLY MECHANISM THAT CATCHES A RECOMPUTED-FORWARD DELETION OR A TAIL TRUNCATION.** A periodic ANCHOR entry – chain head digest AND entry count, recorded to a location the producer can APPEND to but NOT REWRITE (spec.md: , FR- : *"the count is what makes wholesale store deletion a detected absence rather than silence"*) – MUST be written on a declared interval. The anchor's STRENGTH is recorded honestly as `mechanism` (cryptographically or append-only enforced – a git commit history, a write-once object store, or the consuming project's declared equivalent per § . .) or `policy` (procedurally enforced only); `mechanism` MUST NOT be claimed without evidence that rewriting is mechanically prevented, and absent that

evidence the recorded strength is `policy` (`spec.md:` ,
`FR-`). Verification against the anchor detects a tail
truncation that chain-alone verification `PASSES` cleanly on
(`spec.md:` , `FR-` : *"the two results together are
the evidence that the anchor, not the chain, carries this
property"*) – so "the chain verified clean" is NEVER
sufficient on its own; a verdict of tamper-free MUST name
BOTH the chain-walk result AND the anchor cross-check
result. **(D) VERIFICATION THAT CANNOT COMPLETE REFUSES – IT
NEVER REPORTS THE CHAIN INTACT.** A chain truncated mid-
verification, or an anchor location unreachable at
verification time, yields `REFUSAL` (an `UNVERIFIED` status,
per § . . . 's conservative-safe-default-on-an-
unresolvable-signal), never a "clean pass by default"
(`spec.md:` , `FR-`). "Could not verify" and
"verified clean" are never the same reported state. **Honest
boundary** (§ . . .). § . . . detects
deletion, reordering, and tail-truncation of `ALREADY-`
`RECORDED` entries. It does NOT prove the first entry ever
written was itself truthful – the entry's `CONTENT` is only
as trustworthy as the harness that wrote it
(§ . . . (F) machine-written, § . . .
`producer≠verifier` for who may write it), and it does NOT
make...the underlying evidence `CORRECT`, only tamper-EVIDENT.
Where a project already operates a content-addressed,
atomically-committed store for a related purpose (the
§ . . . continuum resume engine),
§ . . . REUSES that mechanism's chaining
primitive rather than inventing a second one – a consuming
project MUST NOT stand up a parallel, divergent chain
implementation when one already satisfies this anchor's
properties (§ . . . extend-don't-duplicate
applied to mechanism reuse, not only to rule text).

Classification: universal (§ . . .) – a platform-
neutral tamper-evidence discipline reusable by any project
maintaining an evidence/verdict ledger; the consuming

project supplies its concrete chain implementation, anchor location, anchor-interval, AND covered-call set (clause A – which state-changing operations this project's evidence chain tracks) as DATA per § . . . Composes § . . . / § . . . / § . . . / § . . . (F) (the verdict entries THIS anchor chains) / § . . . (the event stream this anchor may chain) / § . . . (verification-cannot-complete → refuse, the conservative-safe default) / § . . . (durable/atomic/ append-only construction discipline) / § . . . (reuse its content-addressed store where one already exists) / § . . . (evidence-class-at-closure – a chain entry's own class must match the defect layer it backs) / § . . . (machine-created evidence at every gate – the chain entries this anchor makes tamper-evident ARE the machine-created evidence artefacts § . . . requires; § . . . is the integrity property that keeps those artefacts trustworthy AFTER the moment of PASS, not merely present at it) / § . . . (a genuine tamper finding is itself a destructive-op-class event requiring the § . . . backup-and-investigate protocol) / § . . . Propagation gate CM-COVENANT-114-268-PROPAGATION (literal 11.4.268 present as a block-start, exactly-once per governance file, lockstep content-hash equality across the mirror set per § . . . (B)) + recommended mechanism gates CM-EVIDENCE-CHAIN-DELETE-REORDER-DETECTED (chain-walk verification detects a deleted or reordered entry left internally inconsistent) + CM-EVIDENCE-CHAIN-ANCHOR-CATCHES-RECOMPUTE-AND-TRUNCATION (an anchor record exists on the declared interval, carries head-digest + entry-count + an honest mechanism / policy strength, and anchor-cross-check verification detects a recomputed-forward deletion and a tail truncation that chain-alone verification misses) + CM-EVIDENCE-CHAIN-INCOMPLETE-VERIFICATION-REFUSES (a chain or anchor that cannot be

reached/completed at verification time yields REFUSAL, never a default clean pass) + paired \$. mutations (delete a middle entry and recompute the chain forward from that point using only internal chain data → chain-alone verification MUST still PASS [proving the gap the anchor exists to close] while anchor cross-check MUST FAIL; truncate the tail and verify chain-alone → MUST PASS while anchor cross-check MUST FAIL; make the anchor location unreachable → verification MUST report UNVERIFIED, never PASS; golden-FALSE per \$. . (): an untampered chain with a fresh, reachable, correctly-computed anchor MUST NOT fire any of the three gates). Gate-code = separate work item, NOT claimed shipped (\$. . / \$. .).

Canonical authority: constitution submodule Constitution.md \$. . (pending landing). Non-compliance is a release blocker regardless of context. No escape hatch – no --content-mutation-detection-suffices , --skip-anchor-record , --claim-mechanism-strength-unverified , --chain-truncation-passes-by-default , --reconstructed-chain-is-clean flag.

§ . . – Critic/consensus advisory-only ban AT THE EVIDENCE-ACCEPTANCE SEAM: an ungoverned confidence/consensus signal MAY inform but MUST NEVER substitute for a receipt or adjudicate producer-verifier disagreement; role recorded advisory, never entered into an evidence chain – the CONSTITUTION'S MANDATORY INDEPENDENT-REVIEW-VERDICT FAMILY

(§ . . / . / . / . / . / . / . IS UNWEAKENED (spec-derived, speckit – anti-slop-enforcement, – –)

Compact summary: at the evidence-acceptance / claim-adjudication seam – where a producer's claim of DONE/PASS is accepted or refused, and where a producer-verifier DISAGREEMENT is adjudicated – an ungoverned second opinion (a confidence score, an LLM-judge output, or a consensus aggregation across multiple such opinions, none of them produced under this constitution's own independent-review discipline) is a real, adoptable capability, but it MUST be admitted in an ADVISORY-ONLY role at THAT seam, never as the mechanism that decides accept/refuse there. This project's own operator decision on the point

(spec.md: , A-) states the rule directly:

"Current rules adjudicate producer-verifier disagreement on the recorded receipt, never on confidence or argument quality, and explicitly bar an LLM judge from being the gate."

An ungoverned critic signal that CAN flip an accept/refuse outcome at that seam has re-created a self-certification collapse under a new name – it is a second oracle/gate sharing correlated blind spots with the producer (same model family, same training data, same failure modes as an LLM producing the work it is asked to judge), exactly the class § . . /§ . .

already forbid a producer from occupying. This anchor is **NARROWER** than "no second model may ever review a producer's work" – it does NOT reach, weaken, override, or

substitute for the constitution's SEPARATE, already-mandatory independent-review-verdict family

(§ 4.239 . . . code-review-before-build,
 § 4.237 . . . iterate-to-GO with rock-solid evidence,
 § 4.236 . . . universal every-change review,
 § . . . independent verification agent,
 § . . . Opus- xhigh review substrate,
 § . . . translation context-and-spirit review,
 § . . . independent per-language translation review).

Those seven anchors mandate a STRUCTURALLY-SEPARATED verifier's verdict (§ . 11 . 134 producer≠verifier, § . . . oracle/gate roles) as a GENUINE receipt-generating gate on the work itself – a NO-GO from any of them BLOCKS the build (§ . . .) and RE-ARMS the iterate-to-GO loop (§ . . .) – and every one of the seven remains fully in force, unmodified, unrelaxed. Clause (F) below states this unweakened-list explicitly and mechanically. **The mandate (ALL hold): (A) TWO CLOSED TERM DEFINITIONS, EXCLUDING THE MANDATORY REVIEW FAMILY.** critic = a second (or Nth) model instance, confidence score, or LLM-judge that offers an OPINION on a producer's claim, diff, or artifact WITHOUT itself constituting a structurally-separated, receipt-generating review conducted under § . . . /

§ . . . / § . . . / § . . . /
 § . . . / § . . . / § . . . –

i.e. an ungoverned, ad-hoc, or supplementary second opinion, not the mandatory independent-reviewer verdict those anchors already require and gate on. consensus = a mechanism aggregating multiple such UNGOVERNED opinions (voting, averaging, an LLM-judge score) into one combined signal. Both are ADOPTABLE capability categories a project may wire in – § . . . does not forbid their existence, only their misuse as a gate AT THE EVIDENCE-ACCEPTANCE / CLAIM-ADJUDICATION SEAM. A verdict produced by the constitution's own mandatory review family is NOT a

"critic" for the purposes of this anchor – it is the receipt-generating verifier role § . . . / § . . . already govern, and it continues to gate exactly as those seven anchors require. (B) **A CLAIM WHOSE ONLY SUPPORT IS AN UNGOVERNED CRITIC/CONSENSUS SIGNAL IS REFUSED.** Per spec.md: (FR-): "Given a claim whose only support is a critic consensus or confidence score, the seam **MUST** refuse it" – naming the missing receipt. A recorded receipt – a machine-written § . . . (F) verdict, a captured evidence-chain entry per § . . . , OR a mandatory independent review's own recorded GO/NO-GO under § . . . / § . . . / § . . . / § . . . / § . . . – remains the **ONLY** basis on which producer-verifier disagreement is adjudicated; confidence or argument quality UNBACKED by such a receipt, however produced, is never a substitute. (C) **DETERMINISTIC WITH AND WITHOUT THE UNGOVERNED CRITIC.** Per spec.md: (FR-): the SAME claim, evaluated once with an UNGOVERNED critic step present and once with it absent, **MUST** produce the IDENTICAL accept/refuse outcome. A pipeline in which disabling that ungoverned critic changes the verdict is, by definition, using it as a gate – the deterministic-outcome test is the mechanical proof that (B) is genuinely honoured, not merely stated. (This clause does NOT apply to a mandatory review under the seven-anchor family in clause (A) – skipping a REQUIRED § . . . / § . . . / § . . . review is itself a violation of those anchors, not a deterministic-outcome test to pass; the test in this clause is scoped to the OPTIONAL, ungoverned critic only.) (D) **UNGOVERNED CRITIC OUTPUT IS RECORDED advisory IN A SEPARATE STORE, NEVER ENTERED INTO AN EVIDENCE CHAIN.** Per spec.md: (FR-): ungoverned critic output **MUST** be recorded with a role of advisory and **MUST NOT** appear in any

§ . . . /§ . . . (F) evidence chain – it is recorded in a separate advisory store, never intermixed with the accept/refuse evidence chain itself. A mandatory review's OWN recorded GO/NO-GO under the seven-anchor family continues to be entered into the evidence/verdict record as its own anchors require (§ . . . 's review-completed marker, § . . . 's clean-GO record) – it is not retroactively downgraded to advisory by this clause. **(E) WHY THE UNGOVERNED-CRITIC COLLAPSE MATTERS.** Composes § . . . (A)/(C) and § . . . – an ungoverned critic wired into the evidence-acceptance gate collapses ORACLE and GATE into a single mechanism that is NOT independent of the producer's own failure surface (correlated model errors, shared blind spots), which is precisely the class of collapse § . . . names as maximum-severity. Advisory-only admission avoids re-creating that collapse under the label "review" while retaining the genuine usefulness of an ungoverned critic – surfacing likely issues earlier, feeding § . . . systematic debugging, prioritising a human reviewer's attention – none of which requires it to hold gating authority. The mandatory seven-anchor review family avoids the SAME collapse by a DIFFERENT mechanism entirely – structural separation of the reviewer per § . . . , not advisory-demotion of its verdict – which is exactly why that family's verdicts remain full receipts and this anchor never touches them. **(F) THIS BAN IS SCOPED TO THE...EVIDENCE-ACCEPTANCE / CLAIM-ADJUDICATION SEAM – IT DOES NOT...WEAKEN, OVERRIDE, OR SUBSTITUTE FOR THE MANDATORY INDEPENDENT-REVIEW-VERDICT FAMILY.** § . . . (code-review-agent gate before pre-build + main build), § . . . (iterate-to-GO with rock-solid evidence), § . . . (universal code-review – every change reviewed, always), § . . . (independent verification agent), § . . . (code-review MUST run on the Opus model

at xhigh effort), § . . . (context-and-spirit translation review), and § . . . (independent per-language translation review) each mandate a STRUCTURALLY-SEPARATED verifier (§ . . . producer≠verifier) whose VERDICT genuinely gates, and NONE of these seven anchors is relaxed, narrowed, made advisory, or made optional by § What § . . . forbids is narrower and different: (i) treating an UNGOVERNED second opinion / confidence score – one NOT produced under any of those seven anchors' own independence-plus-evidence discipline – as a substitute receipt (clause B); and (ii) letting such an ungoverned signal, rather than a recorded receipt (including a receipt produced by the seven-anchor family), adjudicate a producer-verifier DISAGREEMENT (the A- scope, spec.md:). A project MUST NOT cite § . . . to justify skipping, weakening, downgrading-to-advisory, or making optional any of the seven listed mandatory reviews – doing so is itself a § . . . (F) violation, exactly the mis-scoping this clause exists to foreclose. **Honest boundary** (§ . . .). § . . . does not forbid using an ungoverned critic/consensus signal to prioritise human review, to surface candidate defects for § . . . investigation, or as ONE input among many an independent human/verifier weighs when authoring (not executing) a gate. It forbids only its use as an automated GATING mechanism substituting for a real, receipt-backed oracle/gate/verifier chain. It does not claim critics are useless – only that their output is advisory context, never a verdict, UNLESS that output is itself produced under the mandatory independent-review-verdict family named in clause (F), in which case it is a receipt and this anchor does not touch it. **Classification: universal** (§ . . .) – a platform-neutral discipline reusable by any project that adopts, or is asked to adopt, a second-model/critic/consensus review capability OUTSIDE

its own mandatory independent-review-verdict family; the consuming project supplies its concrete critic implementation and advisory-record location as DATA per § 11.4.125, and its own set of already-mandatory review-verdict anchors (analogous to this project's § 11.4.125 / 11.4.142) as the fixed, unweakened-by-this-anchor list. Composes § 11.4.145 / 11.4.269 / 11.4.237 / 11.4.240 (F) / § 11.4.240 / § 11.4.240 / § 11.4.240 (the seven mandatory review-verdict anchors clause (F) leaves unweakened) / § 11.4.240 (the separation principle this anchor is a specific instantiation of) / § 11.4.240 (the four-role architecture that names why an ungoverned critic-as-gate is a collapse) / § 11.4.240 (the evidence chain the ungoverned critic's output MUST NOT enter) / § 11.4.240 (the legitimate advisory use – feeding investigation) / § 11.4.240. Propagation gate CM-COVENANT-114-269-PROPAGATION (literal 11.4.269 present as a block-start, exactly-once per governance file, lockstep content-hash equality across the mirror set per § 11.4.240 (B)) + recommended mechanism gate CM-CRITIC-CONSENSUS-ADVISORY-ONLY (a claim supported only by an ungoverned critic/consensus/confidence signal, with no receipt, is refused; the accept/refuse outcome for a claim is identical whether an ungoverned critic step ran or was skipped; every stored ungoverned-critic output carries role advisory and appears in no evidence chain; a mandatory review under § 11.4.240 / 11.4.240 / 11.4.240 / 11.4.240 is NEVER downgraded to advisory or excluded from gating by this gate) + paired § 11.4.240 mutation (wire an ungoverned critic's approve/reject value directly into the pass/fail decision → the gate MUST FAIL;

store an ungoverned critic output with no role field, or role `verdict`, or entered into the evidence chain → the gate MUST FAIL; downgrade a review's own GO/NO-GO record to `advisory`, or exclude it from gating, citing § . . . as the reason → the gate MUST FAIL; golden-FALSE per § . . . (): (i) an ungoverned critic present, its output stored with role `advisory`, absent from the evidence chain, and the deterministic outcome unchanged whether it ran or not → the gate MUST NOT fire; (ii) a § . . . code-review's own NO-GO record blocking the build exactly as § . . . requires → the gate MUST NOT fire). Gate-code = separate work item, NOT claimed shipped (§ . . . / § . . .). **Canonical authority:** constitution submodule `Constitution.md` § . . . (pending landing). Non-compliance is a release blocker regardless of context. No escape hatch – no `--critic-may-gate`, `--consensus-score-is-a-receipt`, `--llm-judge-is-the-gate`, `--critic-output-in-evidence-chain-OK`, `--outcome-may-differ-with-critic`, `--269-overrides-mandatory-review`, `--downgrade-mandatory-review-to-advisory` flag.

§ . . . – **Dependency-existence-verdict register:** every proposed dependency carries a closed-set existence verdict {VERIFIED, AMBIGUOUS, UNVERIFIED} with citable evidence; adoption is gated behind resolution and behind the wiring-sweep result (spec-derived, speckit –anti-slop-enforcement, – –)

Compact summary: a project that adopts an external (or internal, cross-module) dependency to close a coverage gap MUST first record, for that dependency, a closed-set

EXISTENCE verdict – VERIFIED, AMBIGUOUS, or UNVERIFIED – backed by a citable artifact or an explicit statement that none was found (spec.md: , FR-). Existence verification is NECESSARY but NOT SUFFICIENT for adoption (spec.md: , A- : "Existing is not the same as working ... Several verified entries are at initial release versions, one declares a repository that returns , and one's 'tamper-evident' claim is on inspection only 'append-only' – which detects nothing if the appender is the party being checked") – a VERIFIED existence verdict is the FLOOR a dependency clears before it may even be considered adopted, never a claim that it is fit for the purpose it is being adopted for. **The mandate (ALL hold): (A) THE REGISTER, ONE ROW PER CLAIMED DEPENDENCY.** Every dependency a project proposes to adopt gets exactly one register row, keyed by a stable identifier (name + version + source per § . . resolve-by-stable-name). The row carries the verdict + its evidence field (spec.md: , FR- : "MUST carry a verdict of VERIFIED, AMBIGUOUS or UNVERIFIED with a citable artifact or an explicit statement that none was found"). **(B) VERDICT SEMANTICS, EVIDENCE-BASED, NEVER PROMOTED BY OMISSION.** UNVERIFIED is the honest default state – a claim about a dependency's existence with no independent confirmation attempted MUST NOT be silently treated as VERIFIED. AMBIGUOUS is assigned when evidence conflicts or a name collision exists – spec.md: (FR-): "a dependency name that collides with an unrelated project" gets AMBIGUOUS, never VERIFIED – the specific conflicting evidence is recorded, not merely a vague "unsure". VERIFIED requires an independently-reproducible check: a real resolve + checksum/hash match, a real successful call or import, or an equivalent control-needle-proven presence per § . . () (b). **(C) ADOPTION REFUSED WHILE THE VERDICT IS NON-VERIFIED.** Per spec.md: (FR-): "Given a dependency whose verdict is UNVERIFIED or

AMBIGUOUS, adoption MUST be refused." Adoption is admitted ONLY...after the row resolves to VERIFIED (or the AMBIGUOUS conflict is investigated and the row is re-verdicted). (D) **ADOPTION IS GATED BEHIND MAPPING TO A RECORDED UNCOVERED GAP + BEHIND THE WIRING-SWEEP RESULT.** Per spec.md: (FR-): adoption is refused unless the dependency maps to one of the project's own recorded uncovered capability gaps (adopting a dependency for a capability the project already covers is itself refused, naming the covering rule – the § . . duplicate-coverage discipline applied to dependency adoption). Per spec.md: (FR-): *"the adoption step MUST NOT be accepted before the wiring step reports its sweep result"* – a project does not decide "yes, adopt" before it has run the sweep that would reveal whether the dependency's claimed capability is genuinely reachable and wired (§ . . /§ . . (F) investigate-before-adopt, the mirror-image of investigate-before-remove applied to the opposite direction). (E) **ZERO ADOPTED WITHOUT VERIFIED – MONOTONIC.** Per spec.md: (SC-): the count of external dependencies adopted without a VERIFIED existence verdict is ZERO, checked at the SAME seam that already refuses unwired-gate names per § . . (A) (the named-gate ledger), generalised here to dependency-adoption claims. **Honest boundary** (§ . .). § . . 's register proves EXISTENCE/RESOLVABILITY was checked and recorded – it does NOT prove the dependency BEHAVES correctly for the purpose it is adopted for (that stays ordinary integration/contract testing per § . .), and it does NOT prove the dependency is SECURE or well-maintained (that is § . . 's supply-chain territory). A- 's own finding – "existing is not the same as working" – is why this anchor is deliberately narrow: it closes the EXISTENCE-CLAIM bluff, not the fitness-for-purpose question, which stays governed by the constitution's

ordinary testing/review family. **Classification: universal**
 (§ . . .) – a platform-neutral existence-
 verification discipline reusable by any project that
 adopts external or cross-module dependencies to close a
 capability gap; the consuming project supplies its
 register location, its recorded uncovered-gap set, and its
 wiring-sweep mechanism as DATA per § Composes
 § . . . / § . . . / § . . . (stable-
 name...keying) / § . . . (investigate-before-adopt,
 the direction-mirror of investigate-before-remove) /
 § . . . (F) (configured ≠ wired, the same principle
 applied to a dependency's...claimed capability) /
 § . . . () (b) (control-needle-proven presence
 checks) / § (A) (named-gate ledger – this
 anchor generalises its zero-unwired-debt discipline to
 dependency claims) / § . . . (fitness-for-purpose
 stays contract-testing territory, distinct from
 existence) / § . . . (security/supply-chain, also
 distinct from existence) / § . . . (submodule-
 catalogue-first discovery + extend-don't-reimplement – the
 register's VERIFIED-tier check IS the mandated catalogue-
 first discovery...step, applied to any dependency, not only
 own-org submodules) / § . . . (submodule-
 dependency-manifest – an owned-org dependency's `helix-`
`deps.yaml` is itself citable evidence toward a VERIFIED
 verdict) / § . . . (mandatory deep multi-angle
 web research before declaring a capability covered – the
 research pass that PRODUCES a register row's citable
 evidence, or its explicit none-found statement) / § . . .
 Propagation gate `CM-COVENANT-114-270-PROPAGATION` (literal
`11.4.270` present as a block-start, exactly-once per
 governance file, lockstep content-hash equality across the
 mirror set per § . . . (B)) + recommended mechanism
 gate `CM-DEPENDENCY-EXISTENCE-VERDICT-REGISTER` (every claimed
 dependency has a register row carrying a closed-set
 verdict + evidence field; a name collision resolves

AMBIGUOUS never VERIFIED; a shipped change relying on an UNVERIFIED or AMBIGUOUS row is refused; adoption refused when the dependency maps to an already-covered capability or precedes its wiring-sweep result) + paired \$.

mutation (adopt a dependency with no register row → the gate... MUST FAIL; leave a row UNVERIFIED and adopt anyway → the gate MUST FAIL; feed a name-collision fixture and assert the verdict resolves VERIFIED instead of AMBIGUOUS → the gate MUST FAIL; golden-FALSE per

\$. . (): a dependency VERIFIED with real, reproducible, control-needle-proven evidence, mapped to a genuinely-uncovered gap, adopted after its wiring sweep reported a result → the gate MUST NOT fire). Gate-code = separate work item, NOT claimed shipped (\$. . / \$. .). **Canonical authority:** constitution submodule Constitution.md \$. . (pending landing). Non-compliance is a release blocker regardless of context. No escape hatch – no --unverified-adoption-OK , --ambiguous-treated-as-verified , --skip-wiring-sweep-before-adopt , --adopt-already-covered-capability , --existence-proves-fitness flag.

§ . . – Waiver mechanism: the strict FORMALIZATION of allow-with-tracked-debt (unifying § . . (D)'s recorded-deferral, § 4 2.1 . . ()'s explicit-operator-override, and § . . (A)'s deadline-bound quarantine into ONE roster + expiry + tracked-item schema), distinct from refusal (§ . .) and closure (§ . .) vocabulary (spec-derived, speckit –anti-slop-enforcement, –)

Compact summary: every project needs, on rare occasion, a mechanism to ALLOW a change to proceed despite a normally-blocking missing-evidence condition – never a silent, never a permanent bypass, but a tracked, expiring, human-authorized exception. **This is NOT an absent capability being introduced here** – re-verified in this remediation pass: § . . (D) already permits an explicit, documented deferral flag ("The escape hatch is a recorded deferral, not a bypass: the skipped gate remains owed and is caught at the next full run or tag gate" – Constitution.md:); § . . () already requires "an explicit, logged, operator-set flag" for any override of its deploy-readiness seam ("EXPLICIT, RECORDED OVERRIDE ONLY" – Constitution.md:); and § . . (A) already ships a deadline-bound quarantine for one specific gate class (flaky tests, moved to tests/quarantine/ with a tracked § . . stabilisation deadline). Those three are genuine allow-with-known-debt PRECEDENTS this anchor does not discover but FORMALIZES. What § . . genuinely adds – needed: `waiv*` returns ZERO hits anywhere in constitution/Constitution.md outside this draft, confirming the SCHEMA below is new even though the CATEGORY of mechanism is not – is the STRICT, DATA-BEARING, cross-gate schema those three precedents describe only informally and per-

mechanism: a ROSTERED NON-PRODUCER authoriser (\$. . . (D) and \$. . . (A) name no authoriser; \$. . . () names the operator as a role but no roster-resolvable identity schema – \$. . . adds the roster resolution + producer exclusion), a MANDATORY UNELAPSED EXPIRY (\$. . . 's skip flag and \$. . . 's override flag carry no expiry field at all – an indefinite skip/override is currently legal under their text), and a MANDATORY TRACKED ITEM (neither \$. . . (D) nor \$. . . () requires one; \$. . . (A) requires it for its one gate class – \$. . . makes it mandatory cross-gate). \$. . . turns three ad-hoc, per-anchor escape hatches into ONE cross-cutting mechanism any future gate can invoke by reference instead of re-inventing its own – this project's own `scripts/lib/critical_blocker_gate.sh` (built for the \$. . . seam) already implements `waive / override` operations by name, direct evidence this schema generalises code the project has already written, not a speculative addition. **The mandate (ALL hold): (A) THE WAIVER IS THE ONE FORMALIZED SCHEMA FOR AN ALLOW-WITH-KNOWN-DEBT PATH.** A waiver EXISTS at a seam that would otherwise refuse (a missing receipt, an unresolved dependency verdict, an unmet acceptance-criteria scenario, or any other normally-blocking condition this constitution's family of gates enforces) – the SAME class of seam \$. . . (D)'s deferral flag, \$. . . ()'s override flag, and \$. . . (A)'s quarantine already gate. It is invoked explicitly, never inferred, and it never disables the gate permanently – it grants a bounded, tracked exception to ONE candidate. A gate ALREADY implementing an allow-with-known-debt path under \$. . . / \$. . . /\$. . . is NOT required to reimplement one from scratch – it satisfies

§ . . by conforming its existing flag/quarantine mechanism to this clause's roster + expiry + tracked-item schema (clauses B-D below), the strict superset those three anchors' informal mechanisms currently lack. **(B) AUTHORISATION MUST RESOLVE TO A ROSTERED IDENTITY, NEVER THE PRODUCER.** Per spec.md: (FR-): *"Given a waiver whose authorisation value does not resolve to an identity in the declared authoriser roster, the seam MUST refuse it"* – a free-text authorisation ("looks fine") is refused exactly as an absent one is; a waiver naming a genuinely rostered identity is accepted (the false-positive guard – a check that refuses EVERY waiver has replaced a bypass with a deadlock, not fixed anything). The authoriser roster is consumer-supplied DATA per § . . , and the producer of the change under waiver MUST NOT appear as its own waiver's authoriser (composes § . . producer≠verifier – a producer cannot waive their own gate). **(C) EXPIRY IS REQUIRED AND MUST NOT HAVE ELAPSED.** Per spec.md: (FR-): a waiver with NO expiry, or whose expiry HAS elapsed, is refused; a waiver whose expiry is in the future is accepted. An absent expiry is NOT read as "valid forever" – the spec's own framing is exact: *"An expiry absent by omission is not 'valid for every candidate forever' – that is the same downgrade-by-omission the satisfaction path already refuses"* elsewhere in this family (§ . . (D)/§ . . (F) fingerprint-absent-is-refused, the general pattern this clause instantiates for waivers). An elapsed waiver's gate reverts to blocking automatically – no grace period, no silent auto-renewal. **(D) A TRACKED ITEM NAMES THE DEBT; A WAIVER NAMING NONE IS REFUSED.** Per spec.md: (FR-): *"Given a waiver, it MUST name a tracked item for the blocker that remains open, and a waiver naming none MUST be refused"* – because *"a waiver is the only ALLOW that leaves the defect unfixed, so it is the one*

path...on which the owed work must stay traceable." The tracked item is an ordinary § 11.4.15 -stable-id workable item bound to the SAME lifecycle discipline (§ 11.4.15 /§ 11.4.16 /§ 11.4.33 / § 11.4.33) as any other tracked work – a waiver whose tracked item never resolves is itself surfaced by the standing zero-shortcomings audit (§ 11.4.15) as an unresolved-waiver finding, never silently forgotten.

(E) COMPOSITION, NOT SUBSTITUTION – INCLUDING WITH THE THREE PRECEDENT MECHANISMS. A waiver whose expiry lapses UNRESOLVED re-enters the § 11.4.30 Operator-blocked class for its underlying condition (the block returns; nothing about the original blocking condition changed, only the temporary permission to pass did). A waiver's tracked item CAN close via § 11.4.30 Obsolete if the underlying requirement genuinely changed. Neither § 11.4.30 nor § 11.4.30 substitutes for the waiver mechanism itself – a project attempting to express "allow despite missing evidence, tracked, expiring" via either of those two vocabularies is misusing them, per this anchor's own honest-boundary clause. Symmetrically, § 11.4.30 (D)'s deferral flag, § 11.4.30 ()'s override flag, and § 11.4.30 (A)'s quarantine are NOT replaced by § 11.4.30 – each remains the correct, narrower mechanism for its own seam class; § 11.4.30 is the schema they conform TO when a project wants their allow-with-debt behaviour to carry a rostered authoriser, an expiry, and a tracked item, not a rival mechanism competing with them. **Honest boundary** (§ 11.4.30).

§ 11.4.30 formalizes, it does not discover, the ALLOW-with-tracked-debt CATEGORY – § 11.4.30 (D), § 11.4.30 (), and § 11.4.30 (A) already establish that category exists in this constitution; why § 11.4.30 does not make waivers routine or easy is the SCHEMA's own shape (rostered-non-producer authorisation, mandatory unelapsed expiry, mandatory

tracked item) – deliberately strict, because a waiver is the ONE path this constitution's anti-bluff family permits to leave a defect genuinely unfixed while still shipping. It does not weaken any gate a waiver bypasses for OTHER candidates, and it does not authorise an indefinite or permanently-renewed waiver – an expiry that keeps getting pushed forward without the underlying debt closing is itself a § . . . finding (a monotone-decreasing-findings-ledger violation, not a compliant waiver renewal pattern). **Classification: universal** (§ . . .) – a platform-neutral, deliberately-strict allow-with-tracked-debt-discipline reusable by any project whose gate family needs an occasional, bounded, evidence-honest exception; the consuming project supplies its authoriser roster, expiry mechanism, and tracked-item store as DATA per § Composes § . . . / § . . . / § . . . / § . . . (refusal vocabulary – distinct, cited not restated) / § . . . (closure vocabulary) / § . . . / § . . . (stable-id tracked item) / § . . . (closure vocabulary – distinct, cited not restated) / § . . . (D) (the absent-field-is-refused pattern this anchor applies to expiry + authoriser + tracked-item) / § . . . (the tracked item's completion obligation) / § . . . (D) (recorded-deferral precedent this anchor formalizes) / § . . . () (explicit-recorded-override precedent this anchor formalizes) / § . . . (A) (deadline-bound quarantine precedent this anchor formalizes for one gate class) / § . . . (producer≠verifier – the authoriser exclusion) / § . . . (an unresolved waiver is a standing-audit finding) / § . . . Propagation gate CM-COVENANT-114-271-PROPAGATION (literal 11.4.271 present as a block-start, exactly-once per governance file, lockstep content-hash equality across the mirror set per § . . . (B)) + recommended mechanism gate

CM-WAIVER-ROSTERED-EXPIRY-TRACKED (every active waiver resolves to a genuinely rostered, non-producer authoriser identity; a future, unelapsed expiry; and a named § . . . -stable-id tracked item; a waiver missing ANY ONE of the three is invalid and its waived gate reverts to blocking) + paired § . . . mutation (submit a waiver with free-text or absent authorisation → the gate MUST FAIL; submit one with no expiry, or an elapsed expiry → the gate MUST FAIL; submit one naming no tracked item → the gate MUST FAIL; submit one where the authoriser identity equals the change's producer → the gate MUST FAIL; golden-FALSE per § . . . () : a waiver naming a genuinely-rostered non-producer identity, a future expiry, and a real tracked item → the gate MUST NOT fire). Gate-code = separate work item, NOT claimed shipped (§ . . . / § . . .). **Canonical authority:** constitution submodule Constitution.md § . . . (pending landing). Non-compliance is a release blocker regardless of context. No escape hatch – no

--free-text-authorisation-OK , --waiver-without-expiry , --elapsed-waiver-still-valid , --waiver-without-tracked-item , --producer-may-self-waive , --permanent-waiver-renewal flag.

§ . . . – **Dynamic, on-demand skill/extension activation:** the active capability surface is the MINIMUM needed now, everything else stays DISCOVERABLE and one command away (User mandate, – –). Compact summary: verbatim operator mandate ("We MUST use actively only Skills we need at particular moment! ... fully dynamic so token use is ALWAYS at the minimum and any of Skills and the extensions activated / deactivated as they are needed / used!"); forensic case study (measured – – , Claude Code . . .) : a –skill host pool renders , B (~ , tok) of NAMES every turn, , B (~ , tok/turn) with description fields – a . x render-mode cliff. Every project whose agent

platform exposes a capability surface MUST keep the ACTIVE surface at the minimum genuinely needed, everything else activatable on demand: (A) minimal declared `core` set always active, everything else inactive-by-default; (B) discoverability NEVER traded for tokens – a deactivated capability stays findable via an always-available index, § 11.4.11 • 11.4.12 (B) binds unconditionally; (C) fail SAFE never silent (§ 11.4.13) – a failed activation exits non-zero + rolls back, never leaves a dangling/unusable claim; (D) concurrency-safe by reference counting (§ 11.4.14 / § 11.4.15) – one session's deactivation never strips a capability another session is using; (E) honest asymmetry (§ 11.4.16) – measured first-hand that activation hot-loads but deactivation does NOT hot-unload in the live session, so the saving is a minimal session-start baseline plus on-demand growth, never an in-session token-freeing claim; the platform's own `permissions.deny Skill(name)` blocks invocation without removing the rendered entry, saving ZERO tokens; (F) a budget gate guards BOTH the name-only and description-eager render modes; (G) works with AND without the constitution submodule vendored – an explicit resolution order ending in honest, loud degradation, never a silent no-op; (H) PRUNE-TO-CORE is the default (operator decision – § 11.4.17) and is NOT a § 11.4.18 silent removal – structurally reversible+visible by construction (LISTED in the core catalogue, ACTIVATABLE by one command, NOT DELETED – moved/linked not destroyed), with an explicit opt-out at session+project scope, refusing loudly rather than risking content on any unsafe demotion. Honest boundary (§ 11.4.19): reduces per-turn capability-LIST cost only, never a skill's invoked body cost, never a substitute for § 11.4.20 (B) discovery, never a licence to hide a needed capability. Classification: universal (§ 11.4.21). Composes § 11.4.22 / § 11.4.23 / § 11.4.24 / § 11.4.25 / § 11.4.26 / § 11.4.27 /

\$. . (B)/\$. . /\$. . /

\$. . /\$. . /\$. . /

\$. . /\$. . /\$. . /

\$. . /\$. . Propagation gate CM-
 COVENANT-114-272-PROPAGATION (literal 11.4.272) +
 recommended gates CM-SKILL-SURFACE-BUDGET / CM-SKILL-
 ACTIVATION-FAIL-SAFE / CM-SKILL-PRUNE-PRESERVES-CONTENT +
 paired \$. mutation (squeeze the budget below the real
 surface, break an activation target, remove the engine
 without the distinct engine-absent exit code, or make a
 demotion destructive/strip its relocation-safety or other-
 live-session check → the gates MUST FAIL). Gate-code
 beyond the shipped scripts = separate work item, NOT
 claimed shipped (\$. . /\$. .). **Canonical
 authority:** constitution submodule Constitution.md

\$. . . Non-compliance is a release blocker. No
 escape hatch – no --all-skills-always-on , --skip-activation-
 engine , --hide-inactive-skills , --silent-activation-failure ,
 --ignore-surface-budget , --assume-skill-loaded flag.

\$. . – Measuring-instrument verification: a
 census that INFORMS a decision must be control-needed
 before its result is believed (research-derived,
 – –). Compact summary: forensic anchor –
 FIFTEEN distinct ad-hoc-measurement instrument failures in
 one working session (– –), each returning a
 confident, plausible, actionable-looking result that was
 an artefact of the instrument, not a property of the
 system: a block-opener regex missing a second heading form
 manufactured phantom dangling anchors; comm on
 non-lexicographically-sorted input reported all
 anchors missing from a file containing none missing; \$?
 read after a pipeline reported the pipeline's LAST stage,
 turning exit into reported exit ; a field-name
 typo returned <none> for every record; grep -w treated -
 as a word boundary and silently reclassified a record; a
 summary-line grep assumed field order that flips on

failure, reading a real failure as merely quiet;
while read silently dropped a final no-trailing-newline
line; a mutation was applied via the WRONG env knob so the
real (unmutated) tree was tested and the green was
meaningless; a guard-probe used a command the guard does
not target, proving nothing either way; a test fixture
aliased two columns to the same string so a real column-
swap mutation survived / undetected; a positive
control chosen OUTSIDE the instrument's declared scope
reported a working instrument as broken; a positive
control that WAS the very question under test could not
discriminate anything; an awk range mis-escape reported
rows where the live system held (x
wrong), nearly gating a destructive DB recreate; a
denylist with TWO needles was searched with only ONE, so
"zero remaining" was blind to the first while a live
credential was still present in four copy-pasteable lines
per file; \$? was clobbered by a command substitution
sitting earlier in the SAME statement, reporting a failed
pandoc run (exit , nothing written) as exit= . The
gap: \$. . governs guards/gates (block/pass),
\$. . () governs calibrated analyzers –
neither covers the far more common ad-hoc census (a grep,
a set-diff, an exit-code read, a field lookup, a row
count) whose output is a FACT a decision/fix/
classification/report is then built on, written inline,
used once, trusted implicitly, never itself tested. The
mandate: any decision/claim/classification/report-bearing
measurement MUST be control-needed BEFORE acting on or
communicating its result – (a) POSITIVE control (a known-
present value must be found, catching too-narrow patterns/
wrong fields/wrong knobs/silent no-output stages); (b)
NEGATIVE control (a known-absent fabricated needle must
NOT be found, catching too-broad patterns/accidental
substring-or-word-boundary matches); (c) an EMPTY result
is NEVER a finding on its own – genuinely-absent /

instrument-blind / wrong-location / wrong-output-shape are four distinct states an empty result cannot by itself distinguish, reportable only AFTER (a) proves the instrument can see a positive; (d) a result CONTRADICTING directly-observed evidence presumes the INSTRUMENT is at fault first; (e) the instrument that established a FACT is itself part of the evidence – captured claims cite the producing command; (f) when the criterion is a SET (a denylist, a taxonomy, a multi-column key), the control MUST cover EVERY member – needling one member proves only that member is searchable, the instance- failure mode, the most dangerous on the list because both controls PASS while the answer is wrong; (g) capture an exit status BEFORE any other expansion can overwrite it – assign `$?` to a variable on its own line immediately after the command, never read it after a pipeline or a same-statement command substitution. Honest boundary (`$ . . .`): control needles prove an instrument distinguishes present-from-absent, never that it measures the RIGHT property, and never that it is correct for inputs unlike the needles chosen. Classification: universal (`$ . . .`). Composes `$ . . . / $ . . . / $ . . . / $ . . . ( ) / $ . . . / $ . . . / $ . . .`. Propagation gate `CM-COVENANT-114-273-PROPAGATION` (literal `11.4.273`) + recommended gate `CM-CENSUS-CONTROL-NEEDED` (a decision-bearing census cites its positive and negative control) + paired `$ . . .` mutation (accept a decision-bearing census with no control → the gate MUST FAIL). Gate-code = separate work item, NOT claimed shipped (`$ . . . / $ . . .`). **Canonical authority:** constitution submodule `Constitution.md` `$ . . .`. Non-compliance is a release blocker. No escape hatch – no `--skip-control-needle`, `--empty-is-absent`, `--trust-the-grep`, `--census-needs-no-proof`, `--obvious-result-exempt`, `--one-needle-covers-the-set`, `--read-status-after-substitution` flag.

§ . . – Mechanical work belongs in a script, not in an agent's context: extract it, test it, and re-scan for it on a cadence (User mandate, . . .). Compact summary: forensic anchor – in one working session five independent review agents consumed roughly . . . , . . . , . . . tokens EACH, with the dominant cost being MECHANICS (copy a tree to scratch, apply a one-line mutation, run a suite, read the summary, restore, verify the restore byte-identical, compare counts, repeat per mutation) – a deterministic, fixed-input/fixed-output, zero-judgment loop executed one tool call at a time by a language model at roughly four orders of magnitude the cost of a script, with a fresh transcription-error opportunity at every step (the session recorded ten such instrument errors, § . . .). The rule: work that is DETERMINISTIC, REPEATABLE and contains NO JUDGMENT MUST be delegated to a properly written script/utility and invoked, never performed turn-by-turn inside an agent's context – concretely mutation apply/run/restore/verify loops, suite fan-outs + result tabulation, before/after comparisons, scratch-tree setup/teardown, residue/quiescence scans, census/inventory queries, artifact fingerprinting, evidence collection/layout, and any already-fixed recipe. (a) EXTRACT – a mechanical sequence performed by hand more than twice is a tooling defect, not a task; the bar is behavioural (if the steps can be written down exactly, they can be executed exactly, and hand-execution is the wrong instrument). (b) THE EXTRACTED TOOL IS ITSELF GOVERNED – ships documentation (§ . . .), an operator-facing guide where applicable, and tests proving its behaviour + failure paths + exit-code semantics (§ . . .) + its own control needles (§ . . .) with deterministic machine evidence (§ . . . /§ . . .); a fast-but-unproven extraction is a regression, not an optimisation. (c) RE-SCAN ON A CADENCE – every project periodically

scans itself (transcripts/logs/repeated inline recipes/"run these commands in order"..docs) for mechanical work still performed by agents and refactors it out; a STANDING recurring obligation, never a one-off cleanup.

(d) HONEST BOUNDARY (\$. .) – JUDGMENT MUST NOT be scripted: root-cause determination (\$. .), review verdicts (\$. . /\$. . / \$. .), the decision a fix is correct, the choice of what to measure, whether captured evidence actually supports a claim – a script that renders a verdict is itself a bluff gate (\$. .); the division is SCRIPT gathers+executes deterministically and cheaply, AGENT decides what it means; cost is never a reason to move a decision into a script. Classification: universal (\$. .). Composes \$. . / \$. . /\$. . /\$. . / \$. . /\$. . /\$. . /\$. . /\$. . .

Propagation gate CM-COVENANT-114-274-PROPAGATION (literal 11.4.274) + recommended gates CM-MECHANICAL-WORK-EXTRACTED (a hand-performed mechanical sequence used more than twice has an extraction item) / CM-EXTRACTED-TOOL-DOCUMENTED-AND-TESTED (every extracted tool carries docs + tests + deterministic machine evidence) + paired \$. mutation. Gate-code = separate work item, NOT claimed shipped (\$. . /\$. .). **Canonical authority:** constitution submodule Constitution.md \$. . .

Non-compliance is a release blocker. No escape hatch – no --skip-extraction , --agent-can-do-it , --one-off-is-fine , --script-the-review , --extract-without-tests flag.

\$. . – Semantic (Lumen) index + indexing-efficiency gate + universal agent/subagent accessibility (research-derived, – –). Compact summary: every code index the project maintains – the structural CodeGraph index (\$. .) AND the semantic Lumen index, and any future code index – MUST (A) be reachable by EVERY agent AND every dispatched subagent through a

server definition they can actually resolve (committed project config or installed plugin); a config that ENABLES a server nothing DECLARES is a violation; proof = a real tool call from a dispatched subagent returning an index-only fact (\$. . . () unforgeable challenge), un- drivable agents = documented \$. . . SKIP; (B) for the semantic index: embedding model provisioned rootless (\$. . .) and proven by a real embed call returning the declared dimensions, scope GENERATED from the same class source as the structural scope (<constitution>/scripts/lumen/gen_lumenignore.py), health = files AND chunks non-zero + known-symbol hit + negative control + freshness within consumer TTL; (C) be called ready ONLY when pending work == , state complete, file count within consumer tolerance of the accepted count, schema indexes present – every count/empty result control- needed (\$. . .); (D) earn its place by a DETERMINISTIC benchmark (fixed queries + golden answers, index route vs grep+read, $N \geq$ identical canonical hashes per \$. . . , bytes/est-tokens/p /p / correctness per query), reduction computed ONLY over queries both routes answered correctly, wrong answers listed BY ID and never averaged away, thresholds = consumer DATA (\$. . .); (E) route structural→CodeGraph, conceptual→Lumen before grep+read (\$. . .) except measured-wrong query classes (fall back + tracked); (F) root-cause + patch performance defects in the version-keyed runner with RED→GREEN + paired mutation + byte-copy equivalence proof; (G) writes via the single writer entry (\$. . . ()), live reads read-only. Forensic FACTS (genericised): a configured semantic server reported files / no model / never indexed; the structural index cut tokens . % over both-correct fixture queries yet answered of fixture queries WRONG (grep+read of); one uncached full-table file-list query was

. % of resolve time; an FK-cascade stalled the bulk window; a launcher classifying sync by stored rows instead of pending backlog ran a tens-of-millions-reference bulk job.stock + foreground + unsupervised; the vendor reaps locks by age not liveness and a writable open recreates dropped indexes without the lock; an errored instrument's empty result was nearly read as absence.

Latent binding (\$. . pattern): a project with no semantic index is bound only for the indexes it maintains; the semantic clauses bind the moment it adds one. Honest boundary (\$. .): proves reachable/complete/scoped/measured, not every answer correct; never licenses narrowing scope to hit a threshold (\$. .); the reference benchmark harness is still a prototype outside the constitution tree (landing = owed work).

Classification: universal (\$. .). Propagation gate CM-COVENANT-114-275-PROPAGATION (literal 11.4.275) + recommended gates CM-CODE-INDEX-AGENT-ACCESSIBLE / CM-CODE-INDEX-COMPLETE-BEFORE-READY / CM-LUMEN-MODEL-PROVEN / CM-LUMEN-SCOPED-FRESH / CM-CODE-INDEX-BENCH / CM-CODEGRAPH-SCOPE-PROOF / CM-CODEGRAPH-UPGRADE-PROBE-GATED / CM-CODEGRAPH-WRITER-EXCLUSIVE / CM-CODEGRAPH-BULK-BY-BACKLOG / CM-CODEGRAPH-STALL-WATCHDOG + paired \$. mutations (enabled-undeclared server, ready with pending > , reduction without wrong-answer list, unloaded model, bulk-by-rows, dropped liveness check → each gate MUST FAIL; golden-FALSE: unchanged complete reachable index fires none). Gate CODE = separate work item, NOT claimed shipped (\$. . / \$. .). **Canonical authority:** constitution submodule Constitution.md \$. . .

Non-compliance is a release blocker. No escape hatch – no --index-partial-ready , --enabled-is-declared , --main-session-only-index , --skip-model-proof , --reduction-without-wrong-answers , --bulk-by-row-count , --lock-age-is-liveness flag.

\$. . – Review-round budget: every independently-reviewed work item converges within a

declared budget of at most rounds (- , consumer-declared), by front-loading ground truth, class-exhaustive enumeration, real-path tests and a declared threat boundary, with a binding methodology change at round (User mandate, - -). Compact summary: every item passing through the \$. . . / \$. . . /\$. . . review loop carries a round budget `R_max` in the closed range - (operator-mandated ceiling, verbatim: "all work ALWAYS gets done with to reviews max"; absent declaration resolves to); this is NOT the invented-`N` case \$. . . (E) forbids - the consumer's `R_max` within the operator ceiling IS \$. . . 's `N` for the review loop. Forensic root causes (five, genericised from a reference project running items to / +/ / + rounds in one week): () wrong-model ground truth - a fix and its OWN verifying gate shared a simplified model of an external tool's real composition path, caught only when a reviewer read the tool's source and rebuilt the fixture on the real path; () instance-at-a-time closure of a known class - a security gate hardened one command name per round for three consecutive rounds after a whole-file enumeration was explicitly declined for lack of a classifier; () fixer-authored tests that never drive the real code - regression tests re-implementing or text-extracting the guarded logic so the reviewer's own mutations survived the suite; () non-binding escalation signals - the heuristic-tower anchor (\$. . .) invoked ~ times and the stall anchor's (\$. . .) undeclared `N` never stopped a loop; () scope drift - an out-of-scope tangent absorbed into a review's round counter. (A) declared round budget (a round = one independent review verdict; splitting/renaming does not reset the count except per (E)()); a spun-out tracked item inherits the parent's consumed count except for an (E)()-boundary finding, recurrence-linked

per § 1.1.1. (B) ground-truth probe (authoritative source + real-composition-path fixture + recorded behavioural model) required BEFORE the first fix whenever correctness depends on an unowned external system; a gate verified on a simplified model is wrong-layer evidence (§ 1.1.1.1) and cannot GO. (C) class-exhaustive enumeration on first instance – name the class, enumerate every member with a control-needed instrument (§ 1.1.1.2 (b)), close or explicitly scope-out every member; fixing only the reported instance when the class is nameable is itself a violation; building a missing classifier IS the remediation. (D) every remediation-round regression test drives the REAL artifact through its real invocation path (§ 1.1.1.3 (A)) – a test that re-implements or text-extracts the guarded logic is a source-defect (§ 1.1.1.4 (D)); every reviewer-surviving mutation is adopted verbatim and must fail the suite before the next review; the dispatch seam refuses to START a review missing that record, but a review that WAS dispatched and returned a verdict always counts. (E) at round R_{max} + the fixer records a convergence assessment; two-or-more findings sharing a (C)-named class (with $\geq$ non-MINOR) / a reappearing finding / wrong-layer evidence forces a STRUCTURAL round (fresh fixer instance, re-derive (B), re-run (C) over the whole scope, replace rather than layer) unless an operator decision (§ 1.1.1.5) defers it; every review declares its scope/threat boundary before round R_{max} , fixed except by recorded operator decision – narrowing it mid-loop to reach GO is a § 1.1.1.6 fabrication; at R_{max} without GO the item STOPS (never runs $R_{max} + 1$ under the same approach) and escalates to a recorded operator decision (re-scope via § 1.1.1.7 / replace via a structural new item / § 1.1.1.8 waiver for residual LOW only) – parked per § 1.1.1.9 / § 1.1.1.10 (F), parking only that work unit while the autonomous loop (§ 1.1.1.11)

continues everything else, surfaced in every § . . . report until resolved. (F) nothing here lowers § . . . 's zero-finding/zero-warning GO bar, § . . . 's substrate, § . . . (D)'s finding-layer classification, or § . . . 's machine-evidence requirement – the budget bounds rounds, not rigor; reaching `R_max` without GO never yields a GO or a silent close. EXTENDS (not restates, § . . .) § . . . (re-run obligation now bounded per-approach by this anchor, escalating per. (E)() at the ceiling) and § . . . (E) (a one-sentence carve-out: an operator-mandated ceiling recorded with its own verbatim mandate in its own anchor is the one permitted `N-bound` literal). Honest boundary (§ . . .): does not promise every item converges within `R_max` – a genuinely hard problem still produces real findings each round; what it removes is round inflation from re-discovering a wrong model, one member of a known class, a never-real test, an unacted stall signal, or scope drift; (E)() converts an open-ended loop into a bounded, evidence-backed operator decision, never a shallower review. Classification: universal (§ . . .). Propagation gate `CM-COVENANT-114-276-PROPAGATION` (literal `11.4.276`, implemented in this change) + recommended gates `CM-REVIEW-ROUND-BUDGET` / `CM-REVIEW-ROUND3-CONVERGENCE-ASSESSMENT` / `CM-FIX-GROUND-TRUTH-AND-CLASS-INVENTORY` (registered as tracked deferrals `OWED-GATE-..`) + paired § . mutations (round with no operator decision / missing round- classification / a point-fix after a same-class trigger / a class closed with no inventory → each gate MUST FAIL; golden-FALSE: GO at round with a clean round- assessment fires none). Gate CODE for the three recommended gates = separate work item, NOT claimed shipped (§ . . . / § . . .). **Canonical authority:** constitution submodule `Constitution.md` § Non-compliance is a release blocker. No

escape hatch – no `--unbounded-review-rounds` , `--fix-instance-not-class` , `--defer-ground-truth-probe` , `--narrow-threat-model-to-reach-go` , `--reset-round-count-on-refile` ,
`--defer-structural-fix-past-round-3` flag.

EXTENSION landed – – – \$. . /
\$. . . / \$. . indexing mechanism is
consumed BY REFERENCE; the consumer supplies only DATA
(full text in `Constitution.md`). This carrier cites the
three anchors; it is not a second block-start for any of
them (\$. . (B)). \$. . ()-(): the
scope renderer (`<constitution>/scripts/codegraph/
scope_render.py` + `scope_enumerate.js` + `scope_baseline.txt` +
template `scope.example.yaml`), the single writer entry
(`codegraph_safe.sh` + helper/guard/disk tripwire), the
version-keyed runner patcher (`fk_index_patch.py` +
`runner_patches/`), the hazard probe (`fk_cascade_probe.py`),
the progress-proven watchdog (`index_watch.py`), the scope
proof (`codegraph_scope_guard.py`) and `scripts/lumen/
gen_lumenignore.py`. MUST live in the constitution submodule
and be invoked BY REFERENCE (\$. . /
\$. . / \$. .) – a consumer copy/fork/
hand-rolled equivalent is a violation; each tool MUST be
documented (\$. .) and test-first with paired
mutations, and an untested tool never touches a live index
– at the time of writing `index_watch.py` and
`fk_cascade_probe.py` (no test references of their own – the
writer entry's unit test substitutes a stub for the
probe), the runner patches `lockfix1` and `datafrag1` (no
tests of their own – excluded from the default patch set
for that reason), and the \$. . companion guide
of EVERY tool named here (none exists yet) are OWED,
tracked per \$. . ; the consumer supplies ONE
scope DATA file (own-org list, project rubbish,
pathological inputs, re-include negations, reasoned class
overrides, accepted file count + tolerance, efficiency

thresholds) and never drops a baseline class; scope truth = the config the INSTALLED CLI version actually loads (from its own loader, § . .), a config the CLI does not read is dead documentation, the effective scope is GENERATED (hand edits refused, silent fall-back = FAIL); before "ready" the scope guard MUST enumerate through the runner's own discovery code and assert count within tolerance, . . . third-party files, secret-class files, ≥ file per own-org root, each control-needed – database size is NEVER evidence. Where this conflicts with the older clause- filter wording, the extension governs. § . . ()-(): own-org vs third-party is DERIVED by walking .gitmodules recursively and matching remote-URL organisation against the consumer own-org DATA; third-party code nested inside an own-org submodule is third-party; a gitlink change without re-render makes the render stale and the guard FAILs; steps - now mean "edit scope DATA + re-render + re-index only through the single writer entry". § . . ()-(): upgrades are gated on a compatibility probe (loader config filename/keys, bulk-window hazard probe, runner-patch applicability) and REFUSE loudly keeping the prior version when the patched runner cannot be re-proven (§ . .); a runner patch without its own RED→GREEN test + paired mutation + byte-copy equivalence proof never touches a live index; every init/index/sync goes through the ONE writer entry, which MUST refuse while any live process targets the root (real cmdline § 11 4 20 . (D), whatever the lock's age), classify BULK by PENDING backlog never by stored rows, and run bulk only on the proven patched runner, detached, watchdog- and tripwire-supervised; long runs are progress-proven (§ . . (C)) – alive-but-not-advancing is a STALL; cadence is a real timer/hook + constitution-pull hook (§ . .) with a ledger entry each run, a missing validator = FAIL; the live codegraph help diff is

authoritative over the embedded list, and telemetry MUST be off (per-host FACT, captured not assumed). The literals 11.4.78 / 11.4.79 / 11.4.80 are unchanged; no CM-COVENANT-114-78/79/80-PROPAGATION gate exists (none in constitution/scripts/gates/), so no propagation gate is claimed for them.

EXTENSION landed - - - \$. .
host-adaptive V heap budget + \$. . (D)
benchmark run, both live-verified on this project (full text in Constitution.md). \$. . 's item- single writer entry codegraph_safe.sh now computes its Node.js heap budget from measured MemAvailable (never hardcoded, \$. /\$. .) instead of inheriting V 's roughly- -GiB stock default - closing a real bulk-index OOM crash (rc=) on this repo's K-file/ M-edge...tree; landed with new regression tests + live / proc/<pid>/environ verification + a zero-regression full-suite comparison (commit 941841d3e7235a081e3199a4d3073b409d41753b).

\$. . (D)'s deterministic benchmark ran via the now-in-tree constitution/scripts/lumen/lumen_verify.sh (superseding this anchor's own note that the harness lived outside the constitution tree - it now has its own test file): a pre-declared -query golden set, N= byte-identical runs, recall / = . , the failing/unsupported query classes (.sh / .kt , unindexed by Lumen's chunker - confirmed via direct sqlite3 + Lumen's own source, not guessed) reported honestly and tracked, never averaged away (commit ac532a21df78a447c2aceec30f63476348e93077 , docs/research/lumen_benchmark_20260925/RESULTS_v2.md). Gate code for both anchors remains owed, unchanged (\$. . / \$. .).

EXTENSION landed - - (round) -
\$. . heap ceiling corrected to the REAL \$.

%-of-total-RAM bound (was an invented GiB) + CodeGraph now wired as a genuine, project-tracked MCP tool via a new incident-hardened wrapper (full text in `Constitution.md`). Direct operator mandate mid-session: "do not set caps for codegraph and lumen ... indexed space exposed and used by all agents" – audited, and both were genuinely under-delivering. () The earlier round's GiB heap cap was an INVENTED ceiling (~ % of this GiB host's total RAM), not a derivation of $\$ \cdot$; fixed to `MemTotal * 60/100` computed fresh every run – the real, unweakened $\$ \cdot$ bound, mathematically almost never binding an ordinary run (commit `8a5f47ab2e0e7a05d1c1652ba5a3475751652062` , new regression test U). () Audited further and found `.mcp.json` was completely EMPTY – CodeGraph had ZERO agent-facing MCP presence in this repository despite shipping a full query toolset (`codegraph_explore` / `codegraph_node` /etc.) and its own installer; Lumen's own "prefer this over grep" nudge was found to be an UNTRACKED Claude Code PLUGIN hook, not anything this repository owns – neither tool's use was actually guaranteed for any future clone/agent/track. Fixed by registering `.mcp.json` 's `codegraph` server against a new, tested, `$11.4.177` -decoupled wrapper (`constitution/scripts/codegraph/codegraph_mcp_serve.sh`) that resolves the SAME validated patched runner `codegraph_safe.sh` itself uses (never bare stock-on-PATH). **A separate, more serious forensic incident surfaced while building this:** `serve --mcp` 's file watcher WRITES (auto-syncs) the database and the server daemonizes past its own parent's death – a single manual test against this project's own live checkout spawned a stray daemon that collided with an in-flight bulk `sync` , crashing it with "database is locked" (a further ~ minutes of progress lost, on top of the earlier heap-crash loss); recovered via a targeted `REINDEX` (rows, confirmed non-corrupting) + a bulk `sync` resume. Durable fix: the wrapper now ALWAYS passes

`--no-watch` (never optional), making it a true non-writing reader safe to run permanently – a fact PROVEN, not assumed, by a real init'd scratch-project DB-mtime check (unchanged after a tracked-file edit with the server running). Six new regression tests land (`tests/test_codegraph_mcp_serve.sh`), including the incident itself replayed as a repeatable test (T); two unrelated bugs found and fixed during this same investigation before landing (a `.sh` -stub force-fed to `python3` ; a submodule-vs-parent project-root resolution bug). Honest boundary (`$ . .`): `.mcp.json` registration takes effect on a fresh session – this session could not self-restart to verify the MCP tools are end-to-end callable from inside it; that check is a tracked `$ . .` follow-up, not claimed here. Gate code for all of the above remains owed, unchanged (`$ . . /$ . .`).